

Since as Δ_D is moved upward and Δ_W downward the reflux ratio increases, the definition of minimum reflux ratio requires Δ_{D_m} and Δ_{W_m} for the minimum reflux ratio to be located as shown, with Δ_{D_m} at the highest tie-line intersection and Δ_{W_m} at the lowest tie-line intersection. In this case, it is the tie line which, when extended, passes through F , the feed, that determines both, and this is always the case when the xy equilibrium distribution curve is everywhere concave downward.

For some positively deviating mixtures with a tendency to form an azeotrope and for all systems near the critical condition of the more volatile component [66], an enriching-section tie line m in Fig. 9.24*b* gives the highest intersection with $x = z_D$, not that which passes through F . Similarly, as in Fig. 9.24*c* for some negatively deviating mixtures, a stripping-section tie line p gives the lowest intersection with $x = x_W$. These then govern the location of Δ_{D_m} as shown. For the minimum reflux ratio, either Δ_{D_m} is located at the highest intersection of an enriching-section tie line with $x = z_D$, or Δ_{W_m} is at the lowest intersection of a stripping-section tie line with $x = x_W$, consistent with the requirements that Δ_{D_m} , Δ_{W_m} , and F all be on the same straight line and Δ_{D_m} be at the highest position resulting in a pinch. Special considerations are necessary for fractionation with multiple feeds and sidestreams [52].

Once Q_m is determined, the minimum reflux ratio can be computed through Eq. (9.65). Some larger reflux ratio must obviously be used for practical cases, whereupon Δ_D is located above Δ_{D_m} .

Optimum Reflux Ratio

Any reflux ratio between the minimum and infinity will provide the desired separation, with the corresponding number of theoretical trays required varying from infinity to the minimum number, as in Fig. 9.25*a*. Determination of the

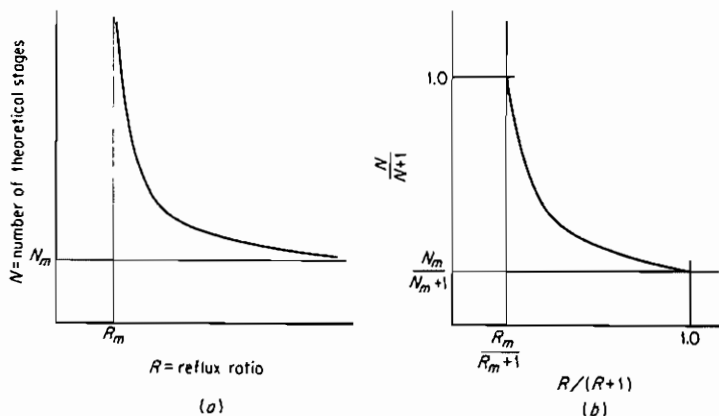


Figure 9.25 Reflux-ratio-stage relation.

number of trays at several values of R , together with the limiting values of N_m and R_m , will usually permit plotting the entire curve with sufficient accuracy for most purposes. The coordinate system of Fig. 9.25b [17] will permit locating the ends of the curve readily by avoiding the awkward asymptotes. There have been several attempts at empirically generalizing the curves of Fig. 9.25 [5, 10, 12, 17, 35], but the resulting charts yield only approximate results. An exact relationship for binary distillations, which can also be applied to multicomponent mixtures, is available [59].

The reflux ratio to be used for a new design should be the optimum, or the most economical, reflux ratio, for which the cost will be the least. Refer to Fig. 9.26. At the minimum reflux ratio the column requires an infinite number of trays, and consequently the fixed cost is infinite, but the operating costs (heat for the reboiler, condenser cooling water, power for reflux pump) are least. As R increases, the number of trays rapidly decreases, but the column diameter increases owing to the larger quantities of recycled liquid and vapor per unit quantity of feed. The condenser, reflux pump, and reboiler must also be larger. The fixed costs therefore fall through a minimum and rise to infinity again at total reflux. The heating and cooling requirements increase almost directly with reflux ratio, as shown. The total cost, which is the sum of operating and fixed costs, must therefore pass through a minimum at the optimum reflux ratio. This will frequently but not always occur at a reflux ratio near the minimum value ($1.2R_m$ to $1.5R_m$), on the average probably near the lower limit. A less empirical method for estimating the optimum is available [34].

Illustration 9.8 A methanol (A)–water (B) solution containing 50 wt % methanol at 26.7°C is to be continuously rectified at 1 std atm pressure at a rate of 5000 kg/h to provide a distillate containing 95% methanol and a residue containing 1.0 % methanol (by weight). The feed is to be preheated by heat exchange with the residue, which will leave the system at 37.8°C. The distillate is to be totally condensed to a liquid and the reflux returned at the bubble point. The withdrawn distillate will be separately cooled before storage. A reflux ratio of 1.5 times the minimum will be used. Determine (a) quantity of the products, (b) enthalpy of feed and of

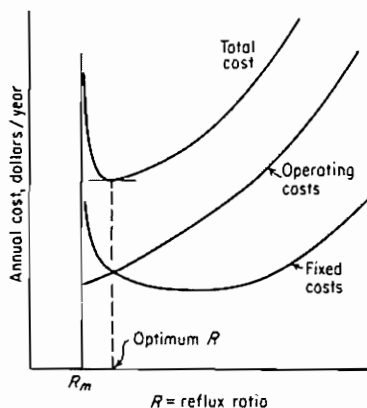


Figure 9.26 Most economical (optimum) reflux ratio.

products, (c) minimum reflux ratio, (d) minimum number of theoretical trays, (e) condenser and reboiler heat loads for specified reflux ratio, (f) number of theoretical trays for specified reflux ratio, and liquid and vapor quantities inside the tower.

SOLUTION (a) Mol wt methanol = 32.04, mol wt water = 18.02. Basis: 1 h. Define quantities in terms of kmol/h.

$$F = \frac{5000(0.50)}{32.04} + \frac{5000(0.50)}{18.02} = 78.0 + 138.8 = 216.8 \text{ kmol/h}$$

$$z_F = \frac{78}{216.8} = 0.360 \text{ mole fraction methanol} \quad M_{av} \text{ for feed} = \frac{5000}{216.8} = 23.1 \text{ kg/kmol}$$

$$x_D = \frac{95/32.04}{95/32.04 + 5/18.02} = \frac{2.94}{3.217} = 0.915 \text{ mole fraction methanol}$$

$$M_{av} \text{ for distillate} = \frac{100}{3.217} = 31.1 \text{ kg/kmol}$$

$$x_W = \frac{1/32.04}{1/32.04 + 99/18.02} = \frac{0.0312}{5.53} = 0.00565 \text{ mole fraction methanol}$$

$$M_{av} \text{ for residue} = \frac{100}{5.53} = 18.08 \text{ kg/kmol}$$

Eq. (9.75):

$$216.8 = D + W$$

Eq. (9.76):

$$216.8(0.360) = D(0.915) + W(0.00565)$$

Solving simultaneously gives

$$D = 84.4 \text{ kmol/h} \quad 84.4(31.1) = 2620 \text{ kg/h}$$

$$W = 132.4 \text{ kmol/h} \quad 132.4(18.08) = 2380 \text{ kg/h}$$

(b) The vapor-liquid equilibrium at 1 std atm pressure is given by Cornell and Montana, *Ind. Eng. Chem.*, 25, 1331 (1933), and by "The Chemical Engineers' Handbook," 4th ed., p. 13-5. Heat capacities of liquid solutions are in the "Handbook," 5th ed., p. 3-136, and latent heats of vaporization of methanol on p. 3-116. Heats of solution are available in "International Critical Tables," vol. V, p. 159, at 19.69°C, which will be used as t_0 , the base temperature for computing enthalpies.

To compute enthalpies of saturated liquids, consider the case of $x = 0.3$ mole fraction methanol, $M_{av} = 22.2$. The bubble point = 78.3°C, heat capacity = 3852 J/kg · K, and the heat of solution = 3055 kJ evolved/kmol methanol.

$$\Delta H_S = -3055(0.3) = -916.5 \text{ kJ/kmol solution. Therefore, Eq. (9.10):}$$

$$H_L = 3.852(78.3 - 19.69)22.2 - 916.5 = 4095 \text{ kJ/kmol}$$

To compute the enthalpy of saturated vapors, consider the case of $y = 0.665$ mole fraction methanol. The dew point is 78.3°C. At this temperature the latent heat of methanol is 1046.7 kJ/kg, that of water is 2314 kJ/kg. The heat capacity of methanol is 2583, of water 2323 J/kg · K. Eq. (9.11):

$$\begin{aligned} H_G &= 0.665[2.583(32.04)(78.3 - 19.69) + 1046.7(32.04)] \\ &\quad + (1 - 0.665)[2.323(18.02)(78.3 - 19.69) + 2314(18.02)] \\ &= 40318 \text{ kJ/kmol} \end{aligned}$$

The enthalpy data of Fig. 9.27 were computed in this manner.

From the vapor-liquid equilibria, the bubble point of the residue is 99°C. Heat capacity of the residue is 4179, of the feed 3852 J/kg · K. Enthalpy balance of the feed preheat exchanger:

$$5000(3852)(t_F - 26.7) = 2380(4179)(99 - 37.8)$$

$$t_F = 58.3^\circ\text{C, temp at which feed enters tower}$$

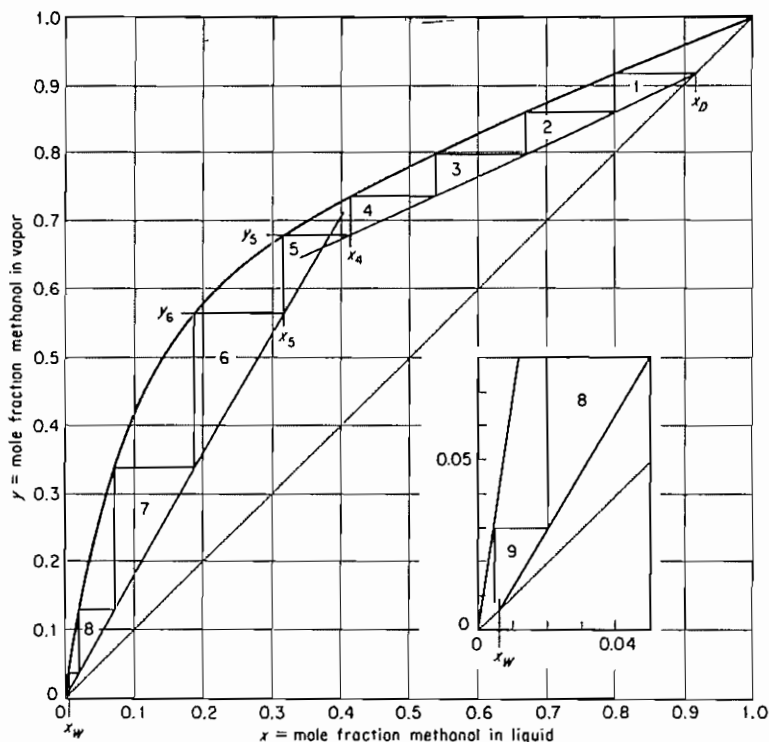


Figure 9.28 xy diagram for Illustration 9.8.

(d) The minimum number of trays was determined on the xy diagram in the manner of the lower part of Fig. 9.23, and 4.9 theoretical stages, including the reboiler, were obtained. $N_m = 4.9 - 1 = 3.9$.

(e) For $R = 1.5(0.685) = 1.029$, Eq. (9.65) becomes

$$1.029 = \frac{Q' - 38\,610}{38\,610 - 3640}$$

$$Q' = 74\,595 = H_D + \frac{Q_C}{D} = 3640 + \frac{Q_C}{84.4} \quad Q_C = 5\,990\,000 \text{ kJ/h} = 1664 \text{ kW}$$

$$\text{Eq. (9.77): } 216.8(2533) = 84.4(74\,595) + 132.4Q''$$

$$Q'' = -43\,403 = H_W - \frac{Q_B}{W} = 6000 - \frac{Q_B}{132.4}$$

$$Q_B = 6\,541\,000 \text{ kJ/h reboiler heat load} = 1817 \text{ kW}$$

(f) In Fig. 9.27, Δ_D at $(x_D = 0.915, Q' = 74\,595)$ and Δ_W at $(x_W = 0.00565, Q'' = -43\,403)$ are plotted. Random lines from the Δ points, as shown, intersect the saturated-vapor and saturated-liquid curves at values of y and x , respectively, corresponding to points on the operating curve (note that for accurate results a large-scale graph and a sharp pencil are needed). These are plotted on Fig. 9.28 to provide the operating curves, which are nearly, but

not exactly, straight. A total of nine theoretical stages including the reboiler, or eight theoretical trays in the tower, are required when the feed tray is the optimum (no. 5) as shown.

At the top of the tower

$$G_1 = D(R + 1) = 84.4(1.029 + 1) = 171.3 \text{ kmol/h}$$

$$L_0 = DR = 84.4(1.029) = 86.7 \text{ kmol/h}$$

At the feed tray, $x_4 = 0.415$, $y_5 = 0.676$, $x_5 = 0.318$, $y_6 = 0.554$ (Fig. 9.28). Eq. (9.64):

$$\frac{L_4}{D} = \frac{L_4}{84.4} = \frac{0.915 - 0.676}{0.676 - 0.415} \quad \text{and} \quad L_4 = 77.2 \text{ kmol/h}$$

Eq. (9.62):

$$\frac{L_4}{G_5} = \frac{77.2}{G_5} = \frac{0.915 - 0.676}{0.915 - 0.415} \quad \text{and} \quad G_5 = 161.5 \text{ kmol/h}$$

Eq. (9.74):

$$\frac{\bar{L}_5}{\bar{W}} = \frac{\bar{L}_5}{132.4} = \frac{0.554 - 0.00565}{0.554 - 0.318} \quad \text{and} \quad \bar{L}_5 = 308 \text{ kmol/h}$$

Eq. (9.72):

$$\frac{\bar{L}_5}{\bar{G}_6} = \frac{308}{\bar{G}_6} = \frac{0.554 - 0.00565}{0.318 - 0.00565} \quad \text{and} \quad \bar{G}_6 = 175.7 \text{ kmol/h}$$

At the bottom of the tower, Eq. (9.66):

$$\bar{L}_{N_r} = \bar{G}_W + W \quad \bar{L}_8 = \bar{G}_W + 132.4$$

Further, $y_W = 0.035$, $x_8 = 0.02$ (Fig. 9.28); Eq. (9.72):

$$\frac{\bar{L}_8}{\bar{G}_W} = \frac{0.035 - 0.00565}{0.02 - 0.00565}$$

Solving simultaneously gives $\bar{G}_W = 127.6$, $\bar{L}_8 = 260 \text{ kmol/h}$.

Reboilers

The heat-exchanger arrangements to provide the necessary heat and vapor return at the bottom of the fractionator may take several forms. Small fractionators used for pilot-plant work may merely require a jacketed kettle, as shown schematically in Fig. 9.29a, but the heat-transfer surface and the corresponding vapor capacity will necessarily be small. The tubular heat exchanger built into the bottom of the tower (Fig. 9.29b) is a variation which provides larger surface, but cleaning requires a shut-down of the distillation operation. This type can also be built with an internal floating head. Both these provide a vapor entering the bottom tray essentially in equilibrium with the residue product, so that the last stage of the previous computations represents the enrichment due to the reboiler.

External reboilers of several varieties are commonly used for large installations, and they can be arranged with spares for cleaning. The kettle reboiler (Fig. 9.29c), with heating medium inside the tubes, provides a vapor to the tower essentially in equilibrium with the residue product and then behaves like a theoretical stage. The vertical thermosiphon reboiler of Fig. 9.29d, with the

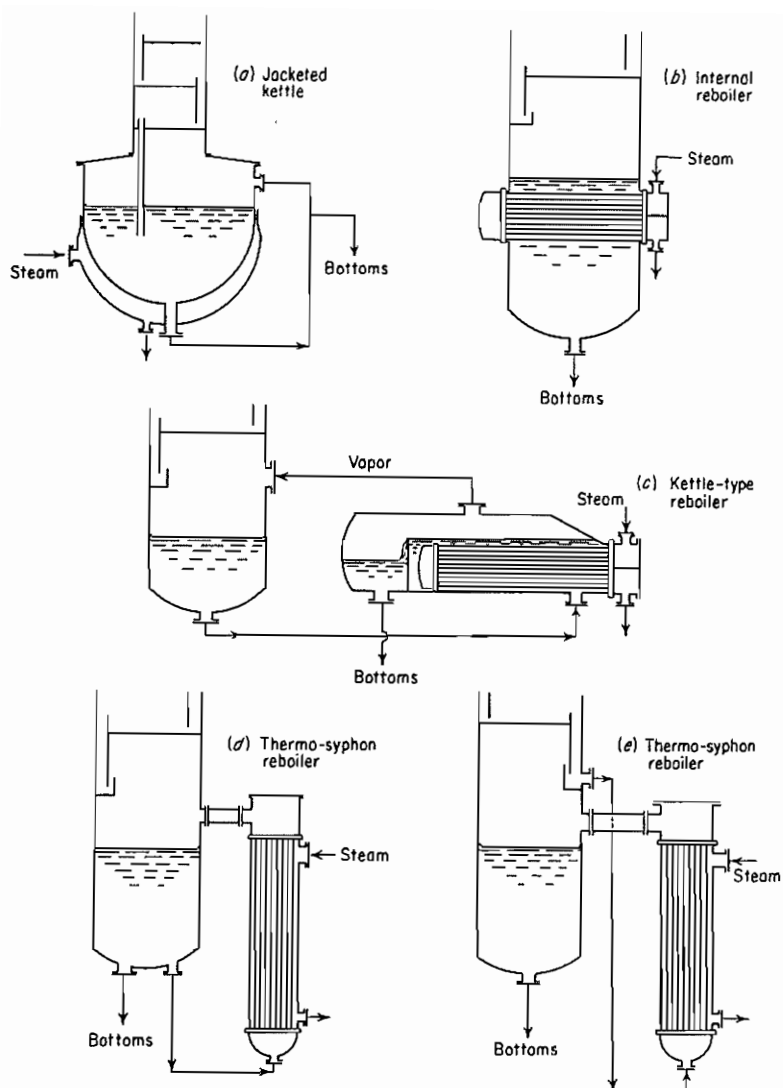


Figure 9.29 Reboiler arrangements (schematic).

heating medium outside the tubes, can be operated so as to vaporize all the liquid entering it to produce a vapor of the same composition as the residue product, in which case no enrichment is provided. However, because of fouling of the tubes, which may occur with this type of operation, it is more customary to provide for only partial vaporization, the mixture issuing from the reboiler comprising both liquid and vapor. The reboiler of Fig. 9.29e receives liquid from the trapout of the bottom tray, which it partially vaporizes. Horizontal reboilers are also known [8]. Piping arrangements [27], a review [37], and detailed design methods [14, 49] are available. It is safest not to assume that a theoretical stage's worth of fractionation will occur with thermosiphon reboilers but instead to provide the necessary stages in the tower itself. In Fig. 9.29, the reservoir at the foot of the tower customarily holds a 5- to 10-min flow of liquid to provide for reasonably steady operation of the reboiler.

Reboilers may be heated by steam, heat-transfer oil, or other hot fluids. For some high-boiling liquids, the reboiler may be a fuel-fired furnace.

Use of Open Steam

When a water solution in which the nonaqueous component is the more volatile is fractionated, so that the water is removed as the residue product, the heat required can be provided by admission of steam directly to the bottom of the tower. The reboiler is then dispensed with. For a given reflux ratio and distillate composition, more trays will usually be required in the tower, but they are usually cheaper than the replaced reboiler and its cleaning costs.

Refer to Fig. 9.30. While the enriching section of the tower is unaffected by the use of open steam and is not shown, nevertheless the overall material and enthalpy balances are influenced. Thus, in the absence of important heat loss,

$$F + \bar{G}_{N_p+1} = D + W \quad (9.86)$$

$$Fz_F = Dz_D + Wx_W \quad (9.87)$$

$$FH_F + \bar{G}_{N_p+1}H_{G, N_p+1} = WH_W + DH_D + Q_C \quad (9.88)$$

where $\bar{G}_{N_p+1}$ is the molar rate of introducing steam. On the Hxy diagram, the Δ_D difference point is located in the usual manner. For the stripping section, Δ_W has its usual meaning, a fictitious stream of size equal to the net flow outward

$$\Delta_W = \bar{L}_m - \bar{G}_{m+1} = W - \bar{G}_{N_p+1} \quad (9.89)$$

of coordinates

$$\Delta_W \left\{ \begin{array}{l} x_{\Delta_W} = \frac{\text{net moles A out}}{\text{net moles out}} = \frac{Wx_W}{W - \bar{G}_{N_p+1}} \\ Q'' = \frac{\text{net heat out}}{\text{net moles out}} = \frac{WH_W - \bar{G}_{N_p+1}H_{G, N_p+1}}{W - \bar{G}_{N_p+1}} \end{array} \right.$$

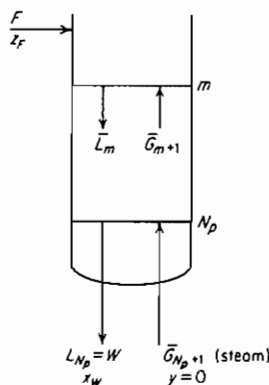


Figure 9.30 Use of open steam.

where H_{G, N_p+1} is the enthalpy of the steam. The point is shown on Fig. 9.31. Thus,

$$\bar{L}_m x_m - \bar{G}_{m+1} y_{m+1} = \Delta_W x_W \quad (9.90)$$

$$\bar{L}_m H_{L_m} - \bar{G}_{m+1} H_{G_{m+1}} = \Delta_W Q'' \quad (9.91)$$

and

$$\frac{\bar{L}_m}{\bar{G}_{m+1}} = \frac{y_{m+1} - x_{\Delta W}}{x_m - x_{\Delta W}} = \frac{H_{G_{m+1}} - Q''}{H_{L_m} - Q''} \quad (9.92)$$

The construction is shown in Fig. 9.31. Equation (9.92) is the slope of a chord (not shown) between points P and T . Here, the steam introduced is shown slightly superheated ($H_{G, N_p+1} >$ saturated enthalpy); had saturated steam been used, $\bar{G}_{N_p+1}$ would be located at point M . Note that the operating curve on the x, y diagram passes through the 45° diagonal at T ($x = x_{\Delta W}$) and through the point $(x_W, y = 0)$ corresponding to the fluids passing each other at the bottom of the tower.

Illustration 9.9 Open steam, initially saturated at 69 kN/m^2 ($10 \text{ lb}_f/\text{in}^2$) gauge pressure, will be used for the methanol fractionator of Illustration 9.8, with the same distillate rate and composition and the same reflux ratio. Assuming that the feed enters the tower at the same enthalpy as in Illustration 9.8, determine the steam rate, bottoms composition, and the number of theoretical trays.

SOLUTION From Illustration 9.8, $F = 216.8 \text{ kmol/h}$, $z_F = 0.360$, $H_F = 2533 \text{ kJ/kmol}$, $D = 84.4$, $z_D = 0.915$, $H_D = 3640$ and $Q_C = 5\,990\,000 \text{ kJ/h}$. From the steam tables, the enthalpy of saturated steam at $69 \text{ kN/m}^2 = 2699 \text{ kJ/kg}$ referred to liquid water at 0°C . On expanding adiabatically through a control valve to the tower pressure, it will be superheated at the same enthalpy. The enthalpy of liquid water at 19.7°C (t_0 for Illustration 9.8) $= 82.7 \text{ kJ/kg}$ referred to 0°C . Therefore $H_{G, N_p+1} = (2699 - 82.7)(18.02) = 47\,146 \text{ kJ/kmol}$.

Eq. (9.86):

$$216.8 + \bar{G}_{N_p+1} = 84.4 + W$$

Eq. (9.87):

$$216.8(0.360) = 84.4(0.915) + Wx_W$$

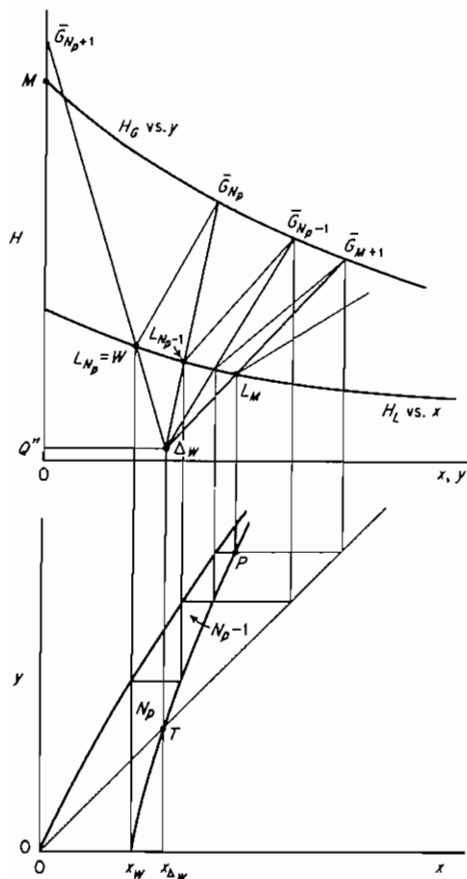


Figure 9.31 Use of open steam.

Eq. (9.88):

$$216.8(2533) + 47\,146\bar{G}_{N_p+1} = WH_W + 84.4(3640) + 5\,990\,000$$

Since the bottoms will be essentially pure water, H_W is tentatively estimated as the enthalpy of saturated water (Fig. 9.27), 6094 kJ/kmol. Solving the equations simultaneously with this value for H_W provides the steam rate as $\bar{G}_{N_p+1} = 159.7$ and $W = 292.1$ kmol/h, with $x_W = 0.00281$. The enthalpy of this solution at its bubble point is 6048, sufficiently close to the 6094 assumed earlier to be acceptable. (Note that had the same interchange of heat between bottoms and feed been used as in Illustration 9.8, with bottoms discharged at 37.8°C, the feed enthalpy would have been changed.) For Δ_W ,

$$x_{\Delta W} = \frac{Wx_W}{W - \bar{G}_{N_p+1}} = \frac{292.1(0.00281)}{292.1 - 159.7} = 0.0062$$

$$\begin{aligned} Q'' &= \frac{WH_W - \bar{G}_{N_p+1}H_{G, N_p+1}}{W - \bar{G}_{N_p+1}} \\ &= \frac{292.1(6048) - 159.7(47\,146)}{292.1 - 159.7} = -43\,520 \text{ kJ/kmol} \end{aligned}$$

The H_{xy} and xy diagrams for the enriching section are the same as in Illustration 9.8. For the stripping section, they resemble Fig. 9.31. The number of theoretical stages is $N_p = 9.5$, and they must all be included in the tower.

Condensers and Reflux Accumulators

Reflux may flow by gravity to the tower, in which case the condenser and reflux drum (accumulator) must be elevated above the level of top tray of the tower. Alternatively, especially in order to obviate the need for elevated platforms and supports required for withdrawing the condenser tube bundle for cleaning, the assemblage may be placed at ground level and the reflux liquid pumped up to the top tray. Kern [28] describes the arrangements.

Reflux accumulators are ordinarily horizontal drums, length/diameter = 4 to 5, with a liquid holding time of the order of 5 min. From entrainment considerations, the allowable vapor velocity through the vertical cross section of the space above the liquid can be specified as [55][†]

$$V = 0.04 \left(\frac{\rho_L - \rho_G}{\rho_G} \right)^{0.5} \quad (9.93)$$

Multiple Feeds

There are occasions when two or more feeds composed of the same substances but of different concentrations require distillation to give the same distillate and residue products. A single fractionator will then suffice for all.

Consider the two-feed fractionator of Fig. 9.32. The construction on the H_{xy} diagram for the sections of the column above F_1 and below F_2 is the same as for a single-feed column, with the Δ_D and Δ_W points located in the usual manner. For the middle section between the feeds, the difference point Δ_M can be located by consideration of material and enthalpy balances either toward the top, as indicated by the envelope shown on Fig. 9.32, or toward the bottom; the net result will be the same. Consider the envelope shown in the figure, with Δ_M representing a fictitious stream of quantity equal to the net flow upward and out

$$G'_{r+1} - L'_r = D - F_1 = \Delta_M \quad (9.94)$$

whose coordinates are

$$\Delta_M \begin{cases} x_{\Delta_M} = \frac{\text{net moles A out}}{\text{net moles out}} = \frac{Dz_D - F_1 z_{F1}}{D - F_1} \\ Q_M = \frac{\text{net heat out}}{\text{net moles out}} = \frac{Q_C + DH_D - F_1 H_{F1}}{D - F_1} \end{cases}$$

Δ_M may be either a positive or negative quantity.

[†] V in Eq. (9.93) is expressed as m/s. For V in ft/s, the coefficient is 0.13.

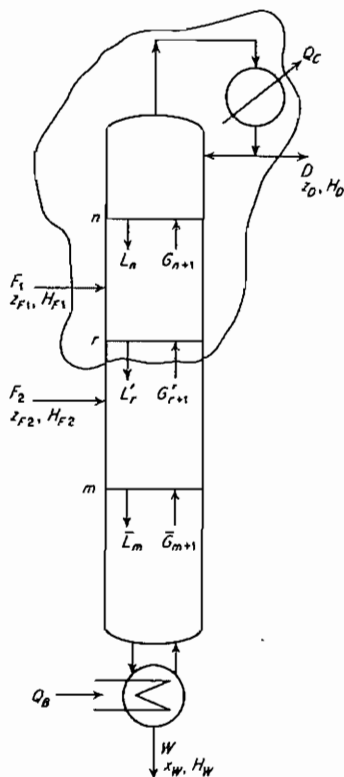


Figure 9.32 Fractionator with two feeds.

Equation (9.94) can be used as a basis for component-A and enthalpy balances

$$G'_{r+1}y_{r+1} - L'_r x_r = \Delta_M x_{\Delta M} \quad (9.95)$$

$$G'_{r+1}H_{G_{r+1}} - L_r H_{L_r} = \Delta_M Q_M \quad (9.96)$$

whence

$$\frac{L'_r}{G'_{r+1}} = \frac{y_{r+1} - x_{\Delta M}}{x_r - x_{\Delta M}} = \frac{H_{G_{r+1}} - Q_M}{H_{L_r} - Q_M} \quad (9.97)$$

Since

$$F_1 + F_2 = D + W = \Delta_D + \Delta_W \quad (9.98)$$

then

$$\Delta_M = F_2 - W \quad (9.99)$$

The construction (both feeds liquid) is shown on Fig. 9.33, where Δ_M lies on the line $\Delta_D F_1$ [Eq. (9.94)] and on the line $\Delta_W F_2$ [Eq. (9.99)]. A solution representing the composited feed, with

$$z_{F,av} = \frac{F_1 z_{F1} + F_2 z_{F2}}{F_1 + F_2} \quad H_{F,av} = \frac{F_1 H_{F1} + F_2 H_{F2}}{F_1 + F_2}$$

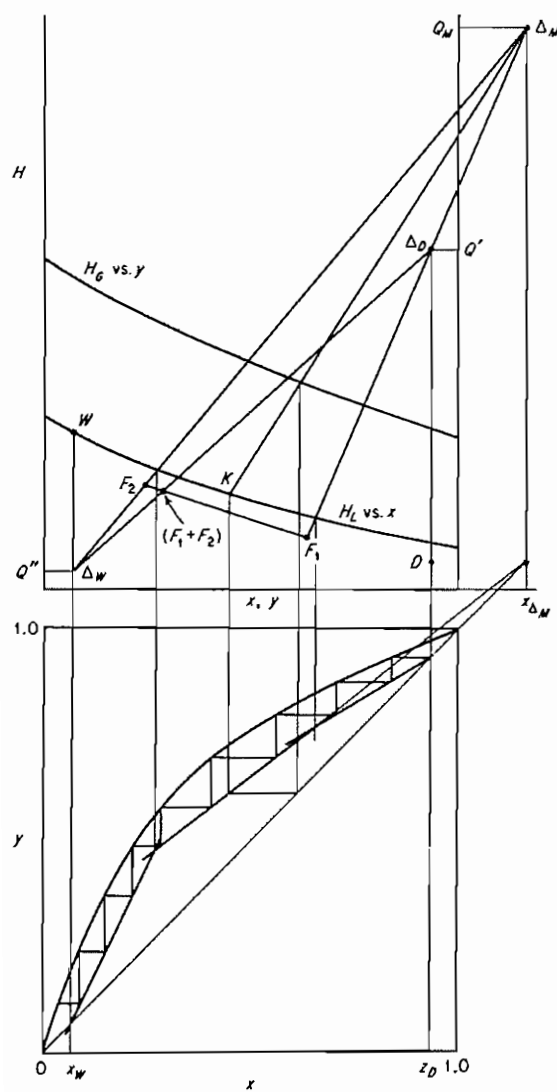


Figure 9.33 Construction for two feeds.

must lie on the line $\Delta_D \Delta_W$ [Eq. (9.98)]. It is also possible for Δ_M to lie below and to the left of Δ_W . The operating curve for the middle section on the xy diagram is located by lines such as $\Delta_M K$, as shown. Trays are best drawn in the usual step fashion on the xy diagram, and for optimum location the feed trays straddle the intersections of the operating curves, as shown.

Side Streams

Side streams are products of intermediate composition withdrawn from the intermediate trays of the column. They are used frequently in the distillation of petroleum products, where intermediate properties not obtainable merely by mixing distillate or bottoms with feed are desired. They are used only infrequently in the case of binary mixtures, and are not treated here (see Prob. 9.17).

Heat Losses

Most fractionators operate above ambient temperature, and heat losses along the column are inevitable since insulating materials have a finite thermal conductivity. The importance of the heat losses and their influence on fractionators will now be considered.

Consider the fractionator of Fig. 9.17. A heat balance for the top n trays of the enriching section (envelope III) which includes the heat loss is [9]

$$G_{n+1}H_{G_{n+1}} = Q_C + DH_D + L_n H_{L_n} + Q_{L_n} \quad (9.100)$$

where Q_{L_n} is the heat loss for trays 1 through n . Defining

$$Q'_L = \frac{Q_C + DH_D + Q_{L_n}}{D} = Q' + \frac{Q_{L_n}}{D} \quad (9.101)$$

we have

$$G_{n+1}H_{G_{n+1}} - L_n H_{L_n} = DQ'_L \quad (9.102)$$

Q'_L is a variable since it depends upon how many trays are included in the heat balance. If only the top tray ($n = 1$) is included, the heat loss is small and Q'_L is nearly equal to Q' . As more trays are included, Q_{L_n} and Q'_L increase, ultimately reaching their largest values when all enriching-section trays are included. Separate difference points are therefore needed for each tray.

For the stripping section up to tray m (envelope IV, Fig. 9.17),

$$\bar{L}_m H_{L_m} + Q_B = WH_W + \bar{G}_{m+1} H_{G_{m+1}} + \bar{Q}_{L_m} \quad (9.103)$$

Letting

$$Q''_L = \frac{WH_W - Q_B + \bar{Q}_{L_m}}{W} = Q'' + \frac{\bar{Q}_{L_m}}{W} \quad (9.104)$$

results in

$$\bar{L}_m H_{L_m} - \bar{G}_{m+1} H_{G_{m+1}} = WQ''_L \quad (9.105)$$

If the heat balance includes only the bottom tray, the heat loss is small and Q''_L nearly equals Q'' . As more trays are included, $\bar{Q}_{L_m}$ and therefore Q''_L increase, reaching their largest values when the balance is made over the entire stripping

section. Separate difference points are needed for each tray. An enthalpy balance for the entire tower is

$$FH_F = D\left(H_D + \frac{Q_C}{D} + \frac{Q_L}{D}\right) + W\left(H_W - \frac{Q_B}{W} + \frac{\bar{Q}_L}{W}\right) \quad (9.106)$$

where Q_L and $\bar{Q}_L$ are the total heat losses for the enriching and stripping sections, respectively. When Eq. (9.106) is solved with the material balances, Eqs. (9.75) and (9.76), the result is

$$\frac{D}{W} = \frac{z_F - x_W}{z_D - z_F} = \frac{H_F - (H_W - Q_B/W + \bar{Q}_L/W)}{(H_D + Q_C/D + Q_L/D) - H_F} \quad (9.107)$$

This is the equation of the line BFT on Fig. 9.34.

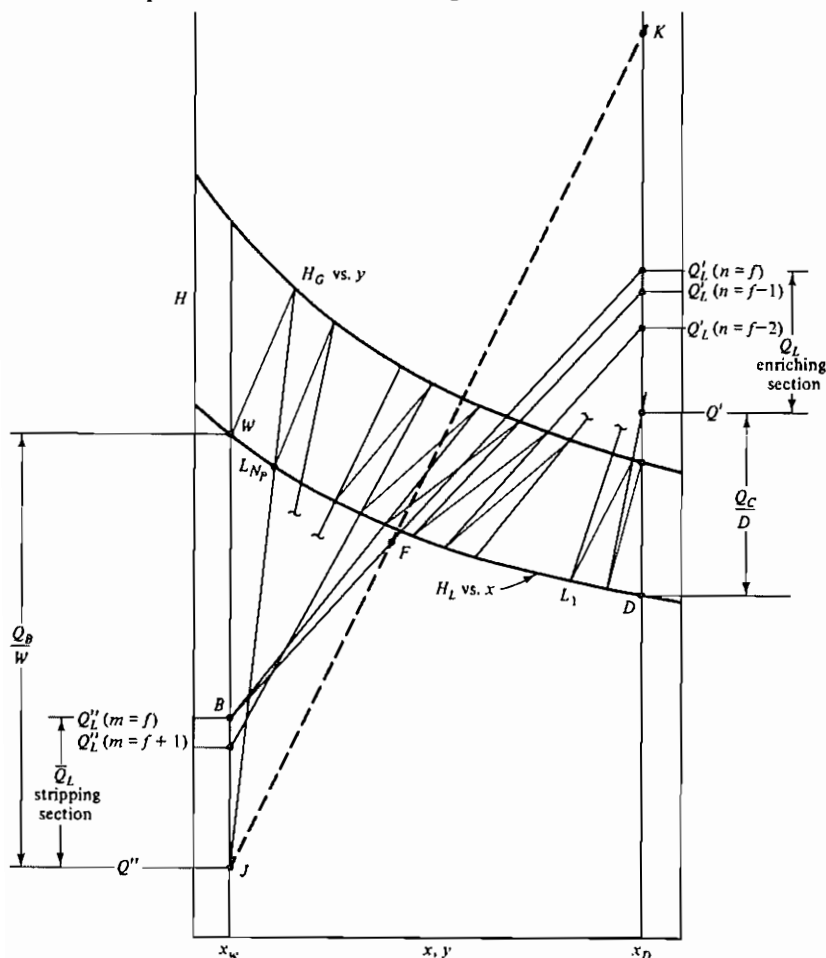


Figure 9.34 Heat losses.

The procedure for design is one of trial and error. For example, as a first trial, heat losses might be neglected and trays calculated with fixed difference points, and after the size of the resulting column has been determined, the first estimate of heat losses for the two column sections can be made by the usual methods of heat-transfer calculations. The heat losses can then be apportioned among the trays and the number of trays redetermined with the appropriate difference points. This leads to a second approximation of the heat loss, and so forth.

As Fig. 9.34 shows, heat losses increase the internal-reflux ratio, and *for a given condenser heat load*, fewer trays for a given separation are required (recall that the higher the enriching-section difference point and the lower the stripping-section difference point, the fewer the trays). However, the reboiler must provide not only the heat removed in the condenser but also the heat losses. Consequently, for the same reboiler heat load as shown on Fig. 9.34 but with complete insulation against heat loss, all the heat would be removed in the condenser, all the stripping trays would use point *J*, and all the enriching trays would use point *K* as their respective difference points. It therefore follows that, *for a given reboiler heat load or heat expenditure*, fewer trays are required for a given separation if heat losses are eliminated. For this reason, fractionators are usually well insulated.

High-Purity Products and Tray Efficiencies

Methods of dealing with these problems are considered following the McCabe-Thiele method and are equally applicable to Ponchon-Savarit calculations.

MULTISTAGE TRAY TOWERS—METHOD OF MCCABE AND THIELE

This method, although less rigorous than that of Ponchon and Savarit, is nevertheless most useful since it does not require detailed enthalpy data. If such data must be approximated from fragmentary information, much of the exactness of the Ponchon-Savarit method is lost in any case. Except where heat losses or heats of solution are unusually large, the McCabe-Thiele method will be found adequate for most purposes. It hinges upon the fact that, as an approximation, the operating lines on the *xy* diagram can be considered straight for each section of a fractionator between points of addition or withdrawal of streams.

Equimolar Overflow and Vaporization

Consider the enriching section of the fractionator of Fig. 9.17. In the absence of heat losses, which can be (and usually are) made very small by thermal insulation for reasons of economy if for no other, Eq. (9.61) can be written

$$\frac{L_n}{G_{n+1}} = 1 - \frac{H_{G_{n+1}} - H_{L_n}}{Q' - H_{L_n}} \quad (9.108)$$

where Q' includes the condenser heat load and the enthalpy of the distillate, per mole of distillate. The liquid enthalpy H_{L_n} is ordinarily small in comparison with Q' since the condenser heat load must include the latent heat of condensation of at least the reflux liquid. If then $H_{G_{n+1}} - H_{L_n}$ is substantially constant, L_n/G_{n+1} will be constant also for a given fractionation [48]. From Eq. (9.11)

$$H_{G_{n+1}} = [y_{n+1}C_{L,A}M_A + (1 - y_{n+1})C_{L,B}M_B](t_{n+1} - t_0) + y_{n+1}\lambda_A M_A + (1 - y_{n+1})\lambda_B M_B \quad (9.109)$$

where t_{n+1} is the temperature of the vapor from tray $n + 1$ and the λ 's are the latent heats of vaporization at this temperature. If the deviation from ideality of liquid solutions is not great, the first term in brackets of Eq. (9.109) is

$$y_{n+1}C_{L,A}M_A + (1 - y_{n+1})C_{L,B}M_B \approx C_{L,M}M_{av} \quad (9.110)$$

From Eq. (9.10),

$$H_{L_n} = C_L(t_n - t_0)M_{av} + \Delta H_S \quad (9.111)$$

$$\text{and} \quad H_{G_{n+1}} - H_{L_n} = C_L M_{av}(t_{n+1} - t_n) + y_{n+1}\lambda_A M_A + (1 - y_{n+1})\lambda_B M_B - \Delta H_S \quad (9.112)$$

For all but unusual cases, the only important terms of Eq. (9.112) are those containing the latent heats. The temperature change between adjacent trays is usually small, so that the sensible-heat term is insignificant. The heat of solution can in most cases be measured in terms of hundreds of kJ/kmol of solution, whereas the latent heats at ordinary pressures are usually in 10^4 kJ/kmol. Therefore, for all practical purposes,

$$H_{G_{n+1}} - H_{L_n} = (\lambda M)_{av} \quad (9.113)$$

where the last term is the weighted average of the molal latent heats. For many pairs of substances, the molal latent heats are nearly identical, so that averaging is unnecessary. If their inequality is the only barrier to application of these simplifying assumptions, it is possible to assign a fictitious molecular weight to one of the components so that the molal latent heats are then forced to be the same (if this is done, the entire computation must be made with the fictitious molecular weight, including operating lines and equilibrium data). This is, however, rarely necessary.†

If it is sufficiently important, therefore, one can be persuaded that, for all but exceptional cases, the ratio L/G in the enriching section of the fractionator is essentially constant. The same reasoning can be applied to any section of a fractionator between points of addition or withdrawal of streams, although each section will have its own ratio.

Consider next two adjacent trays n and r , between which neither addition nor withdrawal of material from the tower occurs. A material balance provides

$$L_{r-1} + G_{n+1} = L_n + G_r \quad (9.114)$$

Since $L_{r-1}/G_r = L_n/G_{n+1}$, it follows that $L_n = L_{r-1}$ and $G_{n+1} = G_r$, which is the *principle of equimolar overflow and vaporization*. The rate of liquid flow from each tray in a section of the tower is constant on a molar basis, but since the average molecular weight changes with tray number, the weight rates of flow are different.

It should be noted that, as shown in the discussion of Fig. 9.12, if the $H_G y$ and $H_L x$ lines on the Hxy diagram are straight and parallel, then in the absence of heat loss the L/G ratio for a given tower section will be constant regardless of the relative size of H_{L_n} and Q' in Eq. (9.108).

The general assumptions involved in the foregoing are customarily called the *usual simplifying assumptions*.

† If the only enthalpy data available are the latent heats of vaporization of the pure components, an approximate Savarit-Ponchon diagram using these and straight H_L vs x and H_G vs y lines will be better than assuming equal latent heats.

Enriching Section; Total Condenser—Reflux at the Bubble Point

Consider a section of the fractionator entirely above the point of introduction of feed, shown schematically in Fig. 9.35a. The condenser removes all the latent heat from the overhead vapor but does not cool the resulting liquid further. The reflux and distillate product are therefore liquids at the bubble point, and $y_1 = x_D = x_0$. Since the liquid, L mol/h, falling from each tray and the vapor, G mol/h, rising from each tray are each constant if the usual simplifying assumptions pertain, subscripts are not needed to identify the source of these streams. The compositions, however, change. The trays shown are theoretical trays, so that the composition y_n of the vapor from the n th tray is in equilibrium with the liquid of composition x_n leaving the same tray. The point (x_n, y_n) , on x, y coordinates, therefore falls on the equilibrium curve.

A total material balance for the envelope in the figure is

$$G = L + D = D(R + 1) \quad (9.115)$$

For component A,

$$Gy_{n+1} = Lx_n + Dx_D \quad (9.116)$$

from which the enriching-section operating line is

$$y_{n+1} = \frac{L}{G} x_n + \frac{D}{G} x_D \quad (9.117)$$

$$y_{n+1} = \frac{R}{R + 1} x_n + \frac{x_D}{R + 1} \quad (9.118)$$

This is the equation of a straight line on x, y coordinates (Fig. 9.35b) of slope $L/G = R/(R + 1)$, and with a y intercept of $x_D/(R + 1)$. Setting $x_n = x_D$

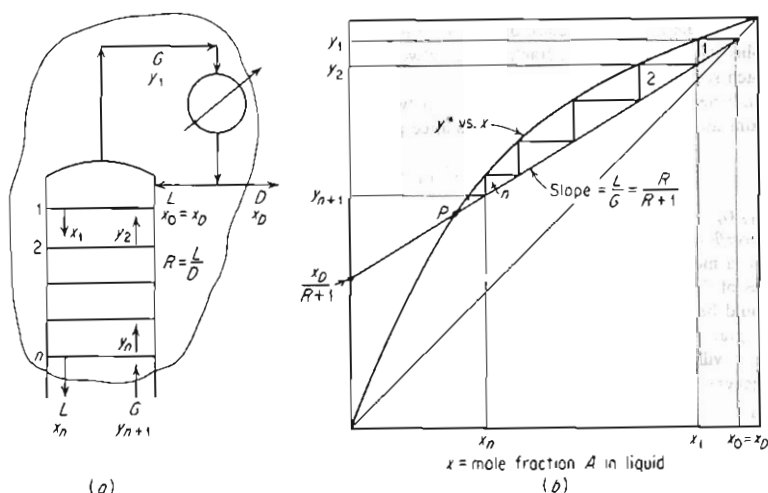


Figure 9.35 Enriching section.

shows $y_{n+1} = x_D$, so that the line passes through the point $y = x = x_D$ on the 45° diagonal. This point and the y intercept permit easy construction of the line. The concentration of liquids and vapors for each tray is shown in accordance with the principles of Chap. 5, and the usual staircase construction between operating line and equilibrium curve is seen to provide the theoretical tray-concentration variation. The construction obviously cannot be carried farther than point P .

In plotting the equilibrium curve of the figure, it is generally assumed that the pressure is constant throughout the tower. If necessary, the variation in pressure from tray to tray can be allowed for after determining the number of real trays, but this will require a trial-and-error procedure. It is ordinarily unnecessary except for operation under very low pressures.

Exhausting Section; Reboiled Vapor in Equilibrium with Residue

Consider next a section of the fractionator below the point of introducing the feed, shown schematically in Fig. 9.36a. The trays are again theoretical trays. The rates of flow $\bar{L}$ and $\bar{G}$ are each constant from tray to tray, but not necessarily equal to the values for the enriching section. A total material balance is

$$\bar{L} = \bar{G} + W \quad (9.119)$$

and, for component A,

$$\bar{L}x_m = \bar{G}y_{m+1} + Wx_W \quad (9.120)$$

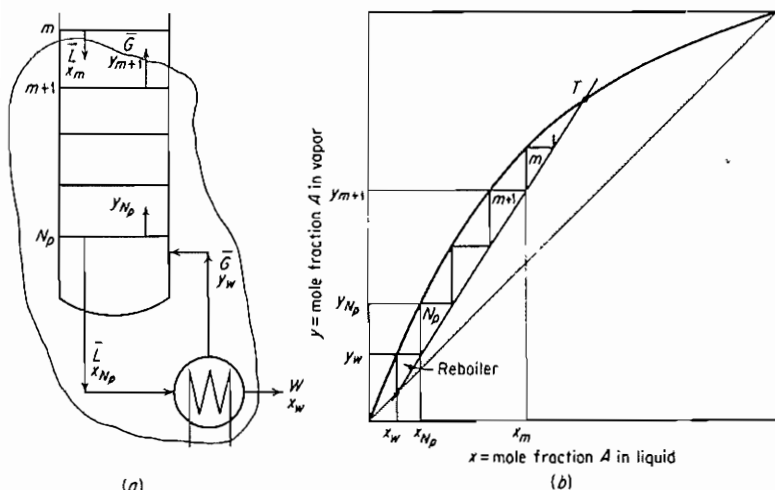


Figure 9.36 Exhausting section.

These provide the equation of the exhausting-section operating line,

$$y_{m+1} = \frac{\bar{L}}{\bar{G}} x_m - \frac{W}{\bar{G}} x_W \quad (9.121)$$

$$y_{m+1} = \frac{\bar{L}}{\bar{L} - \bar{W}} x_m - \frac{W}{\bar{L} - \bar{W}} x_W \quad (9.122)$$

This is a straight line of slope $\bar{L}/\bar{G} = \bar{L}/(\bar{L} - W)$, and since when $x_m = x_W$, $y_{m+1} = x_W$, it passes through $x = y = x_W$ on the 45° diagonal (Fig. 9.36*b*). If the reboiled vapor y_W is in equilibrium with the residue x_W , the first step of the staircase construction represents the reboiler. The steps can be carried no farther than point T .

Introduction of Feed

It is convenient before proceeding further to establish how the introduction of the feed influences the change in slope of the operating lines as we pass from the enriching to the exhausting sections of the fractionator.

Consider the section of the column at the tray where the feed is introduced (Fig. 9.37). The quantities of the liquid and vapor streams change abruptly at this tray, since the feed may consist of liquid, vapor, or a mixture of both. If, for example, the feed is a saturated liquid, $\bar{L}$ will exceed L by the amount of the added feed liquid. To establish the general relationship, an overall material balance about this section is

$$F + L + \bar{G} = G + \bar{L} \quad (9.123)$$

and an enthalpy balance,

$$FH_F + LH_{L,j-1} + \bar{G}H_{G,j+1} = GH_{G,j} + \bar{L}H_{L,j} \quad (9.124)$$

The vapors and liquids inside the tower are all saturated, and the molal enthalpies of all saturated vapors at this section are essentially identical since the temperature and composition changes over one tray are small. The same is true of the molal enthalpies of the saturated liquids, so that $H_{G,j} = H_{G,j+1}$ and

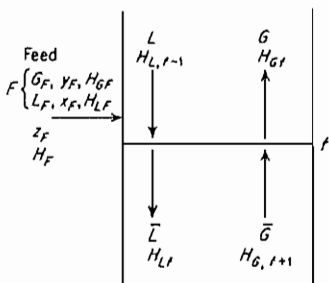


Figure 9.37 Introduction of feed.

$H_{Lj-1} = H_{Lj}$. Equation (9.124) then becomes

$$(\bar{L} - L)H_L = (\bar{G} - G)H_G + FH_F \quad (9.125)$$

Combining this with Eq. (9.123) gives

$$\frac{\bar{L} - L}{F} = \frac{H_G - H_F}{\underbrace{(H_G - H_L)}_{\text{molal latent heat}}} = q \quad (9.126)$$

The quantity q is thus seen to be the heat required to convert 1 mol of feed from its condition H_F to a saturated vapor, divided by the molal latent heat $H_G - H_L$. The feed may be introduced under any of a variety of thermal conditions ranging from a liquid well below its bubble point to a superheated vapor, for each of which the value of q will be different. Typical circumstances are listed in Table 9.1, with the corresponding range of values of q . Combining Eqs. (9.123) and (9.126), we get

$$\bar{G} - G = F(q - 1) \quad (9.127)$$

which provides a convenient method for determining $\bar{G}$.

The point of intersection of the two operating lines will help locate the exhausting-section operating line. This can be established as follows. Rewriting Eqs. (9.117) and (9.121) without the tray subscripts, we have

$$yG = Lx + Dx_D \quad (9.128)$$

$$y\bar{G} = \bar{L}x - Wx_W \quad (9.129)$$

Subtracting gives

$$(\bar{G} - G)y = (\bar{L} - L)x - (Wx_W + Dx_D) \quad (9.130)$$

Further, by an overall material balance,

$$Fz_F = Dx_D + Wx_W \quad (9.76)$$

Substituting this and Eqs. (9.126) and (9.127) in (9.130) gives

$$y = \frac{q}{q-1}x - \frac{z_F}{q-1} \quad (9.131)$$

This, the locus of intersection of operating lines (the q line), is a straight line of slope $q/(q-1)$, and since $y = z_F$ when $x = z_F$, it passes through the point $x = y = z_F$ on the 45° diagonal. The range of the values of the slope $q/(q-1)$ is listed in Table 9.1, and the graphical interpretation for typical cases is shown in Fig. 9.38. Here the operating-line intersection is shown for a particular case of feed as a mixture of liquid and vapor. It is clear that, for a given feed condition, fixing the reflux ratio at the top of the column automatically establishes the liquid/vapor ratio in the exhausting section and the reboiler heat load as well.

Table 9.1 Thermal conditions for the feed

Feed condition	G_F , mol/(area)(time)	L_F , mol/(area)(time)	H_{GF} , energy/mol	H_{LF} , energy/mol	H_F , energy/mol	$q = \frac{H_G - H_F}{H_G - H_L}$	$\frac{q}{q-1}$
Liquid below bubble point	0	F		H_F	$H_F < H_L$	> 1.0	> 1.0
Saturated liquid	0	F		H_F	H_L	1.0	∞
Mixture of liquid and vapor†	G_F $F = G_F + L_F$	L_F	H_G	H_L	$H_G > H_F > H_L$	$1.0 > q > 0$	$\frac{L_F}{L_F - F}$
Saturated vapor	F	0	H_F		H_G	0	0
Superheated vapor	F	0	H_F		$H_F > H_G$	< 0	$1.0 > \frac{q}{q-1} > 0$

† In this case the intersection of the q line with the equilibrium curve gives the compositions of the equilibrium liquid and vapor which constitute the feed. The q line is the flash-vaporization operating line for the feed.

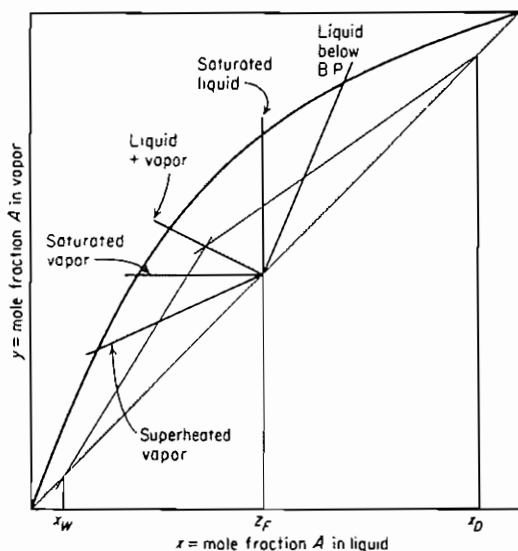


Figure 9.38 Location of q line for typical feed conditions.

Location of the Feed Tray

The q line is useful in simplifying the graphical location of the exhausting line, but the point of intersection of the two operating lines does not necessarily establish the demarcation between the enriching and exhausting sections of the tower. Rather it is the introduction of feed which governs the change from one operating line to the other and establishes the demarcation, and at least in the design of a new column some latitude in the introduction of the feed is available.

Consider the separation shown partially in Fig. 9.39, for example. For a given feed, z_F and the q line are fixed. For particular overhead and residue products, x_D and x_W are fixed. If the reflux ratio is specified, the location of the enriching line DG is fixed, and the exhausting line KC must pass through the q line at E . If the feed is introduced upon the seventh tray from the top (Fig. 9.39a), line DG is used for trays 1 through 6, and, beginning with the seventh tray, the line KC must be used. If, on the other hand, the feed is introduced upon the fourth from the top (Fig. 9.39b), line KC is used for all trays below the fourth. Clearly a transition from one operating line to the other must be made somewhere between points C and D , but anywhere within these limits will serve. The least total number of trays will result if the steps on the diagram are kept as large as possible or if the transition is made at the first opportunity after passing the operating-line intersection, as shown in Fig. 9.39c. In the design of a new column, this is the practice to be followed.

In the adaptation of an existing column to a new separation, the point of introducing the feed is limited to the location of existing nozzles in the column

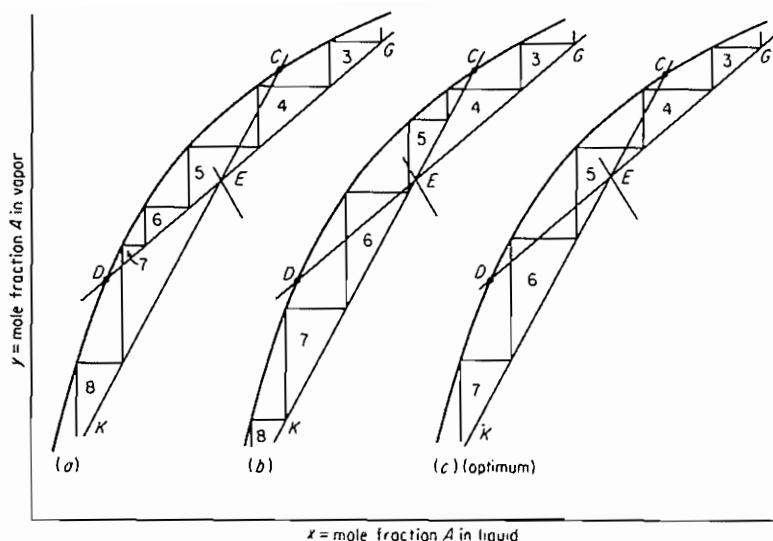


Figure 9.39 Location of feed tray.

wall. The slope of the operating lines (or reflux ratio) and the product compositions to be realized must then be determined by trial and error, in order to obtain numbers of theoretical trays in the two sections of the column consistent with the number of real trays in each section and the expected tray efficiency.

Total Reflux, or Infinite Reflux Ratio

As the reflux ratio $R = L/D$ is increased, the ratio L/G increases, until ultimately, when $R = \infty$, $L/G = 1$ and the operating lines of both sections of the column coincide with the 45° diagonal as in Fig. 9.40. In practice this can be

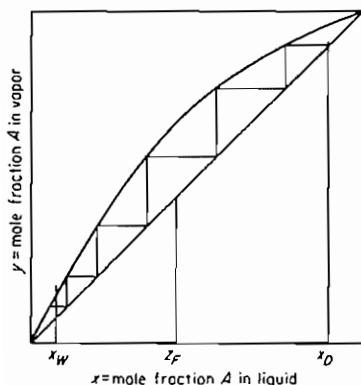


Figure 9.40 Total reflux and minimum trays.

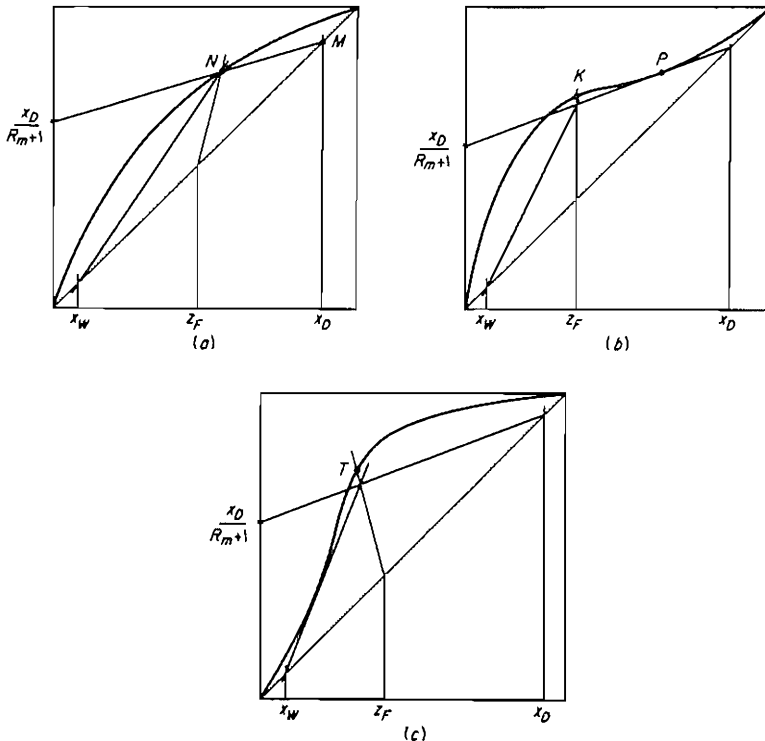
realized by returning all the overhead product back to the column as reflux (total reflux) and reboiling all the residue product, whereupon the forward flow of fresh feed must be reduced to zero. Alternatively such a condition can be interpreted as requiring infinite reboiler heat and condenser cooling capacity for a given rate of feed.

As the operating lines move farther away from the equilibrium curve with increased reflux ratio, the number of theoretical trays required to produce a given separation becomes less, until at total reflux the number of trays is the minimum N_m .

If the relative volatility is constant or nearly so, the analytical expression of Fenske, Eq. (9.85), can be used.

Minimum Reflux Ratio

The minimum reflux ratio R_m is the maximum ratio which will require an infinite number of trays for the separation desired, and it corresponds to the minimum reboiler heat and condenser cooling capacity for the separation. Refer to Fig. 9.41a. As the reflux ratio is decreased, the slope of the enriching



operating line becomes less, and the number of trays required increases. Operating line MN , which passes through the point of intersection of the q line and the equilibrium curve, corresponds to the minimum reflux ratio, and an infinite number of trays would be required to reach point N from either end of the tower. In some cases, as in Fig. 9.41*b*, the minimum-reflux operating line will be tangent to the equilibrium curve in the enriching section, as at point P , while a line through K would clearly represent too small a reflux ratio. It has been pointed out that all systems show concave-upward xy diagrams near the critical condition of the more volatile component [66]. Because of the interdependence of the liquid/vapor ratios in the two sections of the column, a tangent operating line in the exhausting section may also set the minimum reflux ratio, as in Fig. 9.41*c*.

When the equilibrium curve is always concave downward, the minimum reflux ratio can conveniently be calculated analytically [63]. The required relationship can be developed by solving Eqs. (9.118) and (9.131) simultaneously to obtain the coordinates (x_a, y_a) of the point of intersection of the enriching operating line and the q line. When the tray-number designation in Eq. (9.118) is dropped, these are

$$x_a = \frac{x_D(q-1) + z_F(R+1)}{R+q} \quad y_a = \frac{Rz_F + qx_D}{R+q} \quad (9.132)$$

At the minimum reflux ratio R_m , these coordinates are equilibrium values since they occur on the equilibrium curve. Substituting them into the definition of α , Eq. (9.2), gives

$$\frac{R_m z_F + qx_D}{R_m(1-z_F) + q(1-x_D)} = \frac{\alpha[x_D(q-1) + z_F(R_m+1)]}{(R_m+1)(1-z_F) + (q-1)(1-x_D)} \quad (9.133)$$

This conveniently can be solved for R_m for any value of q . Thus, for example:

$$R_m = \left\{ \frac{1}{\alpha-1} \left[\frac{x_D}{x_F} - \frac{\alpha(1-x_D)}{1-x_F} \right] \right. \quad q=1 \text{ (feed liquid at the bubble point)} \quad (9.134)$$

$$\left. \frac{1}{\alpha-1} \left(\frac{\alpha x_D}{y_F} - \frac{1-x_D}{1-y_F} \right) - 1 \right. \quad q=0 \text{ (feed vapor at the dew point)} \quad (9.135)$$

In each case, the α is that prevailing at the intersection of the q line and the equilibrium curve.

Optimum Reflux Ratio

The discussion given under the Ponchon-Savarit method applies.

Illustration 9.10 Redesign the methanol-water fractionator of Illustration 9.8, using the simplifying assumptions of the McCabe-Thiele method. The circumstances are:

Feed. 5000 kg/h, 216.8 kmol/h, $z_F = 0.360$ mole fraction methanol, av mol wt = 23.1, temperature entering the fractionator = 58.3°C

Distillate. 2620 kg/h, 84.4 kmol/h, $x_D = 0.915$ mole fraction methanol, liquid at the bubble point

Residue. 2380 kg/h, 132.4 kmol/h, $x_W = 0.00565$ mole fraction methanol

Reflux ratio = 1.5 times the minimum

SOLUTION Refer to Fig. 9.42. From the txy diagram, the bubble point of the feed at 0.360 mole fraction methanol is 76.0°C, and its dew point is 89.7°C. The latent heats of vaporization at 76.0°C are $\lambda_A = 1046.7$ kJ/kg for methanol and $\lambda_B = 2284$ kJ/kg for water. Specific heat of liquid methanol = 2721, of liquid water = 4187, of feed solution = 3852 J/kg K. If heats of

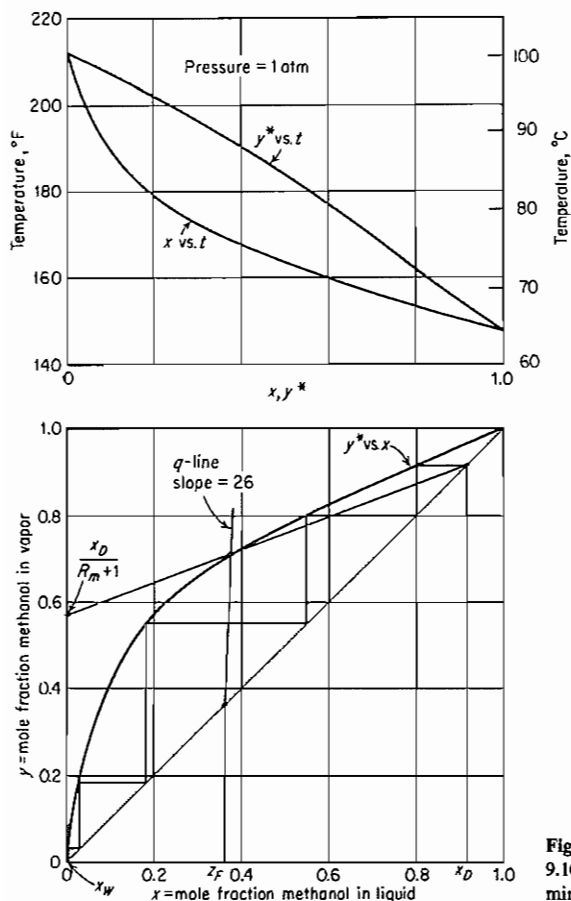


Figure 9.42 Solution to Illustration 9.10. Minimum reflux ratio and minimum trays.

solution are ignored, as is customary with the McCabe-Thiele method, the enthalpy of the feed at 76.0°C (the bubble point) referred to 58.3°C (the feed temperature) is

$$3.852(23.1)(76.0 - 58.3) = 1575 \text{ kJ/kmol}$$

The enthalpy of the saturated vapor at 89.7°C referred to liquids at 58.3°C is

$$\begin{aligned} & 0.36[2.721(32.04)(89.7 - 58.3) + 1046.7(32.04)] \\ & + (1 - 0.36)[4.187(18.02)(89.7 - 58.3) + 2284(18.02)] \\ & = 40\,915 \text{ kJ/kmol} \\ q &= \frac{\text{heat to convert to saturated vapor}}{\text{heat of vaporization}} = \frac{40\,915 - 0}{40\,915 - 1575} = 1.04 \end{aligned}$$

(Essentially the same value is obtained if the heat of solution is considered.)

$$\frac{q}{q-1} = \frac{1.04}{1.04-1} = 26$$

In Fig. 9.42, x_D , x_W , and x_F are located on the 45° diagonal, and the q line is drawn with slope = 26. The operating line for minimum reflux ratio in this case passes through the intersection of the q line and equilibrium curve, as shown.

$$\frac{x_D}{R_m + 1} = \frac{0.915}{R_m + 1} = 0.57$$

$$R_m = 0.605 \text{ mole reflux/mole distillate}$$

The minimum number of theoretical trays is determined using the 45° diagonal as operating line (Fig. 9.42). Theoretical stages to the number of 4.9, including the reboiler, are determined.

For $R = 1.5R_m = 1.5(0.605) = 0.908$ mol reflux/mol distillate, and for equimolar overflow and vaporization:

Eq. (9.49):

$$L = L_0 = RD = 0.908(84.4) = 76.5 \text{ kmol/h}$$

Eq. (9.115):

$$G = D(R + 1) = 85.4(0.908 + 1) = 160.9 \text{ kmol/h}$$

Eq. (9.126):

$$\bar{L} = qF + L = 1.04(216.8) + 76.5 = 302.5 \text{ kmol/h}$$

Eq. (9.127):

$$\bar{G} = F(q - 1) + G = 216.8(1.04 - 1) + 160.9 = 169.7 \text{ kmol/h}$$

$$\frac{x_D}{R + 1} = \frac{0.915}{0.908 + 1} = 0.480$$

Refer to Fig. 9.43. The y intercept 0.480 and enriching and exhausting operating lines are plotted. Steps are drawn to determine the number of theoretical trays, as shown. The exhausting operating line is used immediately after crossing the operating-line intersection, and the feed is therefore to be introduced on the fifth theoretical tray from the top. A total of 8.8 theoretical trays, including the reboiler, is required, and the tower must therefore contain 7.8 theoretical trays. An integral number can be obtained by very slight adjustment of the reflux ratio, but since a tray efficiency must still be applied to obtain the number of real trays, this need not be done.

When the residue composition is very small, it will be necessary to enlarge the scale of the lower left-hand part of the diagram in order to obtain the number of trays. In some cases the graphical determination may still be difficult because of the closeness of the exhausting line and the equilibrium curve. Logarithmic coordinates can then be used to maintain a satisfactory separation of the lines, as in the insert of Fig. 9.43. On such a graph, for very low values of x , the equilibrium curve will be substantially given by $y^* = \alpha x$, which is a straight line of unit slope. The exhausting operating line will, however, be curved and must be plotted from its equation. In this example, the equation is [Eq. (9.121)]

$$y = \frac{302.5}{169.7}x - \frac{132.4}{169.7} = 0.00565 + 1.785x - 0.0044$$

The steps representing the theoretical stages are made in the usual manner on these coordinates, continued down from the last made on the arithmetic coordinates (see also page 422).

The diameter of the tower and the tray design are established through the methods of Chap. 6. Note the substantially different liquid loads in the enriching and exhausting sections. A column of constant diameter for all sections is usually desired for simplicity in construction and lower cost. If the discrepancy between liquid or vapor quantities in the two sections is considerable, however, and particularly if expensive alloy or nonferrous metal is used, different diameters for the two sections may be warranted.

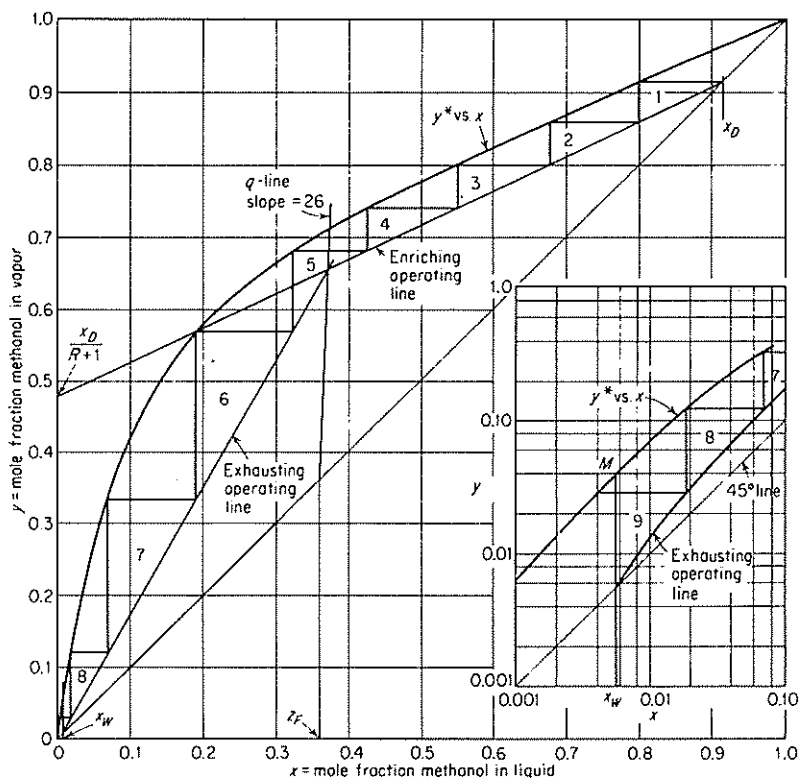


Figure 9.43 Solution to Illustration 9.10. $R = 0.908$ mol reflux/mol distillate product.

Computations for other reflux ratios are easily and quickly made once the diagram and equations have been set up for one value of R . These provide data for determining the most economical R . The following table lists the important quantities for this separation at various values of R :

R	L	G	$\bar{L}$	$\bar{G}$	No. theoretical stages
$R_m = 0.605$	51.0	135.4	277	144.1	∞
0.70	59.0	143.4	285	152.1	11.5
0.80	67.5	151.9	294	160.6	10
0.908	76.5	160.9	303	169.7	8.8
1.029	86.7	171.3	313	180	8.3
2.0	168.8	253	395	262	6.5
4.0	338	422	564	431	5.5
∞	∞	∞	∞	∞	$4.9 = N_m + 1$

Included in the table are the data for the McCabe-Thiele method at $R = 1.029$, the reflux ratio

used in Illustration 9.8 for the exact Ponchon-Savarit calculation. Illustration 9.8 showed nine theoretical stages. It is noteworthy that for either method each at 1.5 times its respective value of R_m , the number of stages is essentially the same, and the maximum flow rates which would be used to set the mechanical design of the tower are sufficiently similar for the same final design to result.

Use of Open Steam

Ordinarily, heat is applied at the base of the tower by means of a heat exchanger (see Fig. 9.29), but when a water solution is fractionated to give the nonaqueous solute as the distillate and the water is removed as the residue product, the heat required may be provided by the use of open steam at the bottom of the tower. The reboiler is then dispensed with. For a given reflux ratio and overhead composition, however, more trays will be required in the tower.

Refer to Fig. 9.44. Overall material balances for both components and for the more volatile substance are

$$F + \bar{G} = D + W \quad (9.136)$$

$$Fz_F = Dx_D + Wx_W \quad (9.137)$$

where $\bar{G}$ is mol/h of steam used, assumed saturated at the tower pressure. The enriching operating line is located as usual, and the slope of the exhausting line $\bar{L}/\bar{G}$ is related to L/G and the feed conditions in the same manner as before. A material balance for the more volatile substance below tray m in the exhausting

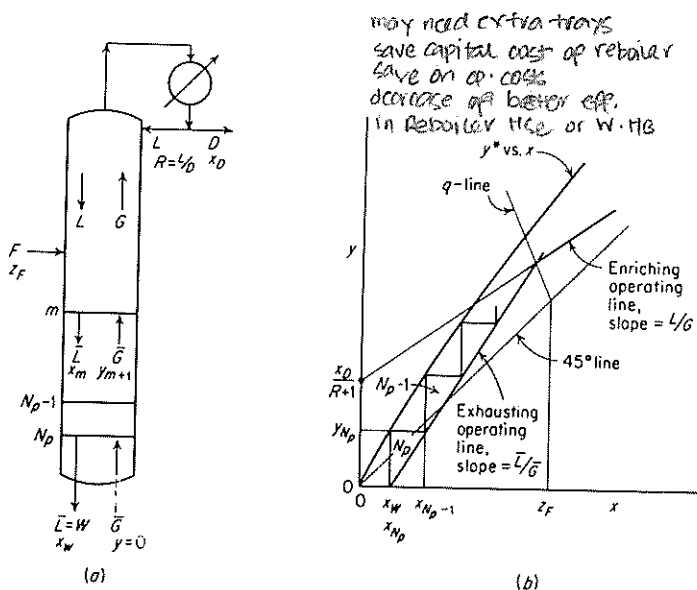


Figure 9.44 Use of open steam.

section is

$$\bar{L}x_m + \bar{G}(0) = \bar{G}y_{m+1} + Wx_W \quad (9.138)$$

and since $\bar{L} = W$ in this case,

$$\frac{\bar{L}}{\bar{G}} = \frac{y_{m+1}}{x_m - x_W} \quad (9.139)$$

The exhausting line therefore passes through the point ($y = 0$, $x = x_W$) as shown in Fig. 9.44*b*. The graphical tray construction must therefore be continued to the x axis of the diagram.

If the steam entering the tower, $\bar{G}_{N_p+1}$, is superheated, it will vaporize liquid on tray N_p to the extent necessary to bring it to saturation, $\bar{G}_{N_p+1}(H_{G, N_p+1} - H_{G, sat})/\lambda M$, where $H_{G, sat}$ is the enthalpy of saturated steam and λM the molar latent heat at the tower pressure.

$$\bar{G} = \bar{G}_{N_p+1} \left(1 + \frac{H_{G, N_p+1} - H_{G, sat}}{\lambda M} \right) \quad (9.140)$$

and

$$\bar{L} = \bar{G} - \bar{G}_{N_p+1} + \bar{L}_{N_p} \quad (9.141)$$

from which the internal $\bar{L}/\bar{G}$ ratio can be computed.

Condensers

Condensers are usually conventional tubular heat exchangers, generally arranged horizontally with the coolant inside the tubes, but also vertically with the coolant on either side of the tubes [27]. The condenser may be placed above the tower for gravity flow of the condensed reflux to the top tray, but it is usually more convenient for purposes of construction and cleaning to place the condenser nearer the ground and to return the reflux from an accumulator drum [55] to the top tray by pumping. This procedure also provides more pressure drop for operation of control valves on the reflux line.

The condenser coolant is most frequently water. The pressure of the distillation must then be high enough for the available cooling water to condense the overhead vapor with an adequate temperature difference to provide reasonably rapid heat-transfer rates. The cost of the fractionator, however, will increase with increased pressure of operation, and in the case of very volatile distillates some low-temperature refrigerant may be used as a coolant.

If the condensate is cooled only to the bubble point, the withdrawn distillate is then usually further cooled in a separate heat exchanger to avoid vaporization loss on storage.

Partial Condensers

It is occasionally desired to withdraw a distillate product in the vapor state, especially if its boiling point is low enough to make complete condensation

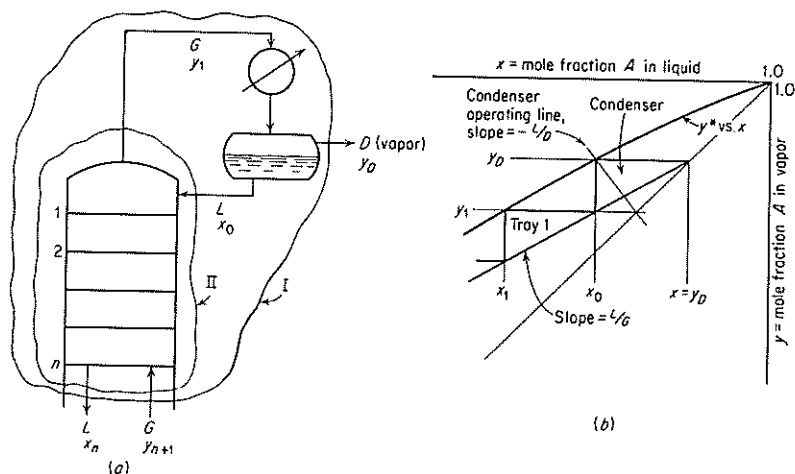


Figure 9.45 Partial equilibrium condenser.

difficult. In this case the overhead vapor is cooled sufficiently to condense the necessary liquid reflux and the residual vapor provides the product, as in Fig. 9.45.

A partial condenser may produce any of several results: (1) If time of contact between vapor product and liquid reflux is sufficient, the two will be in equilibrium with each other and the condenser provides an equilibrium condensation. (2) If the condensate is removed as rapidly as it forms, a differential condensation may occur. (3) If cooling is very rapid, little mass transfer between vapor and condensate results, and the two will have essentially the same composition.

When the first pertains, the condenser acts as one theoretical stage for the separation. The compositions y_D and x_0 can be computed by the methods of equilibrium condensation, as shown in Fig. 9.45b. The enriching operating line is then given as usual by material balances.

$$\text{Envelope I:} \quad G = L + D \quad (9.142)$$

$$Gy_{n+1} = Lx_n + Dy_D \quad (9.143)$$

$$\text{Envelope II:} \quad Gy_{n+1} + Lx_0 = Gy_1 + Lx_n \quad (9.144)$$

In the design of new equipment it is safer to ignore the enrichment which may be obtained by a partial condenser and to include the additional theoretical tray in the column, since it is difficult to ensure that equilibrium condensation will occur.

Cold Reflux

If the overhead vapor is condensed and cooled below its bubble point so that the reflux liquid is cold, vapor G_1 rising from the top tray will be less in quantity than that for the rest of the enriching section since some will be required to condense and heat the reflux to its bubble point. External reflux L_0 will require heat to the extent of $L_0 C_{L0} M_{av}(t_{bp,R} - t_R)$, where $t_{bp,R}$ and t_R are the reflux bubble point and actual temperatures, respectively. An amount of vapor, $L_0 C_{L0} M_{av}(t_{bp,R} - t_R)/(\lambda M)_{av}$ will condense to provide the heat, and the condensed vapor adds to L_0 to provide L , the liquid flow rate below the top tray. Therefore

$$L = L_0 + \frac{L_0 C_{L0} M_{av}(t_{bp,R} - t_R)}{(\lambda M)_{av}} = RD \left[1 + \frac{C_{L0} M_{av}(t_{bp,R} - t_R)}{(\lambda M)_{av}} \right] \quad (9.145)$$

where R is the usual external reflux ratio, L_0/D . Defining an *apparent* reflux ratio R' by

$$R' = \frac{L}{D} = \frac{L}{G - L} \quad (9.146)$$

gives

$$R' = R \left[1 + \frac{C_{L0} M_{av}(t_{bp,R} - t_R)}{(\lambda M)_{av}} \right] \quad (9.147)$$

The enriching operating line becomes

$$y_{n+1} = \frac{R'}{R' + 1} x_n + \frac{x_D}{R' + 1} \quad (9.148)$$

and it is plotted through $y = x = x_D$, with a y intercept at $x_D/(R' + 1)$ and a slope $R'/(R' + 1)$.

Rectification of Azeotropic Mixtures

Minimum- and maximum-boiling azeotropic mixtures of the type shown in Figs. 9.7 and 9.10 can be treated by the methods already described, except that it will be impossible to obtain two products of compositions which fall on opposite sides of the azeotropic composition. In the rectification of a minimum-boiling azeotrope (Fig. 9.7), for example, the distillate product may be as close to the azeotropic composition as desired. But the residue product will be either rich in A or rich in B depending upon whether the feed is richer or leaner in A than the azeotropic mixture. With maximum-boiling mixtures (Fig. 9.10) the residue product will always approach the azeotropic composition. These mixtures can sometimes be separated completely by addition of a third substance, as described later.

Insoluble mixtures which form two-liquid-phase azeotropes, however, can readily be separated completely provided two fractionators are used. This

depends upon the fact that the condensed distillate forms two liquid solutions on opposite sides of the azeotropic composition. Consider the separation of the mixture whose vapor-liquid equilibrium diagram is shown in Fig. 9.46, where the feed has the composition z_F and the solubility limits are $x_{R,I}$ and $x_{R,II}$ at the boiling point. If the feed is introduced into fractionator I of Fig. 9.47, it is evident that the residue product of composition $x_{W,I}$ can be as nearly pure B as desired. The enriching section may contain sufficient trays to produce an overhead vapor approaching the azeotropic composition M , such as vapor $y_{D,I}$. This vapor, when totally condensed to mixture K at its boiling point, will form two insoluble liquids of composition $x_{R,I}$ and $x_{R,II}$, which can be decanted as shown. The layer which is richer in B is returned to the top tray of column I as reflux. The enriching operating line for column I then passes through the point ($y = y_{D,I}$, $x = x_{R,I}$), as shown in Fig. 9.46, and its slope will be the liquid/vapor ratio in the enriching section.

The A-rich layer from the decanter (Fig. 9.47) is sent to the top tray of fractionator II, which contains only a stripping or exhausting section. It is clear from Fig. 9.46 that the residue product composition $x_{W,II}$ can be as nearly pure A as desired (turn the figure upside down to give it its usual appearance). The overhead vapor from tower II will be of composition $y_{D,II}$, which, when totally condensed as mixture N , produces the same two insoluble liquids as the first distillate. Consequently a common condenser can be used for both towers.

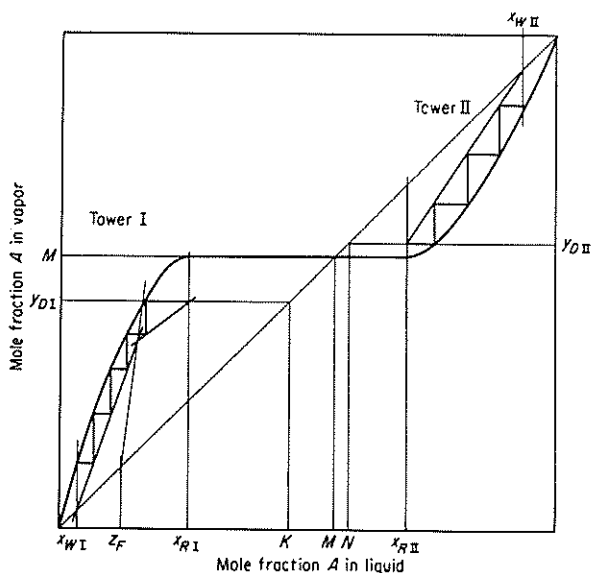


Figure 9.46 Fractionation of partially miscible mixtures.

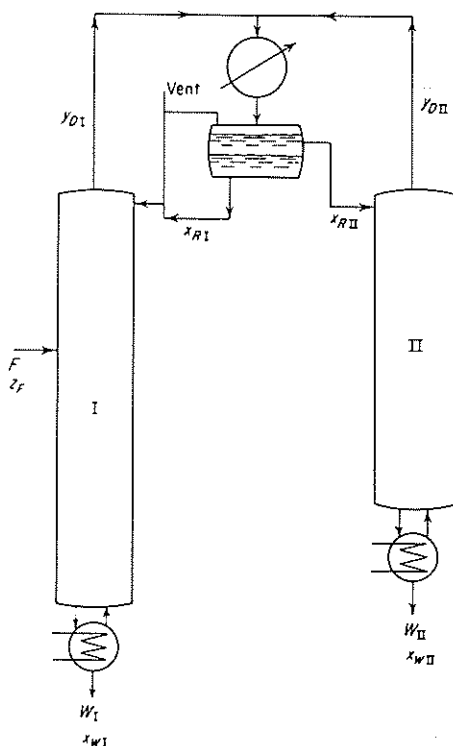


Figure 9.47 Two-tower system for partially miscible mixtures.

In practice it will be desirable to cool the distillate below its bubble point to prevent excessive loss of vapor from the vent of the decanter. This changes the compositions $x_{R, I}$ and $x_{R, II}$ slightly and provides somewhat larger internal reflux ratios. If the feed itself consists of two insoluble liquids, it can be fed to the decanter, whereupon both fractionators then consist of exhausting sections only. When it is desired to remove the last traces of water from a substance such as a hydrocarbon, it is common practice to use only one tower from which the dry hydrocarbon is removed as the residue product. The hydrocarbon-rich layer from the decanter is returned as reflux, but the water layer, which contains very little hydrocarbon, is normally discarded.

The Ponchon-Savarit method can also be used but the necessary enthalpy data are rarely available for these systems.

Multiple Feeds

If two solutions of different concentrations are to be fractionated to give the same products, both can be handled in the same fractionator. The McCabe-Thiele diagram for a two-feed column will resemble the lower part of Fig. 9.33,

with operating lines straight. Each feed is considered separately (as if neither "knew" of the other's presence). The upper operating line is located as usual. The intermediate operating line, for the section of the column between feeds, intersects the enriching line at the q line for the richer feed and has a slope given by the L and G quantities computed through Eqs. (9.126) and (9.127). The lowermost operating line intersects the intermediate one at the q line for the less rich feed. For the least number of stages at a given reflux ratio, the feeds are each introduced at the stage whose construction on the diagram straddles their respective q line.

High-Purity Products

In Illustration 9.10 it was shown that for very low concentrations of more volatile substance in the residue, the graphical construction for stripping-section trays can be completed on logarithmic coordinates. This is readily done, since at very low concentrations the equilibrium curve is essentially linear, given by $y^* = \alpha x$. This plots as a straight 45° line on logarithmic coordinates, and α is taken for very low concentrations. The operating line is curved on such a graph, however, and must be plotted from its equation, (9.121). For very pure distillates a similar logarithmic graph can be constructed by turning the graph paper upside down and re-marking the coordinates as (1—printed marking) [26]. Thus, the printed 0.0001 is marked as 0.9999, etc. On such a graph the equilibrium curve is also straight at 45° for high concentrations, but the operating line is curved and must be plotted from its equation, (9.111).

As an alternative to these methods, calculations can be made analytically using the Kremser equations (5.50) to (5.57) or Fig. 5.16, since even for systems where the McCabe-Thiele assumptions are not generally applicable, the operating lines are straight at the extreme concentrations now under consideration. The *exhausting section* is considered as a stripper for the more volatile component, whence, for a kettle-type reboiler,

$$N_p - m + 1 = \frac{\log \left[\frac{x_m - x_w/\alpha}{x_w - x_w/\alpha} (1 - \bar{A}) + \bar{A} \right]}{\log(1/\bar{A})} \quad (9.149)$$

where x_m is the composition of the liquid leaving tray m , the last equilibrium tray obtained by graphical work, and $\bar{A} = \bar{L}/\alpha G$. For a thermosiphon reboiler which totally vaporizes the liquid, the left-hand side of Eq. (9.149) is $N_p - m$. For open steam the left-hand side is $N_p - m$, and x_w/α is omitted ($y_{N_p+1} = 0$). The *enriching section* is considered as an absorber for the less volatile component

$$n - 1 = \frac{\log \left[\frac{(1 - y_n) - (1 - x_0)/\alpha}{(1 - y_1) - (1 - x_0)/\alpha} \left(1 - \frac{1}{A} \right) + \frac{1}{A} \right]}{\log A} \quad (9.150)$$

where y_n is the vapor leaving tray n , the last equilibrium tray obtained by

graphical construction upward from the feed, x_0 is the reflux composition, and $A = \alpha L/G$.

For many ideal liquids, where α is nearly constant for all concentrations, the analytical calculation of Smoker [58] applies to the entire fractionator.

Tray Efficiencies

Methods for estimating tray efficiencies are discussed in Chap. 6. Murphree vapor efficiencies are most simply used graphically on the xy diagram, whether this is derived from the exact Ponchon-Savarit calculation or with the simplifying assumptions, as shown in Illustration 9.11. Overall efficiencies E_O strictly have meaning only when the Murphree efficiency of all trays is the same and when the equilibrium and operating lines are both straight over the concentrations considered. The latter requirement can be met only over limited concentration ranges in distillation, but nevertheless the empirical correlation of Fig. 6.25 is useful for rough estimates, if not for final designs, with the exception that values of E_O for aqueous solutions are usually higher than the correlation shows.

Illustration 9.11 In the development of a new process, it will be necessary to fractionate 910 kg/h of an ethanol-water solution containing 0.3 mole fraction ethanol, available at the bubble point. It is desired to produce a distillate containing 0.80 mole fraction ethanol, with a substantially negligible loss of ethanol in the residue.

There is available a tower containing 20 identical cross-flow sieve trays, 760 mm diameter, with provision for introducing the feed only on the thirteenth tray from the top. The tower is suitable for use at 1 std atm pressure. The tray spacing is 450 mmHg, the weir on each tray is 530 mm long, extending 60 mm above the tray floor, and the area of the perforations is $0.0462 \text{ m}^2/\text{tray}$.

Determine a suitable reflux ratio for use with this tower, and estimate the corresponding ethanol loss, assuming that open steam at 69 kN/m^2 ($10 \text{ lb}_f/\text{in}^2$) gauge will be used for heating and that adequate condenser capacity will be supplied.

SOLUTION The McCabe-Thiele method will be used. Mol wt ethanol = 46.05, of water = 18.02. $z_F = 0.30$.

Basis: 1 h. The feed composition corresponds to 52.27 wt % ethanol. The feed contains $910(0.5227)/46.05 = 10.33 \text{ kmol}$ ethanol and $910(1 - 0.5227)/18.02 = 24.10 \text{ kmol}$ water, total = 34.43 kmol. If essentially all the ethanol is removed from the residue, the distillate $D = 10.33/0.80 = 12.91 \text{ kmol/h} = 522.2 \text{ kg/h}$. Vapor-liquid equilibrium data are available in "The Chemical Engineers' Handbook," 4th ed., p. 13-5, and in Carey and Lewis, *Ind. Eng. Chem.*, 24, 882 (1932).

Preliminary calculations indicate that the capacity of the tower is governed by the top vapor rate. At tray 1, av mol wt vapor = $0.8(46.05) + 0.2(18.02) = 40.5 \text{ kg/mol}$, temp = 78.2°C . In Chap. 6 notation,

$$\rho_G = \frac{40.5}{22.41} \frac{273}{273 + 78.2} = 1.405 \text{ kg/m}^3 \quad \rho_L = 744.9 \text{ kg/m}^3 \quad \sigma = 0.021 \text{ N/m (est)}$$

Table 6.2: for $t = 0.450 \text{ m}$, $\alpha = 0.0452$, $\beta = 0.0287$. The tower cross-sectional area = $A_t = \pi(0.760)^2/4 = 0.4536 \text{ m}^2$, and for $W = (530/760)T = 0.70T$ (Table 6.1), the downspout area $A_d = 0.0808(0.4536) = 0.0367 \text{ m}^2$. The active area $A_a = 0.4536 - 2(0.0367) = 0.380 \text{ m}^2$. With $A_o = 0.0462 \text{ m}^2$, and if we tentatively take $(L'/G)(\rho_G/\rho_L)^{0.5} = 0.1$, Eq. (6.30) provides $C_F = 0.0746$, and Eq. (6.29) shows $V_F = 1.72 \text{ m/s}$ as the flooding gas velocity based on $A_a = A_t - A_d = 0.4169 \text{ m}^2$.

Calculations will be made for a reflux ratio corresponding to $G = D(R + 1) = 12.91(3 + 1) = 51.64$ kmol/h vapor at the top, or

$$\frac{51.64(22.41)}{3600} \frac{273 + 78.2}{273} = 0.414 \text{ m}^3/\text{s}$$

The vapor velocity is then $0.414/0.4169 = 0.993$ m/s, or 58 percent of the flooding velocity (amply safe). $L = 3(12.91) = 38.73$ kmol/h or 1567 kg/h. Therefore,

$$\frac{L'}{G'} \left(\frac{\rho_G}{\rho_L} \right)^{0.5} = \frac{1567}{0.414(3600)(1.405)} \left(\frac{1.405}{744.9} \right)^{0.5} = 0.0325$$

Since for values of this quantity less than 0.1, Eq. (6.29) nevertheless uses 0.1, the calculated value of C_F is correct.

Since the feed is at the bubble point, $q = 1$, and Eq. (9.126):

$$\bar{L} = L + qF = 38.73 + 1(34.43) = 73.2 \text{ kmol/h}$$

Eq. (9.127):

$$\bar{G} = G + F(q - 1) = G = 51.64 \text{ kmol/h}$$

The enthalpy of saturated steam, referred to 0°C , at 69 kN/m^2 is $2699 \text{ kN} \cdot \text{m/kg}$, and this will be its enthalpy as it enters the tower if expanded adiabatically to the tower pressure. The enthalpy of saturated steam at 1 std atm (neglecting tower pressure drop) is $2676 \text{ kN} \cdot \text{m/kg}$, and the latent heat is $2257 \text{ kN} \cdot \text{m/kg}$. Eq. (9.140):

$$51.64 = \bar{G}_{N_p+1} \left[1 + \frac{(26.99 - 26.76)(18.02)}{2257(18.02)} \right]$$

$$\bar{G}_{N_p+1} = 51.1 \text{ kmol steam/h}$$

Eq. (9.141):

$$73.2 = 51.64 - 51.1 + L_{N_p}$$

$$L_{N_p} = W = 72.7 \text{ kmol residue/h}$$

Tray efficiencies Consider the situation at $x = 0.5$, $y^* = 0.652$, $t = 79.8^\circ\text{C}$, which is in the enriching section. Here (Chap. 6 notation) $\rho_L = 791$, $\rho_G = 1.253 \text{ kg/m}^3$, m (from equilibrium data) $= 0.42$, and by the methods of Chap. 2, $\text{Sc}_G = 0.930$, $D_L = 2.065 \times 10^{-9} \text{ m}^2/\text{s}$. For $L = 38.73 \text{ kmol/h}$, $q = 4.36 \times 10^{-4} \text{ m}^3/\text{s}$, and for $G = 51.64 \text{ kmol/h}$, $V_a = 1.046 \text{ m/s}$. From the tray dimensions, $z = 0.647 \text{ m}$, $Z = 0.542 \text{ m}$, and $h_{w'} = 0.060 \text{ m}$.

$$\text{Eq. (6.61): } N_{iL} = 0.89$$

$$\text{Eq. (6.38): } h_{L_c} = 0.0380 \text{ m}$$

$$\text{Eq. (6.64): } \theta_L = 30.3 \text{ s}$$

$$\text{Eq. (6.62): } N_{iL} = 22.8$$

$$\text{Eq. (6.52): } N_{iOG} = 0.871$$

$$\text{Eq. (6.51): } E_{OG} = 0.581$$

$$\text{Eq. (6.63): } D_E = 1.265 \times 10^{-3} \text{ m}^2/\text{s}$$

$$\text{Eq. (6.59): } \text{Pe} = 7.66$$

$$\text{Eq. (6.57): } E_{MG} = 0.72$$

Entrainment is negligible. Similarly,

x	0	0.1	0.3	0.5	0.7
E_{MG}	0.48	0.543	0.74	0.72	0.72

Tray calculations The operating-line intercept $= x_D/(R + 1) = 0.8/(3 + 1) = 0.2$, $q/(q - 1) = \infty$. The operating line for the enriching section and the q line are located as usual (Fig. 9.48). The exhausting-section operating line, on this scale of plot, for all practical purposes passes

$$\text{EMA: } \frac{y_n - y_{n+1}}{y_1^* - y_{n+1}} \\ = \frac{BG}{AG}$$

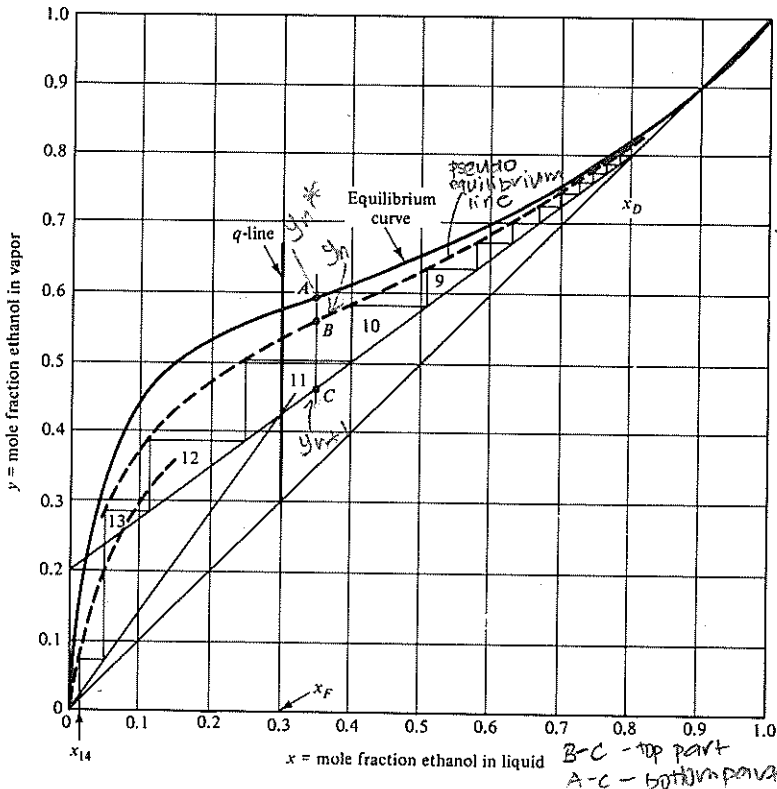


Figure 9.48 Solution to Illustration 9.11.

through the origin. The broken curve is located so that, at each concentration, vertical distances corresponding to lines BC and AC are in the ratio E_{MG} . This curve is used instead of the equilibrium curve to locate real trays, as shown. The feed tray is the thirteenth.

Below tray 14, calculations can be made on logarithmic coordinates as in Illustration 9.10, but that would require trial-and-error location of the operating line since x_w is unknown. It is easier to proceed as follows.

From Fig. 9.48, $x_{14} = 0.0150$, and from this concentration down the equilibrium curve is essentially straight ($\alpha = m = 8.95$), $E_{MG} = 0.48 = \text{const}$, $\bar{A} = \bar{L}/\alpha\bar{G} = 73.2/8.95(51.64) = 0.1584$. Equation (8.16) then provides the overall tray efficiency for the six bottom trays:

$$E_o = \frac{\log[1 + 0.48(1/0.1584 - 1)]}{\log(1/0.1584)} = 0.688$$

The 6 real trays correspond to $6(0.688) = 4.13$ real trays to the bottom of the tower, whence Eq. (9.149) provides, for open steam,

$$4.13 = \frac{\log[(0.015/x_w)(1 - 0.1584) + 0.1584]}{\log(1/0.1584)}$$

$$x_w = 6.25 \times 10^{-6} \text{ mole fraction ethanol in the residue}$$

This corresponds to an ethanol loss of 0.5 kg/day. Larger reflux ratios would reduce this, but

the cost of additional steam will probably make them not worthwhile. Note that this is a matter of economic considerations and that an optimum reflux ratio exists. Smaller values of R , with corresponding reduced steam cost and larger ethanol loss, should be considered, but care must be taken to ensure vapor velocities above the weeping velocity, Eq. (6.46).

CONTINUOUS-CONTACT EQUIPMENT (PACKED TOWERS)

Towers filled with the various types of packings described in Chap. 6 are frequently competitive in cost with trays, and they are particularly useful in cases where pressure drop must be low, as in low-pressure distillations, and where liquid holdup must be small, as in distillation of heat-sensitive materials whose exposure to high temperatures must be minimized. There are also available extremely effective packings for use in bench-scale work, capable of producing the equivalent of many stages in packed heights of only a few feet [40].

The Transfer Unit

As in the case of packed absorbers, the changes in concentration with height produced by these towers are continuous rather than stepwise as for tray towers, and the computation procedure must take this into consideration. Figure 9.49a shows a schematic drawing of a packed-tower fractionator. Like tray towers, it must be provided with a reboiler at the bottom (or open steam may be used if an aqueous residue is produced), a condenser, means of returning reflux and reboiled vapor, as well as means for introducing feed. The last can be accomplished by providing a short unpacked section at the feed entry, with adequate distribution of liquid over the top of the exhausting section (see Chap. 6).

The operating diagram, Fig. 9.49b, is determined exactly as for tray towers, using either the Ponchon-Savarit method or, where applicable, the McCabe-Thiele simplifying assumptions. Equations for operating lines and enthalpy balances previously derived for trays are all directly applicable, except that tray-number subscripts can be omitted. The operating lines are then simply the relation between x and y , the bulk liquid and gas compositions, prevailing at each horizontal section of the tower. As before, the change from enriching- to exhausting-section operating lines is made at the point where the feed is actually introduced, and for new designs a shorter column results, for a given reflux ratio, if this is done at the intersection of the operating lines. In what follows, this practice is assumed.

For packed towers, rates of flow are based on unit tower cross-sectional area, $\text{mol}/(\text{area})(\text{time})$. As for absorbers, in the differential volume dZ , Fig. 9.49a, the interface surface is $a dZ$, where a is the a_i of Chap. 6. The quantity of substance A in the vapor passing through the differential section is Gy $\text{mol}/(\text{area})(\text{time})$, and the rate of mass transfer is $d(Gy)$ $\text{mol A}/(\text{differential volume})(\text{time})$. Similarly, the rate of mass transfer is $d(Lx)$. Even where the usual simplifying assumptions are not strictly applicable, within a section of the column G and L are both sufficiently constant for equimolar counterdiffusion

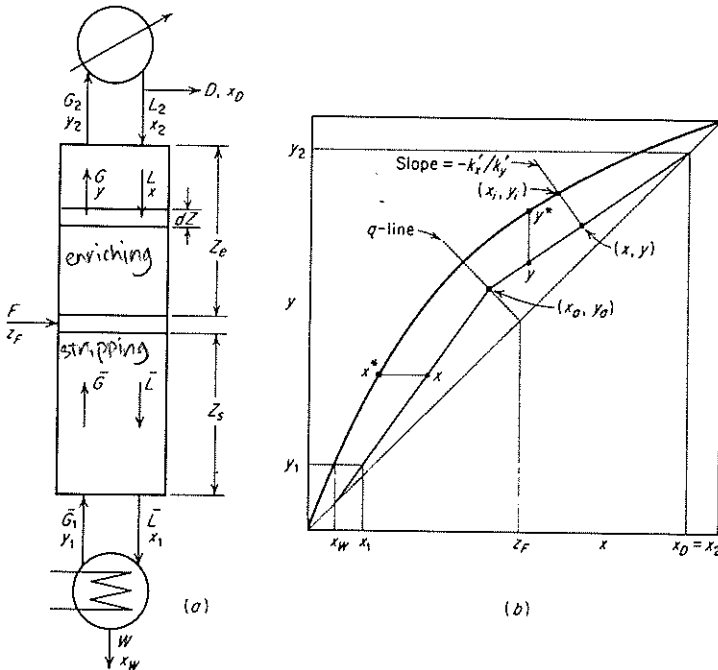


Figure 9.49 Fractionation in a packed tower.

between phases to be practically true: $N_A = -N_B$. Consequently (see Table 3.1), $F_G = k'_y$, $F_L = k'_x$, and the mass-transfer flux is

$$N_A = \frac{d(Gy)}{a dZ} = \underbrace{\left(\frac{k'_y}{a} \right)}_{\text{mass transfer area}} (y_i - y) = \frac{d(Lx)}{a dZ} = \underbrace{\left(\frac{k'_x}{a} \right)}_{\text{dealing with equimolar counter diffusion}} (x - x_i) \quad (9.151)$$

$$\text{Therefore } Z_e = \int_0^{Z_e} dZ = \int_{(Gy)_a}^{(Gy)_2} \frac{d(Gy)}{k'_y a (y_i - y)} = \int_{(Lx)_a}^{(Lx)_2} \frac{d(Lx)}{k'_x a (x - x_i)} \quad (9.152)$$

A similar expression, with appropriate integration limits, applies to the stripping section.

For any point (x, y) on the operating line, the corresponding point (x_i, y_i) on the equilibrium curve is obtained at the intersection with a line of slope $-k'_x/k'_y = -k'_x a / k'_y a$ drawn from (x, y) , as shown in Fig. 9.49b. For $k'_x > k'_y$, so that the principal mass-transfer resistance lies within the vapor, $y_i - y$ is more accurately read than $x - x_i$. The middle integral of Eq. (9.152) is then best used, evaluated graphically as the area under a curve of $1/k'_y a (y_i - y)$ as ordinate, Gy as abscissa, within the appropriate limits. For $k'_x < k'_y$, it is better to use the last

integral. In this manner, variations in G , L , the coefficients, and the interfacial area with location on the operating lines are readily dealt with.

For cases where the usual simplifying assumptions apply, G and L within any section of the tower are both constant, and the heights of transfer units

$$H_{IG} = \frac{G}{k'_y a} \quad H_{IL} = \frac{L}{k'_x a} \quad (9.153)$$

are sometimes sufficiently constant (or else average values for the section can be used), so that Eq. (9.152) can be written

$$Z_e = H_{IG} \int_{y_a}^{y_2} \frac{dy}{y_i - y} = H_{IG} N_{IG} \quad (9.154)$$

$$Z_e = H_{IL} \int_{x_a}^{x_2} \frac{dx}{x - x_i} = H_{IL} N_{IL} \quad (9.155)$$

with similar expressions for Z_e . The numbers of transfer units N_{IG} and N_{IL} are given by the integrals of Eqs. (9.154) and (9.155). It should be kept in mind, however, that the interfacial area a and the mass-transfer coefficients depend upon the mass rates of flow, which, because of changing average molecular weights with concentration, may vary considerably even if the molar rates of flow are constant. The constancy of H_{IG} and H_{IL} should therefore not be assumed without check.

Ordinarily the equilibrium curve for any section of the tower varies in slope sufficiently to prevent overall mass-transfer coefficients and heights of transfer units from being used. If the curve is essentially straight, however, we can write

$$Z_e = H_{IOG} \int_{y_a}^{y_2} \frac{dy}{y^* - y} = H_{IOG} N_{IOG} \quad (9.156)$$

$$Z_e = H_{IOL} \int_{x_a}^{x_2} \frac{dx}{x - x^*} = H_{IOL} N_{IOL} \quad (9.157)$$

where

$$H_{IOG} = \frac{G}{K'_y a} \quad H_{IOL} = \frac{L}{K'_x a} \quad (9.158)$$

Here, $y^* - y$ is an overall *driving force* in terms of vapor compositions, and $x - x^*$ is a similar one for the liquid (see Fig. 9.49b). For such cases, Eqs. (8.48), (8.50), and (8.51) can be used to determine the number of overall transfer units without graphical integration. As shown in Chap. 5, with F 's equal to the corresponding k 's,

$$\frac{1}{K'_y} = \frac{1}{k'_y} + \frac{m}{k'_x} \quad (9.159)$$

$$\frac{1}{K'_x} = \frac{1}{mk'_y} + \frac{1}{k'_x} \quad (9.160)$$

$$H_{IOG} = H_{IG} + \frac{mG}{L} H_{IL} \quad (9.161)$$

$$H_{IOL} = H_{IL} + \frac{L}{mG} H_{IG} \quad (9.162)$$

where, in Eqs. (5.27) and (5.28), $m = m' = m'' =$ slope of a straight equilibrium curve.

Although practically all the meaningful data on mass-transfer coefficients in packings were obtained at relatively low temperatures, the limited evidence is that they can be used for distillation as well, where the temperatures are normally relatively high. As mentioned in Chap. 6, departure has been noted where surface-tension changes with concentration occur. In the case of *surface-tension-positive systems*, where the more volatile component has the lower surface tension, the surface tension increases down the packing, the film of liquid in the packing becomes more stable, and mass-transfer rates become larger. For *surface-tension-negative systems*, where the opposite surface-tension change occurs, results are as yet unpredictable. Research in this area is badly needed.

Illustration 9.12 Determine suitable dimensions of packed sections of a tower for the methanol-water separation of Illustration 9.8, using 38-mm (1.5-in) Raschig rings.

SOLUTION Vapor and liquid quantities throughout the tower, calculated in the manner of Illustration 9.8, with Eqs. (9.62), (9.64), (9.72), and (9.74), are:

x	$t_L, ^\circ\text{C}$	y	$t_G, ^\circ\text{C}$	Vapor		Liquid	
				kmol/h	kg/h	kmol/h	kg/h
Enriching section							
0.915	66	0.915	68.2	171.3	5303	86.7	2723
0.600	71	0.762	74.3	164.0	4684	79.6	2104
0.370	76	0.656	78.7	160.9	4378	76.5	1779
Stripping section							
0.370	76	0.656	78.7	168.6	4585	301	7000
0.200	82	0.360	89.7	161.6	3721	294	6138
0.100	87	0.178	94.7	160.6	3296	293	5690
0.02	96.3	0.032	99.3	127.6	2360	260	4767

The x and y values are those on the operating line (Fig. 9.28). The temperatures are bubble and dew points for the liquids and vapors, respectively. The operating lines intersect at $x = 0.370$, the dividing point between enriching and stripping sections.

The tower diameter will be set by the conditions at the top of the stripping section because of the large liquid flow at this point. Here (Chap. 8 notation),

$$\rho_G = \frac{4585}{168.6(22.41)} \frac{273}{273 + 78.7} = 0.942 \text{ kg/m}^3 \quad \rho_L = 905 \text{ kg/m}^3$$

$$\frac{L'}{G'} \left(\frac{\rho_G}{\rho_L} \right)^{0.5} = \frac{7000}{4185} \left(\frac{0.942}{905} \right)^{0.5} = 0.054$$

In Fig. 6.34, choose a gas-pressure drop of 450 (N/m²)/m, whence the ordinate is 0.0825. Table 6.3: $C_f = 95$, $\mu_L = 4.5 \times 10^{-4}$ kg/m · s, $J = 1$, $g_c = 1$.

$$G' = \left[\frac{0.0825(0.942)(905 - 0.942)}{95(4.5 \times 10^{-4})^{0.1}} \right]^{0.5} = 1.264 \text{ kg/m}^2 \cdot \text{s}$$

The tower cross-sectional area = $4185/3600(1.264) = 0.92 \text{ m}^2$, corresponding to a diameter of 1.1 m . $L' = 7000/3600(0.92) = 1.94 \text{ kg/m}^2 \cdot \text{s}$.

Mass-transfer coefficients will be computed for the same location. Table 6.4: $m = 36.5$, $n = -4.69 \times 10^{-3}$, $p = 0.274$, $a_{AW} = 42.36 \text{ m}^2/\text{m}^3$. Table 6.5: $d_t = 0.0530 \text{ m}$, $\beta = 0.499$, $\varphi_{LW} = 0.0086$, $\varphi_{LW} = 0.0279$, $\varphi_{LW} = 0.0193$, $\varphi_{LW} = 0.00362$ (with $\mu_L = 0.00045 \text{ kg/m} \cdot \text{s}$, $\sigma = 0.029 \text{ N/m}$), $H = 0.91$, $\varphi_{LW} = 0.0175$, $\varphi_{LW} = 0.0211$.

Eq. (6.73): $a_A = 38.0 \text{ m}^2/\text{m}^3 = a$. Table 6.3: $\epsilon = 0.71$; Eq. (6.71): $\epsilon_0 = 0.71 - 0.0211 = 0.689$. By the methods of Chap. 2, $Sc_G = 1.0$. $G = 0.0466 \text{ kmol/m}^2 \cdot \text{s}$, $\mu_G = 2.96 \times 10^{-5} \text{ kg/m} \cdot \text{s}$, and by Eq. (6.70), $F_G = K'_y = 1.64 \times 10^{-3} \text{ kmol/m}^2 \cdot \text{s} \cdot (\text{mole fraction})$.

$$D_L = 4.80 \times 10^{-9} \text{ m}^2/\text{s} \quad Sc_L = 103.5$$

Eq. (6.72): $k_L = 2.66 \times 10^{-4} \text{ kmol/m}^2 \cdot \text{s} \cdot (\text{kmol/m}^3)$. Since the molar density of water at 76°C is 53.83 kmol/m^3 , $F_L = k'_x = 53.82(2.66 \times 10^{-4}) = 0.0143 \text{ kmol/m}^2 \cdot \text{s} \cdot (\text{mole fraction})$.

Similarly values at other concentrations are:

x	G	a	$k'_y \times 10^3$	k'_x
0.915	0.0474	20.18	1.525	0.01055
0.600	0.0454	21.56	1.542	0.00865
0.370	0.0444	21.92	1.545	0.00776
0.370	0.0466	38.00	1.640	0.0143
0.200	0.0447	32.82	1.692	0.0149
0.100	0.0443	31.99	1.766	0.0146
0.02	0.0352	22.25	1.586	0.0150

Figure 9.50 shows the equilibrium curve and operating curves as determined in Illustration 9.8. At $(x = 0.2, y = 0.36)$ on the operating line, line AB was drawn with slope = $-k'_x/k'_y = 0.0149/(1.692 \times 10^{-3}) = -8.8$. Point B then provides the value of y_1 for $y = 0.36$. Similarly the lines were erected at each of the x values of the table. Points such as C and D are joined by the curve CD . Then, at any point M , the corresponding y_1 at N is easily located. The enlarged inset of the figure shows y_1 for the vapor entering the packed section to be in equilibrium with the reboiler liquid x_W . Since $k'_x > k'_y$, the middle integral of Eq. (9.152) will be used. The following values of y_1 were determined in the manner described, with values of G , k'_y , and a obtained from a plot of these against y .

y	y_1	$\frac{1}{k'_y a (y_1 - y)}$	Gy
$y_2 = 0.915$	0.960	634.0	0.0433 = $(Gy)_2$
0.85	0.906	532.8	0.0394
0.80	0.862	481.1	0.0366
0.70	0.760	499.1	0.0314
$y_a = 0.656$	0.702	786.9	0.0292 = $(Gy)_a$
$y_a = 0.656$	0.707	314.7	0.0306 = $(Gy)_a$
0.50	0.639	124.6	0.0225
0.40	0.580	99.6	0.01787
0.30	0.500	89.0	0.01340
0.20	0.390	92.6	0.00888
0.10	0.230	154.5	0.00416
$y_1 = 0.032$	0.091	481.0	0.001124 = $(Gy)_1$

Graphical integration (not shown) according to Eq. (9.152) provides $Z_e = 7.53 \text{ m}$ for the enriching section, $Z_s = 4.54 \text{ m}$ for the stripping section.

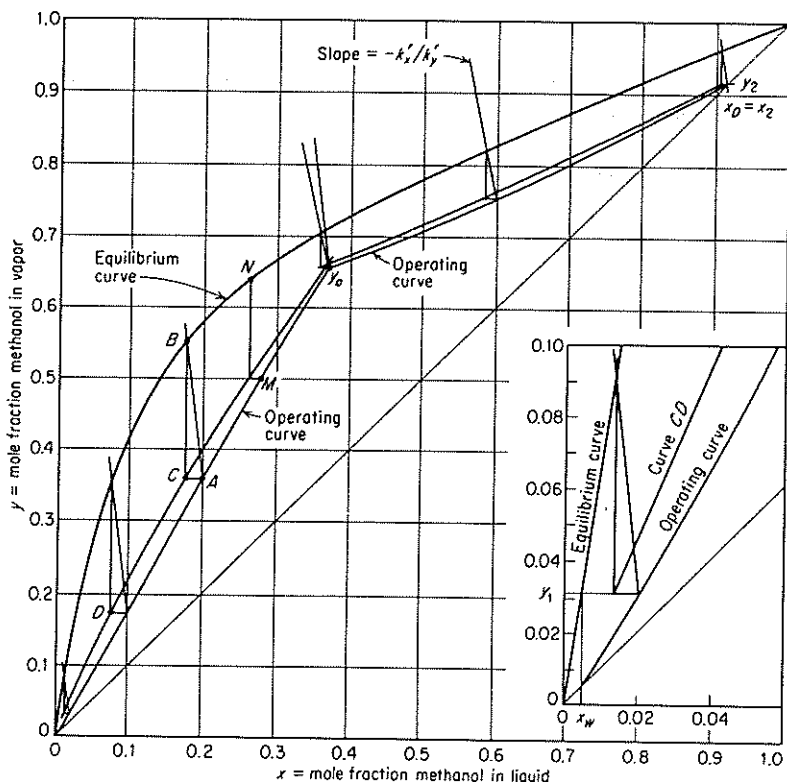


Figure 9.50 Solution to Illustration 9.12.

Berl saddles of the same size, because they provide substantially larger interfacial area, led to packed depths equal to roughly six-tenths of these values, with approximately a 30 percent reduction in gas-pressure drop per meter, at the same mass velocities (although in this instance pressure drop is unimportant). A choice between such packings as these, Pall rings, Intalox saddles, and the like then revolves about their relative cost, not per cubic meter but for the complete tower.

In the above example, H_{IG} for the stripping section varies from $0.0466/(1.640 \times 10^{-3})(38.0) = 0.748$ m at the top to 0.997 m at the bottom. Equation (9.154) with an average $H_{IG} = 0.873$ yields $Z_s = 4.66$ m rather than 4.54, as first calculated. The equilibrium-curve slope varies so greatly that the use of H_{IOG} or overall mass-transfer coefficients is not recommended.

MULTICOMPONENT SYSTEMS

Many of the distillations of industry involve more than two components. While the principles established for binary solutions generally apply to such distillations, new problems of design are introduced which require special consideration.

Consider the continuous separation of a ternary solution consisting of components A, B, and C whose relative volatilities are in that order (A most volatile). In order to obtain the three substances in substantially pure form, either of the schemes of Fig. 9.51 can be used. According to scheme (a), the first column is used to separate C as a residue from the rest of the solution. The residue is necessarily contaminated with a small amount of B and with an even smaller amount of A, although if relative volatilities are reasonably large, the amount of the latter may be exceedingly small. The distillate, which is necessarily contaminated with at least a small amount of C, is then fractionated in a

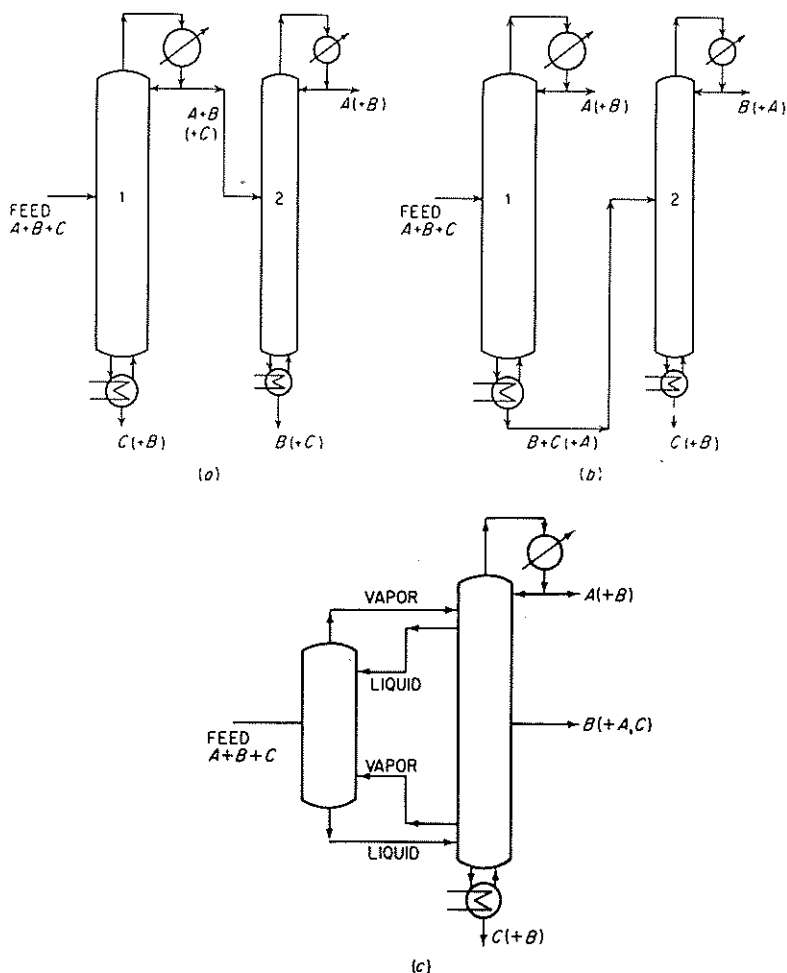


Figure 9.51 Separation of a ternary system.

second column to provide nearly pure A and B. According to scheme (b), the first tower provides nearly pure A directly, and the residue is separated in the second. Still a third scheme is shown in (c). Which of the schemes is used is a matter of cost, and that may depend upon the relative difficulties of the separations [38]. Generally, scheme (b) will be more economical than scheme (a) since it requires only one vaporization of substance A. Scheme (c) will generally be more economical than either of the others [60].

An important principle to be emphasized is that a single fractionator cannot separate more than one component in reasonably pure form from a multicomponent solution and that a total of $n - 1$ fractionators will be required for complete separation of a system of n components. It might at first be thought, for example, that the component of intermediate volatility B would tend to concentrate in reasonably pure form in the central parts of the first tower, from which it might be withdrawn as a *side stream*, thus allowing pure A and pure C to be withdrawn as distillate and residue, respectively. But this cannot occur. The feed tray of column 1, scheme (a), for example, will necessarily contain all three components in proportions not far from those prevailing in the feed itself. Trays immediately above the feed will therefore also contain appreciable quantities of all three substances, with the proportion of C gradually diminishing as we go higher in the enriching section. Similarly, trays immediately below the feed necessarily contain large proportions of all substances, with decreasing amounts of A and B as we penetrate more deeply into the exhausting section. While side streams are indeed sometimes withdrawn from fractionating towers, these streams must be further processed if they are to provide pure products.

The general principles of design of multicomponent fractionators are the same in many respects as those for binary systems, but the dearth of adequate vapor-liquid equilibrium data imposes severe restrictions on their application. These are especially needed for liquids which are not ideal, and the danger of attempting new designs without adequate equilibrium data or pilot-plant study for such solutions cannot be overemphasized. Inadequate methods of dealing with tray efficiencies for multicomponents represent another serious problem still to be solved.

Since the design calculations involve extensive trial and error, high-speed computers are frequently used [1, 24]. Except in extraordinary cases, it is not impossible to carry them out by hand, however, and this may be necessary when only a few designs are to be made. In what follows, hand calculations are assumed and will in any case provide an introduction to the computer methods. The most reliable design procedure is that of Thiele and Geddes [62]. The method assumes that, for a given feed, the number of trays, position of the feed tray, and the liquid/vapor ratio and temperature for each tray are all known at the start, and it proceeds to compute the resulting distillate and residue products. In most cases, of course, at the beginning of a computation the necessary quantities are ordinarily unknown. The scheme outlined here begins by developing, with a minimum of trials, information necessary to use the Thiele-Geddes method, which then provides the final design. High-speed computer techniques have been developed by grouping equations by equilibrium stages and solving

iteratively state to stage, based on the methods of Lewis and Matheson and of Thiele and Geddes. More recently, especially for cases of multiple feeds, a scheme of grouping the equations for each component has been developed.

Specification Limitations

It will be assumed at the beginning that at least the following are established:

1. Temperature, pressure, composition, and rate of flow of the feed
2. Pressure of the distillation (frequently fixed by the temperature of the available cooling water, which must be able to condense the distillate vapor to provide reflux)
3. Feed to be introduced on that tray which will result in the least total number of trays (optimum feed-tray location)
4. Heat losses (even if assumed to be zero)

Under these circumstances it has been shown that there are left to the designer only three additional items to be specified. Having chosen the three, all other characteristics of the fractionator are fixed. The designer can only calculate what they will be, and can arbitrarily assign values to them only temporarily for purposes of trial calculations and later verification. The three can be chosen from the following list, each item counting one:

1. Total number of trays
2. Reflux ratio
3. Reboil ratio, i.e., ratio of vapor produced by the reboiler to residue withdrawn
4. Concentration of one component in one product (a maximum of two may be chosen)
5. Ratio of flow of one component in the distillate to the flow of the same component in the residue, or *split* of the component (a maximum of two may be chosen)
6. Ratio of total distillate to total residue

In what follows, it will be assumed that the reflux ratio and splits of two components are specified. Modifications of the procedures are readily made for other cases. It is clear that, with more than two components in the feed, neither the complete compositions nor the rates of flow of either product are then known.

Key Components

It is convenient first to list the feed components in order of their relative volatility. The more volatile components are called *light*, the less volatile are *heavy*. There will frequently be one component, the *light key component*, which is

present in the residue in important amounts while components lighter than the light key are present in only very small amounts. If all components are present in the residue at important concentrations, then the most volatile is the light key. Similarly there will usually be one component, the *heavy key component*, present in the distillate in important amounts while components heavier than the heavy key are present only in small amounts. If all components are found in the distillate at important concentrations, then the least volatile is the heavy key.

The difficulty of the separation, as measured by the number of trays required for a given reflux ratio, is fixed by the key-component concentrations in the products. It is therefore important to establish which are the keys, and they may or may not be those whose splits have been specified.

Relative volatilities will be computed always with respect to the heavy key

$$\alpha_J = \frac{m_J}{m_{hk}} \quad (9.163)$$

where J represents any component and hk the heavy key. Thus, $\alpha_{hk} = 1$, α 's for components lighter than the heavy key are greater than 1 and for heavier components are less than 1.

Minimum Reflux Ratio

This is the largest reflux ratio requiring an infinite number of trays to separate the key components.

With an infinite number of trays, it is possible to exclude from the distillate all component heavier than the heavy key. Since all components are present at the feed tray, and since it requires several enriching trays to reduce these heavy components to negligible concentrations, the pinch for these components, above which their concentration is zero, lies somewhere above the feed tray. Similarly an infinite number of trays permits exclusion from the residue of all components lighter than the light key, and there is a pinch for these several trays below the feed tray. The situation is therefore different from that for binary distillation, where at minimum reflux ratio there is only one pinch, usually at the feed tray.

Because of the possibility of excluding components from the products, computations at minimum reflux ratio help decide which are the key components. Components between the keys in volatility are found importantly in both products, and they are said to *distribute*. Shiras et al. [54] show that at minimum reflux ratio, approximately

$$\frac{x_{J,D}D}{z_{J,F}F} = \frac{\alpha_J - 1}{\alpha_{lk} - 1} \frac{x_{lk,D}D}{z_{lk,F}F} + \frac{\alpha_{lk} - \alpha_J}{\alpha_{lk} - 1} \frac{x_{hk,D}D}{z_{hk,F}F} \quad (9.164)$$

For $x_{J,D}D/z_{J,F}F$ less than -0.01 or greater than 1.01 , component J will probably not distribute. For $x_{J,D}D/z_{J,F}F$ between 0.01 and 0.99 , component J will undoubtedly distribute. The computations are made with first estimates of what the keys are, to be corrected as necessary.

Many methods of estimating the minimum reflux ratio R_m have been proposed, most of which are tedious to use and not necessarily very accurate. Since the only purpose of obtaining R_m is to estimate the product compositions at R_m and to ensure that the specified R is reasonable, an exact value is not required. Underwood's method [64], which uses constant average α 's and assumes constant L/G , is not exact but provides reasonable values without great effort; it is recommended (refer to the original papers for a derivation, which is lengthy). Two equations must be solved:

$$\sum \frac{\alpha_J z_{JF} F}{\alpha_J - \phi} = F(1 - q) \quad (9.165)$$

and

$$\sum \frac{\alpha_J x_{JD} D}{\alpha_J - \phi} = D(R_m + 1) \quad (9.166)$$

The first of these is written for all J components of the feed and solved for the necessary values of ϕ . These are the values which lie between the α 's of the distributed components. We thus require one more value of ϕ than there are components between the keys, and these will lie between α_{hk} and $\alpha_{lk} = 1$. Equation (9.166) is then written once for each value of ϕ so obtained, including the heavy key and all lighter components in the summations. These are solved simultaneously for R_m and the unknown $x_{J,D}$'s. If $x_{J,D}$ computes to be negative or greater than $z_{J,F}$, component J will not distribute and the keys have been incorrectly chosen.

The α 's are all computed at the average of the distillate dew point and the residue bubble point, which may require a few trial estimates. The method has been extended to columns with multiple feeds [2].

Illustration 9.13 The following feed, at 82°C (180°F), 1035 kN/m² (150 lb_f/in²), is to be fractionated at that pressure so that the vapor distillate contains 98% of the C₃H₈ but only 1% of the C₅H₁₂:

Component	CH ₄	C ₂ H ₆	<i>n</i> -C ₃ H ₈	<i>n</i> -C ₄ H ₁₀	<i>n</i> -C ₅ H ₁₂	<i>n</i> -C ₆ H ₁₄
z_F , mole fraction	0.03	0.07	0.15	0.33	0.30	0.12

Estimate the minimum reflux ratio and the corresponding products.

SOLUTION The components will be identified as C₁, C₂, etc. Subscript numerals on m 's, α 's, etc., are Celsius temperatures if larger than 25; otherwise they represent tray numbers. Liquid solutions are assumed to be ideal. Values of m are taken from Depriester, *Chem. Eng. Prog. Symp. Ser.*, 49(7), 1 (1953). Values of H , all as kJ/kmol referred to the saturated liquid at -129°C (-200°F), are taken from Maxwell, "Data Book on Hydrocarbons," D. Van Nostrand Company, Inc., Princeton, N.J., 1950. For H_G for temperatures below the dew point, the enthalpy is that of the saturated vapor. For H_L for temperatures above the critical, the

enthalpy is that of the gas in solution. Values of m are conveniently plotted as $\log m$ vs. $t^\circ\text{C}$ for interpolation. Enthalpy can be plotted vs. t on arithmetic coordinates.

	30°C	60°C	90°C	120°C
C_1 :				
m	16.1	19.3	21.8	24.0
H_G	12 790	13 910	15 000	16 240
H_L	9 770	11 160	12 790	14 370
C_2 :				
m	3.45	4.90	6.25	8.15
H_G	22 440	24 300	26 240	28 140
H_L	16 280	18 140	19 890	21 630
C_3 :				
m	1.10	2.00	2.90	4.00
H_G	31 170	33 000	35 800	39 000
H_L	16 510	20 590	25 600	30 900
C_4 :				
m	0.35	0.70	1.16	1.78
H_G	41 200	43 850	46 500	50 400
H_L	20 350	25 120	30 000	35 400
C_5 :				
m	0.085	0.26	0.50	0.84
H_G	50 500	54 000	57 800	61 200
H_L	24 200	32 450	35 600	41 400
C_6 :				
m	0.030	0.130	0.239	0.448
H_G	58 800	63 500	68 150	72 700
H_L	27 700	34 200	40 900	48 150

Basis: 1 kmol feed throughout.

Flash vaporization of the feed Use Eq. (9.37). After several trials, assume $G_F/F = 0.333$, $L_F/G_F = 0.667/0.333 = 2.0$.

Component	z_F	m_{82}	$y = \frac{z_F(2 + 1)}{1 + 2/m}$
C_1	0.03	21.0	0.0829
C_2	0.07	5.90	0.1578
C_3	0.15	2.56	0.2530
C_4	0.33	1.00	0.3300
C_5	0.30	0.42	0.1559
C_6	0.12	0.19	0.0312
	1.00		1.0108

Σy is sufficiently close to 1.0. Therefore $q = L_F/F = 0.67$. Tentatively assume C_3 = light key, C_5 = heavy key. Therefore specifications require $x_{lkD}D = 0.98(0.15) = 0.1470$ kmol, $x_{hkD}D = 0.01(0.30) = 0.0030$ kmol. Estimate the average temperature to be 80°C (to be checked).

Component	$z_F F$	m_{80}	α_{80}
C_1	0.03	21.0	$53.2 = 21.0/0.395$
C_2	0.07	5.90	14.94
lk C_3	0.15	2.49	6.30
C_4	0.33	0.95	2.405
hk C_5	0.30	0.395	1.0
C_6	0.12	0.180	0.456

Equation (9.164) is to be used for C_1 , C_2 , and C_6 ; the others distribute at R_m . Use $y_{1,D}$ instead of $x_{1,D}$ since distillate is a vapor.

$$C_1: \quad \frac{y_D D}{z_F F} = \frac{53.2 - 1}{6.30 - 1} \frac{0.1470}{0.15} + \frac{6.30 - 53.2}{6.30 - 1} \frac{0.0030}{0.30} = 9.761$$

Similarly, for C_2 and C_6 the values are 2.744 and -0.0892 , respectively. None of these components will distribute at R_m , and the chosen keys are correct. The distillate contains 0.03 kmol C_1 , 0.07 kmol C_2 , and no C_6 . The distribution of only C_4 to the products is unknown. Equation (9.165):

$$\frac{53.2(0.03)}{53.2 - \phi} + \frac{14.94(0.07)}{14.94 - \phi} + \frac{6.30(0.15)}{6.30 - \phi} + \frac{2.405(0.33)}{2.405 - \phi} + \frac{1.0(0.30)}{1.0 - \phi} + \frac{0.456(0.12)}{0.456 - \phi} = 1(1 - 0.67)$$

This is to be solved for the two values of ϕ lying between α_{C_3} and α_{C_4} , and α_{C_4} and α_{C_5} . Therefore, $\phi = 4.7760$ and 1.4177 . Equation (6.166) is then written for each value of ϕ and for components C_1 through C_5 . For $\phi = 4.776$,

$$\frac{53.2(0.03)}{53.2 - 4.776} + \frac{14.94(0.07)}{14.94 - 4.776} + \frac{6.30(0.1470)}{6.30 - 4.776} + \frac{2.405(y_{C_4D}D)}{2.405 - 4.776} + \frac{1.0(0.0030)}{1.0 - 4.776} = D(R_m + 1)$$

This is repeated with $\phi = 1.4177$, and the two solved simultaneously. $y_{C_4D}D = 0.1306$ kmol C_4 in the distillate, $D(R_m + 1) = 0.6099$. For the distillate, try a dew point = 46°C .

Component	$y_D D$	y_D	m_{46}	α_{46}	$\frac{y_D}{\alpha_{46}}$
C_1	0.03	0.0789	18.0	100	0.000789
C_2	0.07	0.1840	4.2	23.3	0.00789
lk C_3	0.1470	0.3861	1.50	8.34	0.0463
C_4	0.1306	0.3431	0.500	2.78	0.1234
hk C_5	0.0030	0.0079	0.180	1.0	0.0079
$D = 0.3806$		1.0000			0.1863

$m_{hk} = m_{C_5} = 0.1863$ at 46.6°C , and the assumed 46°C is satisfactory. For the residue, try a

bubble point = 113°C. The amount of residue is obtained from $x_{J,W}W = z_{J,F}F - y_{J,D}D$:

Component	$x_{J,W}$	$x_{J,D}$	m_{113}	α_{113}	$\alpha_{113}x_{J,W}$
<i>lk</i> C ₃	0.0030	0.00484	3.65	5.00	0.0242
C ₄	0.1994	0.3219	1.60	2.19	0.7050
<i>hk</i> C ₅	0.2970	0.4795	0.730	1.00	0.4800
C ₆	0.1200	0.1937	0.380	0.521	0.1010
	$W = 0.6194$	1.0000			1.3102

$m_{hk} = m_{C_5} = 1/1.3102 = 0.763$ at 114°C, and the assumed 113°C was close enough.

Average temperature = $(114 + 46.6)/2 = 80.3^\circ\text{C}$ (assumed 80°C is close enough). Therefore,

$$D(R_m + 1) = 0.3806(R_m + 1) = 0.6099$$

$$R_m = 0.58 \text{ mol reflux/mol distillate}$$

Total Reflux

The product compositions change with reflux ratio, and computations at total reflux will help decide the ultimate compositions.

Fenske's equation (9.85) is not limited to binary mixtures and can be applied to the key components to determine the minimum number of trays,

$$N_m + 1 = \frac{\log[(x_{lkD}/x_{hkD})(x_{hkW}/x_{lkW})]}{\log \alpha_{lk,av}} \quad (9.167)$$

where $N_m + 1$ is the total number of theoretical stages including the reboiler (and a partial condenser if credit is taken for its fractionating ability). The equation can then be applied to determine the distribution of other components at total reflux,

$$\frac{x_{J,D}D}{x_{J,W}W} = \alpha_{J,av}^{N_m+1} \frac{x_{hkD}}{x_{hkW}} \quad (9.168)$$

The average α is the geometric mean of the values at the distillate dew point and residue bubble point, which may require a few trials to estimate. Winn [68] suggests a method to reduce the number of trials required.

Having obtained N_m and R_m , reference can be made to any of the several empirical correlations mentioned earlier [5, 10, 11, 17, 35] for an estimate of the number of trays at reflux ratio R . These can be unreliable, however, particularly if the majority of the trays are in the exhausting section of the tower. A relationship which is exact for binary mixtures and can be applied to multicomponents yields better results for that case [59]. The result of such an estimate can be a reasonable basis for proceeding directly to the method of Thiele and Geddes.

Illustration 9.13 continued Compute the number of theoretical trays at local reflux and the corresponding products.

SOLUTION Tentatively assume the same distillate and residue temperatures as obtained for the minimum reflux ratio.

Component	$\alpha_{46.6}$	α_{114}	$\alpha_{av} = (\alpha_{46.6}\alpha_{114})^{0.5}$
C_1	100	31.9	56.4
C_2	23.3	10.43	15.6
$lk\ C_3$	8.34	5.00	6.45
C_4	2.78	2.19	2.465
$hk\ C_5$	1.0	1.0	1.0
C_6	0.415	0.521	0.465

Eq. (9.167):

$$N_m + 1 = \frac{\log[(0.147/0.003)(0.297/0.003)]}{\log 6.45} = 4.55$$

Eq. (9.168) for C_4 :

$$\frac{y_{C_4,D}D}{x_{C_4,W}W} = (2.465)^{4.55} \frac{0.003}{0.297} = 0.611$$

A C_4 balance:

$$y_{C_4,D}D + x_{C_4,W}W = z_{C_4,F}F = 0.33$$

Solving simultaneously gives

$$y_{C_4,D}D = 0.1255 \text{ kmol} \quad x_{C_4,W}W = 0.2045 \text{ kmol}$$

Similarly,

$$C_1: \quad x_{C_1,W}W \approx 0 \quad y_{C_1,D}D = 0.03$$

$$C_2: \quad x_{C_2,W}W \approx 0 \quad y_{C_2,D}D = 0.07$$

$$C_6: \quad y_{C_6,D}D = 0.00003 \text{ (negligible; assume 0)} \quad x_{C_6,W}W = 0.12$$

Therefore, at total reflux,

Component	$y_D D$	$x_W W$
C_1	0.03	Nil
C_2	0.07	Nil
C_3	0.1470	0.0030
C_4	0.1255	0.2045
C_5	0.0030	0.2970
C_6	Nil	0.12
	<u>0.3755 = D</u>	<u>0.6245 = W</u>

The distillate dew point computes to be 46.3°C and the residue bubble point 113°C. The assumed 46 and 114°C are close enough.

Product Compositions

For components between the keys, a reasonable estimate of their distribution at reflux ratio R can be obtained by linear interpolation of $x_{j,D}D$ between R_m and

total reflux according to $R/(R + 1)$. For components lighter than the light key and heavier than the heavy key, unless there is clear indication of important distribution at total reflux and unless R is to be very large, it is best at this time to assume they do not distribute. Even very small amounts of light components in the residue or of heavy components in the distillate enormously disturb the computations to follow.

Illustration 9.13 continued A reflux ratio $R = 0.8$ will be used. Estimate the product compositions at this reflux ratio.

SOLUTION Since C_1 and C_2 do not enter the residue (nor C_6 the distillate) appreciably at either total reflux or minimum reflux ratio, it is assumed that they will not at $R = 0.8$. C_3 and C_5 distributions are fixed by the specifications. Only that of C_4 remains to be estimated.

R	$\frac{R}{R + 1}$	y_{C_4D}
∞	1.0	0.1255
0.8	0.445	?
0.58	0.367	0.1306

Using a linear interpolation, we get

$$y_{C_4D} = \frac{1 - 0.445}{1 - 0.367} (0.1306 - 0.1255) + 0.1255 = 0.1300$$

Therefore, for the distillate,

Component	$y_D D$	y_D	$\frac{y_D}{\alpha_{46.6}}$	$x_0 = \frac{y_D/\alpha}{\sum y_D/\alpha}$
C_1	0.03	0.0789	0.00079	0.00425
C_2	0.07	0.1842	0.0079	0.0425
lk C_3	0.1470	0.3870	0.0464	0.2495
C_4	0.1300	0.3420	0.1230	0.6612
hk C_5	0.0030	0.0079	0.0079	0.0425
$D = 0.3800$		1.0000	0.1860	1.0000

$m_{C_3} = 0.1860$; $t = 46.6^\circ\text{C}$ distillate dew point. The last column of the table shows the liquid reflux in equilibrium with the distillate vapor [Eq. (9.30)]. For the residue,

Component	$x_W W$	x_W	$\alpha_{114} x_W$	$y_{N_r+1} = \frac{\alpha x_W}{\sum \alpha x_W}$
lk C_3	0.003	0.00484	0.0242	0.01855
C_4	0.200	0.3226	0.7060	0.5411
hk C_5	0.2970	0.4790	0.4790	0.3671
C_6	0.1200	0.1935	0.0955	0.0732
$W = 0.6200$		1.0000	1.3047	1.0000

$m_{C_3} = 1/1.3047 = 0.767$; $t = 114^\circ\text{C}$, residue bubble point. The last column shows the reboiler vapor in equilibrium with the residue [Eq. (9.25)].

Feed-Tray Location

Just as with binary mixtures, the change from enriching to exhausting section should be made as soon as greater enrichment is thereby produced. This ultimately can be determined only by trial in the Thiele-Geddes calculation, but in the meantime some guidance is required. The following assumes constant L/G , neglects interference of components other than the keys, and assumes that the optimum feed tray occurs at the intersection of operating lines of the key components [48].

Equation (9.116), the operating line for the enriching section, omitting the tray-number designations, can be written for each key component

$$x_{lk} = y_{lk} \frac{G}{L} - \frac{D}{L} x_{lkD} \quad (9.169)$$

$$x_{hk} = y_{hk} \frac{G}{L} - \frac{D}{L} x_{hkD} \quad (9.170)$$

Elimination of L provides

$$y_{lk} = \frac{x_{lk}}{x_{hk}} \left(y_{hk} - \frac{D}{G} x_{hkD} \right) + \frac{D}{G} x_{lkD} \quad (9.171)$$

Similarly, Eq. (9.120) for the exhausting section yields

$$y_{lk} = \frac{x_{lk}}{x_{hk}} \left(y_{hk} + \frac{W}{G} x_{hkW} \right) - \frac{W}{G} x_{lkW} \quad (9.172)$$

At the operating-line intersection, y_{lk} is the same from Eqs. (9.171) and (9.172) as are y_{hk} and x_{lk}/x_{hk} . Equating the right-hand sides of the two expressions then produces

$$\left(\frac{x_{lk}}{x_{hk}} \right)_{\text{intersection}} = \frac{W x_{lkW} / \bar{G} + D x_{lkD} / G}{W x_{hkW} / \bar{G} + D x_{hkD} / G} \quad (9.173)$$

Combining Eqs. (9.76) and (9.127) for the light key produces

$$\frac{W}{\bar{G}} x_{lkW} + \frac{D}{G} x_{lkD} = \frac{GFz_{lkF} - DF(1-q)x_{lkD}}{[G - F(1-q)]G} \quad (9.174)$$

and a similar result is obtained for the heavy key. Putting these into Eq. (9.173) then yields, with $G/D = R + 1$,

$$\left(\frac{x_{lk}}{x_{hk}} \right)_{\text{intersection}} = \frac{z_{lkF} - x_{lkD}(1-q)/(R+1)}{z_{hkF} - x_{hkD}(1-q)/(R+1)} \quad (9.175)$$

Recall from the treatment of binaries that the feed tray f is the uppermost step on the exhausting-section operating line and that it rarely happens that the feed step exactly coincides with the operating-line intersection. The feed-tray location is then given by

$$\left(\frac{x_{lk}}{x_{hk}} \right)_{f-1} \geq \left(\frac{x_{lk}}{x_{hk}} \right)_{\text{intersection}} \geq \left(\frac{x_{lk}}{x_{hk}} \right)_f \quad (9.176)$$

A method of choosing the feed tray based on minimizing the resulting irreversibility (or increasing the entropy) has been proposed [53]. Ultimately, the best feed-tray location is obtained through the Thiele-Geddes calculation.

Lewis and Matheson Calculation [31]

This establishes the first (and usually the final) estimate of the number of trays required. Although it is possible to allow for variations in the L/G ratio with tray number by tray-to-tray enthalpy balances at the same time, since the products are not yet firmly established, it is best to omit this refinement.

With constant L/G , the McCabe-Thiele operating lines, which are merely material balances, are applicable to each component. Thus, for the enriching section, Eq. (9.116) is

$$y_{j,n+1} = \frac{L}{G} x_{j,n} + \frac{D}{G} z_{j,D} \quad (9.177)$$

This is used alternately with dew-point (equilibrium) calculations to compute tray by tray for each component from the top down to the feed tray. For components heavier than the heavy key, there will be no $z_{j,D}$ available, and these components cannot be included.

For the exhausting section, it is convenient to solve Eq. (9.120) for x :

$$x_{j,m} = y_{j,m+1} \frac{\bar{G}}{L} + \frac{W}{L} x_{j,w} \quad (9.178)$$

This is used alternately with bubble-point (equilibrium) calculations to compute tray by tray for each component from the bottom up to the feed tray. There are no values of $x_{j,w}$ for components lighter than the light key, and they cannot be included.

If, in the first estimate of product compositions, it is found that one product contains all components (lk = most volatile of hk = least volatile component of the feed), the computation can be started with this product and continued past the feed tray to the other end of the column. The operating-line equations appropriate to each section must of course be used. This will avoid the composition corrections discussed later.

Illustration 9.13 continued Determine the number of theoretical trays and the location of the feed tray for $R = 0.8$.

SOLUTION For locating the feed tray, Eq. (9.175):

$$\left(\frac{x_{lk}}{x_{hk}} \right)_{\text{intersection}} = \frac{0.15 - 0.3870(1 - 0.67)/1.8}{0.30 - 0.0079(1 - 0.67)/1.8} = 0.264$$

Enriching section:

$$D = 0.3800 \quad L = RD = 0.8(0.3800) = 0.3040 \quad G = L + D = 0.3040 + 0.3800 = 0.6840$$

$$\frac{L}{G} = \frac{0.3040}{0.6840} = 0.445 \quad \frac{D}{G} = \frac{0.3800}{0.6840} = 0.555$$

Eq. (9.177):

$$C_1: \quad y_{n+1} = 0.445x_n + 0.555(0.0789) = 0.445x_n + 0.0438$$

Similarly,

$$C_2: y_{n+1} = 0.455x_n + 0.1022$$

$$C_3: y_{n+1} = 0.445x_n + 0.2148$$

$$C_4: y_{n+1} = 0.445x_n + 0.1898$$

$$C_5: y_{n+1} = 0.445x_n + 0.0044$$

Estimate $t_1 = 57^\circ\text{C}$.

Component	x_0	y_1 [Eq. (9.177), $n = 0$]	α_{57}	$\frac{y_1}{\alpha}$	$x_1 = \frac{y_1/\alpha}{\sum y_1/\alpha}$
C_1	0.00425	0.0457	79.1	0.00058	0.00226
C_2	0.0425	0.1211	19.6	0.00618	0.0241
$lk\ C_3$	0.2495	0.3259	7.50	0.0435	0.1698
C_4	0.6612	0.4840	2.66	0.1820	0.7104
$hk\ C_5$	0.0425	0.0233	1.00	0.0239	0.0933
	1.0000	1.0000		0.2562	1.0000

$m_{C_3} = 0.2562$, $t_1 = 58.4^\circ\text{C}$. Liquid x_1 's in equilibrium with y_1 . $(x_{lk}/x_{hk})_1 = 0.1698/0.0933 = 1.82$. Tray 1 is not the feed tray. Estimate $t_2 = 63^\circ\text{C}$.

Component	y_2 [Eq. (9.177), $n = 1$]	α_{63}	$\frac{y_2}{\alpha}$	$x_2 = \frac{y_2/\alpha}{\sum y_2/\alpha}$
C_1	0.0448	68.9	0.00065	0.00221
C_2	0.1129	17.85	0.00632	0.0214
$lk\ C_3$	0.2904	6.95	0.0418	0.1419
C_4	0.5060	2.53	0.02000	0.6787
$hk\ C_5$	0.0459	1.00	0.0459	0.1557
	1.0000		0.2947	1.0000

$m_{C_3} = 0.2947$, $t_2 = 65^\circ\text{C}$. $(x_{lk}/x_{hk})_2 = 0.1419/0.1557 = 0.910$. The tray calculations are continued downward in this manner. The results for trays 5 and 6 are:

Component	x_5	x_6
C_1	0.00210	0.00204
C_2	0.0195	0.0187
$lk\ C_3$	0.1125	0.1045
C_4	0.4800	0.4247
$hk\ C_5$	0.3859	0.4500
$t, ^\circ\text{C}$	75.4	79.2
x_{lk}/x_{hk}	0.292	0.232

Applying Eq. (9.176), it is seen that tray 6 is the feed tray. Exhausting section:

$$\bar{L} = L + qF = 0.3040 + 0.67(1) = 0.9740 \quad \bar{G} = \bar{L} - W = 0.9740 - 0.6200 = 0.3540$$

$$\frac{\bar{G}}{\bar{L}} = \frac{0.3540}{0.9740} = 0.364 \quad \frac{W}{\bar{L}} = \frac{0.6200}{0.9740} = 0.636$$

Eq. (9.178):

$$C_3: \quad x_m = 0.364y_{m+1} + 0.636(0.00484) = 0.364y_{m+1} + 0.00308$$

Similarly,

$$C_4: \quad x_m = 0.364y_{m+1} + 0.2052$$

$$C_5: \quad x_m = 0.364y_{m+1} + 0.3046$$

$$C_6: \quad x_m = 0.364y_{m+1} + 0.1231$$

Estimate $t_{N_p} = 110^\circ\text{C}$.

Component	y_{N_p+1}	x_{N_p} [Eq. (9.178), $m = N_p$]	α_{110}	αx_{N_p}	$y_{N_p} = \frac{\alpha x_{N_p}}{\sum \alpha x_{N_p}}$
<i>lk</i> C ₃	0.01855	0.00983	5.00	0.0492	0.0340
C ₄	0.5411	0.4023	2.20	0.885	0.6118
<i>hk</i> C ₅	0.3671	0.4382	1.0	0.4382	0.3028
C ₆	0.0732	0.1497	0.501	0.0750	0.0514
	1.0000	1.0000		1.4474	1.0000

$m_{C_3} = 1/1.447 = 0.691$, $t_{N_p} = 107.9^\circ\text{C}$. y_{N_p} in equilibrium with x_{N_p} : $(x_{lk}/x_{hk})_{N_p} = 0.00983/0.4382 = 0.0224$. N_p is not the feed tray. In like fashion, x_{N_p-1} is obtained from y_{N_p} with Eq. (9.178), and the computations continued up the tower. The results for trays $N_p - 7$ to $N_p - 9$ are:

Component	x_{N_p-7}	x_{N_p-8}	x_{N_p-9}
<i>lk</i> C ₃	0.0790	0.0915	0.1032
C ₄	0.3994	0.3897	0.3812
<i>hk</i> C ₅	0.3850	0.3826	0.3801
C ₆	0.1366	0.1362	0.1355
$t, ^\circ\text{C}$	95.2	94.1	93.6
x_{lk}/x_{hk}	0.205	0.239	0.272

Application of Eq. (9.176) shows tray $N_p - 8$ as the feed tray. The data for $N_p - 9$ are discarded. Then $N_p - 8 = 6$, and $N_p = 14$ trays.

Composition Corrections

The previous computations provide two feed-tray liquids, computed from opposite directions, which do not agree. In most cases, the number of trays will be satisfactory nevertheless, and for purposes of subsequent calculations (Thiele-Geddes), temperatures and L/G ratios need only be roughly estimated if computer methods are used. If better estimates near the feed tray particularly are desired, the following, due to Underwood [63], is reasonably satisfactory.

Starting with the feed-tray liquid as computed from the enriching section, which shows no heavy components, the mole fractions are all reduced proportionately so that their sum, plus the mole fractions of the missing components (as shown in the feed-tray composition as computed from the exhausting section) is unity. The bubble point is then recalculated. The justification for this

somewhat arbitrary procedure is that the *relative* concentrations of the light components in the vapor will remain about the same so long as their relative concentrations in the liquid are unchanged.

The concentrations of the missing heavy components on trays above the feed tray are next estimated. For these, $z_{J,D}$ should be very small, and Eq. (9.177) shows

$$\frac{y_{J,n+1}}{y_{hk,n+1}} = \frac{Lx_{J,n}/G + Dz_{J,D}/G}{Lx_{hk,n}/G + Dz_{hk,D}/G} \approx \frac{x_{J,n}}{x_{hk,n}} \quad (9.179)$$

Since
$$\frac{y_{J,n+1}}{y_{hk,n+1}} = \frac{m_{J,n+1}x_{J,n+1}}{m_{hk,n+1}x_{hk,n+1}} = \frac{\alpha_{J,n+1}x_{J,n+1}}{x_{hk,n+1}} \quad (9.180)$$

then
$$x_{J,n} = \frac{\alpha_{J,n+1}x_{J,n+1}x_{hk,n}}{x_{hk,n+1}} \quad (9.181)$$

The liquid mole fractions on tray n as previously determined are then reduced proportionately to accommodate those of the heavy components, and a new bubble point is calculated. This is continued upward until the concentrations on upper trays as previously computed are no longer changed importantly.

Missing light components are added to trays below the feed in much the same manner. Thus, Eq. (9.178) for these provides

$$\frac{y_{J,m+1}}{y_{lk,m+1}} = \frac{x_{J,m} - Wx_{J,W}/\bar{L}}{x_{lk,m} - Wx_{lk,W}/\bar{L}} \approx \frac{x_{J,m}}{x_{lk,m}} \quad (9.182)$$

where advantage is taken of the fact that $x_{J,W}$ must be small for these components. As before,

$$\frac{y_{J,m+1}}{y_{lk,m+1}} = \frac{m_{J,m+1}x_{J,m+1}}{m_{lk,m+1}x_{lk,m+1}} = \frac{\alpha_{J,m+1}x_{J,m+1}}{\alpha_{lk,m+1}x_{lk,m+1}} \approx \frac{\alpha_{Jm}x_{J,m+1}}{\alpha_{lkm}x_{lk,m+1}} \quad (9.183)$$

The last approximation results from assuming that the ratio of α 's stays reasonably constant with small temperature changes. Then

$$x_{J,m+1} = \frac{x_{J,m}\alpha_{lkm}x_{lk,m+1}}{\alpha_{J,m}x_{lkm}} \quad (9.184)$$

The mole fractions of the components on the lower tray are then proportionately reduced to accommodate the light components, and the bubble point is recalculated.

Illustration 9.13 continued Compositions at the feed tray as previously determined are listed in the first two columns. The old x_6 's are reduced to accommodate the C_6 from $N_p - 8$, and the new x_6 's are determined. The new bubble point is 86.4°C.

Component	x_{N_p-8}	Old x_6	$x_6(1 - 0.1362)$	New x_6	$\alpha_{86.4}$
C_1		0.00204	0.00176	0.00176	46.5
C_2		0.0187	0.0162	0.0162	13.5
$lk\ C_3$	0.0915	0.1045	0.0903	0.0903	5.87
C_4	0.3897	0.4247	0.3668	0.3668	2.39
$hk\ C_5$	0.3826	0.4500	0.3887	0.3887	1.00
C_6	0.1362			0.1362	0.467
	1.0000	1.0000	0.8638	1.0000	

For tray 5, x_{C_6} is estimated, through Eq. (9.181),

$$x_{C_6, 5} = \frac{0.467(0.1362)(0.3859)}{0.3887} = 0.0631$$

where 0.3859 is the concentration of the heavy key on tray 5 as previously calculated. The old x_5 's are all reduced by multiplying them by $1 - 0.0631$, whence their new sum plus 0.0631 for $C_6 = 1$. The new tray 5 has a bubble point of 80°C , and its equilibrium vapor can be obtained from the bubble-point calculation, as usual. In similar fashion, the calculations are continued upward. Results for the top trays and the distillate are:

Component	x_2	y_2	x_1	y_1	x_D	y_D
C_1	0.00221	0.0444	0.00226	0.0451	0.00425	0.0789
C_2	0.0214	0.1111	0.0241	0.1209	0.0425	0.1842
C_3	0.1418	0.2885	0.1697	0.3259	0.2495	0.3870
C_4	0.6786	0.5099	0.7100	0.4840	0.6611	0.3420
C_5	0.1553	0.0458	0.0932	0.0239	0.0425	0.0079
C_6	0.00262	0.00034	0.00079	0.00009	0.00015	0.00001
$t, ^\circ\text{C}$	66		58.5		46.6	

To correct tray $N_p - 7 = \text{tray } 7$, use Eq. (9.184) to determine the concentrations of C_1 and C_2 on tray 7. $x_{ik, m+1}$ is taken from the old $N_p - 7$.

$$x_{C_1, 7} = \frac{0.00176(5.87)(0.0790)}{46.5(0.0903)} = 0.000194 \quad x_{C_2, 7} = \frac{0.0162(5.87)(0.0790)}{13.5(0.0903)} = 0.00615$$

The old x_{N_p-7} must be reduced by multiplying by $1 - 0.000194 - 0.00615$. The adjusted values together with those above constitute the new x_7 's. The new bubble point is 94°C .

The calculations are continued downward in the same fashion. The new tray 6 has $x_{C_1} = 0.000023$, $x_{C_2} = 0.00236$. It is clear that concentrations for these components are reducing so rapidly that there is no need to go further.

Liquid/Vapor Ratios

With the corrected temperatures and compositions, it is now possible to estimate the L/G ratios on the trays reasonably well.

For the enriching section, Eq. (9.59) can be solved simultaneously with Eq. (9.56) to provide

$$G_{n+1} = \frac{Q_C + D(H_D - H_{L_n})}{H_{G_{n+1}} - H_{L_n}} \quad (9.185)$$

Equation (9.56) then provides L_n ; G_n is computed through another application of Eq. (9.185), and hence L_n/G_n is obtained. Similarly for the exhausting section, Eqs. (9.66) and (9.69) provide

$$\bar{G}_{m+1} = \frac{Q_B + W(H_{L_m} - H_W)}{H_{G_{m+1}} - H_{L_m}} \quad (9.186)$$

and Eq. (9.66) gives the liquid rate. Usually it is necessary to compute only a few such ratios in each section and interpolate the rest from a plot of L/G vs. tray number.

It must be remembered that if the enthalpies are computed from temperatures and concentrations taken from the Lewis-Matheson data, even though

corrected in the manner previously shown, the L/G ratios will still be estimates only, since the data upon which they are based assume constant L/G in each section.

Illustration 9.13 continued Compute the condenser and reboiler heat loads and the L/G ratios.

SOLUTION The condenser heat load is given by Eq. (9.54). Values of x_0 , y_D , and y_1 are taken from the corrected concentrations previously obtained.

Component	H_D , vapor, 46.6°C	$y_D H_D$	H_{L0} , 46.6°C	$x_0 H_{L0}$	H_{G1} , 58.8°C	$y_1 H_{G1}$
C ₁	13 490	1 065	10 470	44.4	13 960	629
C ₂	23 380	4 305	17 210	732	24 190	2 926
C ₃	32 100	12 420	18 610	4 643	37 260	10 840
C ₄	42 330	14 470	22 790	15 100	43 500	21 050
C ₅	52 570	415	27 100	1 151	53 900	1 291
C ₆	61 480	0.7	31 050	4.7	63 500	9.5
		32 680 = H_D		21 675 = H_{L0}		36 745 = H_{G1}

Eq. (9.54):

$$Q_C = 0.3800[(0.8 + 1)(36 745) - 0.8(21 675) - 32 680] \\ = 6130 \text{ kJ/kmol feed}$$

In similar fashion, $H_W = 39 220$, $H_F = 34 260$, and Q_B [Eq. (9.55)] = 8606 kJ/kmol feed.

For tray $n = 1$, $G_1 = L_0 + D = D(R + 1) = 0.3800(1.8) = 0.684$ kmol. With x_1 and y_2 from corrected compositions:

Component	H_{G2} , 66°C	$H_{G2} y_2$	H_{L1} , 58.5°C	$H_{L1} x_1$
C ₁	14 070	624.5	11 160	25.4
C ₂	24 610	2 740	17 910	432
C ₃	33 800	9 770	20 470	3 473
C ₄	44 100	22 470	24 900	17 680
C ₅	54 780	2 510	29 500	2 750
C ₆	64 430	21.9	33 840	26.7
		38 136 = H_{G2}		24 380 = H_{L1}

Eq. (9.185):

$$G_2 = \frac{6130 + 0.38(32 680 - 24 380)}{38 136 - 24 380} = 0.675 \text{ kmol}$$

$$L_2 = G_2 - D = 0.675 - 0.380 = 0.295 \text{ kmol}$$

$$\frac{L_2}{G_2} = \frac{0.295}{0.675} = 0.437$$

Similarly, the calculations can be made for other trays in the enriching section. For tray $N_p = 14$:

Component	$H_{G, 15},$ 113.3°C	$y_{15}H_{G, 15}$	$H_{L, 14},$ 107.9°C	$x_{14}H_{L, 14}$
C ₁		Nil		Nil
C ₂		Nil		Nil
C ₃	38 260	709	29 310	287.7
C ₄	49 310	26 700	31 870	12 790
C ₅	60 240	22 100	37 680	16 510
C ₆	71 640	5 245	43 500	6 470
		54 750 = $H_{G, 15}$		36 058 = $H_{L, 14}$

Similarly,

$$H_{L, 13} = 36\,790 \quad \text{and} \quad H_{G, 14} = 52\,610$$

Eq. (9.186):

$$\bar{G}_{15} = \frac{8606 + 0.62(36\,058 - 39\,220)}{54\,750 - 39\,220} = 0.358$$

$$\bar{L}_{14} = W + \bar{G}_{15} = 0.620 + 0.358 = 0.978$$

$$\bar{G}_{14} = \frac{8606 + 0.62(36\,790 - 39\,220)}{52\,610 - 36\,790} = 0.448$$

$$\frac{\bar{L}_{14}}{\bar{G}_{14}} = \frac{0.978}{0.448} = 2.18$$

and similarly for other exhausting-section trays. Thus:

Tray no.	$\frac{L}{G}$ or $\frac{\bar{L}}{\bar{G}}$	$t, ^\circ\text{C}$	Tray no.	$\frac{\bar{L}}{\bar{G}}$	$t, ^\circ\text{C}$
Condenser	0.80	46.6	8	3.25	96.3
1	0.432	58.4	9	2.88	97.7
2	0.437	66	10	2.58	99
3	0.369	70.4	11	2.48	100
4	0.305	74	12	2.47	102.9
5	0.310	80.3	13	2.42	104.6
6	1.53	86.4	14	2.18	107.9
7	4.05	94.1	Reboiler	1.73	113.5

These values are not final. They scatter erratically because they are based on the temperatures and concentrations computed with the assumption of constant L/G .

Method of Thiele and Geddes

With number of trays, position of the feed tray, temperatures, and L/G ratios, the Thiele-Geddes method allows one to compute the products which will result, thus ultimately permitting a check of all previous calculations. The following is Edmister's [11] variation of the original [62].

All the equations which follow apply separately for each component, and component designations are omitted. For the enriching section, consider first the condenser.

$$G_1 y_1 = L_0 x_0 + D z_D \quad (9.187)$$

$$\frac{G_1 y_1}{D z_D} = \frac{L_0 x_0}{D z_D} + 1 \quad (9.188)$$

For a total condenser, $x_0 = z_D = x_D$, and $L_0/D = R$. Therefore

$$\frac{G_1 y_1}{D z_D} = R + 1 = A_0 + 1 \quad (9.189)$$

For a partial condenser behaving like a theoretical stage, $z_D = y_D$, $z_D/x_0 = y_D/x_0 = m_0$, and

$$\frac{G_1 y_1}{D z_D} = \frac{R}{m_0} + 1 = A_0 + 1 \quad (9.190)$$

A_0 is therefore either R or R/m_0 , depending upon the type of condenser. For tray 1,

$$\frac{L_1 x_1}{G_1 y_1} = \frac{L_1}{G_1 m_1} = A_1 \quad (9.191)$$

where A_1 is the absorption factor for tray 1. Then

$$L_1 x_1 = A_1 G_1 y_1 \quad (9.192)$$

For tray 2,

$$G_2 y_2 = L_1 x_1 + D z_D \quad (9.193)$$

$$\frac{G_2 y_2}{D z_D} = \frac{L_1 x_1}{D z_D} + 1 = \frac{A_1 G_1 y_1}{D z_D} + 1 = A_1(A_0 + 1) + 1 = A_0 A_1 + A_1 + 1 \quad (9.194)$$

Generally, for any tray,

$$\frac{G_n y_n}{D z_D} = \frac{A_{n-1} G_{n-1} y_{n-1}}{D z_D} + 1 = A_0 A_1 A_2 \cdots A_{n-1} + A_1 A_2 \cdots A_{n-1} + \cdots + A_{n-1} + 1 \quad (9.195)$$

and for the feed tray,

$$\frac{G_f y_f}{D z_D} = A_0 A_1 A_2 \cdots A_{f-1} + A_1 A_2 \cdots A_{f-1} + \cdots + A_{f-1} + 1 \quad (9.196)$$

Consider next the exhausting section. For a kettle-type reboiler,

$$\frac{\bar{G}_{N_p+1} y_{N_p+1}}{W x_W} = \frac{\bar{G}_{N_p+1} m_W}{W} = S_W \quad (9.197)$$

where S_W is the reboiler stripping factor. Since

$$\bar{L}_{N_p} x_{N_p} = \bar{G}_{N_p+1} y_{N_p+1} + W x_W \quad (9.198)$$

then

$$\frac{\bar{L}_{N_p} x_{N_p}}{W x_W} = \frac{\bar{G}_{N_p+1} y_{N_p+1}}{W x_W} + 1 = S_W + 1 \quad (9.199)$$

For a thermosiphon reboiler, $y_{N_p+1} = x_W$, and

$$S_W = \frac{\bar{G}_{N_p+1}}{W} \quad (9.200)$$

S_W is therefore defined by either Eq. (9.197) or (9.200), depending upon the type of reboiler. For the bottom tray,

$$\frac{\bar{G}_{N_p} y_{N_p}}{\bar{L}_{N_p} x_{N_p}} = \frac{\bar{G}_{N_p} m_{N_p}}{\bar{L}_{N_p}} = S_{N_p} \quad (9.201)$$

$$\bar{G}_{N_p} y_{N_p} = S_{N_p} \bar{L}_{N_p} x_{N_p} \quad (9.202)$$

Then
$$\bar{L}_{N_p-1}x_{N_p-1} = \bar{G}_{N_p}y_{N_p} + Wx_W \quad (9.203)$$

and
$$\frac{\bar{L}_{N_p-1}x_{N_p-1}}{Wx_W} = \frac{\bar{G}_{N_p}y_{N_p}}{Wx_W} + 1 = \frac{S_{N_p}\bar{L}_{N_p}x_{N_p}}{Wx_W} + 1 = S_{N_p}(S_W + 1) + 1 = S_{N_p}S_W + S_{N_p} + 1 \quad (9.204)$$

In general, for any tray,

$$\frac{\bar{L}_m x_m}{Wx_W} = \frac{S_{m+1}\bar{L}_{m+1}x_{m+1}}{Wx_W} + 1 = S_{m+1} \cdots S_{N_p-1}S_{N_p}S_W + S_{m+1} \cdots S_{N_p-1}S_{N_p} + S_{m+1} \cdots S_{N_p-1} + \cdots + S_{m+1} + 1 \quad (9.205)$$

and at the feed tray,

$$\frac{\bar{L}_f x_f}{Wx_W} = S_{f+1} \cdots S_{N_p}S_W + S_{f+1} \cdots S_{N_p} + \cdots + S_{f+1} + 1 \quad (9.206)$$

Edmister [11] provides shortcut methods involving "effective" A 's and S 's for use in Eqs. (9.196) and (9.206), much as for gas absorption.

$$A_f = \frac{\bar{L}_f}{G_f m_f} = \frac{\bar{L}_f x_f}{G_f y_f} \quad (9.207)$$

and
$$\frac{Wx_W}{Dz_D} \equiv \frac{\bar{L}_f x_f}{G_f y_f} \frac{G_f y_f / Dz_D}{\bar{L}_f x_f / Wx_W} = A_f \frac{G_f y_f / Dz_D}{\bar{L}_f x_f / Wx_W} \quad (9.208)$$

Since
$$Fz_D = Wx_W + Dz_D \quad (9.209)$$

$$Dz_D = \frac{Fz_F}{Wx_W / Dz_D + 1} \quad (9.210)$$

For each component, Eqs. (9.196) and (9.206) are then used to compute the numerator and denominator, respectively, of the term in parentheses of Eq. (9.208), whence Wx_W / Dz_D , or the split of the component, is found. Equation (9.210) then provides Dz_D , and Eq. (9.209) provides Wx_W .

The products thus computed are completely consistent with the number of trays, feed-tray position, and reflux ratio used, provided the A 's and S 's (or L/G 's and temperatures) are correct. For checking these, it is necessary to use the general equations (9.195) and (9.205) for each tray in the appropriate section of the tower. To check tray n in the enriching section,

$$\frac{G_n y_n}{Dz_D} \frac{Dz_D}{G_n} = y_n \quad (9.211)$$

and this can be obtained from the data already accumulated. If $\Sigma y_{j,n} = 1$, the temperature is satisfactory. If not, a new temperature is obtained by adjusting the y 's proportionately until they add to unity and computing the corresponding dew point. To check a tray in the exhausting section,

$$\frac{\bar{L}_m x_m}{Wx_W} \frac{Wx_W}{\bar{L}_m} = x_m \quad (9.212)$$

which is obtainable from the data. If $\Sigma x_{j,m} = 1$, the temperature is correct. If not, the x 's are adjusted to add to unity and the bubble point computed. The new temperatures and compositions permit new enthalpy balances to obtain

corrected L/G 's for each tray, and the Thiele-Geddes calculation can be redone. Problems of convergence of this trial-and-error procedure are considered by Holland [24], Lyster et al. [33], and Smith [57].

The true optimum feed-tray location is obtained by trial, altering the location and observing which location produces the smallest number of trays for the desired products.

Illustration 9.13 concluded Reestimate the products through the Thiele-Geddes method.

SOLUTION The temperature and L/G profiles as previously estimated will be used. Computations will be shown for only one component, C_4 ; all other components are treated in the same manner.

Using the tray temperatures to obtain m , $A = L/mG$ for the enriching trays and $S = m\bar{G}/\bar{L}$ for the exhausting trays are computed. Since a partial condenser is used, $A_0 = R/m_0$. For the kettle-type reboiler, $S_W = \bar{G}_{15}m_W/W$. Then, for C_4 :

Tray	Condenser	1	2	3	4	5	6 = f
m	0.50	0.66	0.75	0.81	0.86	0.95	1.07
A	1.600	0.655	0.584	0.455	0.355	0.326	1.431

Tray	7 = $f + 1$	8	9	10	11	12	13	14	Reboiler
m	1.22	1.27	1.29	1.30	1.32	1.40	1.45	1.51	1.65
S	0.301	0.390	0.447	0.503	0.532	0.566	0.599	0.693	0.954

With these data

Eq. (9.196):

$$\frac{G_0 y_f}{D z_D} = 1.5778$$

Eq. (9.206):

$$\frac{\bar{L}_f x_f}{W x_W} = 1.5306$$

Eq. (9.208):

$$\frac{W x_W}{D z_D} = 1.431 \frac{1.5778}{1.5306} = 1.475$$

Eq. (9.210):

$$D y_D = \frac{0.33}{1.475 + 1} = 0.1335$$

Eq. (9.209):

$$W x_W = 0.33 - 0.1335 = 0.1965$$

Similarly:

Component	$\frac{G_f y_f}{D z_D}$	$\frac{L_f x_f}{W x_W}$	$\frac{W x_W}{D y_D}$	$D y_D$	$W x_W$
C ₁	1.0150	254×10^6	288×10^{-10}	0.03	Nil
C ₂	1.0567	8750	298×10^{-5}	0.07	Nil
C ₃	1.1440	17.241	0.0376	0.144 7	0.0053
C ₄	1.5778	1.5306	1.475	0.133 5	0.1965
C ₅	15.580	1.1595	45.7	0.006 43	0.29357
C ₆	1080	1.0687	7230	0.000 016 6	0.11998
				0.3846 = D	0.6154 = W

These show that $0.1447(100)/0.15 = 96.3\%$ of the C₃ and 2.14% of the C₅ are in the distillate. These do not quite meet the original specifications. The temperatures and L/G 's must still be corrected, however. Thus for tray 2 and component C₄, Eq. (9.195) yields $G_2 y_2 / D z_D = 2.705$. From the enthalpy balances, $G_2 = 0.675$. Therefore, by Eq. (9.211),

$$y_2 = \frac{2.705(0.1335)}{0.675} = 0.5349$$

Similarly:

Component	$\frac{G_2 y_2}{D z_D}$	y_2	Adjusted y_2
C ₁	1.0235	0.0454	0.0419
C ₂	1.1062	0.1147	0.1059
C ₃	1.351	0.2896	0.2675
C ₄	2.705	0.5349	0.4939
C ₅	10.18	0.0970	0.0896
C ₆	46.9	0.00115	0.00106
		1.0828	1.0000

Since Σy_2 does not equal 1, the original temperature is incorrect. After adjusting the y 's to add to unity as shown, the dew point = 71°C instead of the 66°C used. Similarly all tray compositions and temperatures must be corrected, new L/G 's obtained by enthalpy balances with the new compositions, temperatures, and L/G 's, and the Thiele-Geddes calculations repeated.

Multicomponent fractionation finds its most complex applications in the field of petroleum refining. Petroleum products such as gasoline, naphthas, kerosenes, gas oils, fuel oils, and lubricating oils are each mixtures of hundreds of hydrocarbons, so many that their identity and actual number cannot readily be established. Fortunately, it is not usually specific substances that are desired in these products but *properties*, so that specifications can be made as to boiling range, specific gravity, viscosity, and the like. Fractionators for these products cannot be designed by the detailed methods just described but must be based upon laboratory studies in small-scale equipment. As an example which illustrates the complex nature of some of these separations, consider the schematic diagram of a topping plant for the initial distillation of a crude oil (Fig.

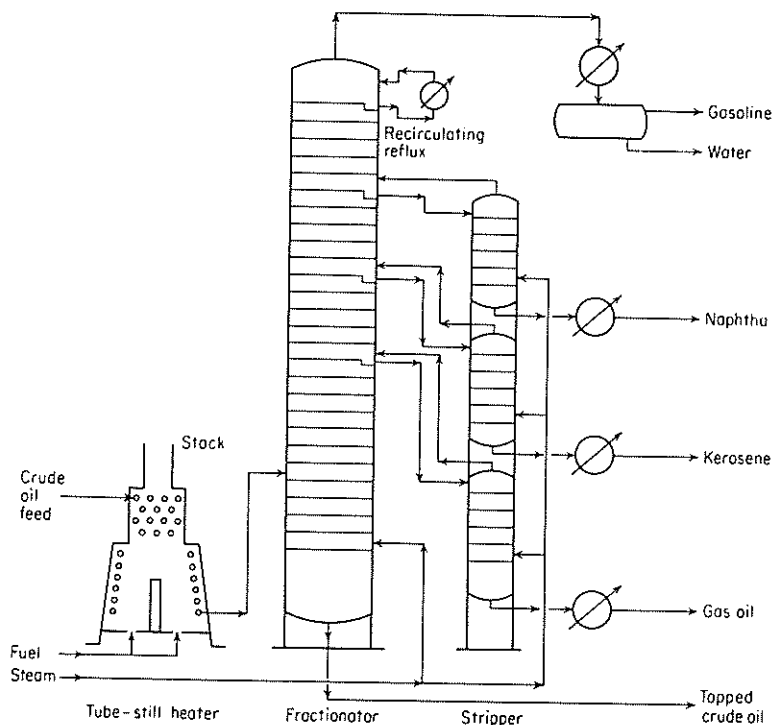


Figure 9.52 Schematic arrangement, petroleum topping plant.

9.52). The crude oil, after preliminary heat exchange with several of the products from the plant, is passed through the tubes of a gas-fired furnace, the tube-still heater. Here a portion of the oil is vaporized, somewhat larger in amount than that ultimately to be taken as vaporized products. The mixture of liquid and vapor then enters the large tray tower. Open steam is introduced at the bottom to strip the last traces of volatile substances from the residue product, and this steam passes up the column, where it lowers the effective pressure and hence the temperature of the distillation. The steam and most volatile substances (crude gasoline) leaving the top tray are condensed, the water separated, and the gasoline sent to storage. Reflux in the scheme shown here is provided by withdrawing a portion of the liquid from the top tray and returning it after it has been cooled. The cold liquid condenses some of the rising vapors to provide internal reflux. Several trays down from the top of the tower, a side stream may be withdrawn which will contain the hydrocarbons characteristic of a desired naphtha product. Since the components of the more volatile gasoline are also present at this point, the liquid is steam-stripped in a short auxiliary tray tower, the steam and vaporized gasoline being sent back to the main fractionator. The stripped naphtha is then withdrawn to storage. In similar fashion kerosene and

gas oil cuts may be withdrawn, but each must be separately steam-stripped. The individual steam strippers are frequently built into a single shell, as shown, for economy, so that from the outside the multipurpose nature of the smaller tower is not readily evident. The withdrawn products must ordinarily be further processed before they are considered finished.

The design, method of operation, and number of products from topping units of this sort may vary considerably from refinery to refinery. Indeed, any individual unit will be built for maximum flexibility of operation, with, for example, multiple nozzles for introducing the feed at various trays and multiple side-stream withdrawal nozzles, to allow for variations in the nature of the feed and in the products to be made.

Azeotropic Distillation

This is a special case of multicomponent distillation [23] used for separation of binary mixtures which are either difficult or impossible to separate by ordinary fractionation. If the relative volatility of a binary mixture is very low, the continuous rectification of the mixture to give nearly pure products will require high reflux ratios and correspondingly high heat requirements, as well as towers of large cross section and numbers of trays. In other cases the formation of a binary azeotrope may make it impossible to produce nearly pure products by ordinary fractionation. Under these circumstances a third component, sometimes called an *entrainer*, may be added to the binary mixture to form a new low-boiling azeotrope with one of the original constituents, whose volatility is such that it can easily be separated from the other original constituent.

As an example of such an operation, consider the flowsheet of Fig. 9.53 for the azeotropic separation of acetic acid–water solutions, using butyl acetate as entrainer [39]. Acetic acid can be separated from water by ordinary methods, but only at great expense because of the low relative volatility of the constituents despite their fairly large difference in boiling points at atmospheric pressure (nbp acetic acid = 118.1°C, nbp water = 100°C). Butyl acetate is only slightly soluble in water and consequently forms a heteroazeotrope with it (bp = 90.2°C). Therefore if at least sufficient butyl acetate is added to the top of the distillation column (1) to form the azeotrope with all the water in the binary feed, the azeotrope can readily be distilled from the high-boiling acetic acid, which leaves as a residue product. The heteroazeotrope on condensation forms two insoluble liquid layers, one nearly pure water but saturated with ester, the other nearly pure ester saturated with water. The latter is returned to the top of the column as reflux and is the source of the entrainer in the column. The former can be stripped of its small entrainer content in a second small column (2). The separation of the heteroazeotrope from acetic acid is readily done, so that relatively few trays are required in the principal tower. On the other hand, heat must be supplied, not only to vaporize the water in the overhead distillate, but to vaporize the entrainer as well. The operation can also be done batchwise, in which case sufficient entrainer is charged to the still kettle, together with the feed, to azeotrope the water. The azeotrope is then distilled overhead.

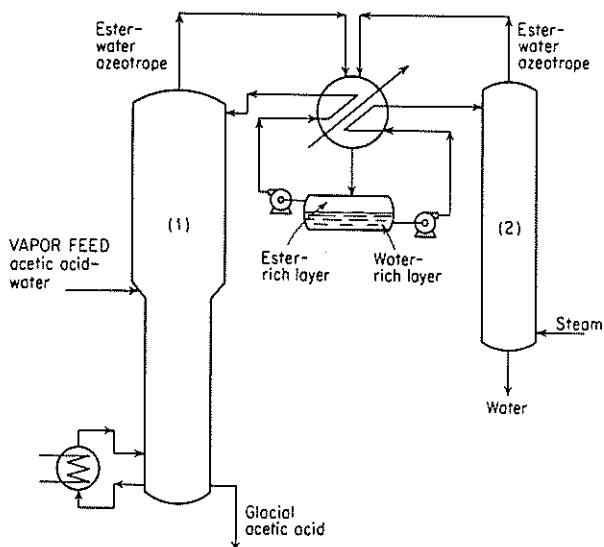


Figure 9.53 Azeotropic distillation of acetic acid–water with butyl acetate.

Sometimes the new azeotrope which is formed contains all three constituents. The dehydration of ethanol–water mixture with benzene as added substance is an example. Dilute ethanol–water solutions can be continuously rectified to give at best mixtures containing 89.4 mole percent ethanol at atmospheric pressure, since this is the composition of the minimum-boiling azeotrope in the binary system. By introducing benzene into the top of a column fed with an ethanol–water mixture, the ternary azeotrope containing benzene (53.9 mol %), water (23.3 mol %), ethanol (22.8 mol %), boiling at 64.9°C , is readily separated from the ethanol ($\text{bp} = 78.4^{\circ}\text{C}$), which leaves as a residue product. In this case also the azeotropic overhead product separates into two liquid layers, one rich in benzene which is returned to the top of the column as reflux, the other rich in water which is withdrawn. Since the latter contains appreciable quantities of both benzene and ethanol, it must be rectified separately. The ternary azeotrope contains nearly equal molar proportions of ethanol and water, and consequently dilute ethanol–water solutions must be given a preliminary rectification to produce substantially the alcohol-rich binary azeotrope which is used as a feed.

In still other cases the new azeotrope which is formed does not separate into two insoluble liquids, and special means for separating it, such as liquid extraction, must be provided, but this is less desirable.

It is clear that the choice of entrainer is a most important consideration. The added substance should preferably form a low-boiling azeotrope with only one of the constituents of the binary mixture it is desired to separate, preferably the

constituent present in the minority so as to reduce the heat requirements of the process. The new azeotrope must be of sufficient volatility to make it readily separable from the remaining constituent and so that inappreciable amounts of entrainer will appear in the residue product. It should preferably be lean in entrainer content, to reduce the amount of vaporization necessary in the distillation. It should preferably be of the heterogeneous-liquid type, which then simplifies greatly the recovery of the entrainer. In addition, a satisfactory entrainer must be (1) cheap and readily available, (2) chemically stable and inactive toward the solution to be separated, (3) noncorrosive toward common construction materials, (4) nontoxic, (5) of low latent heat of vaporization, (6) of low freezing point to facilitate storage and outdoor handling, and (7) of low viscosity to provide high tray efficiencies.

The general methods of design for multicomponent distillation apply, the principal difficulty being the paucity of vapor-liquid equilibrium data for these highly nonideal mixtures [23]. Computer programs are available [4].

Extractive Distillation

This is a multicomponent-rectification method similar in purpose to azeotropic distillation. To a binary mixture which is difficult or impossible to separate by ordinary means a third component, termed a *solvent*, is added which alters the relative volatility of the original constituents, thus permitting the separation. The added solvent is, however, of low volatility and is itself not appreciably vaporized in the fractionator.

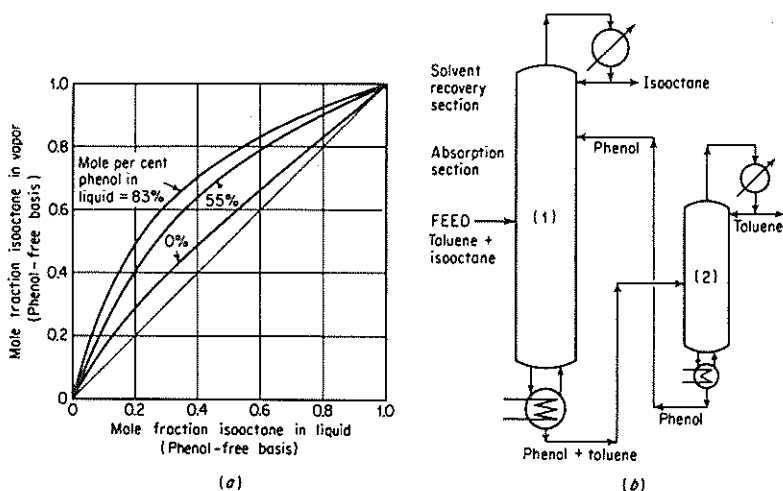


Figure 9.54 Extractive distillation of toluene-isooctane with phenol [Vapor-liquid equilibria from *Drickamer, Brown, and White, Trans. AIChE* 41, 555 (1945).]

As an example of such an operation, consider the process of Fig. 9.54. The separation of toluene ($bp = 110.8^{\circ}\text{C}$) from paraffin hydrocarbons of approximately the same molecular weight is either very difficult or impossible, due to low relative volatility or azeotropism, yet such a separation is necessary in the recovery of toluene from certain petroleum hydrocarbon mixtures. Using isooctane ($bp = 99.3^{\circ}\text{C}$) as an example of a paraffin hydrocarbon, Fig. 9.54*a* shows that isooctane in this mixture is the more volatile, but the separation is obviously difficult. In the presence of phenol ($bp = 181.4^{\circ}\text{C}$), however, the isooctane relative volatility increases, so that, with as much as 83 mole percent phenol in the liquid, the separation from toluene is relatively easy. A flowsheet for accomplishing this is shown in Fig. 9.54*b*, where the binary mixture is introduced more or less centrally into the extractive distillation tower (1), and phenol as the solvent is introduced near the top so as to be present in high concentration upon most of the trays in the tower. Under these conditions isooctane is readily distilled as an overhead product, while toluene and phenol are removed as a residue. Although phenol is relatively high-boiling, its vapor pressure is nevertheless sufficient for its appearance in the overhead product to be prevented. The solvent-recovery section of the tower, which may be relatively short, serves to separate the phenol from the isooctane. The residue from the tower must be rectified in the auxiliary tower (2) to separate toluene from the phenol, which is recycled, but this is a relatively easy separation. In practice, the paraffin hydrocarbon is a mixture rather than the pure substance isooctane, but the principle of the operation remains the same.

Such a process depends upon the difference in departure from ideality between the solvent and the components of the binary mixture to be separated. In the example given, both toluene and isooctane separately form nonideal liquid solutions with phenol, but the extent of the nonideality with isooctane is greater than that with toluene. When all three substances are present, therefore, the toluene and isooctane themselves behave as a nonideal mixture and their relative volatility becomes high. Considerations of this sort form the basis for the choice of an extractive-distillation solvent. If, for example, a mixture of acetone ($bp = 56.4^{\circ}\text{C}$) and methanol ($bp = 64.7^{\circ}\text{C}$), which form a binary azeotrope, were to be separated by extractive distillation, a suitable solvent could probably be chosen from the group of aliphatic alcohols. Butanol ($bp = 117.8^{\circ}\text{C}$), since it is a member of the same homologous series but not far removed, forms substantially ideal solutions with methanol, which are themselves readily separated. It will form solutions of positive deviation from ideality with acetone, however, and the acetone-methanol vapor-liquid equilibria will therefore be substantially altered in ternary mixtures. If butanol forms no azeotrope with acetone, and if it alters the vapor-liquid equilibrium of acetone-methanol sufficiently to destroy the azeotrope in this system, it will serve as an extractive-distillation solvent. When both substances of the binary mixture to be separated are themselves chemically very similar, a solvent of an entirely different chemical nature will be necessary. Acetone and furfural, for example, are useful as extractive-distillation solvents for separating the hydrocarbons butene-2 and *n*-butane.

Generally the requirements of a satisfactory extractive-distillation solvent are [16, 51, 61]:

1. High selectivity, or ability to alter the vapor-liquid equilibria of the original mixture sufficiently to permit its easy separation, with, however, use of only small quantities of solvent.
2. High capacity, or ability to dissolve the components in the mixture to be separated. It frequently happens that substances which are incompletely miscible with the mixture are very selective; yet if sufficiently high concentrations of solvent cannot be obtained in the liquid phase, the separation ability cannot be fully developed.
3. Low volatility in order to prevent vaporization of the solvent with the overhead product and to maintain high concentration in the liquid phase. Nonvolatile salts can be especially useful [15].
4. Separability; the solvent must be readily separated from the mixture to which it is added, and particularly it must form no azeotropes with the original substances.
5. The same considerations of cost, toxicity, corrosive character, chemical stability, freezing point, and viscosity apply as for entrainers for azeotropic distillation.

The general methods of design for multicomponent distillation apply. Since in most cases all the components of the feed streams are found in the bottoms, the method of Lewis and Matheson can be used, starting at the bottom and computing to the top. There will be an optimum solvent-circulation rate: small solvent rates require many trays, but the column diameters are small; large ratios require fewer trays but larger column diameters and greater solvent-circulation costs.

Comparison of Azeotropic and Extractive Distillation

It is generally true that adding an extraneous substance such as an entrainer or solvent to a process is undesirable. Since it can never be completely removed, it adds an unexpected impurity to the products. There are inevitable losses (ordinarily of the order of 0.1 percent of the solvent-circulation rate), and they may be large since solvent/feed ratios must frequently be greater than 3 or 4 to be effective. An inventory and source of supply must be maintained. Solvent recovery costs can be large, and new problems in choices of materials of construction are introduced. It follows that these processes can be considered only if, despite these drawbacks, the resulting process is less costly than conventional distillation.

Extractive distillation is generally considered to be more desirable than azeotropic distillation since (1) there is a greater choice of added component because the process does not depend upon the accident of azeotrope formation and (2) generally smaller quantities of solvent must be volatilized. However, the latter advantage can disappear if the volatilized impurity is a minor constituent

of the feed and the azeotrope composition is favorable, i.e., low in solvent composition. Thus, in the dehydration of ethanol from an 85.6 mole percent ethanol-water solution, azeotroping the water with *n*-pentane as entrainer is more economical than using ethylene glycol as an extractive distillation solvent [3]. The reverse might be true for a more dilute alcohol feed.

LOW-PRESSURE DISTILLATION

Many organic substances cannot be heated to temperatures which even approach their normal boiling points without chemical decomposition. If such substances are to be separated by distillation, the pressure and the corresponding temperature must be kept low. The time of exposure of the substances to the distillation temperature must also be kept to a minimum, since the extent of thermal decomposition will thereby be reduced. For distillation under absolute pressures of the order of 7 to 35 kN/m² (1 to 5 lb_f/in²), packed towers can be used, bubble-cap and sieve trays can be designed with pressure drops approaching 350 N/m² (0.05 lb_f/in², or 2.6 mmHg), and other simpler designs, such as the shower tray of Fig. 6.18, for which the pressure drops are of the order of 103 N/m² (0.015 lb_f/in², or 0.75 mmHg), are possible. Mechanically stirred spray and wetted-wall columns provide even smaller pressure drops [45].

In the distillation of many natural products, such as the separation of vitamins from animal and fish oils, as well as the separation of many synthetic industrial products such as plasticizers, the temperature may not exceed perhaps 200 to 300°C, where the vapor pressures of the substances may be a fraction of a millimeter of mercury. The conventional equipment is, of course, wholly unsuitable for such separations, not only because the pressure drop would result in high temperatures at the bottom of columns but also because of the long exposure time to the prevailing temperatures resulting from high holdup.

Molecular Distillation

This is a form of distillation at very low pressure conducted industrially at absolute pressures of the order of 0.3 to 3 N/m² (0.003 to 0.03 mmHg), suitable for the heat-sensitive substances described above.

The rate at which evaporation takes place from a liquid surface is given by the *Langmuir equation*

$$N_A = 1006 \bar{p}_A \left(\frac{1}{2\pi M_A R' T} \right)^{0.5} \quad (9.213)$$

where R' is the gas constant. At ordinary pressures the net rate of evaporation is, however, very much less than this, because the evaporated molecules are reflected back to the liquid after collisions occurring in the vapor. By reducing the absolute pressure to values used in molecular distillation, the mean free path of the molecules becomes very large, of the order of 1 cm. If the condensing surface is then placed at a distance from the vaporizing liquid surface not exceeding a few centimeters, very few molecules will return to the liquid, and the

net rate of evaporation of each substance of a binary mixture will approach that given by Eq. (9.213). The vapor composition, or the composition of the distillate, will now be different from that given by ordinary equilibrium vaporization, and the ratio of the constituents in the distillate will approach

$$\frac{N_A}{N_B} = \frac{\text{moles of A}}{\text{moles of B}} = \frac{\bar{p}_A/M_A^{0.5}}{\bar{p}_B/M_B^{0.5}} \quad (9.214)$$

If this ratio is to be maintained, however, the surface of the liquid must be rapidly renewed, since otherwise the ratio of constituents in the surface will change as evaporation proceeds. The vigorous agitation or boiling present during ordinary distillations is absent under conditions of molecular distillation, and in most devices the liquid is caused to flow in a thin film over a solid surface, thus continually renewing the surface but at the same time maintaining low holdup of liquid.

Figure 9.55 shows a device which is used industrially for accomplishing a molecular distillation [19, 65]. The degassed liquid to be distilled is introduced continuously at the bottom of the inner surface of the rotor, a rotating conical-shaped surface. The rotor may be as large as 1.5 m in diameter at the top and may revolve at speeds of 400 to 500 r/min. A thin layer of liquid to be distilled, 0.05 to 0.1 mm thick, then spreads over the inner surface and travels rapidly to the upper periphery under the action of centrifugal force. Heat is supplied to the liquid through the rotor by radiant electrical heaters, and the vaporized material is condensed upon the water-cooled louver-shaped condenser. This is maintained at temperatures sufficiently low to prevent reevaporation or reflection of the vaporized molecules. The residue liquid is caught in the collection gutter at the top of the rotor, and the distillate is drained from the collection troughs on the condenser. Each product is pumped from the still body, which is evacuated to the low pressures necessary for molecular distillation, and the time of

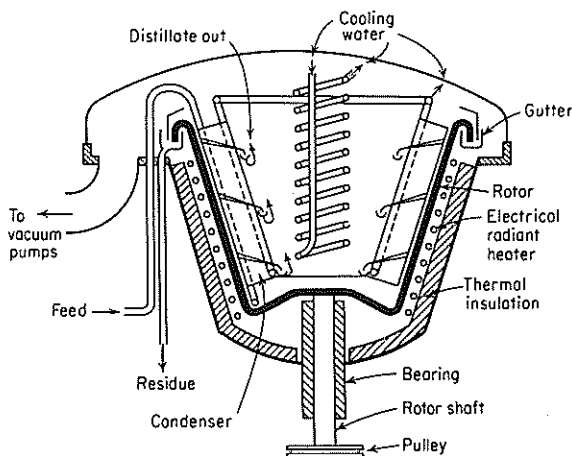


Figure 9.55 Schematic section, centrifugal molecular still of Hickman [19].

residence of the substances in the still may be as low as 1 s or less. Such a device is capable of handling 5×10^{-5} to $25 \times 10^{-5} \text{ m}^3/\text{s}$ (50 to 250 gal/h) of liquid to be distilled and gives a separation 80 to 95 percent of that indicated by Eq. (9.214). Multiple distillations are necessary for multistage separation effects [20, 32].

Falling-film stills are constructed of two vertical concentric tubes. The inner tube is heated internally, and the liquid to be distilled flows down the outer wall of the inner tube in a thin film. The outer tube is the condenser, and the annular space is under vacuum. Some thin-film arrangements involve continuous wiping of the film by a rotating blade.

NOTATION FOR CHAPTER 9

Any consistent set of units may be used.

a	specific interfacial area, L^2/L^3
A	more volatile component of a binary mixture absorption factor = L/mG , dimensionless
B	less volatile component of a binary mixture
C	molar heat capacity at constant pressure, $\text{FL}/\text{mole } T$
d	differential operator
D	distillate rate, mole/Θ for tray towers, $\text{mole}/\text{L}^2\Theta$ for packed towers
E_O	overall tray efficiency, fractional
E_{MG}	Murphree vapor tray efficiency, fractional
F	feed rate, mole/Θ for tray towers, $\text{mole}/\text{L}^2\Theta$ for packed towers
	quantity of feed, batch distillation, mole
	(with subscript G or L) mass-transfer coefficient, $\text{mole}/\text{L}^2\Theta$
g_c	conversion factor, $\text{ML}/\text{F}\Theta^2$
$\bar{G}$	vapor rate, enriching section, mole/Θ for tray towers, $\text{mole}/\text{L}^2\Theta$ for packed towers
$\bar{G}$	vapor rate, stripping section, mole/Θ for tray towers, $\text{mole}/\text{L}^2\Theta$ for packed towers
H	molar enthalpy, FL/mole
H_{IG}	height of a gas transfer unit, L
H_{IL}	height of a liquid transfer unit, L
H_{IOG}	height of an overall gas transfer unit, L
H_{IOL}	height of an overall liquid transfer unit, L
ΔH_S	integral heat of solution per mole of solution, FL/mole
I, J	components I, J
k'_x	liquid mass-transfer coefficient, $\text{mole}/\text{L}^2\Theta(\text{mole fraction})$
k'_y	gas mass-transfer coefficient, $\text{mole}/\text{L}^2\Theta(\text{mole fraction})$
K'_x	overall liquid mass-transfer coefficient, $\text{mole}/\text{L}^2\Theta(\text{mole fraction})$
K'_y	overall gas mass-transfer coefficient, $\text{mole}/\text{L}^2\Theta(\text{mole fraction})$
L	liquid rate, enriching section, mole/Θ for tray towers, $\text{mole}/\text{L}^2\Theta$ for packed towers
$\bar{L}$	liquid rate, exhausting section, mole/Θ for tray towers, $\text{mole}/\text{L}^2\Theta$ for packed towers
L_0	external reflux rate, mole/Θ for tray towers, $\text{mole}/\text{L}^2\Theta$ for packed towers
m	equilibrium distribution coefficient, $y^*/x, y/x^*$, dimensionless
M	molecular weight, M/mole
N	mass-transfer flux, $\text{mole}/\text{L}^2\Theta$
N_m	minimum number of trays
N_p	number of theoretical trays
N_{IG}	number of gas transfer units
N_{IL}	number of liquid transfer units
N_{IOG}	number of overall gas transfer units
N_{IOL}	number of overall liquid transfer units

p	vapor pressure, F/L^2
$\bar{p}$	partial pressure, F/L^2
q	quantity defined by Eq. (9.127)
Q	net rate of heating, FL/Θ
Q'	quantity defined by Eq. (9.60), FL/mole
Q''	quantity defined by Eq. (9.70), FL/mole
Q_B	heat added at the reboiler, FL/Θ for tray towers, $FL/L^2\Theta$ for packed towers
Q_C	heat removed at the condenser, FL/Θ for tray towers, $FL/L^2\Theta$ for packed towers
Q_L	heat loss, FL/Θ for tray towers, $FL/L^2\Theta$ for packed towers
R	external reflux ratio, mole reflux/mole distillate, mole/mole
R'	gas constant, $FL/\text{mole } T$
R_m	minimum reflux ratio, moles reflux/mole distillate, mole/mole
S	stripping factor, mG/L , dimensionless
t	temperature, T
T	absolute temperature, T
V	allowable vapor velocity, L/Θ
W	residue rate, mole / Θ for tray towers, mole / $L^2\Theta$ for packed towers
x	concentration (of A in binaries) in the liquid, mole fraction
x^*	x in equilibrium with y , mole fraction
y	concentration (of A in binaries) in the gas, mole fraction
y^*	y in equilibrium with x , mole fraction
z	average concentration in a solution or multiphase mixture, mole fraction
Z	height of packing, L
α	relative volatility, defined by Eq. (9.2), dimensionless
α_{JC}	relative volatility of component J with respect to component C, dimensionless
γ	activity coefficient, dimensionless
Δ	difference point, representing a fictitious stream, mole / Θ for tray towers, mole / $L^2\Theta$ for packed towers
λ	molar latent heat of vaporization, FL/mole
ρ	density, M/L^3
Σ	summation
φ	root of Eq. (9.165)

Subscripts

a	operating-line intersection
av	average
A, B	components A, B
bp	bubble point
D	distillate
e	enriching section
f	feed tray
F	feed
G	gas
hk	heavy key component
i	interface; any component
lk	light key component
L	liquid
m	tray m ; minimum
n	tray n
r	tray r
s	stripping section
0	reflux stream; base condition for enthalpy
W	residue
1	bottom of packed tower; tray 1
2	top of tray tower, tray 2

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PROBLEMS

9.1 Solutions of methanol and ethanol are substantially ideal.

(a) Compute the vapor-liquid equilibria for this system at 1 and at 5 atm abs pressure, and plot xy and txy diagrams at each pressure.

(b) For each pressure compute relative volatilities, and determine an average value.

(c) Using Eq. (9.2) with the average volatilities, compare the values of y^* at each value of x so obtained with those computed directly from the vapor pressures.

9.2 A 1000-kg batch of nitrobenzene is to be steam-distilled from a very small amount of a nonvolatile impurity, insufficient to influence the vapor pressure of the nitrobenzene. The operation is to be carried out in a jacketed kettle fitted with a condenser and distillate receiver. Saturated steam at 35 kN/m² (5.1 lb/in²) gauge is introduced into the kettle jacket for heating. The nitrobenzene is charged to the kettle at 25°C, and it is substantially insoluble in water.

Liquid water at 25°C is continuously introduced into the nitrobenzene in the kettle, to maintain a liquid water level. The mixture is distilled at atmospheric pressure. (a) At what temperature does

the distillation proceed? (b) How much water is vaporized? (c) How much steam must be condensed in the kettle jacket? Neglect the heat required to bring the still up to operating temperature. The heat capacity of nitrobenzene is $1382 \text{ J/kg} \cdot \text{K}$, and its latent heat of vaporization can be determined by the methods of Chap. 7.

9.3 A solution has the following composition, expressed as mole fraction: ethane, 0.0025; propane, 0.25; isobutane, 0.185; *n*-butane, 0.560; isopentane, 0.0025. In the following, the pressure is 10 bars ($145 \text{ lb}_f/\text{in}^2$) abs. Use equilibrium distribution coefficients from the Depriester nomograph (see Illustration 9.3).

(a) Calculate the bubble point.

(b) Calculate the dew point.

(c) The solution is flash-vaporized to vaporize 40 mol % of the feed. Calculate the composition of the products.

(d) The solution is differentially distilled to produce a residue containing 0.80 mole fraction *n*-butane. Calculate the complete composition of the residue and the percentage of the feed which is vaporized.

9.4 Vapor-liquid equilibrium data at 1 std atm abs, heats of solution, heat capacities, and latent heats of vaporization for the system acetone-water are as follows:

Mole fraction acetone in liquid, x	Integral heat of solution at 15°C, kJ/kmol soln	Equivalent mole fraction acetone in vapor, y^*	Vapor-liquid temperature, °C	Heat capacity at 17.2°C, kJ/(kg soln) · °C		
0.001	0	0.00	100.0	4.187		
0.01		0.253	91.7	4.179		
0.02	-188.4	0.425	86.6	4.162		
0.05	-447.3	0.624	75.7	4.124		
0.10	-668.7	0.755	66.6	4.020		
0.15	-770	0.798	63.4	3.894		
0.20	-786	0.815	62.2	3.810		
0.30	-719	0.830	61.0	3.559		
0.40	-509	0.839	60.4	3.350		
0.50	-350.1	0.849	60.0	3.140		
0.60	-252.6	0.859	59.5	2.931		
0.70		0.874	58.9	2.763		
0.80		0.898	58.2	2.554		
0.90		0.935	57.5	2.387		
0.95		0.963	57.0	2.303		
1.00		1.00	56.5			
$t, ^\circ\text{C}$		20	37.8	65.6	93.3	100
Heat capacity acetone, kJ/kg · °C		2.22	2.26	2.34	2.43	
Latent heat of vaporization, kJ/kg		1013	976	917	863	850

Compute the enthalpies of saturated liquids and vapors relative to pure acetone and water at 15°C and plot the enthalpy-concentration diagram, for 1 std atm abs. Retain for use in Probs. 9.5, 9.6, and 9.10.

9.5 A liquid mixture containing 60 mol % acetone, 40 mol % water, at 26.7°C (80°F), is to be continuously flash-vaporized at 1 std atm pressure, to vaporize 30 mol % of the feed.

(a) What will the composition of the products and the temperature in the separator be if equilibrium is established?

(b) How much heat, kJ/mol of feed, is required?

Ans.: 13 590.

(c) If the products are each cooled to 26.7°C, how much heat, kJ/mol of feed, must be removed from each?

9.6 A saturated vapor at 1 std atm pressure, containing 50 mol % acetone and 50 mol % water, is subject to equilibrium condensation to yield 50 mol % of the feed as liquid. Compute the equilibrium vapor and liquid compositions, the equilibrium temperature, and the heat to be removed, J/kmol of feed.

9.7 The liquid solution of Prob. 9.5 is differentially distilled at 1 atm pressure to vaporize 30 mol % of the feed. Compute the composition of the composited distillate and the residue. Compare with the results of Prob. 9.5.

Ans.: $y_D = 0.857$.

9.8 A jacketed kettle is originally charged with 30 mol of a mixture containing 40 mol % benzene, 60 mol % toluene. The vapors from the kettle pass directly to a total condenser, and the condensate is withdrawn. Liquid of the same composition as the charge is continuously added to the kettle at the rate of 15 mol/h. Heat to the kettle is regulated to generate vapor at 15 mol/h, so that the total molar content of the kettle remains constant. The mixture is ideal, and the average relative volatility is 2.51. The distillation is essentially differential.

(a) How long will the still have to be operated before vapor containing 50 mol % benzene is produced, and what is the composition of the composited distillate?

Ans.: 1.466 h, 0.555 mole fraction benzene.

(b) If the rate at which heat is supplied to the kettle is incorrectly regulated so that 18 mol/h of vapor is generated, how long will it take until vapor containing 50 mol % benzene is produced?

Ans.: 1.08 h.

9.9 An open kettle contains 50 kmol of a dilute aqueous solution of methanol, mole fraction methanol = 0.02, at the bubble point, into which steam is continuously sparged. The entering steam agitates the kettle contents so that they are always of uniform composition, and the vapor produced, always in equilibrium with the liquid, is led away. Operation is adiabatic. For the concentrations encountered it may be assumed that the enthalpy of the steam and evolved vapor are the same, the enthalpy of the liquid in the kettle is essentially constant, and the relative volatility is constant at 7.6. Compute the quantity of steam to be introduced in order to reduce the concentration of methanol in the kettle contents to 0.001 mole fraction.

Ans.: 20.5 kmol.

Continuous fractionation: Savarit-Ponchon method

9.10 An acetone-water solution containing 25 wt % acetone is to be fractionated at a rate of 10 000 kg/h at 1 std atm pressure, and it is planned to recover 99.5% of the acetone in the distillate at a concentration of 99 wt %. The feed will be available at 26.7°C (80°F) and will be preheated by heat exchange with the residue product from the fractionator, which will in turn be cooled to 51.7°C (125°F). The distilled vapors will be condensed and cooled to 37.8°C (125°F) by cooling water entering at 26.7°C and leaving at 40.6°C. Reflux will be returned at 37.8°C, at a reflux ratio $L_0/D = 1.8$. Open steam, available at 70 kN/m², will be used at the base of the tower. The tower will be insulated to reduce heat losses to negligible values. Physical property data are given in Prob. 9.4. Determine:

(a) The hourly rate and composition of distillate and reflux.

(b) The hourly condenser heat load and cooling water rate.

(c) The hourly rate of steam and residue and the composition of the residue.

(d) The enthalpy of the feed as it enters the tower and its condition (express quantitatively).

(e) The number of theoretical trays required if the feed is introduced at the optimum location.

Use large-size graph paper and a sharp pencil.

Ans.: 13.1.

(f) The rates of flow, kg/h, of liquid and vapor at the top, at $x = 0.6, 0.1, 0.025$, and at the bottom tray. For a tower of uniform diameter, the conditions on which tray control the diameter, if the criterion is a 75% approach to flooding?

9.11 In a certain binary distillation, the overhead vapor condenser must be run at a temperature requiring mild refrigeration, and the available refrigeration capacity is very limited. The reboiler temperature is very high, and the available high-temperature heat supply is also limited. Consequently the scheme shown in Fig. 9.56 is adopted, where the intermediate condenser E and reboiler S operate at moderate temperatures. Condensers C and E are total condensers, delivering liquids at their bubble points. Reboiler S is a total vaporizer, delivering saturated vapor. Heat losses are negligible, $D = W$, the external reflux ratio $R = L_0/D = 1.0$, and other conditions are shown on the figure. The system is one which follows the McCabe-Thiele assumptions (H_Lx and H_Vy curves are straight and parallel). Feed is liquid at the bubble point.

(a) Per mole of feed, compute the rates of flow of all streams, the relative size of the two condenser heat loads, and the relative size of the two vaporizer heat loads.

(b) Establish the coordinates of all difference points. Sketch the Hxy diagram and locate the difference points, showing how they are related to the feed. Show the construction for trays.

(c) Sketch the xy diagram, show all operating lines, and locate their intersections with the 45° diagonal. Show the construction for trays.

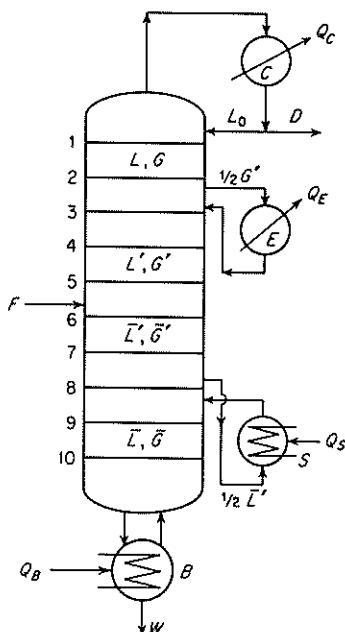


Figure 9.56 Arrangement for Prob. 9.11.

9.12 A distillation column is operated at a temperature below that of the surroundings (as in the distillation of air). Imperfect insulation causes the entire enriching section to be subject to a heat leak inward of Q_E J/s, while the entire stripping section is subject to a heat leak inward of Q_S J/s. Aside from the heat leaks, the system has all the characteristics which satisfy the McCabe-Thiele requirements.

(a) Sketch the Hxy and the xy diagrams, labeling the coordinates of all significant points. Are the operating curves on the xy diagram concave upward or downward?

(b) For a given condenser heat load, explain which is better: to design a new column with the inward heat leak minimized, or to take any possible advantage of a heat leak. Consider the reboiler heat load free, available from the surroundings.

(c) Suppose the column with the inward heat leak is operated at full reboiler heat load, limited by the available heat-transfer surface. The column is then insulated thoroughly, essentially eliminating the heat leak, but without provision to alter the heat load of the reboiler or otherwise to alter the operation. Explain whether the separation obtained will be better than, worse than, or the same as, before the insulation was installed.

9.13 A distillation tower for a binary solution was designed with a partial condenser, using $L_0/D = 1.0$, as in Fig. 9.57a. When built, the piping was installed incorrectly, as in Fig. 9.57b. It was then argued that the error was actually a fortunate one, since the richer reflux would provide a withdrawn distillate richer than planned. The mixture to be distilled has all the characteristics of one obeying McCabe-Thiele assumptions. For a reflux ratio = 1.0, reflux at the bubble point, and the

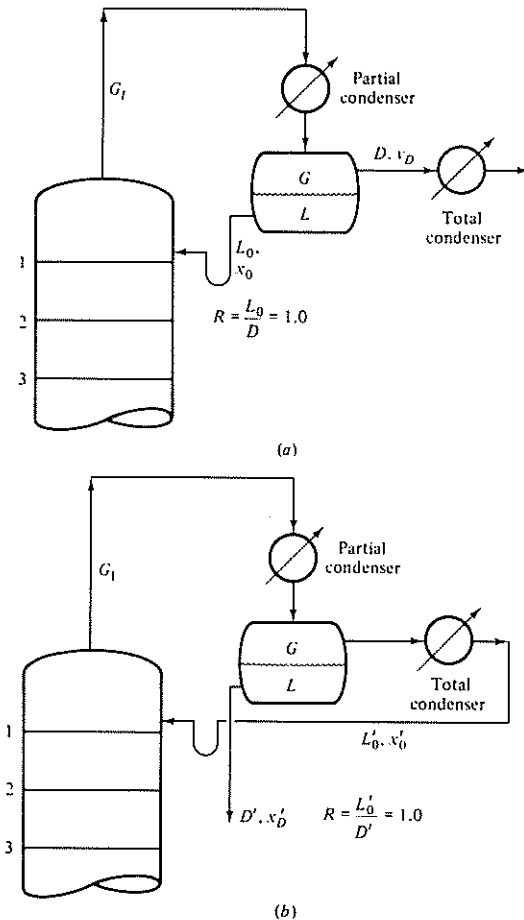


Figure 9.57 Arrangements for Prob. 9.13.

arrangement of Fig. 9.57b:

(a) Sketch the Ponchon-Savarit and xy diagrams for the parts of the columns indicated in the figure. Label the indicated vapor and liquid concentrations.

(b) Establish whether or not the distillate will be richer than originally planned.

9.14 Figure 9.58a shows a few of the top trays of a conventionally arranged fractionator. It has been suggested that a richer distillate might be obtained by the scheme shown in Fig. 9.58b, because of the richer reflux to tray 1. All trays are theoretical. The condenser is a total condenser, delivering liquid at the bubble point in both cases. For both arrangements, $L_0/D = L_0/L_R = 1.0$.

(a) For a binary mixture which in every respect follows the McCabe-Thiele simplifying assumptions, sketch fully labeled Hxy and xy diagrams for the scheme of Fig. 9.58b. Mark on the Hxy diagram the location of every stream on Fig. 9.58b and the required difference points. Show construction lines and tie lines.

(b) Prove that for all otherwise similar circumstances the scheme of Fig. 9.58b in fact delivers a leaner distillate than that of Fig. 9.58a.

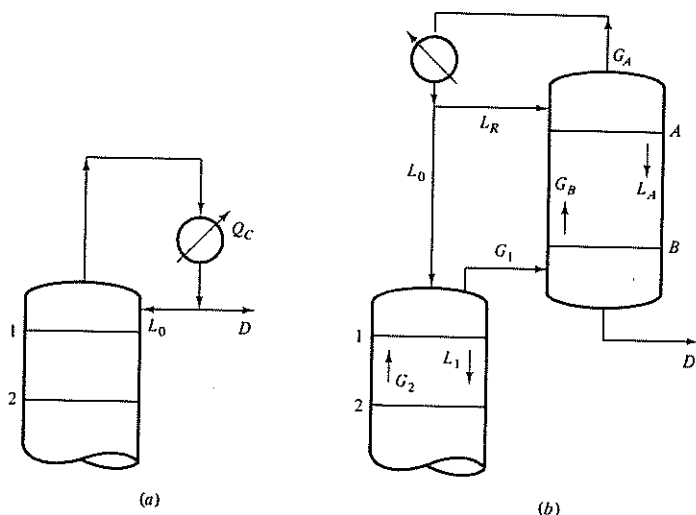


Figure 9.58 Arrangements for Prob. 9.14.

Continuous fractionation: McCabe-Thiele method

9.15 A solution of carbon tetrachloride and carbon disulfide containing 50 wt% each is to be continuously fractionated at standard atmospheric pressure at the rate 4000 kg/h. The distillate product is to contain 95 wt % carbon disulfide, the residue 0.5%. The feed will be 30 mol % vaporized before it enters the tower. A total condenser will be used, and reflux will be returned at the bubble point. Equilibrium data ("The Chemical Engineers' Handbook," 4th ed., p. 13-5), x , y = mole fraction CS_2 :

$t, ^\circ C$	x	y^*	$t, ^\circ C$	x	y^*
76.7	0	0	59.3	0.3908	0.6340
74.9	0.0296	0.0823	55.3	0.5318	0.7470
73.1	0.0615	0.1555	52.3	0.6630	0.8290
70.3	0.1106	0.2660	50.4	0.7574	0.8780
68.6	0.1435	0.3325	48.5	0.8604	0.9320
63.8	0.2585	0.4950	46.3	1.000	1.000

- (a) Determine the product rates, kg/h.
 (b) Determine the minimum reflux ratio.
 (c) Determine the minimum number of theoretical trays, graphically and by means of Eq. (9.85).

(d) Determine the number of theoretical trays required at a reflux ratio equal to twice the minimum and the position of the feed tray.

Ans.: 12.5 theoretical trays.

(e) Estimate the overall tray efficiency of a sieve-tray tower of conventional design and the number of real trays.

(f) Using the distillate temperature as the base temperature, determine the enthalpy of the feed, the products, and the vapor entering the condenser. Determine the heat loads of the condenser and reboiler. Latent and specific heats are available in "The Chemical Engineers' Handbook," 5th ed., pp. 3-116 and 3-129.

9.16 A solution of carbon tetrachloride and carbon disulfide containing 50 wt % of each is to be continuously fractionated to give a distillate and residue analyzing 99.5 and 0.5 wt % carbon disulfide, respectively. Feed will be available at the bubble point, and reflux will be returned at the bubble point.

There is available a sieve-tray tower of conventional design, suitable for use at 1 std atm pressure abs. The diameter is 0.75 m (30 in), and it contains 26 identical cross-flow trays at a spacing of 0.50 m (20 in). A feed nozzle is available only for the tenth tray from the top. Each tray contains a straight weir, 0.46 m (18 in) long, extending 65 mm (2.5 in) from the tray floor. The perforated area is 12% of the active area of the perforated sheet. Adequate condenser and reboiler will be supplied. Equilibrium data are listed in Prob. 9.15.

The listed products represent the minimum purities acceptable, and higher purities at the expense of additional heat load or reduced capacity are not warranted. Estimate the largest feed rate which the column can reasonably be expected to handle.

9.17 Refer to Fig. 9.46, the McCabe-Thiele diagram for separating partially miscible mixtures. Determine the coordinates of the point of intersection of the enriching-section operating line of tower I and the 45° diagonal of the diagram.

9.18 An aqueous solution of furfural contains 4 mol % furfural. It is to be continuously fractionated at 1 std atm pressure to provide solutions containing 0.1 and 99.5 mol % furfural, respectively. Feed is liquid at the bubble point. Equilibrium data ("The Chemical Engineers' Handbook," 4th ed., p. 13-5), x , y = mole fraction furfural:

t , °C	x	y^*	t , °C	x	y^*
100.	0	0	100.6	0.80	0.11
98.56	0.01	0.055	109.5	0.90	0.19
98.07	0.02	0.080	122.5	0.92	0.32
97.90	0.04	0.092	146.0	0.94	0.64
97.90	0.092	0.092	154.8	0.96	0.81
97.90	0.50	0.092	158.8	0.98	0.90
98.7	0.70	0.095	161.7	1.00	1.00

(a) Arrange a scheme for the separation using kettle-type reboilers, and determine the number of theoretical trays required for a vapor boilup rate of 1.25 times the minimum for an infinite number of trays.

(b) Repeat part (a) but use open steam at essentially atmospheric pressure for the tower delivering the water-rich product, with 1.25 times the minimum steam and vapor boilup rates.

(c) Which scheme incurs the greater furfural loss to the water-rich product?

9.19 An aniline-water solution containing 7 wt % aniline, at the bubble point, is to be steam-stripped at a rate of 1000 kg/h in a tower packed with 38-mm (1.5-in) Berl saddles with open steam to remove 99% of the aniline. The condensed overhead vapor will be decanted at 98.5°C, and the water-rich layer returned to the column. The aniline-rich layer will be withdrawn as distillate product. The steam rate will be 1.3 times the minimum. Design the tower for a pressure drop of 400

(N/m²)/m at the top. Note: For concentrations where the equilibrium curve is nearly straight, it should be possible to use overall numbers and heights of transfer units.

Ans.: 6.5 m of packing.

Data: at 98.5°C, the solubility of aniline in water is 7.02 and 89.90 wt % aniline. The vapor-liquid equilibria at 745 mmHg, at which pressure the tower will be operated, are [Griswold et al., *Ind. Eng. Chem.*, 32, 878 (1940)]:

x = mole fraction aniline	0.002	0.004	0.006	0.008	0.010	0.012	Two liquid phases, 98.5°C
y^* = mole fraction aniline	0.01025	0.0185	0.0263	0.0338	0.03575	0.03585	0.0360

9.20 A distillation column is to be fed with the following solution:

$$n\text{-C}_3\text{H}_8 = 0.06 \text{ mole fraction}$$

$$i\text{-C}_4\text{H}_{10} = 0.65$$

$$n\text{-C}_4\text{H}_{10} = 0.26$$

$$n\text{-C}_5\text{H}_{12} = \frac{0.03}{1.00}$$

The distillate is to contain no more than 0.05 mole fraction $n\text{-C}_4\text{H}_{10}$ and essentially no C_5H_{12} . The bottoms will contain no more than 0.05 mole fraction $i\text{-C}_4\text{H}_{10}$ and essentially no C_3H_8 . The overhead vapor will be completely condensed at its bubble point, 37.8°C (100°F), with reflux to the top tray at the rate 2 kmol reflux/kmol distillate withdrawn. Compute the column pressure to be used.

9.21 The following feed at the bubble point at 827 kN/m² (120 lb_f/in²) abs, is to be fractionated at that pressure to provide no more than 7.45% of the $n\text{-C}_5\text{H}_{12}$ in the distillate and no more than 7.6% of the $n\text{-C}_4\text{H}_{10}$ in the bottoms, with a total condenser and reflux returned at the bubble point:

Component	x_F	60°C	90°C	120°C
$n\text{-C}_3\text{H}_8$	0.05			
m		2.39	3.59	5.13
H_G		33 730	36 290	39 080
H_L		20 350	25 800	31 050
$i\text{-C}_4\text{H}_{10}$	0.15			
m		1.21	1.84	2.87
H_G		40 470	43 960	47 690
H_L		24 050	28 800	34 560
$n\text{-C}_4\text{H}_{10}$	0.25			
m		0.80	1.43	2.43
H_G		43 730	46 750	50 360
H_L		24 960	30 010	35 220
$i\text{-C}_5\text{H}_{12}$	0.20			
m		0.39	0.72	1.29
H_G		51 870	55 270	59 430
H_L		29 080	34 610	40 470
$n\text{-C}_5\text{H}_{12}$	0.35			
m		0.20	0.615	1.13
H_G		54 310	57 680	61 130
H_L		29 560	35 590	41 710

The table provides Henry's-law constants and enthalpies expressed as kJ/kmol, from the same sources and using the same base as the data of Illustration 9.13.

Compute the minimum reflux ratio, the minimum number of theoretical trays, and for a reflux ratio $R = 2.58$ estimate the product analyses, condenser and reboiler heat loads, vapor and liquid rates per mole of feed throughout, and the number of theoretical trays.

Ans.: 8 theoretical trays + reboiler.

9.22 The following feed at 60°C (140°F), 2070 kN/m² (300 lb_f/in²) abs, is to be fractionated at that pressure so that the vapor distillate product contains 0.913 percent of the isopentane and the bottoms 0.284 percent of the isobutane:

CH ₄	= 0.50 mole fraction
C ₃ H ₈	= 0.03
<i>i</i> -C ₄ H ₁₀	= 0.10
<i>n</i> -C ₄ H ₁₀	= 0.15
<i>i</i> -C ₅ H ₁₂	= 0.07
<i>n</i> -C ₅ H ₁₂	= 0.05
<i>n</i> -C ₆ H ₁₄	= 0.10
	1.00

Compute: (a) The minimum reflux ratio and (b) the minimum number of trays. At a reflux ratio = 2.0, estimate (c) the product analyses and (d) the number of theoretical trays by the correlations of Gilliland [17], Erbar and Maddox [12], Brown and Martin [5], and Strangio and Treybal [59].

PART THREE

LIQUID-LIQUID OPERATIONS

Liquid extraction, the only operation in this category, is basically very similar to the operations of gas-liquid contact described in the previous part. The creation of a new insoluble liquid phase by addition of a solvent to a mixture accomplishes in many respects the same result as the creation of a new phase by the addition of heat in distillation operations, for example, or by addition of gas in desorption operations. The separations produced by single stages, the use of countercurrent cascades and reflux to enhance the extent of separation—all have their liquid-extraction counterparts. The similarity between them will be exploited in explaining what liquid extraction can accomplish.

Certain differences in the operations nevertheless make it expedient to provide a separate treatment for liquid extraction. The considerably greater change in mutual solubility of the contacted liquid phases which may occur during passage through a cascade of stages requires somewhat different techniques of computation than are necessary for gas absorption or stripping. The considerably smaller differences in density of the contacted phases and their relatively low interfacial tension, compared with the corresponding gas-liquid systems, require consideration in the design of apparatus. The larger number of variables which apparently influence the rate of mass transfer in countercurrent equipment makes the correlation of design data much more difficult. For these reasons separate treatment is desirable, at least for the present.

Despite the increasingly extensive application of liquid extraction, thanks to its great versatility, and the extensive amount of research that has been done, it is nevertheless a relatively immature unit operation. It is characteristic of such operations that equipment types change rapidly, new designs being proposed frequently and lasting through a few applications, only to be replaced by others. Design principles for such equipment are never fully developed, and a reliance must often be put on pilot-plant testing for new installations.

LIQUID EXTRACTION

Liquid extraction, sometimes called solvent extraction, is the separation of the constituents of a liquid solution by contact with another insoluble liquid. If the substances constituting the original solution distribute themselves differently between the two liquid phases, a certain degree of separation will result, and this can be enhanced by use of multiple contacts or their equivalent in the manner of gas absorption and distillation.

A simple example will indicate the scope of the operation and some of its characteristics. If a solution of acetic acid in water is agitated with a liquid such as ethyl acetate, some of the acid but relatively little water will enter the ester phase. Since at equilibrium the densities of the aqueous and ester layers are different, they will settle when agitation stops and can be decanted from each other. Since now the ratio of acid to water in the ester layer is different from that in the original solution and also different from that in the residual water solution, a certain degree of separation will have occurred. This is an example of stagewise contact, and it can be carried out either in batch or in continuous fashion. The residual water can be repeatedly extracted with more ester to reduce the acid content still further, or we can arrange a countercurrent cascade of stages. Another possibility is to use some sort of countercurrent continuous-contact device, where discrete stages are not involved. The use of reflux, as in distillation, may enhance the ultimate separation still further.

In all such operations, the solution which is to be extracted is called the *feed*, and the liquid with which the feed is contacted is the *solvent*. The solvent-rich product of the operation is called the *extract*, and the residual liquid from which solute has been removed is the *raffinate*.

More complicated processes may use two solvents to separate the components of a feed. For example, a mixture of *p*- and *o*-nitrobenzoic acids can be

separated by distributing them between the insoluble liquids chloroform and water. The chloroform preferentially dissolves the para isomer and the water the ortho isomer. This is called *double-solvent*, or *fractional*, extraction.

Fields of Usefulness

Applications of liquid extraction fall into several categories: those where extraction is in direct competition with other separation methods and those where it seems uniquely qualified.

In competition with other mass-transfer operations Here relative costs are important. Distillation and evaporation are direct separation methods, the products of which are composed of essentially pure substances. Liquid extraction, on the other hand, produces new solutions which must in turn be separated, often by distillation or evaporation. Thus, for example, acetic acid can be separated from dilute solution with water, with difficulty by distillation or with relative ease by extraction into a suitable solvent followed by distillation of the extract. For the more dilute solutions particularly, where water must be vaporized in distillation, extraction is more economical, especially since the heat of vaporization of most organic solvents is substantially less than that of water. Extraction may also be attractive as an alternative to distillation under high vacuum at very low temperatures to avoid thermal decomposition. For example, long-chain fatty acids can be separated from vegetable oils by high-vacuum distillation but more economically by extraction with liquid propane. Tantalum and niobium can be separated by very tedious fractional crystallization of the double fluorides with potassium but with relative ease by liquid extraction of the hydrofluoric acid solutions with methyl isobutyl ketone.

As a substitute for chemical methods Chemical methods consume reagents and frequently lead to expensive disposal problems for chemical by-products. Liquid extraction, which incurs no chemical consumption or by-product production, can be less costly. Metal separations such as uranium-vanadium, hafnium-zirconium, and tungsten-molybdenum and the fission products of atomic-energy processes are more economical by liquid extraction. Even lower-cost metals such as copper and inorganic chemicals such as phosphoric acid, boric acid, and the like are economically purified by liquid extraction, despite the fact that the cost of solvent recovery must be included in the final reckoning.

For separations not now possible by other methods In distillation, where the vapor phase is created from the liquid by addition of heat, the vapor and liquid are necessarily composed of the same substances and are therefore chemically very similar. The separations produced then depend upon the vapor pressures of the substances. In liquid extraction, in contrast, the major constituents of the two phases are chemically very different, and this makes separations according to chemical type possible. For example, aromatic and paraffinic hydrocarbons

of nearly the same molecular weight are impossible to separate by distillation because their vapor pressures are nearly the same, but they can readily be separated by extraction with any of a number of solvents, e.g., liquid sulfur dioxide, diethylene glycol, or sulfolane. (Extractive distillation is also useful for such operations, but it is merely extraction of the *vapor* phase with a solvent, whereas liquid extraction is extraction of the *liquid* phase. Frequently the same solvents are useful for both, as might be expected.) Many pharmaceutical products, e.g., penicillin, are produced in mixtures so complex that only liquid extraction is a feasible separation device.

LIQUID EQUILIBRIA

Extraction involves the use of systems composed of at least three substances, and although for the most part the insoluble phases are chemically very different, generally all three components appear at least to some extent in both phases. The following notation scheme will be used to describe the concentrations and amounts of these ternary mixtures, for purposes of discussing both equilibria and material balances.

Notation Scheme

1. A and B are pure, substantially insoluble liquids, and C is the distributed solute. Mixtures to be separated by extraction are composed of A and C, and B is the extracting solvent.
2. The same letter will be used to indicate the quantity of a solution or mixture and the location of the mixture on a phase diagram. Quantities are measured by mass for batch operations, mass/time for continuous operation. Thus,

E = mass/time of solution E , an extract, shown on a phase diagram by point E

R = mass/time of solution R , a raffinate, shown on a phase diagram by point R

B = mass/time of solvent B

Solvent-free (B-free) quantities are indicated by primed letters. Thus,

E' = mass B-free solution/time, shown on a phase diagram by point E

$$E = E'(1 + N_E)$$

3. x = weight fraction C in the solvent-lean (A-rich), or raffinate, liquids

y = weight fraction C in the solvent-rich (B-rich), or extract, liquids

$$x' = x/(1 - x) = \text{mass C/mass non-C in the raffinate liquids}$$

$$y' = y/(1 - y) = \text{mass C/mass non-C in the extract liquids}$$

X = weight fraction C in the raffinate liquids on a B-free basis, mass C/(mass A + mass C)

Y = weight fraction C in the extract liquids on a B-free basis, mass C/(mass A + mass C)

N = weight fraction B on a B-free basis, mass B/(mass A + mass C)

Subscripts identify the solution or mixture to which the concentration terms refer. Stages are identified by number. Thus, x_3 = wt fraction C in the raffinate from stage 3, Y_3 = wt fraction C (B-free basis) in the extract from stage 3, etc. For other solutions identified by a letter on a phase diagram, the same letter is used as an identifying subscript. Thus, x_M = wt fraction C in the mixture M . An asterisk specifically identifies equilibrium concentrations where the condition of equilibrium is emphasized. Thus, y_E^* = wt fraction C in the equilibrium solution E .

- Throughout the discussion of equilibria, material balances, and stagewise calculations, mole fractions, mole ratios, and kilomoles may be consistently substituted for weight fractions, weight ratios, and kilograms, respectively.

Equilateral-Triangular Coordinates

These are used extensively in the chemical literature to describe graphically the concentrations in ternary systems. It is the property of an equilateral triangle that the sum of the perpendicular distances from any point within the triangle to the three sides equals the altitude of the triangle. We can therefore let the altitude represent 100 percent composition and the distances to the three sides the percentages or fractions of the three components. Refer to Fig. 10.1. Each apex of the triangle represents one of the pure components, as marked. The perpendicular distance from any point such as K to the base AB represents the percentage of C in the mixture at K , the distance to the base AC the percentage of B, and that to the base CB the percentage of A. Thus $x_K = 0.4$. Any point on a side of the triangle represents a binary mixture. Point D , for example, is a binary containing 80 percent A, 20 percent B. All points on the line DC represent mixtures containing the same ratio of A to B and can be considered as

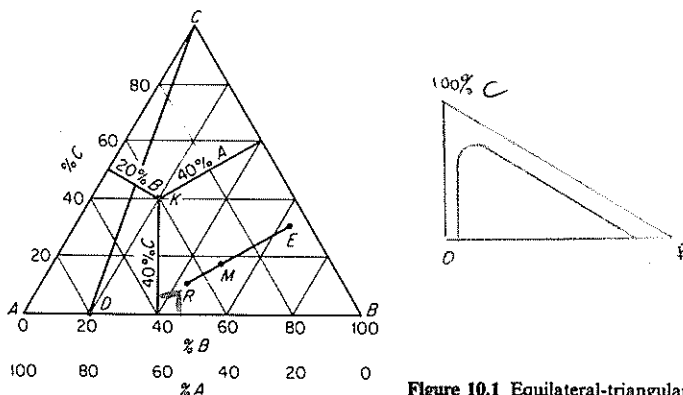


Figure 10.1 Equilateral-triangular coordinates.

mixtures originally at D to which C has been added. If R kg of a mixture at point R is added to E kg of a mixture at E , the new mixture is shown on the straight line RE at point M , such that

$$\frac{R}{E} = \frac{\text{line } ME}{\text{line } RM} = \frac{x_E - x_M}{x_M - x_R} \quad \text{inverse lever arm rule} \quad (10.1)$$

Alternatively the composition corresponding to point M can be computed by material balances, as will be shown later. Similarly, if a mixture at M has removed from it a mixture of composition E , the new mixture is on the straight line EM extended in the direction away from E and located at R so that Eq. (10.1) applies.

Equation (10.1) is readily established. Refer to Fig. 10.2, which again shows R kg of mixture at R added to E kg of mixture at E . Let M represent the kilograms of new mixture as well as the composition on the figure. Line RL = weight fraction C in $R = x_R$, line MO = weight fraction C in $M = x_M$, and line ET = weight fraction C in $E = x_E$. A total material balance,

$$R + E = M$$

A balance for component C ,

$$R(\text{line } RL) + E(\text{line } ET) = M(\text{line } MO)$$

$$Rx_R + Ex_E = Mx_M$$

Eliminating M , we get

$$\frac{R}{E} = \frac{\text{line } ET - \text{line } MO}{\text{line } MO - \text{line } RL} = \frac{x_E - x_M}{x_M - x_R}$$

But line $ET - \text{line } MO = \text{line } EP$, and line $MO - \text{line } RL = \text{line } MK = \text{line } PS$. Therefore

$$\frac{R}{E} = \frac{\text{line } EP}{\text{line } PS} = \frac{\text{line } ME}{\text{line } RM}$$

The following discussion is limited to those types of systems which most frequently occur in liquid-extraction operations. For a complete consideration of the many types of systems which may be encountered, the student is referred to one of the more comprehensive texts on the phase rule [51].

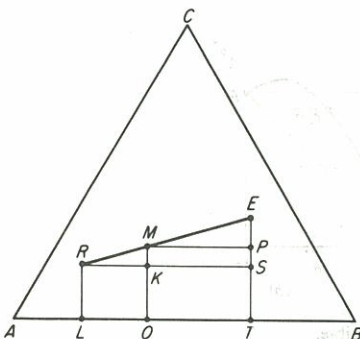


Figure 10.2 The mixture rule.

conveniently shown on the distribution diagram (Fig. 10.3*b*), where the point (E , R) lies above the diagonal $y = x$. The ratio y^*/x , the *distribution coefficient*, is in this case greater than unity. The concentrations of C at the ends of the tie lines, when plotted against each other, give rise to the distribution curve shown. Should the tie lines on Fig. 10.3*a* slope in the opposite direction, with C favoring A at equilibrium, the distribution curve will lie below the diagonal. The distribution curve may be used for interpolating between tie lines when only a few have been experimentally determined. Other methods of interpolation are also available [72].

Effect of temperature To show the effect of temperature in detail requires a three-dimensional figure, as in Fig. 10.4*a*, where temperature is plotted vertically and the isothermal triangles are seen to be sections through the prism. For most systems of this type, the mutual solubility of A and B increases with increasing temperature, and above some temperature t_4 , the critical solution temperature, they dissolve completely. The increased solubility at higher temperatures influences the ternary equilibria considerably, and this is best shown by projection of the isotherms onto the base triangle, as in Fig. 10.4*b*. Not only does the area of heterogeneity decrease at higher temperatures, but the slopes of the tie lines may also change. Liquid-extraction operations, which depend upon the formation of insoluble liquid phases, must be carried on at temperatures below t_4 . Other, less common, temperature effects are also known [51, 72].

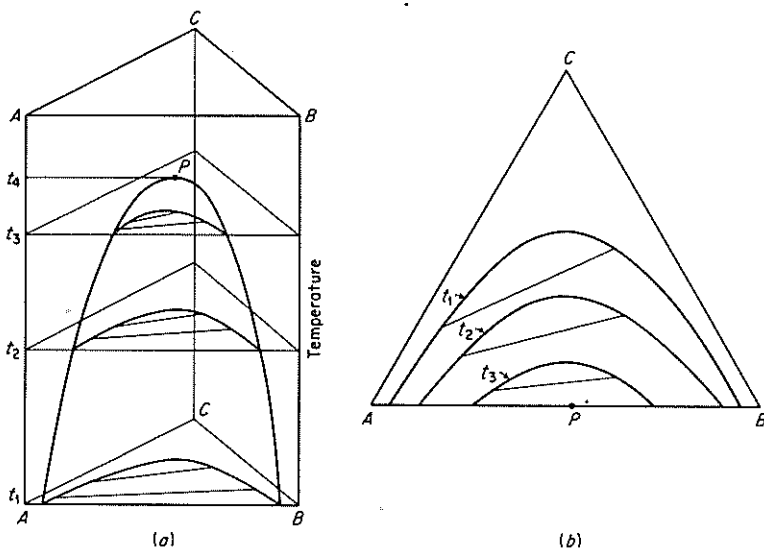


Figure 10.4 Effect of temperature on ternary equilibria.

Effect of pressure Except at very high pressures, the influence of pressure on the liquid equilibrium is so small that it can generally be ignored. All the diagrams shown are therefore to be considered as having been plotted at sufficiently high pressure to maintain a completely condensed system, i.e., above the vapor pressures of the solutions. However, should the pressure be sufficiently reduced to become less than the vapor pressure of the solutions, a vapor phase will appear and the liquid equilibrium will be interrupted. Such an effect on a binary solubility curve of the type *APB* of Fig. 10.4*a* is shown in Fig. 9.8.

Systems of Three Liquids—Two Pairs Partially Soluble

This type is exemplified by the system chlorobenzene (A)–water (B)–methyl ethyl ketone (C), where A and C are completely soluble, while the pairs A–B and B–C show only limited solubility. Refer to Fig. 10.5*a*, a typical isotherm. At the prevailing temperature, points *K* and *J* represent the mutual solubilities of A and B and points *H* and *L* those of B and C. Curves *KRH* (A-rich) and *JEL* (B-rich) are the ternary solubility curves, and mixtures outside the band between these curves form homogeneous single-phase liquid solutions. Mixtures such as *M*, inside the heterogeneous area, form two liquid phases at equilibrium at *E* and *R*, joined on the diagram by a tie line. The corresponding distribution curve is shown in Fig. 10.5*b*.

Effect of temperature Increased temperature usually increases the mutual solubilities and at the same time influences the slope of the tie lines. Figure 10.6 is typical of the effect that can be expected. Above the critical solution temperature of the binary B–C at t_3 , the system is similar to the first type discussed. Other temperature effects are also possible [51, 72].

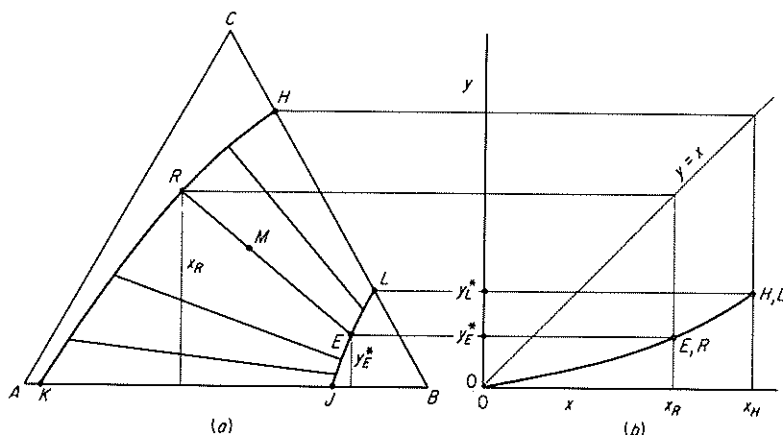


Figure 10.5 System of three liquids, A–B and B–C partially miscible.

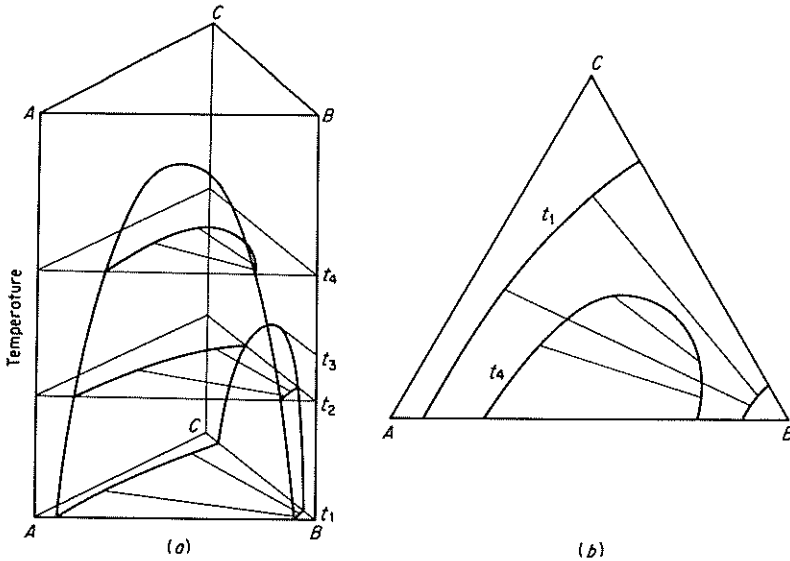


Figure 10.6 Effect of temperature on ternary equilibria.

Systems of Two Partially Soluble Liquids and One Solid

When the solid does not form compounds such as hydrates with the liquids, the system will frequently have the characteristics of the isotherm of Fig. 10.7, an example of which is the system naphthalene (C)–aniline (A)–isooctane (B). Solid C dissolves in liquid A to form a saturated solution at *K* and in liquid B to give the saturated solution at *L*. A and B are soluble only to the extent shown at *H*

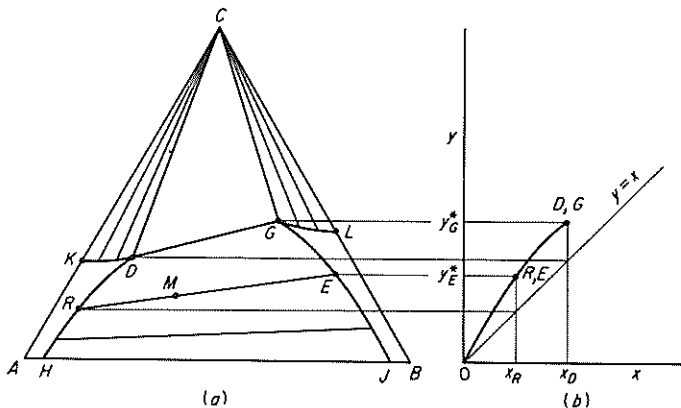


Figure 10.7 System of two partially soluble liquids A and B and one solid C.

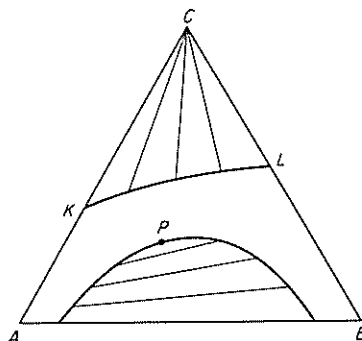


Figure 10.8 System of two partially soluble liquids A and B and one solid C.

and *J*. Mixtures in the regions *AKDH* and *BLGJ* are homogeneous liquid solutions. The curves *KD* and *GL* show the effect of adding A and B upon the solubilities of the solid. In the region *HDGJ* two liquid phases form: if C is added to the insoluble liquids *H* and *J* to give a mixture *M*, the equilibrium liquid phases will be *R* and *E*, joined by a tie line. All mixtures in the region *CDG* consist of three phases, solid C, and saturated liquid solutions at *D* and *G*. Liquid-extraction operations are usually confined to the region of the two liquid phases, which is that corresponding to the distribution curve shown.

Increased temperature frequently changes these systems to the configuration shown in Fig. 10.8.

Other Coordinates

Because the equilibrium relationship can rarely be expressed algebraically with any convenience, extraction computations must usually be made graphically on a phase diagram. The coordinate scales of equilateral triangles are necessarily

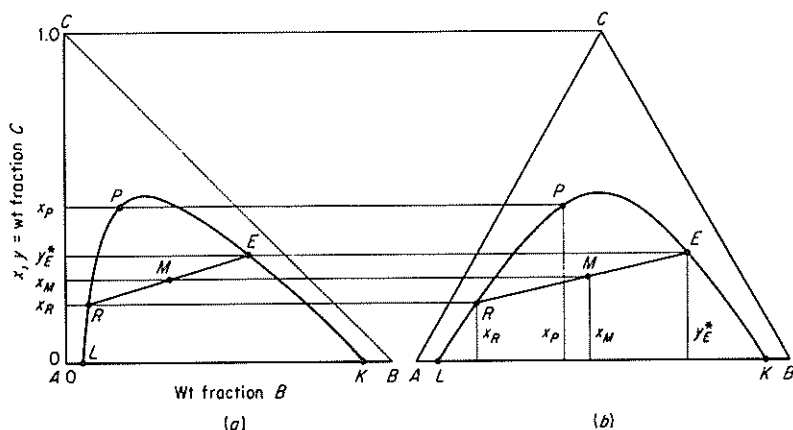


Figure 10.9 Rectangular coordinates.

always the same, and in order to be able to expand one concentration scale relative to the other, rectangular coordinates can be used. One of these is formed by plotting concentrations of B as abscissa against concentrations of C (x and y) as ordinate, as in Fig. 10.9a. Unequal scales can be used in order to expand the plot as desired. Equation (10.1) applies for mixtures on Fig. 10.9a, regardless of any inequality of the scales.

Another rectangular coordinate system involves plotting as abscissa the weight fraction C on a B-free basis, X and Y in the A- and B-rich phases, respectively, against N , the B-concentration on a B-free basis, as ordinate, as shown in the upper part of Fig. 10.10. This has been plotted for a system of two partly miscible pairs, such as that of Fig. 10.5.

The similarity of such diagrams to the enthalpy-concentration diagrams of Chap. 9 is clear. In extraction, the two phases are produced by addition of solvent, in distillation by addition of heat, and solvent becomes the analog of heat. This is emphasized by the ordinate of the upper part of Fig. 10.10. Tie lines such as QS can be projected to X, Y coordinates, as shown in the lower figure, to produce a solvent-free distribution graph similar to those of distillation. The mixture rule on these coordinates (see the upper part of Fig. 10.10) is

$$\frac{M'}{N'} = \frac{Y_N - Y_P}{Y_P - Y_M} = \frac{X_N - X_P}{X_P - X_M} = \frac{\text{line } NP}{\text{line } PM} \quad \left. \begin{array}{l} \text{must be} \\ \text{solvent} \\ \text{free amount} \end{array} \right\} (10.2)$$

where M' and N' are the B-free weights of these mixtures.

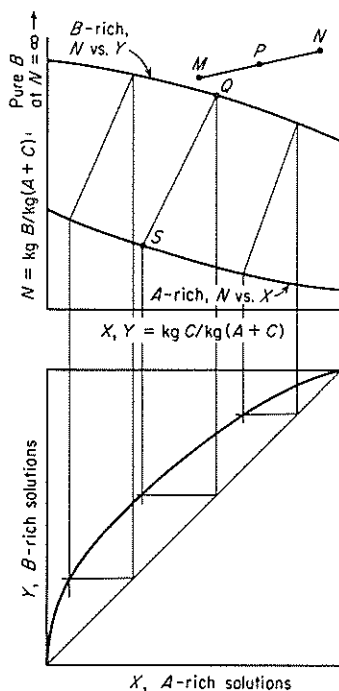


Figure 10.10 Rectangular coordinates, solvent-free basis, for a system of two partly miscible liquid pairs.

Multicomponent Systems

The simplest system of four components occurs when two solutes distribute between two solvents, e.g., the distribution of formic and acetic acids between the partly soluble solvents water and carbon tetrachloride. Complete display of such equilibria requires a three-dimensional graph [72], which is difficult to work with, but frequently we can simplify this to the distribution (xy) curves, one for each solute, such as that for a single solute in Fig. 10.3*b*. More than four components cannot be conveniently dealt with graphically.

Choice of Solvent

There is usually a wide choice of liquids to be used as solvents for extraction operations. It is unlikely that any particular liquid will exhibit all the properties considered desirable for extraction, and some compromise is usually necessary. The following are the quantities to be given consideration in making a choice:

1. *Selectivity.* The effectiveness of solvent B for separating a solution of A and C into its components is measured by comparing the ratio of C to A in the B-rich phase to that in the A-rich phase at equilibrium. The ratio of the ratios, the separation factor, or selectivity, β is analogous to the relative volatility of distillation. If E and R are the equilibrium phases,

$$\beta = \frac{(\text{wt fraction C in } E)/(\text{wt fraction A in } E)}{(\text{wt fraction C in } R)/(\text{wt fraction A in } R)} = \frac{y_E^*(\text{wt fraction A in } R)}{x_R(\text{wt fraction A in } E)} \quad (10.3)$$

For all useful extraction operations the selectivity must exceed unity, the more so the better. If the selectivity is unity, no separation is possible.

Selectivity usually varies considerably with solute concentration, and in systems of the type shown in Fig. 10.3 it will be unity at the plait point. In some systems it passes from large values through unity to fractional values, and these are analogous to azeotropic systems of distillation.

2. *Distribution coefficient.* This is the ratio y^*/x at equilibrium. While it is not necessary that the distribution coefficient be larger than 1, large values are very desirable since less solvent will then be required for the extraction.
3. *Insolubility of solvent.* Refer to Fig. 10.11. For both systems shown, only those A-C mixtures between D and A can be separated by use of the solvents B or B', since mixtures richer in C will not form two liquid phases with the solvents. Clearly the solvent in Fig. 10.11*a*, which is the more insoluble of the two, will be the more useful. In systems of the type shown in Figs. 10.5*a* and 10.7*a*, if the solubility of C in B is small (point L near the B apex), the

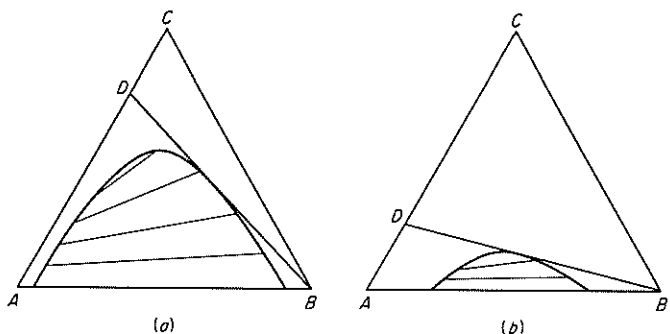


Figure 10.11 Influence of solvent solubility on extraction.

capacity of solvent B to extract C is small and large amounts of solvent are then required.

4. *Recoverability.* It is always necessary to recover the solvent for reuse, and this must ordinarily be done by another of the mass-transfer operations, most frequently distillation. If distillation is to be used, the solvent should form no azeotrope with the extracted solute and mixtures should show high relative volatility for low-cost recovery. That substance in the extract, either solvent or solute, which is present as the lesser quantity should be the more volatile in order to reduce heat costs. If the solvent must be volatilized, its latent heat of vaporization should be small.
5. *Density.* A difference in densities of the saturated liquid phases is necessary, both for stagewise and continuous-contact equipment operation. The larger this difference the better. In systems of the type shown in Fig. 10.3, the density difference for equilibrium phases will become less as C concentrations increase and will be zero at the plait point. It may reverse in sign before reaching the plait point, in which case continuous-contact equipment cannot be specified to operate at the concentrations at which the density difference passes through zero.
6. *Interfacial tension.* The larger the interfacial tension, the more readily coalescence of emulsions will occur but the more difficult the dispersion of one liquid in the other will be. Coalescence is usually of greater importance, and interfacial tension should therefore be high. Interfacial tension between equilibrium phases in systems of the type shown in Fig. 10.3 falls to zero at the plait point.
7. *Chemical reactivity.* The solvent should be stable chemically and inert toward the other components of the system and toward the common materials of construction.
8. *Viscosity, vapor pressure, and freezing point.* These should be low for ease in handling and storage.
9. The solvent should be *nontoxic, nonflammable*, and of *low cost*.

EQUIPMENT AND FLOWSHEETS

As with the gas-liquid operations, liquid extraction processes are carried out in stagewise equipment which can be arranged in multistage cascades and also in differential-contact apparatus. We shall consider the methods of calculation and the nature of the equipment and its characteristics separately for each type.

STAGewise CONTACT

Extraction in equipment of the stage type can be carried on according to a variety of flowsheets, depending upon the nature of the system and the extent of separation desired. In the discussion which follows it is to be understood that each state is a *theoretical* or *equilibrium* stage, such that the effluent extract and raffinate solutions are in equilibrium with each other. Each stage must include facilities for contacting the insoluble liquids and separating the product streams. A combination of a mixer and a settler may therefore constitute a stage, and in multistage operation these may be arranged in cascades as desired. In counter-current multistage operation it is also possible to use towers of the multistage type, as described earlier.

Single-Stage Extraction

This may be a batch or a continuous operation. Refer to Fig. 10.12. The flowsheet shows the extraction stage. Feed of mass F (if batch) or F mass/time (if continuous) contains substances A and C at x_F weight fraction C. This is contacted with mass S_1 (or mass/time) of a solvent, principally B, containing y_S weight fraction C, to give the equilibrium extract E_1 and raffinate R_1 , each measured in mass or mass/time. Solvent recovery then involves separate removal of solvent B from each product stream (not shown).

The operation can be followed in either of the phase diagrams as shown. If the solvent is pure B ($y_S = 0$), it will be plotted at the B apex, but sometimes it has been recovered from a previous extraction and therefore contains a little A and C as well, as shown by the location of S . Adding S to F produces in the extraction stage a mixture M_1 which, on settling, forms the equilibrium phases E_1 and R_1 joined by the tie line through M_1 . A total material balance is

$$F + S_1 = M_1 = E_1 + R_1 \quad (10.4)$$

and point M_1 can be located on line FS by the mixture rule, Eq. (10.1), but it is usually more satisfactory to locate M_1 by computing its C concentration. Thus, a C balance provides

$$Fx_F + S_1y_S = M_1x_{M1} \quad (10.5)$$

from which x_{M1} can be computed. Alternatively, the amount of solvent to

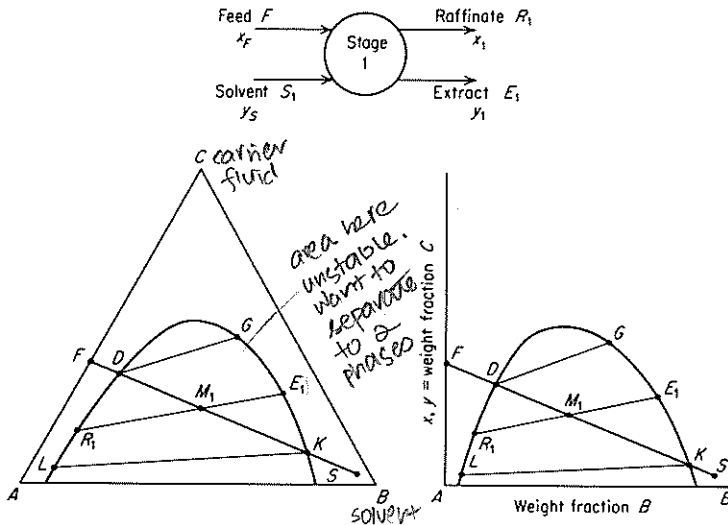


Figure 10.12 Single-stage extraction.

provide a given location for M_1 on the line FS can be computed:

$$\frac{S_1}{F} = \frac{x_F - x_{M_1}}{x_{M_1} - y_S} \quad (10.6)$$

The quantities of extract and raffinate can be computed by the mixture rule, Eq. (10.1), or by the material balance for C:

$$E_1 y_1 + R_1 x_1 = M_1 x_{M_1} \quad (10.7)$$

$$E_1 = \frac{M_1 (x_{M_1} - x_1)}{y_1 - x_1} \quad (10.8)$$

and R_1 can be determined through Eq. (10.4).

Since two insoluble phases must form for an extraction operation, point M_1 must lie within the heterogeneous liquid area, as shown. The minimum amount of solvent is thus found by locating M_1 at D , which would then provide an infinitesimal amount of extract at G , and the maximum amount of solvent is found by locating M_1 at K , which provides an infinitesimal amount of raffinate at L . Point L also represents the raffinate with the lowest possible C concentration, and if a lower value were required, the recovered solvent S would have to have a lower C concentration.

Computations for systems of two insoluble liquid pairs, or with a distributed solute which is a solid, are made in exactly the same manner, and Eqs. (10.4) to (10.8) all apply.

All the computations can also be made on a solvent-free basis, as in the upper part of Fig. 10.13, if the nature of this diagram makes it convenient. If

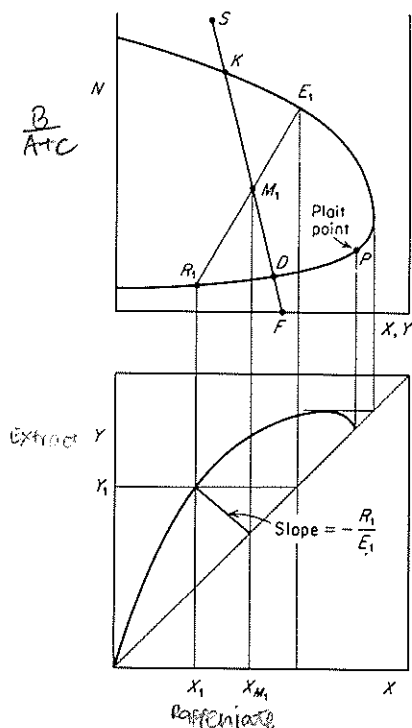


Figure 10.13 Single-stage extraction, solvent-free coordinates.

solvent S is pure B , its N value is infinite and the line FS is then vertical. Products E_1 and R_1 lie on a tie line through M_1 representing the entire mixture in the extractor. Material balances for use with this diagram must be made on a B -free basis. Thus,

$$F' + S' = M'_1 = E'_1 + R'_1 \quad (10.9)$$

where the primes indicate B -free weight (the feed is normally B -free, and $F = F'$). A balance for C is

$$F'X_F + S'Y_S = M'_1X_{M1} = E'_1Y_1 + R'_1X_1 \quad (10.10)$$

$$\text{and for } B \quad F'N_F + X'N_S = M'_1N_{M1} = E'_1N_{E1} + R'_1N_{R1} \quad (10.11)$$

(ordinarily, $N_F = 0$ since the feed contains no B). The coordinates of M'_1 can be computed, point M'_1 located on line FS , the tie line located, and the B -free weights, E'_1 and R'_1 , computed:

$$E'_1 = \frac{M'_1(X_{M1} - X_1)}{Y_1 - X_1} \quad (10.12)$$

and R'_1 is obtained through Eq. (10.9). The total weights of the saturated extract and raffinate are then

$$E_1 = E'_1(1 + N_{E1}) \quad R_1 = R'_1(1 + N_{R1}) \quad (10.13)$$

If the solvent is pure B, whence $N_S = \infty$, these equations still apply, with the simplification that $S' = 0$, $Y_S = 0$, $S'N_S = B$, and $F' = M'_1$. Minimum and maximum amounts of solvent correspond to putting M_1 at D and K on the figure, as before.

Equations (10.9) and (10.10) lead to

$$\frac{R'_1}{E'_1} = \frac{Y_1 - X_{M1}}{X_{M1} - X_1} \quad (10.14)$$

When the equilibrium extract and raffinate are located on the lower diagram of Fig. 10.13, Eq. (10.14) is seen to be that of the operating line shown, of slope $-R'_1/E'_1$. The single stage is seen to be analogous to the flash vaporization of distillation, with solvent replacing heat. If pure B is used as solvent, the operating line on the lower figure passes through the 45° line at X_F , which completes the analogy (see Fig. 9.14).

Multistage Crosscurrent Extraction

This is an extension of single-stage extraction, wherein the raffinate is successively contacted with fresh solvent, and may be done continuously or in batch. Refer to Fig. 10.14, which shows the flow sheet for a three-stage extraction. A

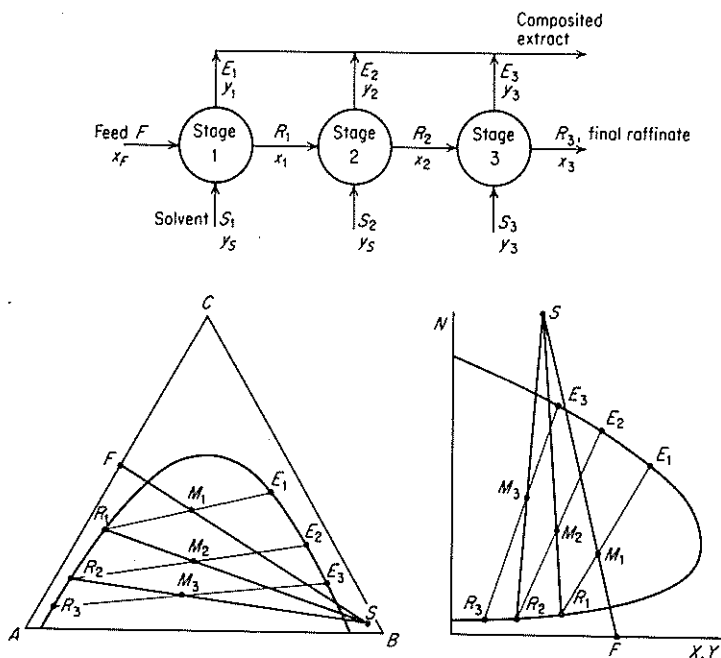


Figure 10.14 Crosscurrent extraction.

single final raffinate results, and the extracts can be combined to provide the composited extract, as shown. As many stages as desired can be used.

Computations are shown on triangular and on solvent-free coordinates. All the material balances for a single stage of course now apply to the first stage. Subsequent stages are dealt with in the same manner, except of course that the feed to any stage is the raffinate from the preceding stage. Thus, for any stage n :

$$\text{Total balance:} \quad R_{n-1} + S_n = M_n = E_n + R_n \quad (10.15)$$

$$\text{C balance:} \quad R_{n-1}x_{n-1} + S_n y_S = M_n x_{Mn} = E_n y_n + R_n x_n \quad (10.16)$$

For the solvent-free coordinates,

$$\text{A + C balance:} \quad R'_{n-1} + S'_n = M'_n = E'_n + R'_n \quad (10.17)$$

$$\text{C balance:} \quad R'_{n-1}X_{n-1} + S'_n Y_S = M'_n X_{Mn} = E'_n Y_n + R'_n X_n \quad (10.18)$$

$$\text{B balance:} \quad R'_{n-1}N_{R_{n-1}} + S'_n N_S = M'_n N_{Mn} = E'_n N_{E_n} + R'_n N_{R_n} \quad (10.19)$$

from which the quantities can be calculated for either type of graph.

Unequal amounts of solvent can be used in the various stages, and even different temperatures, in which case each stage must be computed with the help of a phase diagram at the appropriate temperature. For a given final raffinate concentration, the greater the number of stages the less total solvent will be used.

Illustration 10.1 If 100 kg of a solution of acetic acid (C) and water (A) containing 30% acid is to be extracted three times with isopropyl ether (B) at 20°C, using 40 kg of solvent in each stage, determine the quantities and compositions of the various streams. How much solvent would be required if the same final raffinate concentration were to be obtained with one stage?

SOLUTION The equilibrium data at 20°C are listed below [*Trans. AIChE*, 36, 628 (1940), with permission]. The horizontal rows give the concentrations in equilibrium solutions. The system is of the type shown in Fig. 10.9, except that the tie lines slope downward toward the B apex. The rectangular coordinates of Fig. 10.9a will be used, but only for acid concentrations up to $x = 0.30$. These are plotted in Fig. 10.15.

Wt % acetic acid, 100x	Water layer		Isopropyl ether layer		
	Water	Isopropyl ether	Acetic acid, 100y*	Water	Isopropyl ether
0.69	98.1	1.2	0.18	0.5	99.3
1.41	97.1	1.5	0.37	0.7	98.9
2.89	95.5	1.6	0.79	0.8	98.4
6.42	91.7	1.9	1.93	1.0	97.1
13.30	84.4	2.3	4.82	1.9	93.3
25.50	71.1	3.4	11.40	3.9	84.7
36.70	58.9	4.4	21.60	6.9	71.5
44.30	45.1	10.6	31.10	10.8	58.1
46.40	37.1	16.5	36.20	15.1	48.7

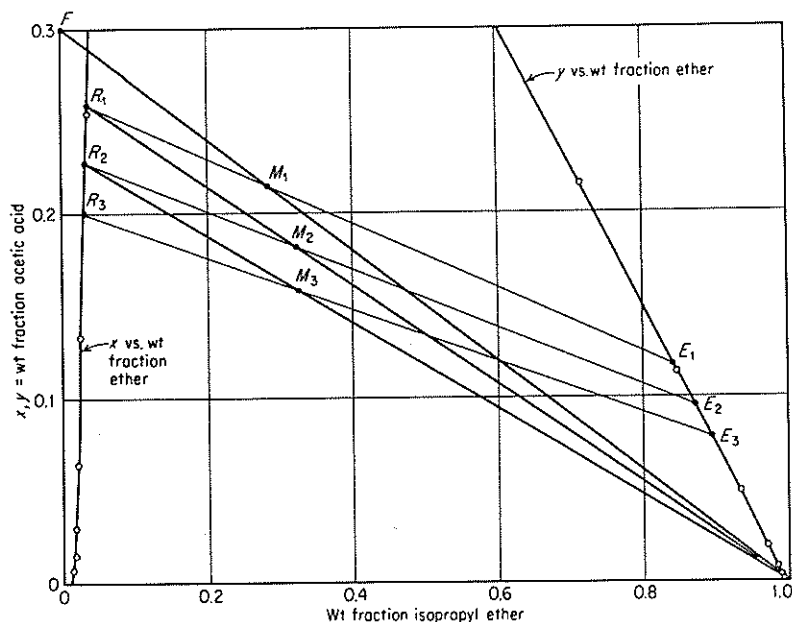


Figure 10.15 Solution to Illustration 10.1.

Stage 1 $F = 100$ kg, $x_F = 0.30$, $y_S = 0$, $S_1 = B_1 = 40$ kg. Eq. (10.4):

$$M_1 = 100 + 40 = 140 \text{ kg}$$

Eq. (10.5):

$$100(0.30) + 40(0) = 140x_{M1} \quad x_{M1} = 0.214$$

Point M_1 is located on line FB . With the help of a distribution curve, the tie line passing through M_1 is located as shown, and $x_1 = 0.258$, $y_1 = 0.117$ wt fraction acetic acid. Eq. (10.8):

$$E_1 = \frac{140(0.214 - 0.258)}{0.117 - 0.258} = 43.6 \text{ kg}$$

Eq. (10.4):

$$R_1 = 140 - 43.6 = 96.4 \text{ kg}$$

Stage 2 $S_2 = B_2 = 40$ kg. Eq. (10.15):

$$M_2 = R_1 + B_2 = 96.4 + 40 = 136.4 \text{ kg}$$

Eq. (10.16):

$$96.4(0.258) + 40(0) = 136.4x_{M2} \quad x_{M2} = 0.1822$$

Point M_2 is located on line R_1B and the tie line R_2E_2 through M_2 , $x_2 = 0.227$, $y_2 = 0.095$. Eq. (10.8) becomes

$$E_2 = \frac{M_2(x_{M2} - x_2)}{y_2 - x_2} = \frac{136.4(0.1822 - 0.227)}{0.095 - 0.227} = 46.3 \text{ kg}$$

Eq. (10.15):

$$R_2 = M_2 - E_2 = 136.4 - 46.3 = 90.1 \text{ kg}$$

Stage 3 In a similar manner, $B_3 = 40$, $M_3 = 130.1$, $x_{M3} = 0.1572$, $x_3 = 0.20$, $y_3 = 0.078$, $E_3 = 45.7$, and $R_3 = 84.4$. The acid content of the final raffinate is $0.20(84.4) = 16.88 \text{ kg}$.

The composited extract is $E_1 + E_2 + E_3 = 43.6 + 46.3 + 45.7 = 135.6 \text{ kg}$, and its acid content $= E_1 y_1 + E_2 y_2 + E_3 y_3 = 13.12 \text{ kg}$.

If an extraction to give the same final raffinate concentration, $x = 0.20$, were to be done in one stage, the point M would be at the intersection of tie line $R_3 E_3$ and line BF of Fig. 10.15, or at $x_M = 0.12$. The solvent required would then be, by Eq. (10.6), $S_1 = 100(0.30 - 0.12)/(0.12 - 0) = 150 \text{ kg}$, instead of the total of 120 required in the three-stage extraction.

Insoluble liquids When the extraction solvent and feed solution are insoluble and remain so at all concentrations of the distributed solute occurring in the operation, the computations can be simplified. For this purpose, the equilibrium concentrations are plotted as in Fig. 10.16, $x' = x/(1 - x)$ against $y' = y/(1 - y)$. Since the liquids A and B are insoluble, there are $A \text{ kg}$ of this substance in all raffinate. Similarly, the extract from each stage contains all the solvent B fed to that stage. A solute-C balance about any stage n is then

$$A x'_{n-1} + B_n y'_S = B_n y'_n + A x'_n \quad (10.20)$$

$$-\frac{A}{B_n} = \frac{y'_S - y'_n}{x'_{n-1} - x'_n} \quad (10.21)$$

This is the operating-line equation for stage n , of slope $-A/B_n$, passing through points (x'_{n-1}, y'_S) and (x'_n, y'_n) . The construction for a three-stage extraction is shown in Fig. 10.16, where for each stage a line is drawn of slope appropriate to that stage. Each operating line intersects the equilibrium curve at the raffinate and extract compositions. No raffinate of concentration smaller than that in equilibrium with the entering solvent is possible.

Solute concentrations for these cases are sometimes also conveniently expressed as mass/volume. Equations (10.20) and (10.21) then apply with x' and y' expressed as mass solute/volume, and A and B as volume/time (or volumes for batch operations).

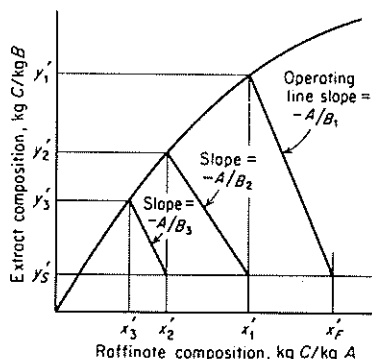


Figure 10.16 Crosscurrent extraction with an insoluble solvent.

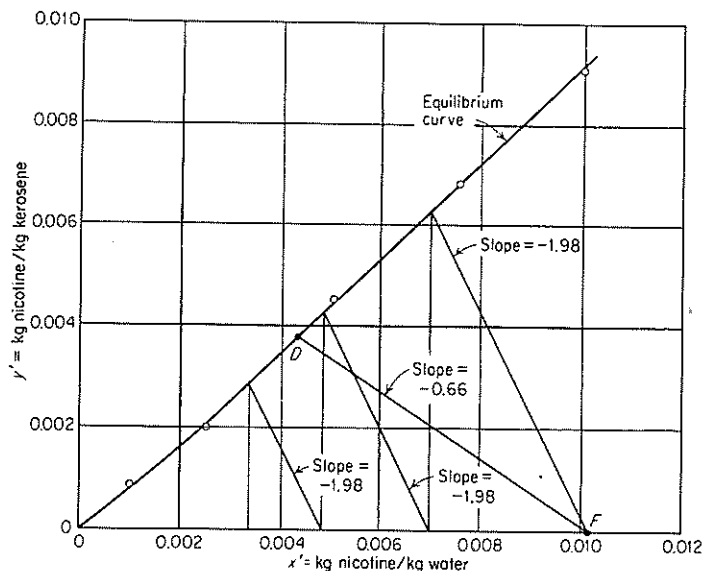


Figure 10.17 Solution to Illustration 10.2.

Illustration 10.2 Nicotine (C) in a water (A) solution containing 1% nicotine is to be extracted with kerosene (B) at 20°C. Water and kerosene are essentially insoluble. (a) Determine the percentage extraction of nicotine if 100 kg of feed solution is extracted once with 150 kg solvent. (b) Repeat for three theoretical extractions using 50 kg solvent each.

SOLUTION Equilibrium data are provided by Claffey et al., *Ind. Eng. Chem.*, 42, 166 (1950), and expressed as kg nicotine/kg liquid, they are as follows:

$x' = \frac{\text{kg nicotine}}{\text{kg water}}$	0	0.001 011	0.00246	0.00502	0.00751	0.00998	0.0204
$y'^* = \frac{\text{kg nicotine}}{\text{kg kerosene}}$	0	0.000 807	0.001961	0.00456	0.00686	0.00913	0.01870

(a) $x_F = 0.01$ wt fraction nicotine, $x'_F = 0.01/(1 - 0.01) = 0.0101$ kg nicotine/kg water. $F = 100$ kg. $A = 100(1 - 0.01) = 99$ kg water. $A/B = 99/150 = 0.66$.

Refer to Fig. 10.17, which shows the equilibrium data and the point F representing the composition of the feed. From F , line FD is drawn of slope -0.66 , intersecting the equilibrium curve at D , where $x'_1 = 0.00425$ and $y'_1 = 0.00380$ kg nicotine/kg liquid. The nicotine removed from the water is therefore $99(0.0101 - 0.00425) = 0.580$ kg, or 58% of that in the feed.

(b) For each stage, $A/B = 99/50 = 1.98$. The construction is started at F , with operating lines of slope -1.98 . The final raffinate composition is $x'_3 = 0.0034$, and the nicotine extracted is $99(0.0101 - 0.0034) = 0.663$ kg, or 66.3% of that in the feed.

Continuous Countercurrent Multistage Extraction

The flowsheet for this type of operation is shown in Fig. 10.18. Extract and raffinate streams flow from stage to stage in countercurrent and provide two

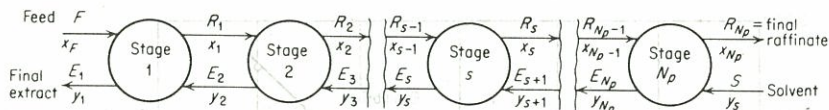


Figure 10.18 Countercurrent multistage extraction.

final products, raffinate R_{N_p} and extract E_1 . For a given degree of separation, this type of operation requires fewer stages for a given amount of solvent, or less solvent for a fixed number of stages, than the crosscurrent methods described above.

The graphical treatment is developed in Fig. 10.19 on rectangular coordinates. Construction on the equilateral triangle is identical with this. A total material balance about the entire plant is

$$F + S = E_1 + R_{N_p} = M \quad (10.22)$$

Point M can be located on line FS through a balance for substance C ,

$$Fx_F + Sy_S = E_1y_1 + R_{N_p}x_{N_p} = Mx_M \quad (10.23)$$

$$x_M = \frac{Fx_F + Sy_S}{F + S} \quad (10.24)$$

Equation (10.22) indicates that M must lie on line $R_{N_p}E_1$, as shown. Rearrangement of Eq. (10.22) provides

$$R_{N_p} - S = F - E_1 = \Delta_R \quad \text{2 straight lines} \quad (10.25)$$

where Δ_R , a difference point, is the net flow outward at the last stage N_p .

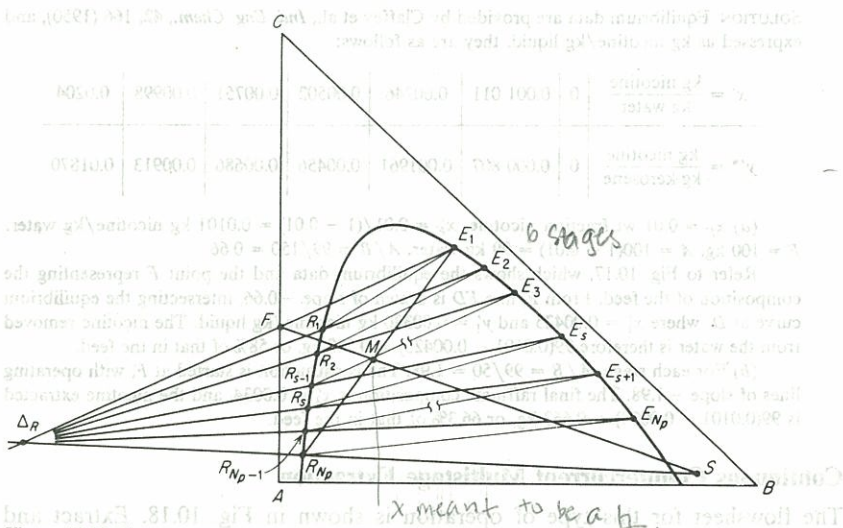


Figure 10.19 Countercurrent multistage extraction.

are extended to line SR_{N_p} , to give intersections with line SR_{N_p} as shown. The intersection farthest from S (if on the left-hand side of the diagram) or nearest S (if on the right) represents the difference point for minimum solvent, as at point Δ_{R_m} (Fig. 10.20). The actual position of Δ_R must be farther from S (if on the left) or nearer to S (if on the right) for a finite number of stages. The larger the amount of solvent, the fewer the number of stages. Usually, but not in the instance shown, the tie line which when extended passes through F , that is, tie line JK , will locate Δ_{R_m} for minimum solvent.

When the number of stages is very large, the construction indicated in Fig. 10.21 may be more convenient. A few lines are drawn at random from point Δ_R to intersect the two branches of the solubility curve as shown, where the intersections do not now necessarily indicate streams between two actual adjacent stages. The C concentrations x_s and y_{s+1} corresponding to these are plotted on x, y coordinates as shown to provide an operating curve. Tie-line data provide the equilibrium curve y^* vs. x , and the theoretical stages are stepped off in the manner used for gas absorption and distillation.

Figure 10.22 shows the construction for solvent-free coordinates. The B-free material balance for the entire plant is

$$F' + S' = E'_1 + R'_{N_p} = M' \quad (10.28)$$

where ordinarily $F = F'$ since the feed is usually B-free. M' is therefore on the line FS at X_M calculated by a C balance

$$F'X_F + S'Y_S = M'X_M \quad (10.29)$$

If pure B is used as solvent, $S' = 0$, $S'Y_S = 0$, $F' = M'$, $X_M = X_F$, and point M is vertically above F .

Line $E_1R_{N_p}$ must pass through M . Then

$$R'_{N_p} - S' = F' - E'_1 = \Delta'_R \quad (10.30)$$

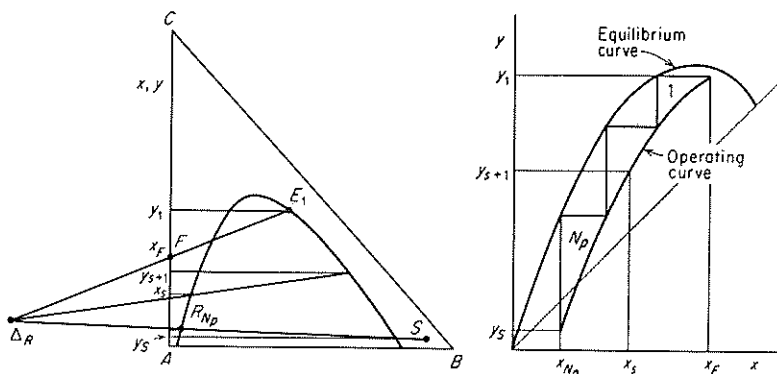


Figure 10.21 Transfer of coordinates to distribution diagram.

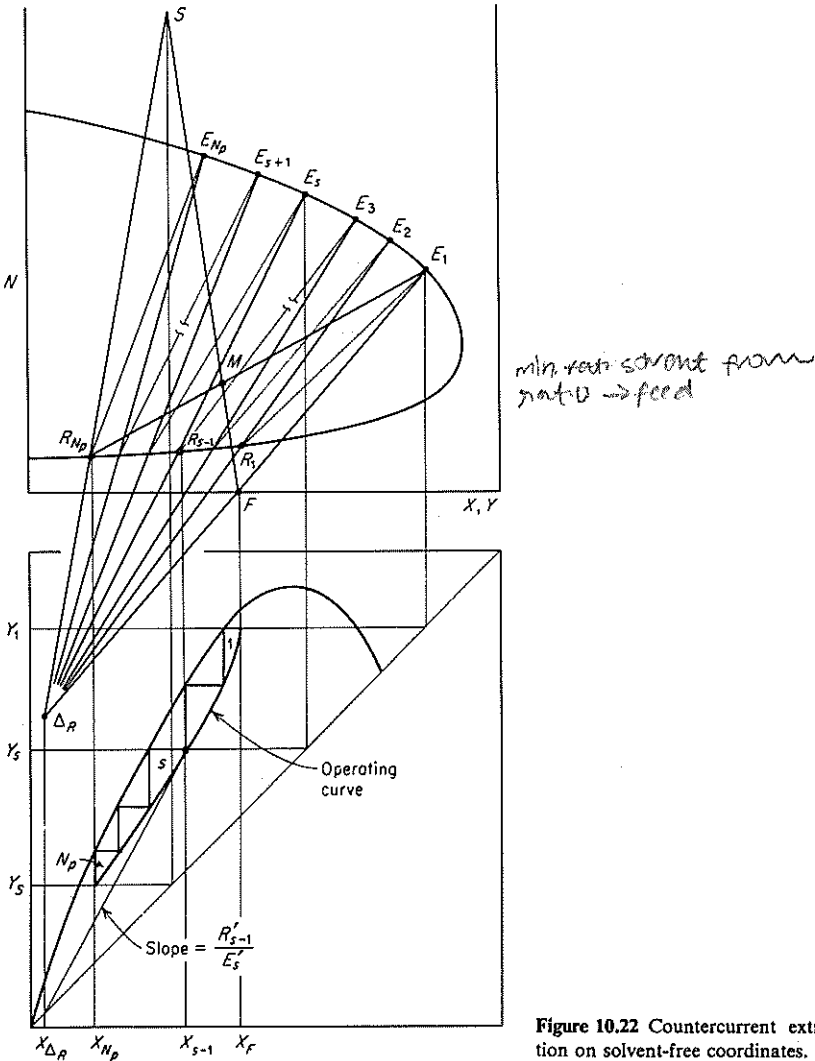


Figure 10.22 Countercurrent extraction on solvent-free coordinates.

The balance for stages s through N_p is

$$R'_{N_p} - S' = R'_{s-1} - E'_s = \Delta'_R \quad (10.31)$$

where Δ'_R is the difference in solvent-free flow, out minus in, at stage N_p , and the constant difference in solvent-free flows of the streams between any two adjacent stages. Line $E_s R'_{s-1}$ extended, where s is any stage, must therefore pass through Δ'_R on the graph.

The graphical construction is then as follows. After locating points F , S , M , E_1 , R_{N_p} , and Δ_R , a tie line from E_1 provides R_1 , line $\Delta_R R_1$ extended locates E_2 , etc. If the solvent is pure B ($N_S = \infty$), line $R_{N_p} \Delta_R$ is vertical.

A C balance, stages s through N_p , following Eq. (10.30) is

$$R'_{s-1}X_{s-1} - E'_sY_s = \Delta'_R X_{\Delta R} \quad (10.32)$$

A B balance is

$$R'_{s-1}N_{R_{s-1}} - E'_sN_{E_s} = \Delta'_R N_{\Delta R} \quad (10.33)$$

Elimination of Δ'_R between Eqs. (10.30) to (10.32) provides

$$\frac{R'_{s-1}}{E'_s} = \frac{Y_s - X_{\Delta R}}{X_{s-1} - X_{\Delta R}} = \frac{N_{E_s} - N_{\Delta R}}{N_{R_{s-1}} - N_{\Delta R}} \quad (10.34)$$

Thus, the ratio of flows R'_{s-1}/E'_s can be obtained from the ratio of line lengths $E_s \Delta_R / R_{s-1} \Delta_R$ on the upper part of Fig. 10.22, or as the slope of the cord from (X_{s-1}, Y_s) to $X_{\Delta R}$ on the 45° diagonal of the lower diagram, as shown. The difference-point coordinates are

$$\Delta'_R \begin{cases} N_{\Delta R} = \frac{\text{difference in B flow, out} - \text{in, at stage } N_p}{\text{net flow out, B-free}} \\ X_{\Delta R} = \frac{\text{difference in C flow, out} - \text{in, at stage } N_p}{\text{net flow out, B-free}} \end{cases}$$

and Δ'_R is analogous to the Δ_W of distillation (see Chap. 9).

As the solvent/feed ratio is increased, Δ_R on Fig. 10.22 moves lower, and the minimum solvent is determined by the lowest point of intersection of all extended tie lines with the line SR_{N_p} . A practical Δ_R must be located below this, corresponding to larger amounts of solvent.

Solvent recovery by extraction While most processes use distillation or evaporation to recover the solvent from the product solutions of liquid extraction, it is not uncommon to recover solvent by liquid extraction. A typical example is the recovery of penicillin from the acidified fermentation broth in which it occurs by extraction with amyl acetate as solvent, followed by stripping of the penicillin from the solvent by extracting it into an aqueous buffer solution. The amyl acetate is then returned to the first extraction. The calculations of such solvent-recovery operations are made in the same manner as those for the first extraction.

Illustration 10.3 If 8000 kg/h of an acetic acid (C)–water (A) solution, containing 30% acid, is to be countercurrently extracted with isopropyl ether (B) to reduce the acid concentration to 2% in the solvent-free raffinate product, determine (a) the minimum amount of solvent which can be used and (b) the number of theoretical stages if 20 000 kg/h of solvent is used.

SOLUTION The equilibrium data of Illustration 10.1 are plotted on triangular coordinates in Fig. 10.23. The tie lines have been omitted for clarity.

(a) $F = 8000$ kg/h; $x_F = 0.30$ wt fraction acetic acid, corresponding to point F on the figure. R'_N is located on the AC base at 2% acid, and line BR'_N intersects the water-rich solubility curve at R_N , as shown. In this case the tie line J , which when extended passes through F , provides the conditions for minimum solvent, and this intersects line $R_N B$ on the right of the figure nearer B than any other lower tie line. Tie line J provides the minimum E_1 as shown at $y_1 = 0.143$. Line $E_1 m R_N$ intersects line FB at M_m , for which $x_M = 0.114$. Eq. (10.24), with $y_S = 0$ and $S = B$:

$$B_m = \frac{Fx_F}{x_m} - F = \frac{8000(0.30)}{0.114} - 8000 = 13\,040 \text{ kg/h, min solvent rate}$$

(b) For $B = 20\,000$ kg solvent/h [Eq. (10.24) with $y_S = 0$ and $S = B$],

$$x_M = \frac{Fx_F}{F + B} = \frac{8000(0.30)}{8000 + 20\,000} = 0.0857$$

and point M is located as shown on line FB . Line $R_N M$ extended provides E_1 at $y_1 = 0.10$. Line FE_1 is extended to intersect line $R_N B$ at Δ_R . Random lines such as OKL are drawn to provide y_{s+1} at K and x_s at L , as follows:

y_{s+1}	0	0.01	0.02	0.04	0.06	0.08	0.10 = y_1
x_s	0.02	0.055	0.090	0.150	0.205	0.250	0.30 = x_F

These are plotted on Fig. 10.24 as the operating curve, along with the tie-line data as the equilibrium curve. There are required 7.6 theoretical stages. The weight of extract can be

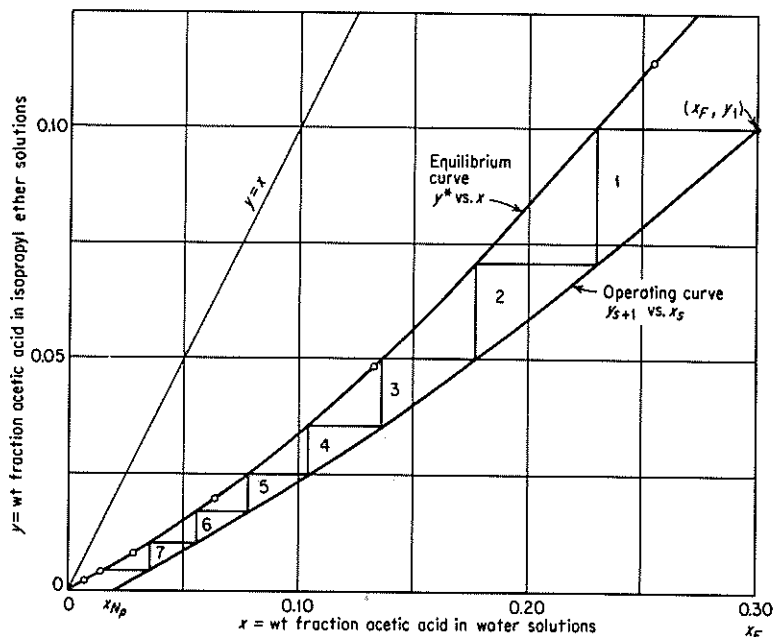


Figure 10.24 Solution to Illustration 10.3.

obtained by an acid balance,

$$E_1 = \frac{M(x_M - x_{N_p})}{y_1 - x_{N_p}} = \frac{28\,000(0.0857 - 0.02)}{0.10 - 0.02} = 23\,000 \text{ kg/h}$$

and

$$R_{N_p} = M - E_1 = 28\,000 - 23\,000 = 5000 \text{ kg/h}$$

Insoluble liquids When the liquids A and B are insoluble over the range of solute concentrations encountered, the stage computation is made more simply on x' , y' coordinates. For this case, the solvent content of all extracts and the A content of all raffinates are constant. An overall plant balance for substance C is

$$By'_S + Ax'_F = Ax'_{N_p} + By'_1 \quad (10.35)$$

or
$$\frac{A}{B} = \frac{y'_1 - y'_S}{x'_F - x'_{N_p}} \quad (10.36)$$

which is the equation of a straight line, the operating line, of slope A/B , through points (y'_1, x'_F) , (y'_S, x'_{N_p}) . For stages 1 through s , similarly,

$$\frac{A}{B} = \frac{y'_1 - y'_{s+1}}{x'_F - x'_s} \quad (10.37)$$

and Fig. 10.25 shows the construction for stages. x' and y' can also be expressed as mass/volume, with A and B as volume/time.

For the special case where the equilibrium curve is of constant slope $m' = y'^*/x'$, Eq. (5.50) applies,

$$\frac{x'_F - x'_{N_p}}{y'_F - y'_S/m'} = \frac{(m'B/A)^{N_p+1} - m'B/A}{(m'B/A)^{N_p+1} - 1} \quad (10.38)$$

where $m'B/A$ is the extraction factor. This can be used in conjunction with Fig. 5.16, with $(x'_{N_p} - y'_S/m')/(x'_F - y'_S/m')$ as ordinate and $m'B/A$ as parameter.

Equation (8.16) for overall efficiency E_O can be used for extraction under the following conditions:

1. Replace E_{MGE} with E_{ME} , and $1/A$ with $m'B/A$
2. Replace E_{MGR} with E_{MR} and A with $A/m'B$

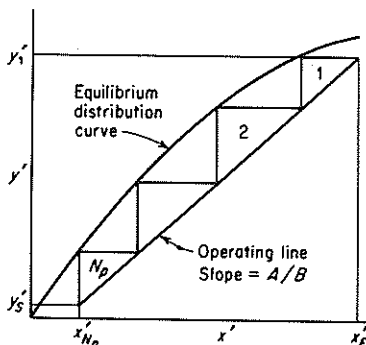


Figure 10.25 Countercurrent extraction, insoluble solvents.

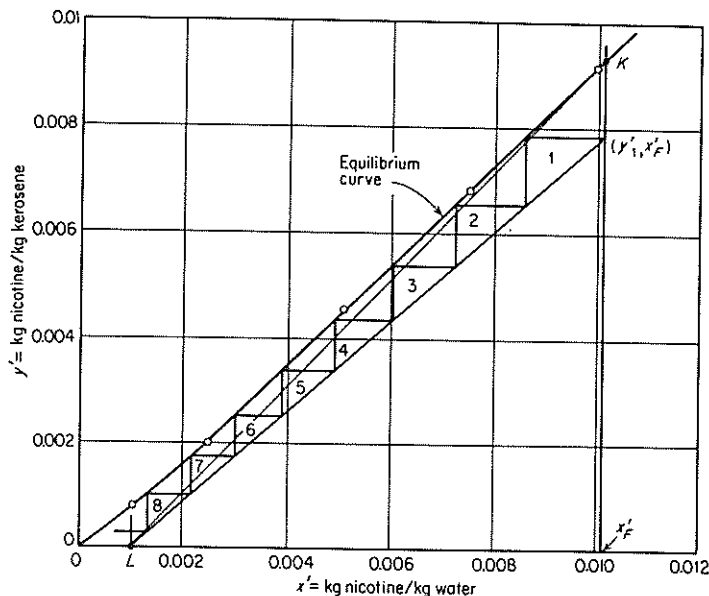


Figure 10.26 Solution to Illustration 10.6.

Illustration 10.4 If 1000 kg/h of a nicotine (C)–water (A) solution containing 1% nicotine is to be countercurrently extracted with kerosene at 20°C to reduce the nicotine content to 0.1%, determine (a) the minimum kerosene rate and (b) the number of theoretical stages required if 1150 kg of kerosene is used per hour.

SOLUTION The equilibrium data of Illustration 10.2 are plotted in Fig. 10.26.

(a) $F = 1000$ kg/h, $x_F = 0.01$, $A = 1000(1 - 0.01) = 990$ kg water/h, $y_S = 0$.

$$x'_F = \frac{0.01}{1 - 0.01} = 0.0101 \text{ kg nicotine/kg water}$$

$$x_{N_p} = 0.001 \quad x'_{N_p} = \frac{0.001}{1 - 0.001} = 0.001001 \text{ kg nicotine/kg water}$$

The operating line starts at point L ($y' = 0$, $x' = 0.001001$) and for infinite stages passes through K on the equilibrium curve at x'_F . Since $y_K = 0.0093$,

$$\frac{A}{B_m} = \frac{0.0093 - 0}{0.0101 - 0.001001} = 1.021$$

and $B_m = A/1.021 = 990/1.021 = 969$ kg kerosene/h.

(b) $B = 1150$ kg/h, $A/B = 990/1150 = 0.860$. Eq. (10.36):

$$\frac{y'_1}{x'_F - x'_{N_p}} = \frac{y'_1}{0.0101 - 0.001001} = 0.860$$

$$y'_1 = 0.00782 \text{ kg nicotine/kg kerosene}$$

The operating line is drawn through (y'_1, x'_F) , and 8.3 theoretical stages are determined graphically.

Alternatively, at the dilute end of the system, $m' = dy'/dx' = 0.798$, and

$$\frac{m'B}{A} = \frac{0.798(1150)}{990} = 0.928$$

At the concentrated end, $m' = 0.953$, and $m'B/A = 0.953(1150)/990 = 1.110$. The average is $[0.928(1.110)]^{0.5} = 1.01$. $x'_{N_e}/x'_F = 0.001001/0.0101 = 0.099$, and Fig. 5.16 indicates 8.4 theoretical stages.

Continuous Countercurrent Extraction with Reflux

Whereas in ordinary countercurrent operation the richest possible extract product leaving the plant is at best only in equilibrium with the feed solution, the use of reflux at the extract end of the plant can provide a product even richer, as in the case of the rectifying section of a distillation column. Reflux is not needed at the raffinate end of the cascade since, unlike distillation, where heat must be carried in from the reboiler by a vapor reflux, in extraction the solvent (the analog of heat) can enter without a carrier stream.

An arrangement for this is shown in Fig. 10.27. The feed to be separated into its components is introduced at an appropriate place into the cascade, through which extract and raffinate liquids are passing countercurrently. The concentration of solute C is increased in the extract-enriching section by countercurrent contact with a raffinate liquid rich in C. This is provided by removing the solvent from extract E_1 to produce the solvent-free stream E' , part of which is removed as extract product P'_E and part returned as reflux R_0 . The raffinate-stripping section of the cascade is the same as the countercurrent extractor of Fig. 10.18, and C is stripped from the raffinate by countercurrent contact with solvent.

Graphical determination of stages required for such operations is usually inconvenient to carry out on triangular coordinates [72] because of crowding, and only the use of N, X, Y coordinates will be described. For the stages in the raffinate stripping section (Fig. 10.27) the developments for simple countercurrent extraction, Eqs. (10.31) to (10.34) apply, and this section of the plant needs no further consideration.

In the extract-enriching section, an A + C balance about the solvent separator is

$$E'_1 = E' = P'_E + R_0 \quad (10.39)$$

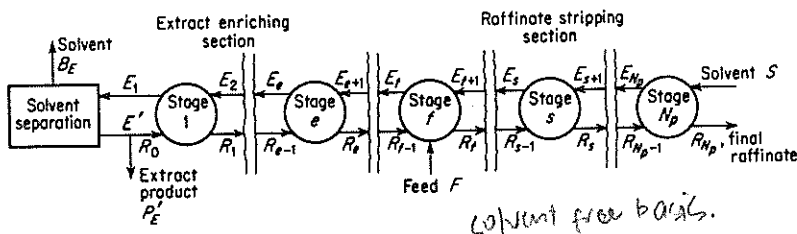


Figure 10.27 Countercurrent extraction with reflux.

Let Δ_E represent the net rate of flow outward from this section. Then, for its A + C content

$$\Delta'_E = P'_E \quad (10.40)$$

and for its C content

$$X_{\Delta E} = X_{PE} \quad (10.41)$$

while for its B content

$$B_E = \Delta'_E N_{\Delta E} \quad (10.42)$$

The point Δ_E is plotted on Fig. 10.28, which is drawn for a system of two partly miscible component pairs. For all stages through e , an A + C balance is

$$E'_{e+1} = P'_E + R'_e = \Delta'_E + R'_e \quad (10.43)$$

or

$$\Delta'_E = E'_{e+1} - R'_e \quad (10.44)$$

A C balance is

$$\Delta'_E X_{\Delta E} = E'_{e+1} Y_{e+1} - R'_e X_{R_e} \quad (10.45)$$

and a B balance is

$$\Delta'_E N_{\Delta E} = E'_{e+1} N'_{E_{e+1}} - R'_e N_{R_e} \quad (10.46)$$

Since e is any stage in this section, lines radiating from point Δ_E cut the solubility curves of Fig. 10.28 at points representing extract and raffinate flowing between any two adjacent stages. Δ_E is therefore a difference point, constant for all stages in this section, whose coordinates mean

$$\Delta'_E \begin{cases} N_{\Delta E} = \frac{\text{difference in B flow, out} - \text{in}}{\text{net flow out, B-free}} = \frac{\text{B out}}{(\text{A} + \text{C}) \text{ out}} \\ X_{\Delta E} = \frac{\text{difference in C flow, out} - \text{in}}{\text{net flow out, B-free}} = \frac{\text{C out}}{(\text{A} + \text{C}) \text{ out}} \end{cases}$$

Alternating tie lines and lines from Δ_E then establish the stages, starting with stage 1 and continuing to the feed stage. Figure 10.28 also shows the raffinate-stripping stages, developed as before for countercurrent extraction.

Solving Eq. (10.43) with (10.45) and (10.46) produces the internal reflux ratio at any stage,

$$\frac{R'_e}{E'_{e+1}} = \frac{N_{\Delta E} - N_{E_{e+1}}}{N_{\Delta E} - N_{R_e}} = \frac{X_{\Delta E} - Y_{e+1}}{X_{\Delta E} - X_{R_e}} = \frac{\text{line } \Delta_E E_{e+1}}{\text{line } \Delta_E R_e} \quad (10.47)$$

This can be computed from line lengths on the upper diagram of Fig. 10.28 or from the slope of a chord on the lower diagram, as shown. The external reflux ratio is

$$\frac{R'_0}{P'_E} = \frac{R_0}{P_E} = \frac{N_{\Delta E} - N_{E1}}{N_{E1} - N_{PE,0}} \quad (10.48)$$

which can be used to locate Δ_E for any specified reflux ratio.

difference points, and the optimum location of the feed stage is represented by the tie line which crosses line $\Delta_R F \Delta_E$, as shown. The similarity of Fig. 10.28 to the enthalpy-concentration diagram of distillation is clear, and the two become completely analogous when solvent S is pure B .

The higher the location of Δ_E (and the lower Δ_R), the larger the reflux ratio and the smaller the number of stages. When R_0/P'_E is infinite (infinite reflux ratio or total reflux), $N_{\Delta E} = \infty$ and the minimum number of stages results. The capacity of the plant falls to zero, feed must be stopped, and solvent B_E is recirculated to become S . The configuration is shown in Fig. 10.29 on N, X, Y coordinates. The corresponding XY diagram uses the 45° diagonal as operating lines for both sections of the cascade.

An infinity of stages is required if a line radiating from either Δ_E or Δ_R coincides with a tie line, and the greatest reflux ratio for which this occurs is the *minimum reflux ratio*. This may be determined as in Fig. 10.30. Tie lines to the left of J are extended to intersect line SR_{N+1} , and those to the right of J are extended to line $P'_E E_1$. Points Δ_{Em} and Δ_{Rm} for the minimum reflux ratio are established by selecting the intersections farthest from $N = 0$, consistent with the requirement that Δ_R , Δ_E , and F must always lie on the same straight line. Frequently tie line J , which when extended passes through F , will establish the minimum reflux ratio, and always if the XY equilibrium distribution curve is everywhere concave downward.

Illustration 10.5 A solution containing 50% ethylbenzene (A) and 50% styrene (C) is to be separated at 25°C at a rate of 1000 kg/h into products containing 10% and 90% styrene, respectively, with diethylene glycol (B) as solvent. Determine (a) the minimum number of theoretical stages, (b) the minimum extract reflux ratio, and (c) the number of theoretical stages and the important flow quantities at an extract reflux ratio of 1.5 times the minimum value.

SOLUTION Equilibrium data of Boobar et al., *Ind. Eng. Chem.*, 43, 2922 (1951), have been converted to a solvent-free basis and are tabulated below.

Hydrocarbon-rich solutions		Solvent-rich solutions	
X , kg styrene kg hydrocarbon	N , kg glycol kg hydrocarbon	Y^* , kg styrene kg hydrocarbon	N , kg glycol kg hydrocarbon
0	0.00675	0	8.62
0.0870	0.00817	0.1429	7.71
0.1883	0.00938	0.273	6.81
0.288	0.01010	0.386	6.04
0.384	0.01101	0.480	5.44
0.458	0.01215	0.557	5.02
0.464	0.01215	0.565	4.95
0.561	0.01410	0.655	4.46
0.573	0.01405	0.674	4.37
0.781	0.01833	0.833	3.47
1.00	0.0256	1.00	2.69

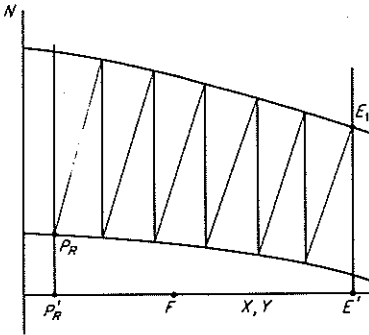


Figure 10.29 Total reflux.

These are plotted on the solvent-free coordinate system of Fig. 10.31. The tie lines corresponding to these points are not drawn in, for clarity. $F = 1000$ kg/h, $X_F = 0.5$ wt fraction styrene, $X_{P'E} = 0.9$, $X_{R,N_p} = 0.1$, all on a solvent-free basis. Point E_1 is located as shown. $N_{E1} = 3.10$.

(a) Minimum theoretical stages are determined by drawing a tie line from E_1 , a vertical line from R_1 , a tie line from E_2 , etc., until the raffinate product is reached, as shown. The minimum number of stages, corresponding to the constructed tie lines, is 9.5.

(b) The tie line which when extended passes through F provides the minimum reflux ratios, since it provides intersections Δ_{Em} and Δ_{Rm} farthest from the line $N = 0$. From the plot,

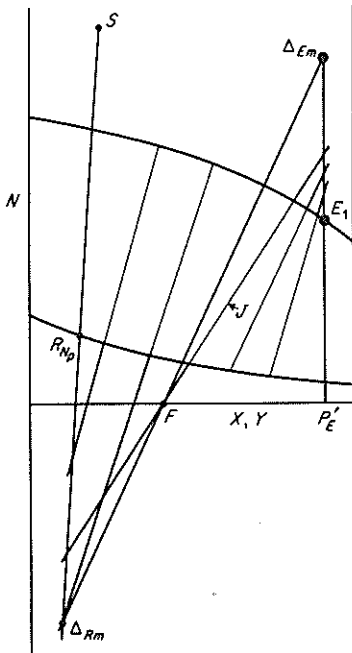


Figure 10.30 Determination of minimum reflux ratio.

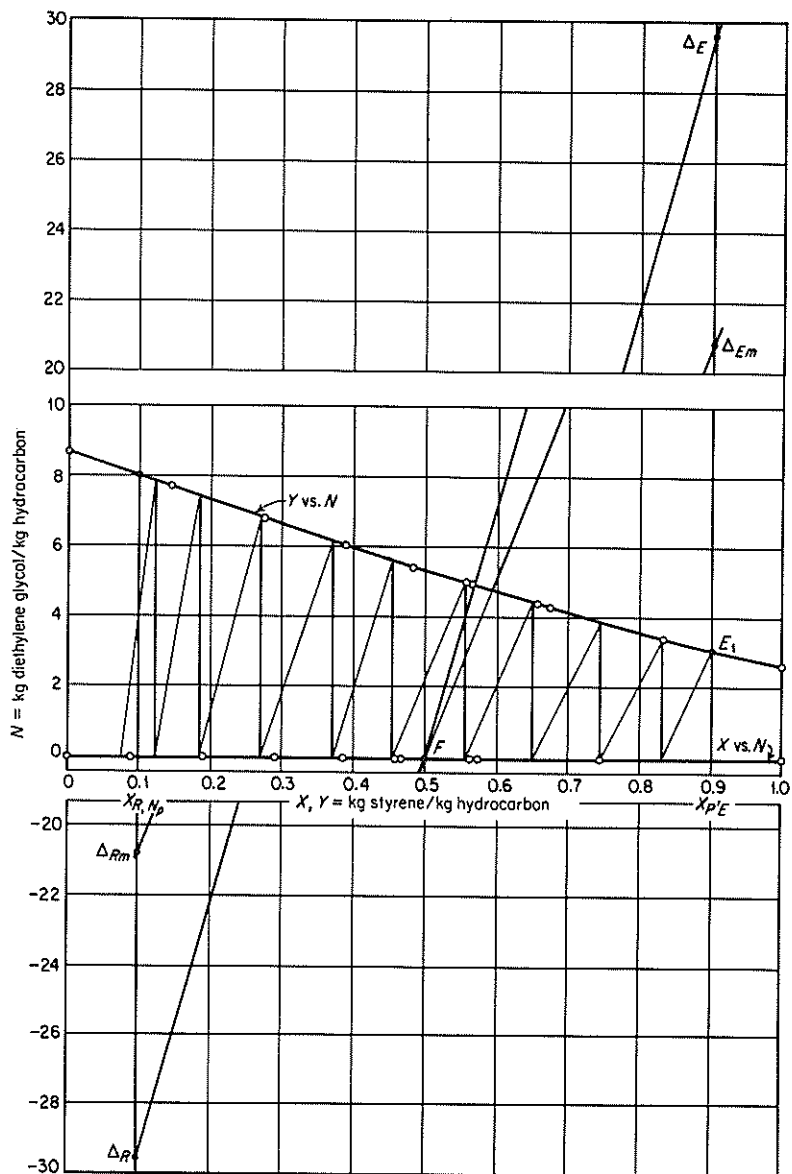


Figure 10.31 Solution to Illustration 10.5. Minimum theoretical stages at total reflux and location of difference points.

$$N_{\Delta E_m} = 20.76. \text{ Eq. (10.48):}$$

$$\left(\frac{R_0}{P'_E} \right)_m = \frac{20.76 - 3.1}{3.1} = 5.70 \text{ kg reflux/kg extract product}$$

(c) For $R_0/P'_E = 1.5(5.70) = 8.55 \text{ lb reflux/kg extract product [Eq. (10.48)],}$

$$8.55 = \frac{N_{\Delta E} - 3.1}{3.1}$$

and $N_{\Delta E} = 29.6$. Point Δ_E is plotted as shown. A straight line from Δ_E through F intersects line $X = 0.10$ at Δ_R . $N_{\Delta R} = -29.6$. Random lines are drawn from Δ_E for concentrations to the right of F , and from Δ_R for those to the left, and intersections of these with the solubility curves provide the coordinates of the operating curve (Fig. 10.32). The tie-line data plotted directly provide the equilibrium curve. The number of theoretical stages is seen to be 15.5, and the feed is to be introduced into the seventh from the extract-product end of the cascade.

From Fig. 10.31, $X_R, N_p = 0.10$, $N_R, N_p = 0.0082$. On the basis of 1 h, an overall plant balance is

$$F = 1000 = P'_E + R'_{N_p}$$

A C balance is

$$FX_F = 500 = P'_E(0.9) + R'_{N_p}(0.1)$$

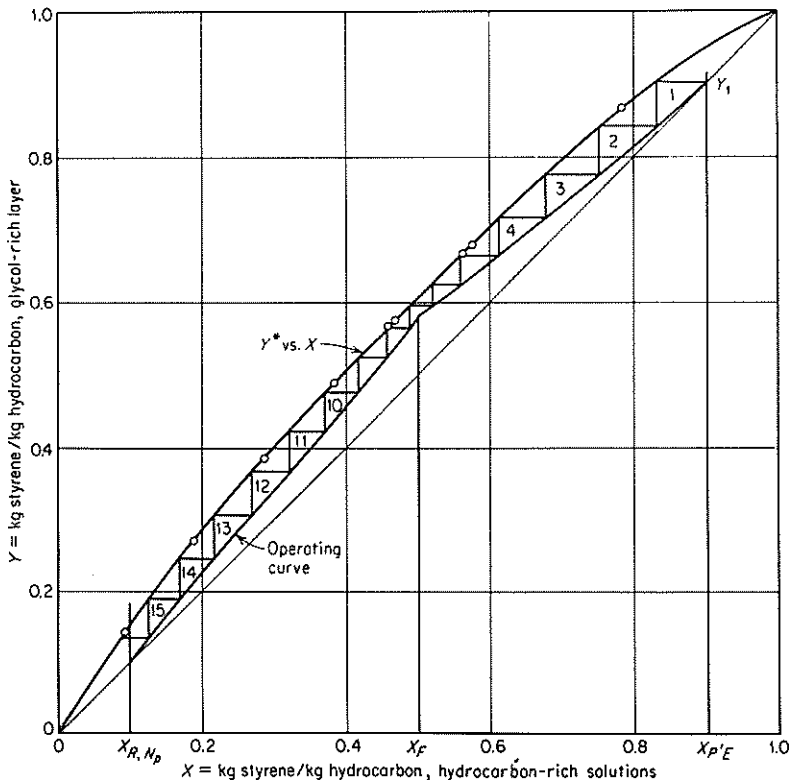


Figure 10.32 Solution to Illustration 10.5.

Solving simultaneously, we get $P'_E = R'_{N_r} = 500 \text{ kg/h}$

$$R_0 = R'_0 = 8.55P'_E = 8.55(500) = 4275 \text{ kg/h}$$

Eq. (10.39):

$$E'_1 = R'_0 + P'_E = 4275 + 500 = 4775 \text{ kg/h}$$

$$B_E = E'_1 N_{E1} = 4775(3.10) = 14\,800 \text{ kg/h}$$

$$E_1 = B_E + E'_1 = 14\,800 + 4775 = 19\,575 \text{ kg/h}$$

$$R_{N_r} = R'_{N_r}(1 + N_{R, N_r}) = 500(1.0082) = 504 \text{ kg/h}$$

$$S = B_E + R'_{N_r} N_{R, N_r} = 14\,800 + 500(0.0082) = 14\,804 \text{ kg/h}$$

Economic Balances

Several types of economic balance can be made for the various flowsheets just described [71, 72]. For example, the amount of solute extracted for a fixed solvent/feed ratio increases with increased number of stages, and therefore the value of the unextracted solute can be balanced against the cost of the extraction equipment required to recover it. The amount of solvent per unit of feed, or reflux ratio in the case of the last flowsheet described, is also subject to economic balance. For a fixed extent of extraction, the number of stages required decreases as solvent rate or reflux ratio increases. Since the capacity of the equipment for handling the larger liquid flow must at the same time increase, the cost of equipment must then pass through a minimum. The extract solutions become more dilute as solvent rate is increased, and consequently the cost of solvent removal increases. The total cost, which is the sum of investment and operating costs, must pass through a minimum at the optimum solvent rate or reflux ratio. In all such economic balances, the cost of solvent recovery will always be a major item and usually must include consideration of recovery from the saturated raffinate product as well as the extract.

Fractional Extraction—Separation of Two Solutes

If a solution contains two extractable solutes, both can be removed from the solution by countercurrent extraction with a suitable solvent, but it is impossible to produce any great degree of separation of the solutes by this method unless the ratio of their distribution coefficients is very large.

Separation to any extent, however, can be achieved, so long as their distribution coefficients are different, by the techniques of fractional extraction. The simplest flowsheet for this is shown in Fig. 10.33. Here solutes B and C, which constitute the feed, are introduced into a countercurrent cascade where partly miscible solvents A and D flow countercurrently. At the feed stage, both solutes distribute between the solvents, with solute B favoring solvent A, solute C favoring solvent D. In the section to the left of the feed stage, solvent A preferentially extracts the B from D, and D leaves this section with a solute content rich in C. In the section to the right of the feed stage, solvent D preferentially extracts the C from A, and A leaves with a solute content rich in B.

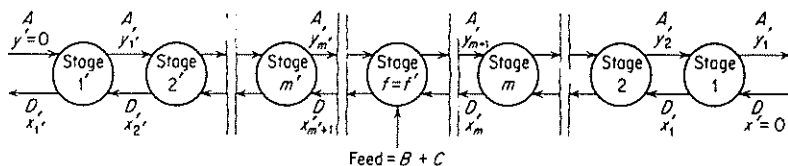


Figure 10.33 Fractional extraction.

This operation will be considered only in its simplest form, where the solvents A and D are essentially immiscible and enter free of solutes and where solutes B and C distribute independently; i.e., the distribution of each is uninfluenced by the presence of the other. A and D will represent the weight rates of flow of these solvents, mass/time, and concentrations will be expressed as

$$y' = \text{mass solute/mass A } (y'_B \text{ for solute B, } y'_C \text{ for solute C})$$

$$x' = \text{mass solute/mass D } (x'_B \text{ for solute B, } x'_C \text{ for solute C})$$

Consider the stages $1'$ through m' . For *either* solute, a solute material balance is

$$Dx'_{m'+1} = Ay'_{m'} + Dx'_{1'} \quad (10.51)$$

or

$$\frac{D}{A} = \frac{y'_{m'} - 0}{x'_{m'+1} - x'_{1'}} \quad (10.52)$$

and we need only add the subscripts B or C to the concentrations to apply these to particular solutes. Similarly, for stages 1 through m , for *either* solute, the balance is

$$Ay'_{m+1} = Dx'_m + Ay'_1 \quad (10.53)$$

$$\frac{D}{A} = \frac{y'_{m+1} - y'_1}{x'_m - 0} \quad (10.54)$$

Equations (10.52) and (10.54) represent the operating lines for each solute, one for each section of the cascade, of slope D/A . Figure 10.34 shows the distribution diagrams for the solutes, each of which distributes independently of the other. The operating lines are also drawn on the figure, and the stages stepped off in the usual manner, beginning at stages 1 and $1'$.

To determine the number of stages required in each section of the cascade, a match of concentrations and stage numbers at the point of feed introduction must be made. At the feed stage (which is numbered f and f') for *either* solute, f must be the same, f' must be the same, and the concentration must be the same when computed from each end of the cascade. The match can be established by plotting the concentrations of both B and C in one of the solvents against stage number, as in Fig. 10.35. The requirements listed above are then met where the rectangle $HJKL$ can be drawn in the manner shown.

It can be seen from Fig. 10.35 that the concentrations of both solutes is greatest at the feed stage, least at stages 1 and $1'$. Sufficient rates of solvent flow must be used to ensure that solute solubilities are not exceeded at the feed stage. Changing the solvent ratio changes the total number of stages and the relative

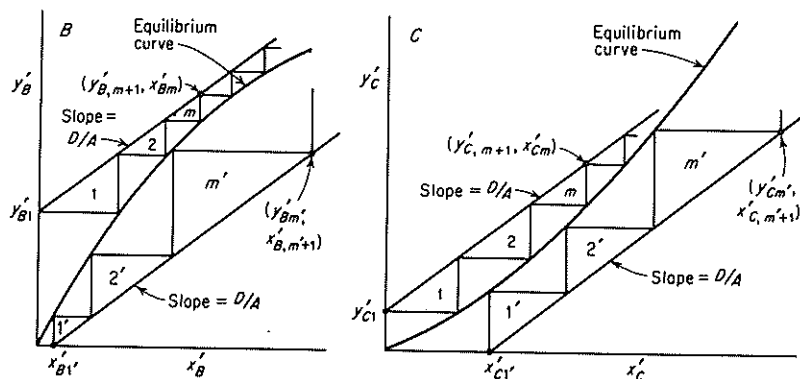


Figure 10.34 Fractional extraction, operating lines and stages.

position of the feed stage, and there is always a solvent ratio for which the total number of stages is the least.

The number of stages can be reduced by use of solute reflux at either or both ends of the cascade. Computation of this effect, treatment of cases where solute distributions are interdependent or solvent miscibility varies, and other special flowsheets are beyond the scope of this book but have been thoroughly worked out [56, 72].

Illustration 10.6 A mixture containing 40% *p*-chloronitrobenzene (B) and 60% *o*-chloronitrobenzene (C) is to be separated at the rate of 100 kg/h into products containing 80 and 15%,

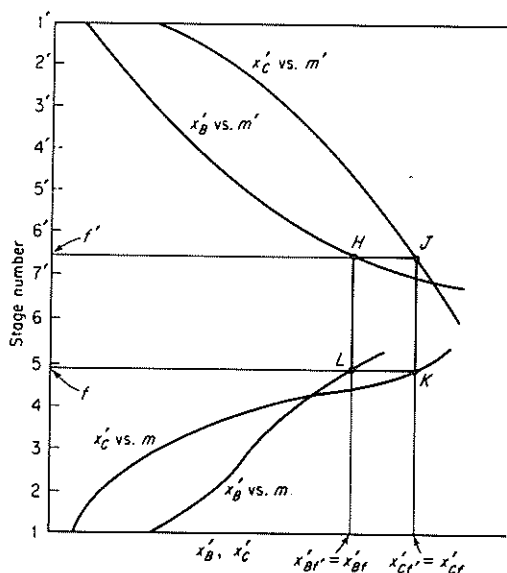


Figure 10.35 Feed-stage matching.

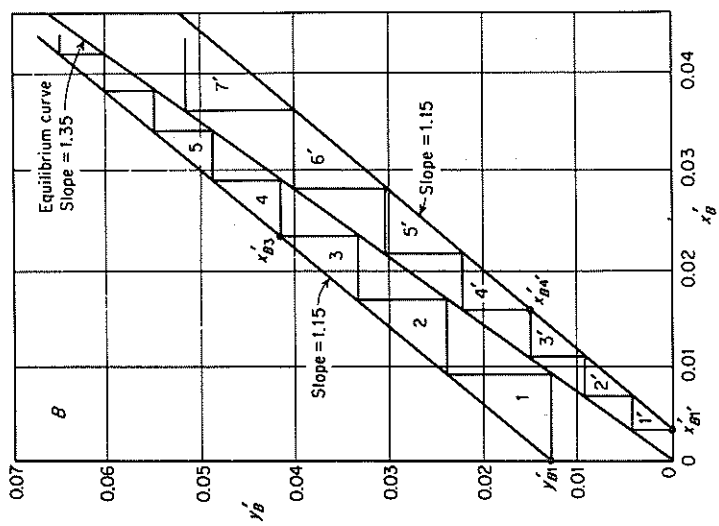
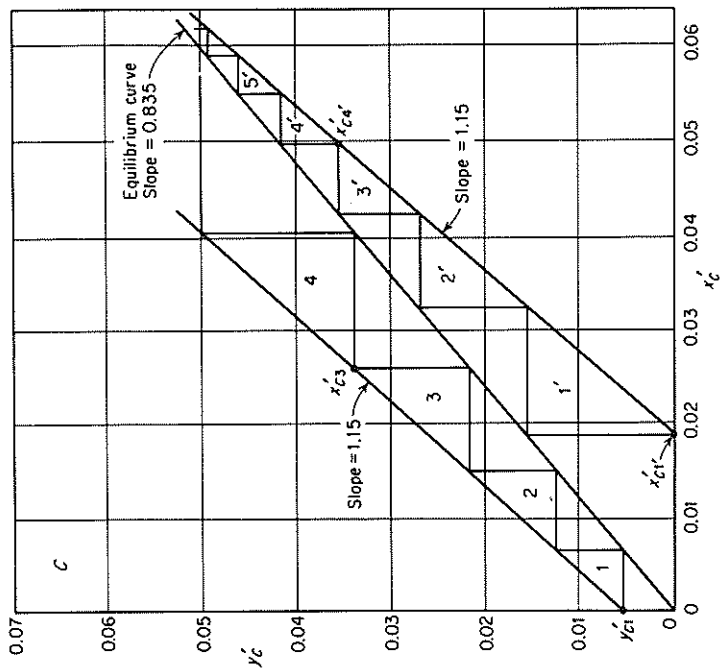


Figure 10.36 Solution to Illustration 10.6.

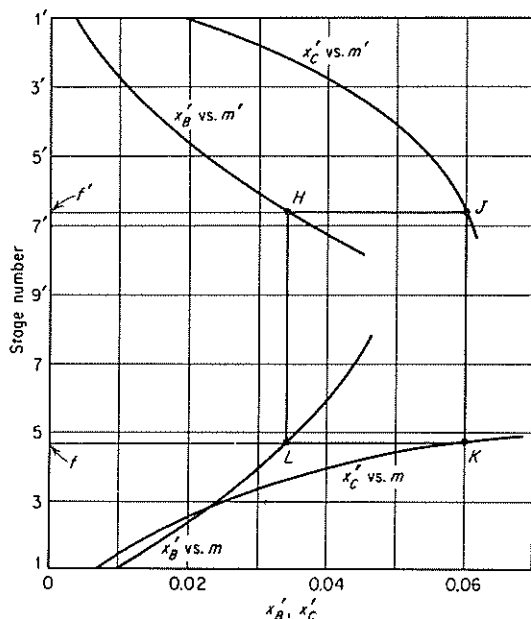


Figure 10.37 Solution to Illustration 10.6, feed-stage matching.

respectively, of the para isomer after removal of solvents. The insoluble solvents to be used are 2400 kg/h heptane (A) and 2760 kg/h aqueous methanol (D). The distribution coefficients are constant and independent, $y_B^*/x_B = 1.35$ and $y_C^*/x_C = 0.835$. Determine the number of theoretical stages required and the position of the feed stage.

SOLUTION The para isomer (B) favors the heptane (A), and the ortho (C) favors the methanol (D). *Basis:* 1 h. Feed = 100 kg = 40 kg B, 60 kg C. Let W = kg A-rich product, Z = kg D-rich product, after solvent removal. Then B and C balances are, respectively,

$$0.80W + 0.15Z = 40 \quad 0.20W + 0.85Z = 60$$

from which $W = 38.5$ kg (30.8 kg B, 7.7 kg C), and $Z = 61.5$ kg (9.23 kg B, 52.27 kg C).

$$x_{B1}' = \frac{9.23}{2760} = 0.00334 \quad x_{C1}' = \frac{52.27}{2760} = 0.01895$$

$$y_{B1}' = \frac{30.8}{2400} = 0.01283 \quad y_{C1}' = \frac{7.7}{2400} = 0.00321$$

$D/A = 2760/2400 = 1.15$. The stages are constructed on Fig. 10.36, and the feed-stage match is shown on Fig. 10.37. From the latter, the feed stage is $f' = 6.6$ and $f = 4.6$. Therefore the total number of ideal stages is $6.6 + 4.6 - 1 = 10.2$ and the feed stage is the 4.6th from the solvent-D inlet.

Multicomponent Systems

For most systems containing more than four components, the display of equilibrium data and the computation of stages is very difficult. Some of our most important separations involve hundreds of components whose identity is

not even firmly established, as in the extraction of petroleum lubricating oils. In such cases, the stage requirements are best obtained in the laboratory, without detailed study of the equilibria. There are available small-scale mixer-settler extractors of essentially 100 percent stage efficiency which can be used to study extraction processes [72], but they must be run continuously to steady state. Countercurrent flowsheets, which are necessarily for continuous operation, can be simulated with small batches of feed and solvent in separatory funnels, however, and this is frequently most convenient and least costly.

Suppose, for example, it is desired to determine what products will be obtained from a five-stage countercurrent extraction of a complex feed flowing at F kg/h with a solvent flowing at S kg/h. Refer to Fig. 10.38. Here each circle represents a separatory funnel which, when adequately shaken and the liquids allowed to settle, will represent one theoretical stage. Into funnel a are placed suitable amounts of feed and solvent in the ratio F/S , the mixture is shaken,

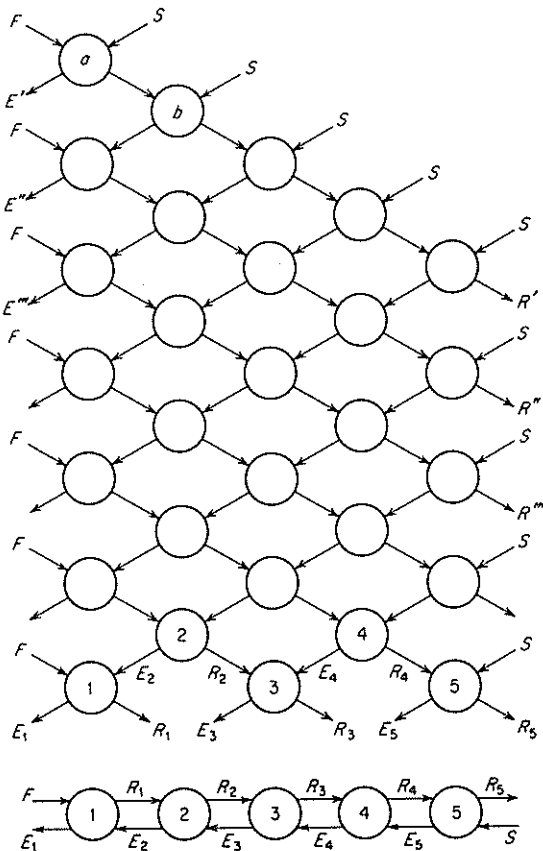


Figure 10.38 Batch simulation of a countercurrent cascade.

settled, and the two phases separated. Extract layer E' is removed and raffinate R_a is contacted in funnel b with precisely the same amount of solvent S as used in funnel a . Raffinates and extracts, feeds and solvents, are moved and contacted in the manner indicated in the figure, care being taken each time to use precisely the same amount of feed and solvent at streams marked F and S as in funnel a .

Clearly extract E' and raffinate R' will not be the same as those produced by a continuous countercurrent extraction. Subsequent raffinates R'' , R''' , and extracts E'' , E''' , etc., will approach the desired result, however. One can follow the approach to steady state by observing any conveniently measured property of the raffinates and extracts (density, refractive index, etc.). The properties of R' , R'' , etc., will approach a constant value, as will those of E' , E'' , etc. If this has occurred by the time extract E_1 and raffinate R_5 in the upper part of Fig. 10.38 is reached, funnels 1 through 5 represent in every detail the stages of the continuous cascade just below. Thus, for example, raffinate R_3 from funnel 3 will have all the properties and relative volume of raffinate R_3 of the continuous plant.

In a similar manner, flowsheets involving reflux and fractional extraction can also be studied [72].

Stage Efficiency

As in gas-liquid contact, the performance of individual extraction stages can be described in terms of the approach to equilibrium actually realized by the effluent extract and raffinate streams. The Murphree stage efficiency can be expressed in terms of extract compositions as E_{ME} or in terms of raffinate compositions as E_{MR} . Applying Eq. (5.39) to stage m of the countercurrent cascade of Fig. 10.39, for example, gives

$$E_{ME} = \frac{y_m - y_{m+1}}{y_m^* - y_{m+1}} \quad E_{MR} = \frac{x_{m-1} - x_m}{x_{m-1} - x_m^*} \quad (10.55)$$

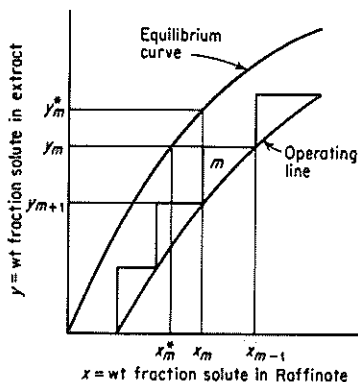


Figure 10.39 Murphree stage efficiency.

where x_m and y_m represent the actual average effluent compositions and y_{m+1} and x_{m-1} those of the streams entering the stage. Any other consistent set of concentration units can equally well be used. The overall stage efficiency E_o of a cascade is defined simply as the ratio of the number of theoretical stages to the number of real stages required to bring about a given concentration change.

STAGE-TYPE EXTRACTORS

Stage-type equipment of two major types is used: (1) single-stage mixer-settlers and multistage cascades constructed from them and (2) sieve-tray, multistage towers.

A *mixer-settler* is a single-stage device, ordinarily consisting of two parts, a mixer for contacting the two liquid phases to bring about mass transfer and a settler for their mechanical separation. Operation may be continuous or batch-wise.

Mixers are of two types: flow mixers and mixing vessels. *Flow mixers*, or *line mixers*, are devices of very small volume placed in a pipeline, such as a series of orifices or mixing nozzles [72, 76] through which the two liquids to be contacted are pumped cocurrently. The loss in mechanical energy corresponding to the pressure drop is in part used to disperse the liquids in each other. The resulting dispersion then passes to the settler. These devices are useful only for continuous operation and are limited in application: the degree of dispersion produced by a given device depends upon the flow rate, and since the specific interfacial area decays rapidly downstream from the mixer [40, 59] while the holding time is very short, mass transfer can be expected to be poor.

Agitated Vessels

These are of the type shown in the right-hand portion of Fig. 6.4*b* and in Fig. 6.4*c*. If there is a gas-liquid surface, they should be baffled to prevent formation of a vortex. They may be covered, operated full, in which case they may be baffled or not. For continuous operation, the liquids to be contacted may enter at the bottom and leave at the top; in some cascade arrangements the light and heavy liquids enter through the side wall near the top and bottom of the vessel, respectively, and leave through a port in the wall opposite the impeller. For batch operation, the mixing vessel itself may act as the settler after agitation is stopped.

Impellers are usually of the flat-blade turbine type, Fig. 6.3*b* and *c*, either centrally located in the vessel or nearer the bottom entry of the liquids. The impeller-tank-diameter ratio d_i/T is ordinarily in the range 0.25 to 0.33.

Dispersions The mixture of liquids produced consists of droplets of one liquid dispersed in a continuum of the other. Drop diameters are usually in the range 0.1 to 1 mm diameter. If they are too small, difficulty in subsequent settling will

result. Ordinarily the liquid flowing at the smaller volume rate will be dispersed in the other, but some control can be exercised. For continuous operation the vessel should first be filled with the liquid to be continuous, and with agitation in progress the two liquids can then be introduced in the desired ratio. For batch operation, the liquid in which the impeller is immersed when at rest is usually continuous. It is ordinarily difficult to maintain ϕ_D , the volume fraction dispersed phase, at values much above 0.6 to 0.7, and attempts to increase it beyond this will frequently result in inversion of the dispersion (continuous phase becomes dispersed) [37, 49, 83].

Power for agitation Power for agitation of two-liquid-phase mixtures by disk flat-blade turbines is given by curves *c* (baffles) and *g* (unbaffled, full, no vortex) of Fig. 6.5, provided suitable average density ρ_M and viscosity μ_M are used [35]:

$$\rho_M = \rho_C \phi_C + \rho_D \phi_D \quad (10.56)$$

$$\text{Baffled vessels:} \quad \mu_M = \frac{\mu_C}{\phi_C} \left(1 + \frac{1.5 \mu_D \phi_D}{\mu_C + \mu_D} \right) \quad (10.57)$$

Unbaffled vessels, no gas-liquid interface, no vortex:

$$\mu_M = \begin{cases} \frac{\mu_w}{\phi_w} \left(1 + \frac{6 \mu_o \phi_o}{\mu_o + \mu_w} \right) & \phi_w > 0.4 \\ \frac{\mu_o}{\phi_o} \left(1 - \frac{1.5 \mu_w \phi_w}{\mu_o + \mu_w} \right) & \phi_w < 0.4 \end{cases} \quad (10.58)$$

$$(10.59)$$

where the subscripts *o* and *w* refer to organic and aqueous liquids, respectively [the difference for Eqs. (10.58) and (10.59) may very well be related to which liquid is dispersed]. The viscosity of a mixture can exceed that of either constituent. For other types of turbines, see Ref. 46.

Dispersed-phase holdup At agitator-power levels which are relatively low, the holdup ϕ_D of the dispersed phase in the vessel will be less than its fraction of the feed mixture, $\phi_{DF} = q_{DF}/q_{CF} + q_{DF}$. The value of ϕ_D will approach that of ϕ_{DF} as agitator power increases. For ϕ_D/ϕ_{DF} in the range 0.5 to 0.95, the following provide estimates for the indicated circumstances [81]. Exact prediction cannot be expected because the range of the studies thus far completed is relatively limited. For continuous flow:

Baffled vessels, impeller power/vessel volume $> 105 \text{ W/m}^3 = 2.2 \text{ ft lb}_f/\text{ft}^3 \cdot \text{s}$:

$$\frac{\phi_D}{\phi_{DF}} = 0.764 \left(\frac{P q_D \mu_C^2}{v_L \sigma^3 g_c} \right)^{0.300} \left(\frac{\mu_C^3}{q_D \rho_C^2 \sigma g_c} \right)^{0.178} \left(\frac{\rho_C}{\Delta \rho} \right)^{0.0741} \left(\frac{\sigma^3 \rho_C g_c^3}{\mu_C^2 g} \right)^{0.276} \left(\frac{\mu_D}{\mu_C} \right)^{0.136} \quad (10.60)$$

Unbaffled vessels, full, no gas-liquid surface, no vortex:

$$\frac{\phi_D}{\phi_{DF}} = 3.39 \left(\frac{P q_D \mu_C^2}{v_L \sigma^3 g_c} \right)^{0.247} \left(\frac{\mu_C^3}{q_D \rho_C^2 \sigma g_c} \right)^{0.427} \left(\frac{\rho_C}{\Delta \rho} \right)^{0.430} \left(\frac{\sigma^3 \rho_C g_c^3}{\mu_C^2 g} \right)^{0.401} \left(\frac{\mu_D}{\mu_C} \right)^{0.0987} \quad (10.61)$$

The quantities in parentheses are dimensionless. If ϕ_D/ϕ_{DF} computes to exceed 1.0, it should be taken as 1.0. Some anomalies can be expected for water dispersed in hydrocarbons.

Specific interfacial area and drop size Interfacial area in extraction vessels has been measured most generally by a light-transmittance technique [10]. It varies considerably with location in the vessel [58, 81]. Intense turbulence at the impeller tip results in small drop sizes there, and *a* is consequently large. The drops coalesce as they flow to the more quiescent regions removed from the impeller, and

the specific area becomes smaller. Average specific area and drop size are related by

$$a = \frac{6\phi_D}{d_p} \quad (6.9)$$

There have been many studies in baffled vessels but relatively few in unbaffled ones. The measurements have been made in the absence of mass transfer, which will probably have an influence through coalescence rates of the drops, and usually with an organic liquid dispersed in an aqueous phase. The following are recommended: for baffled vessels [6]

$$\frac{d_p}{d_p^o} = 1 + 1.18\phi_D \left(\frac{\sigma^2 g_c^2}{d_p^2 \mu_c^2 g} \right) \left(\frac{\mu_c^4 g}{\Delta \rho \sigma^3 g_c^3} \right)^{0.62} \left(\frac{\Delta \rho}{\rho_c} \right)^{0.05} \quad (10.62)$$

where d_p^o is given by

$$\frac{d_p^{o3} \rho_c^2 g}{\mu_c^2} = 29 \left(\frac{v_L^2 \rho_c^2 \mu_c g^4}{P^3 g_c^3} \right)^{0.32} \left(\frac{\sigma^3 \rho_c g^2}{\mu_c^4 g} \right)^{0.14} \quad (10.63)$$

The groups in parentheses are dimensionless. For unbaffled vessels, full, no gas-liquid surface, no vortex [81]†

$$d_p = 10^{-2.066 + 0.732\phi_D} \left(\frac{\mu_c}{\rho_c} \right)^{0.0473} \left(\frac{P g_c}{v_L \rho_M} \right)^{-0.204} \left(\frac{\sigma g_c}{\rho_c} \right)^{0.274} \quad (10.64)$$

Mass-transfer coefficients There have been relatively few measurements of overall mass-transfer coefficients for liquid-liquid systems in agitated vessels [72, 76] and still fewer studies of k_C for the continuous phase [33, 42, 58], far too few for useful generalizations. On the other hand there have been fairly extensive studies of mass transfer from solid particles. For small solid particles, most closely related in size at least to the droplets in a well-agitated vessel, the recommended relation is [36]

$$\text{Sh}_C = \frac{k_{LC} d_p}{D_C} = 2 + 0.47 \left[d_p^{4/3} \left(\frac{P g_c}{v_L} \right)^{1/3} \frac{\rho_c}{\mu_c} \right]^{0.62} \left(\frac{d_p}{T} \right)^{0.17} \left(\frac{\mu_c}{\rho_c D_C} \right)^{0.36} \quad (10.65)$$

The group in square brackets is a form of particle Reynolds number, as shown in Chap. 11. Values of k_{LC} for liquid particles are known to be larger than those for solids [42, 58], and it is suspected that drop coalescence and breakup may be responsible. In any event, Eq. (10.65) can be expected to provide conservative values.

For small circulating drops, of the sort encountered here, the dispersed-phase coefficient is given by [15]

$$k_{LD} = -\frac{d_p}{6\theta} \ln \left[\frac{3}{8} \sum_{n=1}^{\infty} B_n^2 \exp \left(-\frac{\lambda_n 64 D_D \theta}{d_p^2} \right) \right] \quad (10.66)$$

where θ is the residence time in the vessel. B_n and λ_n are eigenvalues given by Table 10.1. Then, in the absence of any chemical reaction or interfacial resistance (see Chap. 5)

$$\frac{1}{K_{LD}} = \frac{1}{k_{LD}} + \frac{1}{m_{CD} k_{LC}} \quad (10.67)$$

where m_{CD} is the distribution coefficient, dc_C/dc_D .

† The coefficient in Eq. (10.64) is for SI units. For units of feet, pounds force, pounds mass, and seconds the coefficient is $10^{-1.812 + 0.732\phi_D}$.

Table 10.1 Eigenvalues for circulating drops †

$k_C d_p / D_C$	λ_1	λ_2	λ_3	B_1	B_2	B_3
3.20	0.262	0.424		1.49	0.107	
5.33	0.386					
8.00	0.534					
10.7	0.680	4.92		1.49	0.300	
16.0	0.860	5.26		1.48	0.382	
21.3	0.982	5.63		1.47	0.428	
26.7	1.082	5.90	15.7	1.49	0.495	0.205
53.3	1.324	7.04	17.5	1.43	0.603	0.298
107	1.484	7.88	19.5	1.39	0.603	0.384
213	1.560	8.50	20.8	1.31	0.588	0.396
320	1.60	8.62	21.3	1.31	0.583	0.391
∞	1.656	9.08	22.2	1.29	0.596	0.386

† From E. L. Elzinga and J. T. Banchero: *Chem. Eng. Prog. Symp. Ser.*, 55(29), 149 (1959), with permission.

Stage efficiency In view of the uncertainties in the mass transfer data, it is reasonable to use a simplified version of the number of transfer units [compare Eq. (8.43)]. For the dispersed phase,

$$N_{IOD} = \int_{c_{2D}}^{c_{1D}} \frac{dc_D}{c_D - c_D^*} \quad (10.68)$$

where the dispersed-phase concentration changes from c_{1D} to c_{2D} and c_D^* is the dispersed-phase concentration in equilibrium with that in the continuous phase. It has been established that the liquids in a well-agitated vessel are thoroughly back-mixed, so that everywhere the concentrations are constant at the effluent values [58]. Then

$$N_{IOD} = \frac{1}{c_{2D} - c_{2D}^*} \int_{c_{2D}}^{c_{1D}} dc_D = \frac{c_{1D} - c_{2D}}{c_{2D} - c_{2D}^*} \quad (10.69)$$

and

$$N_{IOD} = \frac{Z}{H_{IOD}} = \frac{Z}{V_D / K_{LD} a} \quad (10.70)$$

The Murphree dispersed-phase efficiency E_{MD} is given by Eq. (5.39) adapted to the present situation:

$$\begin{aligned} E_{MD} &= \frac{c_{1D} - c_{2D}}{c_{1D} - c_{2D}^*} = \frac{c_{1D} - c_{2D}}{(c_{1D} - c_{2D}) + (c_{2D} - c_{2D}^*)} \\ &= \frac{(c_{1D} - c_{2D}) / (c_{2D} - c_{2D}^*)}{(c_{1D} - c_{2D}) / (c_{2D} - c_{2D}^*) + 1} = \frac{N_{IOD}}{N_{IOD} + 1} \end{aligned} \quad (10.71)$$

In the absence of slow chemical reactions, such as may occur during extraction of metals from ore-leach liquors, and in the absence of interfacial resistance, stage efficiencies of well-agitated vessels can be expected to be high, and relatively short (60 s) holding times are ordinarily ample. The preceding equations, in any event, can be expected to provide rough estimates only, and experimental studies with new systems are essential before designs are completed.

Recycling Especially when the ratio of liquid flows is small, the interfacial area, with the minority liquid dispersed, may be small. Recycling some of the settled liquid back to the mixer may then improve the stage efficiency. Recycling that liquid favored by the solute distribution (frequently the minority liquid) increases the stage efficiency, whereas recycling the other or recycling both reduces it [75].

Illustration 10.7 A covered, baffled vessel, 0.5 m (19.7 in) diameter, 0.5 m deep is to be used for continuous extraction of benzoic acid from water ($3 \times 10^{-3} \text{ m}^3/\text{s} = 6.36 \text{ ft}^3/\text{min}$) into toluene ($3 \times 10^{-4} \text{ m}^3/\text{s}$) as solvent. Liquids are to flow cocurrently upward and will fill the vessel; there will be no gas-liquid interface and hence no vortex. A six-blade, flat-blade disk turbine, 0.15 m (6 in) diameter is to be located axially and centrally, turning at 13.3 r/s (800 r/min). The liquid properties are:

	Water solution	Toluene solution
Density ρ	998 kg/m ³	865 kg/m ³
Viscosity μ	$0.95 \times 10^{-3} \text{ kg/m} \cdot \text{s}$	$0.59 \times 10^{-3} \text{ kg/m} \cdot \text{s}$
Diffusivity D	$2.2 \times 10^{-9} \text{ m}^2/\text{s}$	$1.5 \times 10^{-9} \text{ m}^2/\text{s}$
Interfacial tension, $\sigma = 0.022 \text{ N/m}$		
Distribution coefficient = $c_{\text{toluene}}/c_{\text{water}} = 20.8$		

Estimate the stage efficiency to be expected.

SOLUTION In view of the flow ratio, the toluene will be dispersed. Vessel volume = $\pi(0.5)^2(0.5)/4 = 0.09817 \text{ m}^3$.

$\phi_{DF} = 3 \times 10^{-4} / (3 \times 10^{-3} + 3 \times 10^{-4}) = 0.0909$ vol fraction toluene in the feed mixture. $q_D = 3 \times 10^{-4} \text{ m}^3/\text{s}$ toluene.

$$\frac{\mu_C^3}{q_D \rho_C^2 \sigma g_c} = \frac{(0.95 \times 10^{-3})^3}{(3 \times 10^{-4})(998)^2(0.022)(1)} = 1.304 \times 10^{-10}$$

$$\frac{\rho_C}{\Delta \rho} = \frac{998}{998 - 865} = 7.504$$

$$\frac{\sigma^3 \rho_C g_c^3}{\mu_C^4 g} = \frac{(0.022)^3(998)(1)}{(0.95 \times 10^{-3})^4(9.807)} = 1.330 \times 10^9$$

$$\frac{\mu_D}{\mu_C} = \frac{0.59 \times 10^{-3}}{0.95 \times 10^{-3}} = 0.621$$

Tentatively take $\phi_D/\phi_{DF} = 0.9$; $\phi_D = 0.9(0.0909) = 0.0818$. Eq. (10.56):

$$\rho_M = 998(1 - 0.0818) + 865(0.0818) = 987 \text{ kg/m}^3$$

Eq. (10.58):

$$\mu_M = \frac{(0.95 \times 10^{-3})}{1 - 0.0818} \left[1 + \frac{6(0.59 \times 10^{-3})(0.0818)}{0.59 \times 10^{-3} + 0.95 \times 10^{-3}} \right] = 1.067 \times 10^{-3} \text{ kg/m} \cdot \text{s}$$

$$\text{Impeller Reynolds number} = \frac{d_i^2 N \rho_M}{\mu_M} = \frac{(0.15)^2(13.3)(987)}{1.067 \times 10^{-3}} = 277 \text{ 000}$$

Fig. 6.5, curve g:

$$\text{Po} = 0.72$$

$$P = \frac{\text{Po} \rho_M N^3 d_i^5}{g_c} = \frac{0.72(987)(13.3)^3(0.15)^5}{1} = 127 \text{ W impeller power output}$$

$$\frac{P q_D \mu_C^2}{v_L \sigma^3 g_c^2} = \frac{127(3 \times 10^{-4})(0.95 \times 10^{-3})^2}{0.09817(0.022)^3(1)} = 0.0329$$

Eq. (10.61): $\phi_D/\phi_{DF} = 0.907$. This is sufficiently close to the estimated 0.9 to be acceptable.

$$\phi_D = 0.907(0.0909) = 0.0824 \quad \frac{\mu_C}{\rho_C} = \frac{0.95 \times 10^{-3}}{998} = 9.52 \times 10^{-7} \text{ m/s}^2$$

$$\frac{P_{gC}}{v_L \rho_M} = \frac{127(1)}{0.0982(987)} = 1.311 \quad \frac{\sigma g_c}{\rho_c} = \frac{0.022(1)}{998} = 2.204 \times 10^{-5}$$

Eq. (10.64):

$$d_p = 2.55 \times 10^{-4} \text{ m} \quad \dot{a} = \frac{6\phi_D}{d_p} = 1940 \text{ m}^2/\text{m}^3$$

$$Sc_C = \frac{\mu_C}{\rho_C D_C} = \frac{0.95 \times 10^{-3}}{998(2.2 \times 10^{-9})} = 433$$

Eq. (10.65):

$$Sh_C = \frac{k_{LC} d_p}{D_C} = 65.3$$

$$k_{LC} = \frac{65.3 D_C}{d_p} = 5.45 \times 10^{-4} \text{ kmol/m}^2 \cdot \text{s} \cdot (\text{kmol/m}^3)$$

$$\theta = \frac{v_L}{q_C + q_D} = 29.8 \text{ s}$$

Table (10.1):

	$\lambda_1 = 1.359$	$\lambda_2 = 7.23$	$\lambda_3 = 17.9$
	$B_1 = 1.42$	$B_2 = 0.603$	$B_3 = 0.317$
n	1	2	3
$B_n^2 \exp(-\lambda_n 64 D_D \theta / d_p^2)$	2.18×10^{-26}	0	0

Eq. (10.66):

$$k_{LD} = 8.57 \times 10^{-5} \text{ kmol/m}^2 \cdot \text{s} \cdot (\text{kmol/m}^3)$$

$$m_{CD} = \frac{1}{20.8} = 0.0481$$

Eq. (10.67):

$$K_{LD} = 2.01 \times 10^{-5} \text{ kmol/m}^2 \cdot \text{s} \cdot (\text{kmol/m}^3)$$

$$Z = 0.5 \text{ m} \quad V_D = \frac{3 \times 10^{-4}}{\pi(0.5)^2/4} = 1.528 \times 10^{-3} \text{ m/s}$$

Eq. (10.70):

$$N_{iOD} = \frac{0.5}{1.528 \times 10^{-3} / [(2.01 \times 10^{-5})(1940)]} = 12.76$$

Eq. (10.71):

$$E_{MD} = 0.93 \quad \text{Ans.}$$

Emulsions and Dispersions

The mixture of liquids issuing from any mixing device, an emulsion, consists of small droplets of one liquid dispersed throughout a continuum of the other. The

stability, or permanence, of the emulsion is of the utmost importance in liquid extraction, since it is necessary to separate the phases at each extraction stage. Stable emulsions, those which do not settle and coalesce rapidly, must be avoided. For an emulsion to "break," or separate into its phases in bulk, both sedimentation and coalescence of the dispersed phase must occur.

The rate of sedimentation of a quiescent emulsion is the more rapid if the size of the droplets and the density difference of the liquids are large and the viscosity of the continuous phase is small. Stable emulsions, those which settle only over long periods of time, are usually formed when the diameter of the dispersed droplets is of the order of 1 to 1.5 μm , whereas dispersions of particle diameter 1 mm or larger usually sediment rapidly.

Coalescence of the settled droplets is the more rapid the higher the interfacial tension. Interfacial tension is ordinarily low for liquids of high mutual solubility and is lowered by the presence of emulsifying or wetting agents. In addition, high viscosity of the continuous phase hinders coalescence by reducing the rate at which the residual film between drops is removed. Dust particles, which usually accumulate at the interface between liquids, also hinder coalescence.

In an unstable emulsion, after agitation has stopped, the mixture settles and coalesces rapidly into two liquid phases unless the viscosity is high. The appearance of a sharply defined interface between the phases (primary break) is usually very rapid, but one of the phases, ordinarily that in the majority, may remain clouded by a very fine fog or haze, a dispersion of the other phase. The cloud will eventually settle and leave the clouded phase clear (secondary break), but this may take a considerable time. The primary break of an unstable emulsion is usually so rapid that merely stopping agitation for a very short time, a matter of minutes, is sufficient to bring it about. In continuous multistage operation, it is usually impractical to hold the mixture between stages long enough to attain the secondary break.

Settlers

In continuous extraction, the dispersion issuing from the mixer must be passed to a settler, or decanter, where at least the primary break occurs. Typical gravity settler arrangements are shown in Fig. 10.40. The simplest design (Fig. 10.40a) is perhaps the most common: the entering dispersion is prevented from disturbing the vessel contents excessively by the inlet "picket-fence" baffle. The drops then settle in the main part of the vessel, where the velocity must be low enough to prevent turbulent disturbances. At larger flow rates for a given vessel, the dispersion-band thickness increases, and ultimately unsettled dispersion issues from the outlets, which is of course to be avoided. Although a great many fundamental studies have been made [1, 28], design methods are still empirical and arbitrary. There have been three approaches to the design: (1) provision of sufficient residence time based on laboratory observation of settling, (2) estimation of the rate of flow to produce a suitable dispersion-band thickness, and (3) calculation of the time to settle individual drops through a clear liquid above

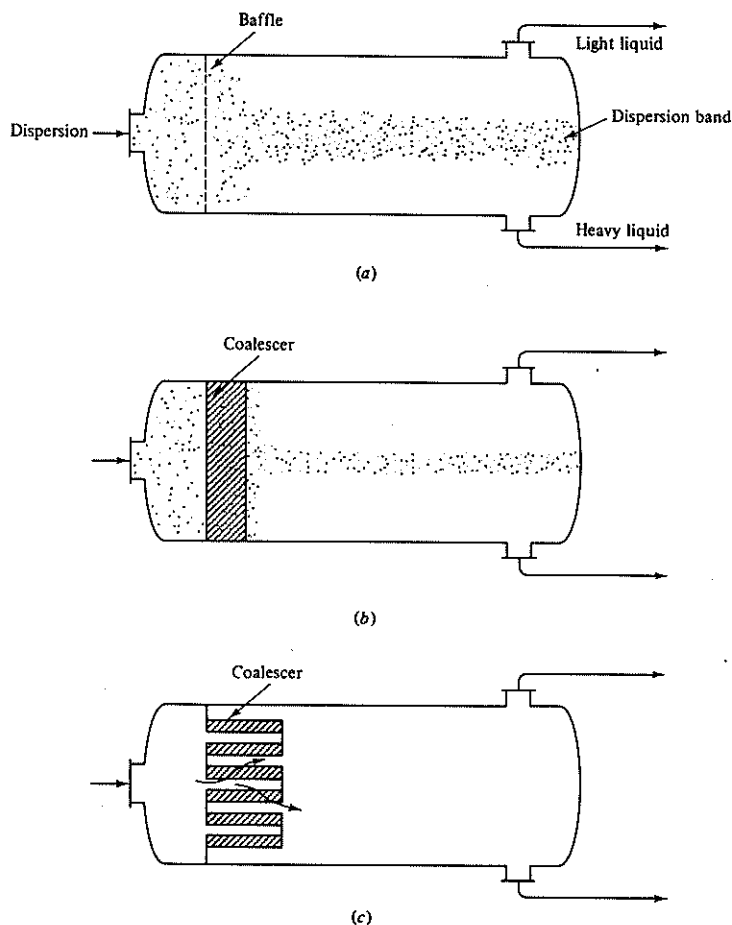


Figure 10.40 Gravity settlers (schematic): (a) simple, (b) and (c) with coalescer.

and below the dispersion band. None of these methods is very reliable [12]. In the absence of any experience with the dispersion at hand, a first estimate of settler size can be made from the empirical expression†

$$T_S = 8.4(q_C + q_D)^{0.5} \quad (10.72)$$

developed on the basis of typical settling rates for droplets of dispersed phase [18] and a length/diameter ratio for the vessel equal to 4.

† Equation (10.72) is satisfactory only for SI units. For T_S in feet and q in gal/min the coefficient is 0.22.

Settler auxiliaries The dispersion may be passed through a *coalescer* in order to increase the size of the droplets and hence their settling rate. Typically, these are relatively thin beds of substances of extended surface and relatively large voids, such as excelsior (especially for coalescing the salt brines accompanying crude petroleum), steel wool, fiber glass, polypropylene cloth, Raschig rings, and the like [20, 66]. One of the best coalescers is a mixture of cotton fibers and glass wool or of cotton and Dynel fibers [53]. A simple arrangement is shown in Fig. 10.40*b*, and a more elaborate but very effective one in *c* [72]. *Separator membranes* are porous membranes, preferentially wet by the continuous phase, which prevent the passage of the nonwetting dispersed phase at low pressure drop but freely pass the continuous liquid. They can be placed in the outlet from the decanter [72].

Mixer-Settler Cascades

A continuous multistage extraction plant contains the required number of stages arranged according to the desired flowsheet. Each stage will consist of at least a mixer and a settler, as in the countercurrent plant of Fig. 10.41. The liquids are generally pumped from one stage to the next, but occasionally gravity flow can be arranged if sufficient headroom is available. Many arrangements have been designed to reduce the amount of interstage piping and the corresponding cost [72]. Figure 10.42 is one such: the mixing vessels are immersed in the large circular settling tanks, heavy liquid flows by gravity, light liquid by air lift, and recycling of settled light liquid to the mixer is accomplished by overflow. Figure 10.43 is typical of several designs of so-called *box extractors*, designed to permit dispensing with intermediate piping. The mixers and settlers are rectangular in cross section, hence "box," and are arranged in alternate positions for adjacent stages, as shown in the plan view of Fig. 10.43. Another arrangement places the stages one above the other in a vertical stack, with mixing impellers on a common shaft, utilizing the impellers as pumps as well as mixing devices [73].

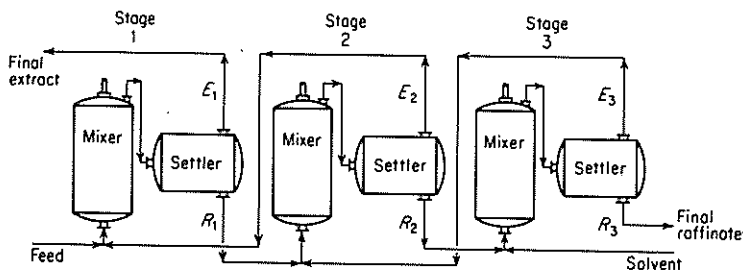


Figure 10.41 Flowsheet of three-stage countercurrent mixer-settler extraction cascade.

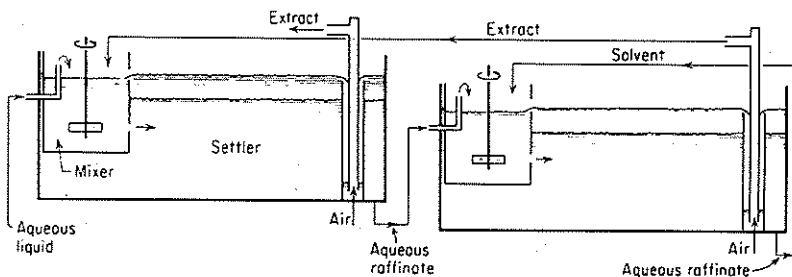


Figure 10.42 Kerr-McGee uranium extractor.

Sieve-Tray (Perforated-Plate) Towers

These multistage, countercurrent towers are very effective, both with respect to liquid-handling capacity and extraction efficiency, particularly for systems of low interfacial tension which do not require mechanical agitation for good dispersion. Their mass-transfer effectiveness results because (1) axial mixing of

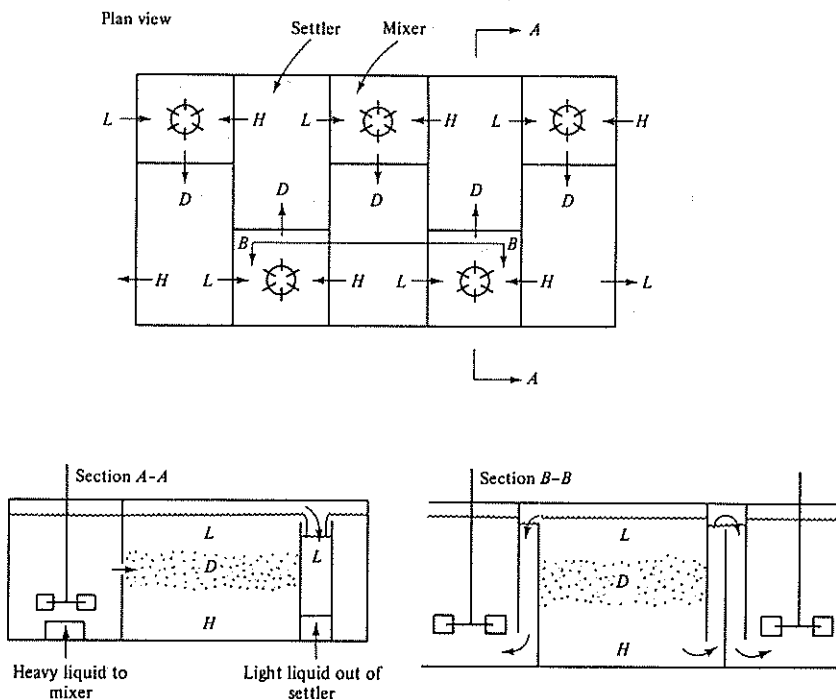


Figure 10.43 Box-type mixer-settler cascade (schematic): D = dispersion, H = heavy liquid, L = light liquid.

the continuous phase is confined to the region between trays and does not spread throughout the tower from stage to stage and (2) the dispersed-phase droplets coalesce and are formed again at each tray, destroying the tendency to establish concentration gradients within the drops which persist for the entire tower height. A tower of simple design is shown in Fig. 10.44, where the general arrangement of plates and downspouts is much the same as for gas-liquid contact except that no weir is required. The figure shows the arrangement for light liquid dispersed. Light liquid passes through the perforations, and the bubbles rise through the heavy continuous phase and coalesce into a layer, which accumulates beneath each plate. The heavy liquid flows across each plate through the rising droplets and passes through the downspouts to the plate below. By turning the tower as shown upside down, the downspouts become "upspouts" and carry the light liquid from plate to plate, while the heavy liquid flows through the perforations and is dispersed into drops. As an alternative, the

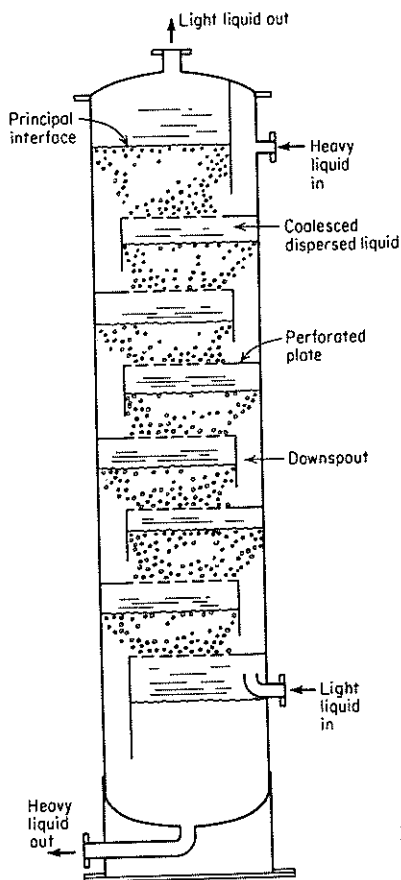


Figure 10.44 Sieve-tray extraction tower, arranged for light liquid dispersed.

heavy liquid can be dispersed in one part of the tower and the light liquid in the other, while the principal interface is maintained in the central portion of the tower. The cross-flow trays of Fig. 10.44 are suitable for relatively small tower diameters (up to roughly 2 m). For large towers, multiple downspouts can be arranged at intervals across the tray.

Sieve-Tray Hydraulics

The flow capacity of the sieve-tray tower depends upon drop-formation characteristics of the system, drop terminal velocities, dispersed-phase holdup, and pressure drop. With these established, the design can be developed and mass-transfer rates estimated.

Drop formation The liquid to be dispersed flows through perforations of diameter 3 to 8 mm, set 12 to 20 mm apart. If the drop liquid preferentially wets the material of the plate, the drop size becomes uncontrollably large and it is best to use small nozzles projecting from the surface in order to avoid this difficulty. The holes can be punched in a flat plate (Fig. 10.45) with the burr left in place facing in the direction of drop formation.

In the absence of mass transfer, the effect of which has not been firmly established [8], when the dispersed phase issues from a perforation in a plate which is not preferentially wet by the dispersed liquid, drop size will be uniform at a given velocity through the opening, while drops form at the orifice. At some moderate velocity, a jet of dispersed liquid issues from the opening and drops are formed by breakup of the jet. The velocity through the orifice when jetting begins can be estimated [54], but as a rule of thumb it will be roughly 0.1 m/s (0.3 ft/s) [19]. For velocities below this, drop sizes can be estimated principally from either of two correlations [19, 54], which, to a large extent but not always, agree reasonably well. For present purposes, estimates are sufficient, so that differences are unimportant, and Fig. 10.46 will serve. At velocities greater than about 0.1 m/s, drop sizes are not uniform. In extraction, maximizing the interface surface of all the drops is important for ensuring rapid mass transfer, and the orifice velocity $V_{o, \max}$ which does this can be estimated from [11, 64]

$$V_{o, \max} = 2.69 \left(\frac{d_j}{d_o} \right)^2 \left[\frac{\sigma}{d_j(0.5137\rho_D + 0.4719\rho_C)} \right]^{0.5} \quad (10.73)$$

The orifice/jet diameter ratio is given by

$$\frac{d_o}{d_j} = \begin{cases} 0.485 \left[\frac{d_o}{(\sigma g_c / \Delta \rho g)^{0.5}} \right]^2 + 1 & \text{for } \frac{d_o}{(\sigma g_c / \Delta \rho g)^{0.5}} < 0.785 \\ 1.51 \frac{d_o}{(\sigma g_c / \Delta \rho g)^{0.5}} + 0.12 & \text{for } \frac{d_o}{(\sigma g_c / \Delta \rho g)^{0.5}} > 0.785 \end{cases} \quad (10.74a)$$

$$(10.74b)$$

[Equations (10.73) and (10.74) can be used with any consistent set of units.] The drop diameter corresponding to the above values is $d_d = 2d_j$. The value of σ to be used is intermediate between that for the binary solvent pair and the equilibrium value for the extracting system if that can be estimated (data are scarce). Equations (10.73) and (10.74) are recommended for sieve-tray design, but if the resulting velocity calculates to be less than 0.1 m/s, V_o should be set at least at from 0.1 to 0.15 m/s and the drop diameter should be estimated from Fig. 10.46.

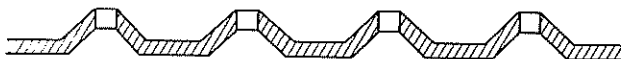


Figure 10.45 Punched perforations for dispersed phase [39].

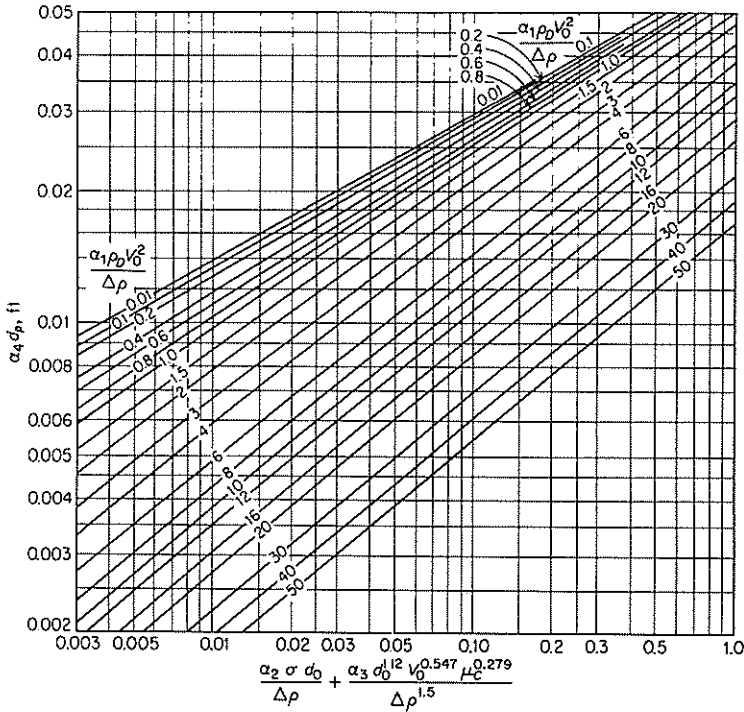


Figure 10.46 Drop diameters for dispersion of insoluble liquids through nozzles and perforations [19].

	α_1	α_2	α_3	α_4
SI	10.76	52 560	1.246×10^6	3.281
lb, lb _f , ft, s	1.0	14 600	3040	1.0

Drop terminal velocity It will be necessary to estimate V_t , the terminal velocity of a liquid drop in a liquid medium. This is the free-fall (or rise, depending on the relative density) velocity of a single isolated drop in the gravitational field. The terminal velocities of small drops, which are essentially spherical, are larger than those of solid spheres of the same diameter and density, owing to the mobility and internal circulation within the drop: the surface velocity is not zero, as it is for a solid. With increasing diameter, there occurs a transition drop size, beyond which the drop shape is no longer spherical (although it is nevertheless described with d_p as the diameter of a sphere of the same volume), and the drop oscillates and distorts. The terminal velocity of the transition size is a maximum, and for larger sizes the velocity falls slowly with increased diameter. Dimensional analysis shows $Re = f(C_D, We)$, where Re is the drop Reynolds number at terminal velocity, We is the drop Weber number, and C_D is the usual drag coefficient. For very pure liquids (no surface-active agents, no mass transfer), and continuous phase viscosities less than $0.005 \text{ kg/m} \cdot \text{s}$ (5 cP), the Hu-Kintner correlation [24] (Fig. 10.47) provides the functional relation. U is defined as

$$U = \frac{4 Re^4}{3 C_D We^3} = \frac{\rho C^2 (\sigma g_c)^3}{g \mu_c^4 \Delta \rho} \quad (10.75)$$

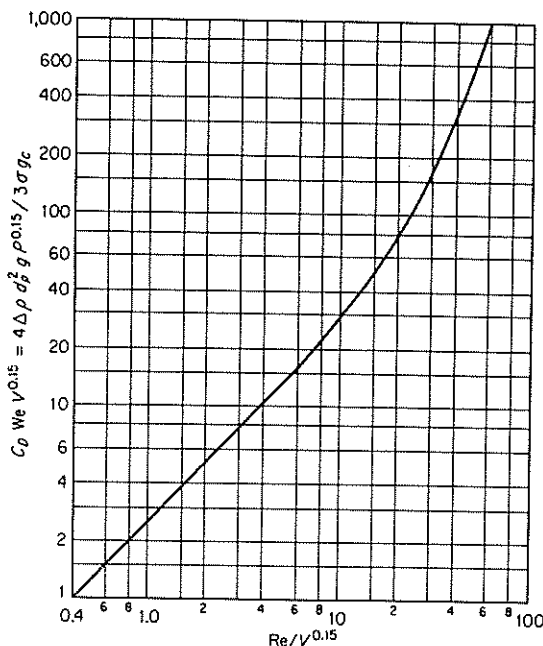


Figure 10.47 Terminal settling velocities of single liquid drops in infinite liquid media [24].

and the transition maximum velocity occurs at an ordinate of Fig. 10.47 equal to approximately 70, or when

$$d_{p, \text{trans}} = 7.25 \left(\frac{\sigma g_c}{g \Delta \rho U^{0.15}} \right)^{0.5} \quad (10.76)$$

For larger continuous phase viscosities, but not exceeding $0.030 \text{ kg/m} \cdot \text{s}$ (30 cP), the ordinate of Fig. 10.47 should be multiplied by $(\mu_w/\mu_c)^{0.14}$, where μ_w is the viscosity of water [30]. Minute amounts of impurities which are frequently present in practice can alter the terminal velocities profoundly (usually to lower values), and the effect of mass transfer is unknown.†

Downspouts The tower will flood if excessive quantities of dispersed-phase droplets are carried through the downspouts (or upspouts) entrained in the continuous phase. The velocity of liquid in the downspouts V_d should therefore be less than the terminal velocity of all except the smallest

† The curve of Fig. 10.47 should not be extrapolated to lower values. For values of the ordinate below 1.0, the equation of Klee and Treybal [34] will give more correct results for $\mu_c < 0.002 \text{ kg/m} \cdot \text{s}$ (2 cP):

$$\text{Re} = 22.2 C_D^{-5.18} \text{We}^{-0.169} \quad (10.77a)$$

This is conveniently rearranged to

$$V_t = \frac{0.8364 \Delta \rho^{0.5742} d_p^{0.7037} g^{0.5742}}{\rho_c^{0.4446} (\sigma g_c)^{0.01873} \mu_c^{0.11087}} \quad (10.77b)$$

which is applicable for any consistent set of units for values of d_p less than $d_{p, \text{trans}}$.

droplets, e.g., those smaller than 0.6 to 0.8 mm [39], and the downspout cross section A_d is set accordingly. The downspouts are extended well beyond the depth of coalesced dispersed liquid on the tray, to prevent dispersed liquid from flowing through the downspout. Downspouts are installed flush with the tray from which they lead, and no weirs like those needed for gas-liquid towers should be used.

Coalesced liquid on the trays The depth h of coalesced dispersed liquid accumulating on each tray is determined by the pressure drop required for counterflow of the liquids [9]:

$$h = h_C + h_D \quad (10.78)$$

where h_C and h_D are the contributions of each of the liquids. The contribution from the dispersed liquid h_D is that necessary to cause flow through the perforations of the tray h_o plus that necessary to overcome interfacial tension h_o ,

$$h_D = h_o + h_o \quad (10.79)$$

The value of h_o can be computed from the usual orifice equation with a coefficient of 0.67,

$$h_o = \frac{(V_o^2 - V_n^2)\rho_D}{2g(0.67)^2 \Delta\rho} \quad (10.80)$$

h_o , which is important only when the dispersed phase flows slowly, can be computed from

$$h = \frac{6\sigma g_c}{d_p \Delta\rho g} \quad (10.81)$$

where d_p is the drop diameter produced at perforation velocities $V_o = 0.03$ m/s (0.1 ft/s). At perforation velocities where jets of dispersed liquid issue from the perforations, h_o can be omitted.

The head required for flow of the continuous phase h_C includes losses owing to (1) friction in the downspout, ordinarily negligible, (2) contraction and expansion upon entering and leaving the downspout, equal to 0.5 and 1.0 velocity heads, respectively, and (3) the two abrupt changes in direction each equivalent to 1.47 velocity heads. The value of h_C is therefore substantially equal to 4.5 velocity heads, or

$$h_C = \frac{4.5 V_D^2 \rho_C}{2g \Delta\rho} \quad (10.82)$$

Should h calculate to be small, say 50 mm or less, there is danger that not all perforations will operate unless the tray is installed perfectly level. Under these circumstances, it is best to increase the value of h_C by placing a restriction in the bottom of the downspouts (or upper end of upspouts), the effect of which can be computed as flow through an orifice.

Dispersed-phase holdup The space between a tray and the coalesced layer on the next tray is filled with a dispersion of the dispersed liquid in the continuous liquid. At flow rates below those causing flooding, it has been established [4, 80] that the ratio of slip velocity V_S to terminal velocity V_t of a single particle ($\phi_D = 0$) is a unique function of dispersed-phase holdup ϕ_D for all vertically moving fluid particulate systems, including gas-solid, liquid-solid, and liquid-liquid systems. Slip velocity is the net relative velocity between the two phases, and if they flow countercurrently,

$$V_S = \frac{V_D}{\phi_D} + \frac{V_C}{1 - \phi_D} \quad (10.83)$$

The correlation of Zenz [84] (Fig. 10.48), derived from fluid-solid systems, provides the $(V_S/V_t, \phi_D)$ function, with V_t for solids given by the curve for $\phi_D = 0$. For sieve-tray extractors, V_D becomes V_n , and since the continuous phase flows horizontally, V_C is taken as zero, so that

$$V_S = \frac{V_n}{\phi_D} \quad (10.84)$$

Illustration 10.8 describes the use of the correlation. The specific interfacial area corresponding to the holdup is given by Eq. (6.9).

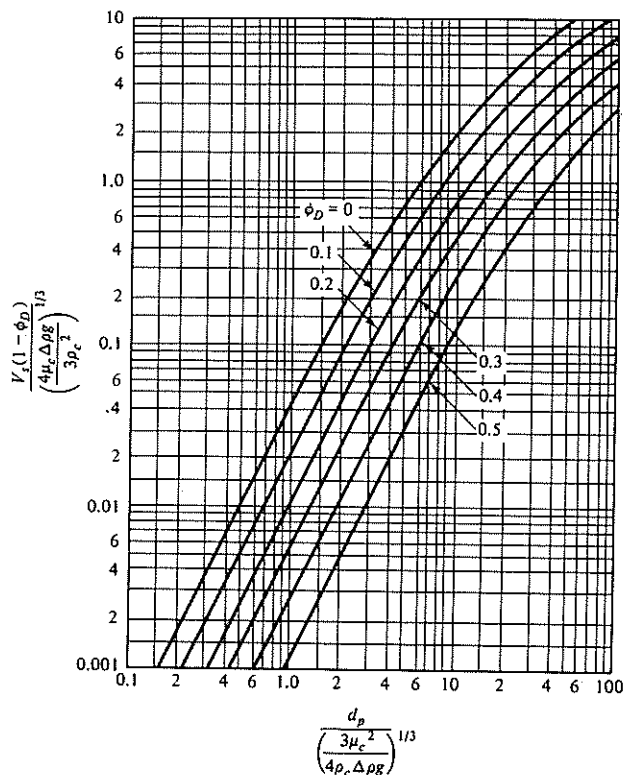


Figure 10.48 Zenz's correlation for fluidized solid particles. [F. Zenz, *Pet. Refiner*, **36** (8), 147 (1957), with permission.]

Sieve-Tray Mass Transfer

Mass transfer occurs during three separate regimes: drop formation and release, drop rise (or fall), and drop coalescence into the layer of coalesced liquid on the tray.

Mass transfer during drop formation There have been a great many studies, for most of which a review is available [21]. There are evidently many influencing phenomena: the rate of drop formation [7]; whether the drops are formed at nozzles or orifices in plates or at the ends of jets; and the presence or absence of interfacial turbulence or surfactants. Although fairly elaborate expressions have been devised to describe some of the data [62, 63], the great divergence of the data at present does not seem to warrant anything more than a simple estimate. The mass-transfer coefficient K_{LDf} can be defined by [76]

$$N_f = K_{LDf}(c_D - c_D^s) \quad (10.85)$$

where N_f is the flux, averaged over the time of drop formation θ_f , and based on the area A_p of the drop at breakaway. If we assume the same mechanism of mass transfer for the liquids on either side of the interface, e.g., surface renewal, penetration theory, or surface stretch, (see Chap. 3), so that

$k_{LCf} = k_{LDf} (D_C/D_D)^{0.5}$, then

$$\frac{1}{K_{LDf}} = \frac{1}{k_{LDf}} + \frac{1}{k_{LCf} m_{CD}} = \frac{1}{k_{LDf}} \left[1 + \frac{1}{m_{CD}} \left(\frac{D_D}{D_C} \right)^{0.5} \right] \quad (10.86)$$

Theoretical treatment of mass transfer during drop formation generally leads to expressions of the form

$$k_{LDf} = \text{const} \left(\frac{D_D}{\pi \theta_f} \right)^{0.5} \quad (10.87)$$

with values of the constant in the range 0.857 to 3.43 but mostly in the range 1.3 to 1.8 except in the presence of interfacial turbulence or surfactants.

Mass transfer during drop rise (or fall) Swarms of drops behave differently from single drops [25]. For small drops ($d_p < d_{p, \text{trans}}$) which circulate [52]

$$k_{LCr} = 0.725 \left(\frac{d_p V_S \rho_C}{\mu_C} \right)^{-0.43} Sc^{-0.58} V_S (1 - \phi_D) \quad (10.88)$$

The dispersed-phase coefficient k_{LDr} is then given by Eq. (10.66) and Table 10.1.

For large drops ($d_p > d_{p, \text{trans}}$) which oscillate as they rise or fall, surface-stretch theory [2] is recommended

$$k_{LDr} = \sqrt{\frac{4 D_D \omega}{\pi}} \left(1 + \delta + \frac{3}{8} \delta^2 \right) \quad (10.89)$$

where $\omega = \text{oscillation frequency} = \frac{1}{2\pi} \sqrt{\frac{192 \sigma g_c b}{d_p^3 (3\rho_D + 2\rho_C)}}$ (10.90)

and† $b = 1.052 d_p^{0.225}$ (10.91)

δ is a dimensionless amplitude factor characteristic of the oscillating system, and may be taken as 0.2 in the absence of more specific data. As with drop formation, the same mechanism is assumed to govern the phases on both sides of the interface, so that an equation similar to that of Eq. (10.86) provides K_{LDr} .

Mass transfer during drop coalescence Experimental measurement of this quantity is very difficult, and results are hard to reproduce. The data for only a few systems have been empirically correlated [65]. The mass-transfer coefficient is apparently an order of magnitude less than that for drop formation, and the area on which it is based is difficult to deduce. For the present, it is recommended that $K_{DC} = 0.1 K_{Df}$, based on the area of the drops A_p .

Stage efficiency Refer to Fig. 10.49a, which shows a schematic section through part of a sieve-tray tower. The equilibrium and operating curves corresponding to the entire extractor are shown in Fig. 10.49b. Curve DBE is drawn between the equilibrium and operating curves, everywhere at a fractional distance such as BC/AC equal to the Murphree dispersed-phase stage efficiency E_{MD} . The steps on the diagram then correspond to real stages, as for gas absorption (Fig. 8.16) and distillation (Fig. 9.48)

$$E_{MD} = \frac{c_{D_{n+1}} - c_{D_n}}{c_{D_{n+1}} - c_{D_n}^*} \quad (10.92)$$

† b is dimensionless, and Eq. (10.91) is written for d_p in meters. For d_p in feet the coefficient is 0.805.

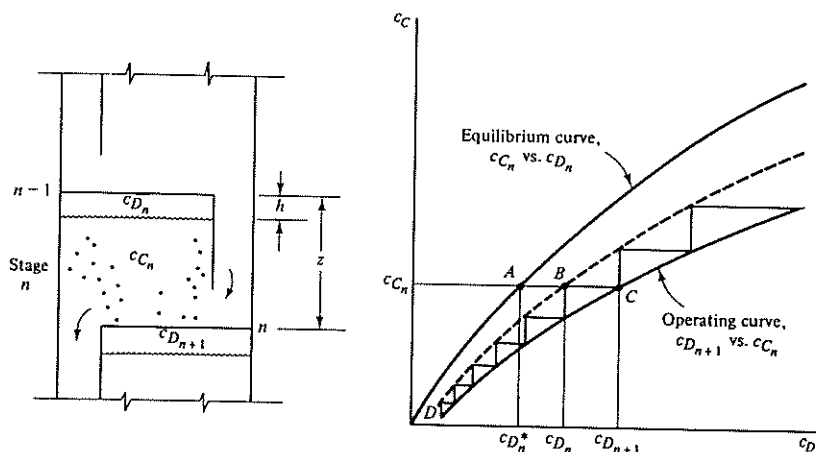


Figure 10.49 Stage efficiency of sieve-tray extractors.

The relatively low velocities prevailing in liquid extractors mean that the continuous phase can be assumed to be thoroughly mixed by the motion of the drops of the dispersed phase and everywhere of solute concentration $c_{C,n}$. The total rate of extraction occurring in stage n is then

$$q_D(c_{D,n+1} - c_{D,n}) = K_{LDf}A_f(c_{D,n+1} - c_{D,n}^*) + K_{LDc}A_r(c_D - c_{D,n}^*) + K_{LDc}(c_{D,n} - c_{D,n}^*) \quad (10.93)$$

Equations (10.92) and (10.93) produce

$$E_{MD} = \frac{K_{LDf}A_f}{q_D} + \frac{K_{LDc}A_r(c_D - c_{D,n}^*)}{q_D(c_{D,n+1} - c_{D,n}^*)} + \frac{K_{LDc}A_p(c_{D,n} - c_{D,n}^*)}{q_D(c_{D,n+1} - c_{D,n}^*)} \quad (10.94)$$

A_p is the surface of the N_o drops issuing from the N_o orifices.

$$A_p = \pi d_p^2 N_o \quad (10.95)$$

The interfacial area of the dispersed-phase holdup in the drop-rise region is

$$A_r = a(Z - h)(A_i - A_d) = \frac{6\phi_D}{d_p}(Z - h)A_n \quad (10.96)$$

The mean driving force for the drop-rise region is strictly the logarithmic average, but for present purposes the arithmetic average is adequate:

$$(c_D - c_{D,n}^*)_M = \frac{(c_{D,n+1} - c_{D,n}^*) + (c_{D,n} - c_{D,n}^*)}{2} \quad (10.97)$$

and, as indicated earlier, K_{LDc} is taken as $0.1K_{LDf}$. Substitution of this and Eqs. (10.95) to (10.97) into (10.94) and noting that $q_D/A_n = V_n$, $V_o = 4q_D/\pi d_o^2 N_o$, and $1 - E_{MD} = (c_{D,n} - c_{D,n}^*)/(c_{D,n+1} - c_{D,n}^*)$ lead to

$$E_{MD} = \frac{\frac{4.4K_{LDf}}{V_o} \left(\frac{d_p}{d_o}\right)^2 + \frac{6K_{LDf}\phi_D(Z - h)}{d_p V_n}}{1 + \frac{0.4K_{LDf}}{V_o} \left(\frac{d_p}{d_o}\right)^2 + \frac{3K_{LDf}\phi_D(Z - h)}{d_p V_n}} \quad (10.98)$$

Skelland [62] has outlined a similar, more elaborate, procedure, but Eq. (10.98) is believed more than adequate in view of the many uncertainties involved. In any event, it must be understood that we can at best estimate E_{MD} only roughly. The stage efficiency will usually be far less than that obtained for gas-liquid contact, owing to the lower phase velocities resulting from lower density differences and higher viscosities.

Illustration 10.8 Isopropyl ether is to be used to extract acetic acid from water in a sieve-tray tower, ether dispersed. The design conditions agreed upon are those of Illustration 10.3 and Fig. 10.24, stage 2:

	Water solution (continuous)	Isopropyl ether solution (dispersed)
Flow rate	8000 kg/h = 17 600 lb/h	20 000 kg/h = 44 100 lb/h
Acid concentration	$x_{C_s} = 0.175$ mass fraction	$x_{D_{A+1}} = 0.05$ mass fraction
Density	1009 kg/m ³ = 63.0 lb/ft ³	730 kg/m ³ = 45.6 lb/ft ³
Viscosity	3.1×10^{-3} kg/m · s = 3.1 cP	0.9×10^{-3} kg/m · s = 0.9 cP
Acid diffusivity	1.24×10^{-9} m ² /s = 4.8×10^{-5} ft ² /h	1.96×10^{-9} m ² /s = 7.59×10^{-5} ft ² /h
Interfacial tension = 0.013 N/m = 13 dyn/cm		
Distribution coefficient = $m = \frac{\Delta x_C}{\Delta x_D} = 2.68$		

SOLUTION Mol wt acetic acid = 60.1.

$$c_{C_s} = \frac{0.175(1009)}{60.1} = 2.94 \text{ kmol/m}^3 \quad c_{D_{A+1}} = \frac{0.05(730)}{60.1} = 0.608 \text{ kmol/m}^3$$

$$m_{CD} = 2.68 \frac{1009}{730} = 3.704 \frac{\text{kmol/m}^3 \text{ in ether}}{\text{kmol/m}^3 \text{ in water}}$$

Perforations Set $d_o = 6 \text{ mm} = 0.006 \text{ m}$, arranged in triangular pitch on 15-mm centers. $\rho_D = 730 \text{ kg/m}^3$; $q_D = 20\,000/730(3600) = 7.61 \times 10^{-3} \text{ m}^3/\text{s}$; $\sigma = 0.013 \text{ N/m}$; $g = 9.807 \text{ m/s}^2$; $\rho_c = 1.0$ $\rho_C = 1009 \text{ kg/m}^3$; $\Delta\rho = 1009 - 730 = 279 \text{ kg/m}^3$.

$$\frac{d_o}{\left(\frac{\sigma g_c}{\Delta\rho g}\right)^{0.5}} = 2.75$$

Eq. (10.74b):

$$\frac{d_o}{d_f} = 4.28 \quad d_f = 1.402 \times 10^{-3} \text{ m}$$

Eq. (10.73):

$$V_{o, \max} = 0.0153 \text{ m/s}$$

Since this is less than 0.1, use $V_o = 0.1 \text{ m/s}$. Perforation area = $q_D/V_o = 7.61 \times 10^{-2} \text{ m}^2$; $N_o = \text{perforation area}/(\pi d_o^2/4) = 2690$. Eq. (6.31):

$$\text{Plate area for perforations} = \frac{7.61 \times 10^{-2}}{0.907(0.006/0.015)^2} = 0.5244 \text{ m}^2$$

Downspout Set the continuous-phase velocity V_d equal to the terminal velocity of a dispersed-phase drop, $d_p = 0.7 \text{ mm} = 0.0007 \text{ m}$. $\mu_C = 0.0031 \text{ kg/m} \cdot \text{s}$. Eq. (10.75): $U = 8\,852\,000$. Fig.

10.47: ordinate = 1.515; abscissa = $0.62 = d_p V_t \rho_C / \mu_C U^{0.15}$; $V_t = 0.03$ m/s = V_d ; $q_C = 8000/(1009)(3600) = 2.202 \times 10^{-3}$ m³/s. $A_d = q_C / V_d = 0.0734$ m². Refer to Table 6.2. Allow-ing for supports and unperforated area,

$$A_t = \frac{0.5244}{0.65} = 0.807 \text{ m}^2 \quad T = \left(0.807 \frac{4}{\pi}\right)^{0.5} = 1.0 \text{ m} \quad A_n = A_t - A_d = 0.734 \text{ m}^2$$

Drop size Fig. 10.46:

$$\text{Abscissa } (\alpha_2 = 52\,560, \alpha_3 = 1.24 \times 10^6) = 0.0639$$

$$\text{Parameter } (\alpha_1 = 10.76) = 0.282 \quad \text{ordinate} = 0.024 = 3.281 d_p$$

Therefore

$$d_p = 7.31 \times 10^{-3} \text{ m}$$

Coalesced layer $V_n = q_D / A_n = 0.01037$ m/s. Eq. (10.80): $h_o = 2.94 \times 10^{-3}$ m. h_o is unim-portant at the present value of V_o ; therefore $h_D = h_o = 2.94 \times 10^{-3}$ m. Eq. (10.82): $h_C = 7.47 \times 10^{-4}$ m; Eq. (10.78): $h = 3.69 \times 10^{-3}$ m. Since this is very shallow, increase it by placing an orifice in the bottom of the downspout. If V_R is the velocity through this restriction and h_R the corresponding depth of coalesced layer that results, the orifice equation provides

$$h_R = \frac{(V_R^2 - V_d^2) \rho_C}{2g_c(0.67)\Delta_p}$$

Set $h_R = 0.045$ m, whence $V_R = 0.332$ m/s. This corresponds to a restriction area = $q_C / V_R = 6.633 \times 10^{-3}$ m², in turn corresponding to a circular hole 92 mm in diameter. Then $h = 0.045 + 0.00369 = 0.049$ m = 4.9 cm, which is satisfactory.

Set the tray spacing at $Z = 0.35$ m, and lead the downspout apron to within 0.1 m of the tray below.

Dispersed-phase holdup Eq. (10.84): $V_S \phi_D = 0.01037$ m/s. For $d_p = 7.31 \times 10^{-3}$ m, $U = 8\,852\,000$; ordinate of Fig. 10.47 = 165.2, abscissa = $30 = d_p V_t \rho_C / \mu_C U^{0.15}$; then $V_t = 0.1389$ m/s.

Fig. 10.48: abscissa = 52.6; ordinate = $44.65 V_S(1 - \phi_D)$ for solid particles.

ϕ_D = parameter, Fig. 10.48 (1)	Ordinate, Fig. 10.48 (2)	$V_S(1 - \phi_D)$ for solids (3)	V_S for solids (4)	$\frac{V_S}{V_t}$ (5)	V_S for liquids (6)	$V_S \phi_D$ for liquids (7)
0	8.8	0.1971	0.1971	1.0	0.1389	0
			= V_t , solids		= V_t	
0.1	5.9	0.1321	0.1468	0.7448	0.1035	0.01035
0.2	4.3	0.0963	0.1204	0.6109	0.0849	0.1698
0.3	3.0	0.0672	0.0960	0.4871	0.0677	0.2031

Example for $\phi_D = 0.1$, ordinate Fig. 10.48 = 5.9; column 3 = $5.9/44.65$; column 4 = $0.1321/(1 - 0.1)$; column 5 = $0.1468/0.1971$; column 6 = $0.7448(0.1389)$; column 7 = $0.1035(0.1)$.

Interpolation in column 7 for $V_S \phi_D = 0.01037$ provides $\phi_D = 0.10$.

Mass transfer

$$\theta_f = \frac{\pi d_p^3 / 6}{q_D / N_o} = 0.2498 \text{ s} \quad D_D = 1.96 \times 10^{-9} \text{ m}^2/\text{s}$$

Eq. (10.87) with $\text{const} = 1.5$; $k_{LDf} = 7.5 \times 10^{-5} \text{ m/s}$; Eq. (10.86): $K_{LDf} = 5.6 \times 10^{-5} \text{ m/s}$ or $\text{kmol/m}^2 \cdot \text{s} \cdot (\text{kmol/m}^3)$.

The ordinate of Fig. 10.47 for the drops is larger than 70, hence $d_p > d_{p, \text{trans}}$, and the mass-transfer coefficient during drop rise is given by Eq. (10.89).

Eq. (10.91): $b = 0.348$. Eq. (10.90): $\omega = 3.66 \text{ s}^{-1}$. Eq. (10.89) with $\delta = 0.2$: $k_{LDr} = 1.053 \times 10^{-4} \text{ m/s}$

$$\frac{1}{K_{LDr}} = \frac{1}{k_{LDr}} \left[1 + \frac{1}{m_{CD}} \left(\frac{D_D}{D_C} \right)^{0.5} \right]$$

$$K_{LDr} = 7.86 \times 10^{-5} \text{ m/s or kmol/m}^2 \cdot \text{s} \cdot (\text{kmol/m}^3)$$

Eq. (10.98):

$$E_{MD} = 0.175 \quad \text{Ans.}$$

DIFFERENTIAL (CONTINUOUS-CONTACT) EXTRACTORS

When the liquids flow countercurrently through a single piece of equipment, the equivalent of as many theoretical stages may be had as desired. In such devices, the countercurrent flow is produced by virtue of the difference in densities of the liquids, and if the motivating force is the force of gravity, the extractor usually takes the form of a vertical tower, the light liquid entering at the bottom, the heavy liquid at the top. As an alternative, a larger centrifugal force can be generated by rotating the extractor rapidly, in which case the counterflow is radial with respect to the axis of revolution.

There are several characteristics common to countercurrent extractors which have an important bearing upon their design and performance. It is typical that only one of the liquids can be pumped through the device at any desired rate. The maximum rate for the other will depend, among other things, upon the density difference of the liquids. If an attempt is made to exceed this rate for the second liquid, the extractor will reject one of the liquids, and it is said to be *flooded*. The same is true, of course, for gas-liquid contactors, but since the density difference is much smaller than for a gas and liquid, the flooding velocities of extractors are much lower. For a given volumetric rate of liquids to be handled, the extractor cross section must be large enough to ensure that the flooding velocities are not reached. The more open the cross section, the greater the flow rates before flooding occurs. Internal structures, packing, mechanical agitators, and the like normally reduce the velocities at which flooding occurs.

The differential extractors are also subject to *axial mixing* (see Chap. 6); it severely reduces extraction rates because of the deterioration of the concentration differences between phases which is the driving force for mass transfer. This is illustrated in Fig. 10.50, where the real (axial-mixing) concentration profiles show a substantially smaller concentration difference than those for plug flow. If the flow ratio of the liquids is not unity, and it very rarely is, dispersing the liquid flowing at the lower rate will lead to small numbers of dispersed-phase drops, small interfacial areas, and small mass-transfer rates. On the other hand, if the majority liquid is dispersed, the axial-mixing problem is exacerbated. The

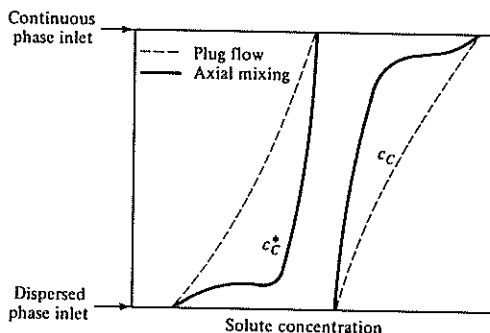


Figure 10.50 Effect of axial mixing on concentration profiles in towers subject to axial mixing.

problem becomes more acute the greater the flow ratio, and this difficulty is common to all differential extractors.

Reviews of calculation methods to account for axial mixing [41] (simplified procedures have been devised [47, 78]) and of axial-mixing data [26] are available.

Spray Towers

These, the simplest of the differential-contact devices, consist merely of an empty shell with provisions at the extremities for introducing and removing the liquids. Because the shell is empty, the extreme freedom of liquid movement makes these towers the worst offenders as far as axial mixing is concerned, so much so that it is difficult to obtain much more than the equivalent of a single stage with them. It is not recommended that they be used. Horizontal baffles (both the segmental and the disk-and-doughnut type) have been used to reduce the axial mixing but with little improvement in results. Spray towers have also been fairly extensively studied for direct-contact heat exchange between two liquids, but the deleterious effects of axial mixing are just as severe in this service as for extraction.

Packed Towers

Towers filled with the same random packings used for gas-liquid contact (Chap. 6) have also been used for liquid extractors. The packing serves to reduce axial mixing somewhat and to jostle and distort the drops of dispersed phase. A typical packed tower is shown schematically in Fig. 10.51, arranged for light liquid dispersed. The void space in the packing is largely filled with the continuous heavy liquid, which flows downward. The remainder of the void space is filled with droplets of light liquid formed at the lower distributor, which rise through the heavy liquid and coalesce at the top into a bulk layer, forming an interface as shown. To maintain the interface in this position, the pressure of liquid in the tower at the bottom must be balanced by a corresponding pressure

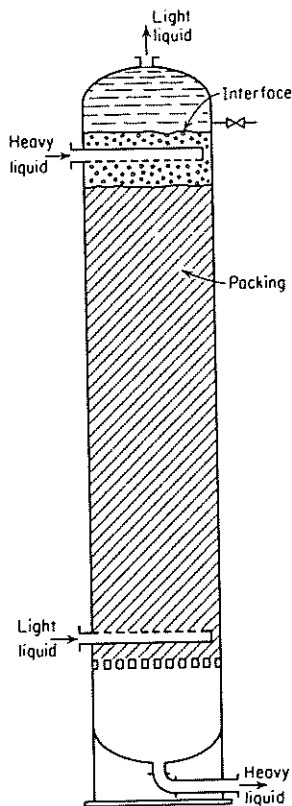


Figure 10.51 Packed extraction tower, light liquid dispersed.

in the bottom outlet pipe, as set by a control valve. If the pressure drop through that valve is reduced, the weight of the contents of the tower adjusts to a lower value, the light liquid becomes continuous, the heavy liquid dispersed, and the interface falls to below the light-liquid distributor. The interface positions are best regulated by a liquid-level control instrument activating the bottom outlet valve. The valve just above the interface in Fig. 10.51 is for periodic removal of scum and dust particles (*rag*) which accumulate at the interface.

The nature of the liquid flow in such towers requires that the choice of packing and arrangement of dispersed-phase distributor be given careful attention. If the dispersed liquid preferentially wets the packing, it will pass through in rivulets on the packing, not as droplets, and the interfacial area produced will be small. For this reason, the packing material should be preferentially wetted by the continuous phase. Usually, ceramics are preferentially wet by aqueous liquids and carbon and plastics by organic liquids. The packing should be sufficiently small, no greater than one-eighth the tower diameter, for the packing

density to be fully developed yet larger than a certain critical size (see below) [48, 72, 76]. Where the material of the packing support is not wet by the dispersed droplets and the distributor is placed outside the packing, the drops will have difficulty in entering the packing and premature flooding results. For this reason it is always desirable to embed the dispersed-phase distributor in the packing, as in Fig. 10.51.

Correlations for estimating flooding rates are available, and data for mass-transfer coefficients and axial mixing have been summarized [23, 43, 72, 76]. Although axial mixing is less severe than in spray towers, mass-transfer rates are poor. It is recommended instead that sieve-tray towers be used for systems of low interfacial tension and mechanically agitated extractors for those of high interfacial tension.

Mechanically Agitated, Countercurrent Extractors

The extraction towers previously described are very similar to those used for gas-liquid contact, where density differences between liquid and gas are of the

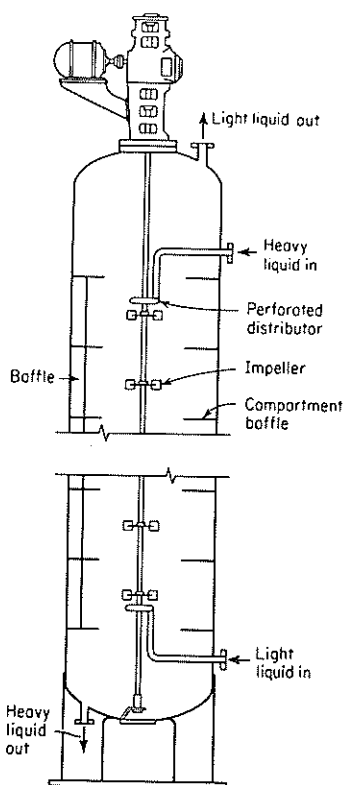


Figure 10.52 Mixco Lightnin CMContractor. (Mixing Equipment Co.).

order of 800 kg/m^3 (50 lb/ft^3) or more, available to provide the energy for good dispersion of one fluid into the other. In liquid extractors, where density differences are likely to be one-tenth as large or less, good dispersion of systems of high interfacial tension is impossible in such towers, and mass-transfer rates are poor. For such systems, dispersion is best brought about by mechanical agitation of the liquids, whereupon good mass-transfer rates are developed. Some examples of such extractors follow. Except for pulse columns, they are proprietary devices for which complete design procedures are not publicly available, and the manufacturers are best consulted.

Mixco Lightnin CMContactor (Oldshue-Rushton Extractor) Refer to Fig. 10.52. This device uses flat-blade disk turbine impellers (Fig. 6.3c) to disperse and mix the liquids and horizontal compartmenting plates to reduce axial mixing. There have been a few mass-transfer and somewhat more extensive axial-mixing studies [5, 13, 16, 17, 27, 44, 45]

Rotating-Disk Contactor (RDC) This (Fig. 10.53) is a somewhat similar device, except that the vertical baffles are omitted and agitation results from rotating disks, which usually turn at much higher speeds than turbine-type impellers [29, 50, 67].

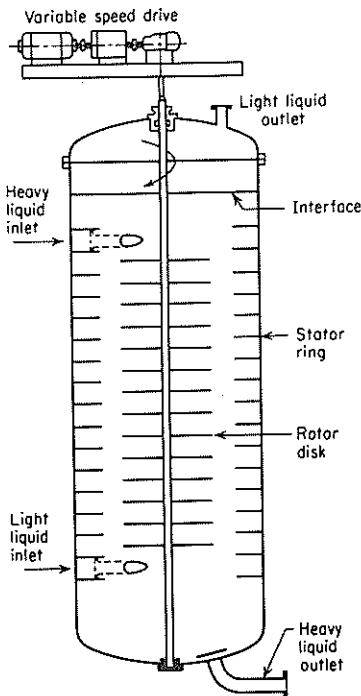


Figure 10.53 Rotating-Disk Contactor (RDC). (General American Transportation Corp.)

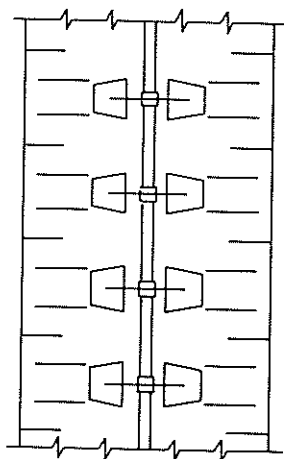


Figure 10.54 Scheibel extractor.

Scheibel Extractor [22, 56, 57] There have been several designs, of which the most recent is shown in Fig. 10.54. The impellers are of the turbine type, and the doughnut-type baffles surrounding them are supported by vertical tie rods, not shown. Earlier designs included sections of knit, wire-mesh packing alternating with sections containing an impeller.

Karr Reciprocating-Plate Extractor [31, 32] This follows an early design of van Dijk [77], who suggested moving the plates of a perforated-plate extractor of the type shown in Fig. 10.55 up and down. The Karr design uses plates of much larger free area, fitting loosely in the tower shell and attached to a vertical, central shaft. They are moved vertically up and down over a short distance.

Treybal Extractor [73] This is in reality a vertical stack of mixer-settlers. The mixers are in a vertical line, and the impellers for the stages on a common shaft. They not only mix but also pump, so that throughput rates are high. Since there is no axial mixing, mass-transfer rates are high.

Graesser Extractor [60] This is a horizontal shell fitted with a series of rotating disks on a central horizontal shaft. C-shaped buckets between the disks shower the liquids, one in the other, as they flow countercurrently and horizontally through the extractor. It has been used especially in Europe.

Pulsed columns [38, 77, 82] A rapid (0.5 to 4 s^{-1}) reciprocating pulse of short amplitude (5 to 25 mm) is hydraulically transmitted to the liquid contents. Since the extractors have no moving parts, they have found extensive (and exclusive) use in processing radioactive solutions in atomic-energy work, where they can be put behind heavy radiation shields without requiring maintenance. The most common arrangement is that of Fig. 10.55; the perforated plates, which have no

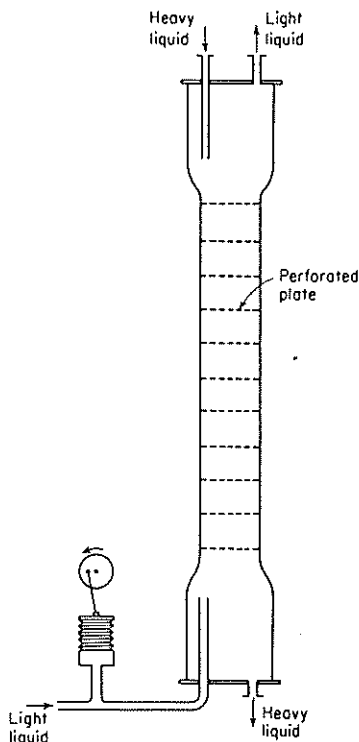


Figure 10.55 Pulsed column.

downspouts, are drilled with very small holes so that ordinarily flow will not occur. The pulsing superimposed upon the liquids alternately forces light and heavy liquids through the perforations. Packed columns, indeed any type of extractor, can also be pulsed. Although the mass-transfer rates are thereby improved at the expense of substantial energy costs, the flow capacities become smaller.

Centrifugal Extractors

The most important of these is the *Podbielniak extractor* [68, 69, 70] (Fig. 10.56). The cylindrical drum containing perforated concentric shells is rapidly rotated on the horizontal shaft (30 to 85 r/s). Liquids enter through the shaft: heavy liquid is led to the center of the drum, light liquid to the periphery. The heavy liquid flows radially outward, displacing the light liquid inwardly, and both are led out through the shaft. These extractors are especially useful for liquids of very small density difference and where very short residence times are essential, as in some pharmaceutical applications, e.g., extraction of penicillin from nutrient broth.

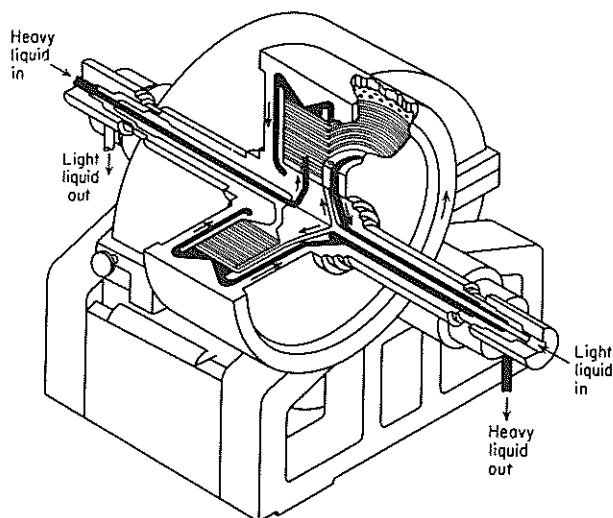


Figure 10.56 Podbielniak centrifugal extractor (schematic). (Podbielniak, Inc.)

The *Luwesta extractor* [14] and the *Rotabel* [3], extractors revolving about a vertical shaft, are used more extensively in Europe than in the United States.

Design

Since the change in concentration with height of either liquid as it passes through the extractor is differential, the height of the tower is expressed not in terms of stages or steps but in terms of transfer units.

Consider the continuous-contact tower of Fig. 10.57. Although the raffinate is shown flowing downward as if it were denser, in some instances the solvent-rich, or extract, phase will be denser and will enter at the top. In either case, in what follows subscript 1 will always represent that end of the tower where the raffinate enters and extract leaves, while subscript 2 will indicate where extract enters and raffinate leaves. We are presently unconcerned with which phase is dispersed and which is continuous. If the extractor is fed along the side, Fig. 10.57 and the relationships which follow apply separately to each section above and below the feed inlet.

Throughout this discussion, *unless otherwise specified*, x and y will refer to solute concentrations expressed as mole fractions in the raffinate and extract, respectively, and rates of flow of raffinate R and of extract E will be expressed as $\text{mol}/(\text{cross-sectional area})(\text{time})$. Except in special cases, the transfer of solute usually results in changes of mutual solubility of the contacted liquids, so that in general all components of the systems transfer from one phase to the other. The F -type mass-transfer coefficients are capable of handling this problem, but in

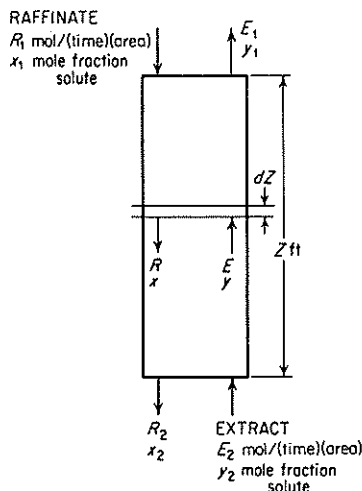


Figure 10.57 Continuous-contact tower.

reality our knowledge of mass-transfer rates in extractors is so poorly developed that we can ordinarily consider only the transfer of a single solute. This discussion is therefore limited to cases where the liquids are insoluble, only solute is transferred, and mutual solubility unchanged. In practice, however, the expressions developed are used for all cases.

It then follows that the equations derived for gas absorption apply. Thus, for the raffinate (which corresponds to the gas in gas absorption), we have the counterparts of Eqs. (8.23) to (8.27)

$$Z = \int_{x_2}^{x_1} \frac{R(1-x)_{iM} dx}{F_R a (1-x)(x-x_i)} = H_{iR} \int_{x_2}^{x_1} \frac{(1-x)_{iM} dx}{(1-x)(x-x_i)} = H_{iR} N_{iR} \quad (10.99)$$

$$H_{iR} = \frac{R}{F_R a} = \frac{R}{k_R a (1-x)_{iM}} \quad (10.100)$$

$$N_{iR} = \int_{x_2}^{x_1} \frac{(1-x)_{iM} dx}{(1-x)(x-x_i)} = \int_{x_2}^{x_1} \frac{dx}{x-x_i} + \frac{1}{2} \ln \frac{1-x_2}{1-x_1} \quad (10.101)$$

where x_i = interface concentration of solute

F_R, k_R = transfer coefficients for raffinate phase

H_{iR} = raffinate height of transfer unit

N_{iR} = number of raffinate transfer units

$(1-x)_{iM}$ = logarithmic mean of $1-x$ and $1-x_i$

The interface concentration corresponding to any bulk raffinate concentration x is found through Eq. (5.21) adapted to the present situation

$$\frac{1-x_i}{1-x} = \left(\frac{1-y}{1-y_i} \right)^{F_E/F_R} \quad (10.102)$$

This equation is plotted, for any value of (x, y) on the operating curve, to determine its intersection with the equilibrium curve at (x_i, y_i) , just as for gas absorption. As an approximation, a straight line of slope $-k_R/k_E$ is plotted from (x, y) on the operating line to obtain the intersection at (x_i, y_i) .

Similarly, we have the corresponding expressions for the extract (corresponding to the liquid in gas absorption).

In reality, we know so little about the mass-transfer coefficients that the above expressions are of little use. For practical reasons, even though not strictly applicable unless the equilibrium curve is of constant slope, it is usually necessary to deal with overall coefficients and transfer units. These we can take from their gas-absorption counterparts:

$$Z = H_{IOR} N_{IOR} = H_{IOE} N_{IOE} \quad (10.103)$$

packed towers
$$H_{IOR} = \frac{R}{F_{OR} a} = \frac{R}{K_R a (1-x)_{*M}} \quad (10.104)$$

$$H_{IOE} = \frac{E}{F_{OE} a} = \frac{E}{K_E a (1-y)_{*M}} \quad (10.105)$$

$$N_{IOR} = \int_{x_2}^{x_1} \frac{(1-x)_{*M} dx}{(1-x)(x-x^*)} = \int_{x_2}^{x_1} \frac{dx}{x-x^*} + \frac{1}{2} \ln \frac{1-x_2}{1-x_1} \quad (10.106)$$

$$N_{IOE} = \int_{y_2}^{y_1} \frac{(1-y)_{*M} dy}{(1-y)(y^*-y)} = \int_{y_2}^{y_1} \frac{dy}{y^*-y} + \frac{1}{2} \ln \frac{1-y_1}{1-y_2} \quad (10.107)$$

$$(1-x)_{*M} = \frac{(1-x^*) - (1-x)}{\ln[(1-x^*)/(1-x)]} \quad (10.108)$$

$$(1-y)_{*M} = \frac{(1-y) - (1-y^*)}{\ln[(1-y)/(1-y^*)]} \quad (10.109)$$

where x^* is the concentration in equilibrium with y and y^* that in equilibrium with x .

Concentrations in Eqs. (10.103) to (10.109) are in mole fractions. If x and y are expressed as weight fractions, for convenience in use with the stage-calculation operating diagrams in terms of weight fractions,

$$N_{IOR} = \int_{x_2}^{x_1} \frac{dx}{x-x^*} + \frac{1}{2} \ln \frac{1-x_2}{1-x_1} + \frac{1}{2} \ln \frac{x_2(r-1)+1}{x_1(r-1)+1} \quad (10.110)$$

$$N_{IOE} = \int_{y_2}^{y_1} \frac{dy}{y^*-y} + \frac{1}{2} \ln \frac{1-y_1}{1-y_2} + \frac{1}{2} \ln \frac{y_1(r-1)+1}{y_2(r-1)+1} \quad (10.111)$$

where r is the ratio of molecular weights of nonsolute to solute. For weight-ratio concentrations,

$$N_{IOR} = \int_{x_2'}^{x_1'} \frac{dx'}{x'-x'^*} + \frac{1}{2} \ln \frac{1+rx_2'}{1+rx_1'} \quad (10.112)$$

$$N_{IOE} = \int_{y_2'}^{y_1'} \frac{dy'}{y'^*-y'} + \frac{1}{2} \ln \frac{1+ry_1'}{1+ry_2'} \quad (10.113)$$

Dilute Solutions

For dilute solutions, only the integral terms of the above equations for N_{IOE} and N_{IOR} are important. If in addition the equilibrium curve and operating line are straight over the concentration range encountered, it is readily shown in the manner of Chap. 8 that logarithmic averages of the terminal concentration differences are applicable

$$N_{IOR} = \frac{x_1 - x_2}{(x - x^*)_M} \quad N_{IOE} = \frac{y_1 - y_2}{(y^* - y)_M} \quad (10.114)$$

The equivalent expressions in terms of mass-transfer coefficients are

$$R(x_1 - x_2) = E(y_1 - y_2) = K_R a Z (x - x^*)_M = K_E a Z (y^* - y)_M \quad (10.115)$$

If in addition the equivalent of Henry's law applies, so that the equilibrium-distribution curve is a straight line passing through the origin ($m = y^*/x = y/x^* = \text{const}$), a procedure exactly similar to that used previously in the case of gas absorption gives

$$N_{IOR} = \frac{\ln \left[\frac{x_1 - y_2/m}{x_2 - y_2/m} \left(1 - \frac{R}{mE} \right) + \frac{R}{mE} \right]}{1 - R/mE} \quad (10.116)$$

Figure 8.20 represents a graphical solution, provided $(x_2 - y_2/m)/(x_1 - y_2/m)$ is considered the ordinate and mE/R the parameter. Similarly for the extract-enriching section of a tower used with reflux, the same circumstances provide

$$N_{IOE} = \frac{\ln \left[\frac{y_2 - mx_1}{y_1 - mx_1} \left(1 - \frac{mE}{R} \right) + \frac{mE}{R} \right]}{1 - mE/R} \quad (10.117)$$

which is also solved graphically in Fig. 8.20 provided $(y_1 - mx_1)/(y_2 - mx_1)$ is the ordinate and R/mE the parameter.

For these dilute solutions Eqs. (10.114), (10.116), and (10.117) can be used with concentrations in terms of weight fractions, in which case m must be defined in these terms as well while E and R are measured as mass/(area)(time). Weight ratios can also be used. Equations (10.114) and (10.115) are frequently used with concentrations expressed as c mol/volume, in which case the extract and raffinate flow rates are measured in terms of V volume/(area)(time) and the appropriate mass-transfer coefficients are K_{LE} and K_{LR} mol/(area)(time) $\cdot \Delta c$. The requirements of constant m (in whatever units it is measured) and straight operating lines still apply, of course.

Illustration 10.9 Determine the number of transfer units N_{IOR} for the extraction of Illustration 10.3 if 20 000 kg/h of solvent is used.

SOLUTION Define x and y in terms of weight fractions acetic acid. $x_1 = x_F = 0.30$; $y_2 = 0$; $x_2 = 0.02$; $y_1 = 0.10$. The operating diagram is already plotted in Fig. 10.23. From this plot, values of x and x^* are taken from the operating line and equilibrium curve at various values of

y , as follows:

x	0.30	0.25	0.20	0.15	0.10	0.05	0.02
x^*	0.230	0.192	0.154	0.114	0.075	0.030	0
$\frac{1}{x - x^*}$	14.30	17.25	20.75	27.8	40.0	50.0	50.0

The area under a curve of x as abscissa against $1/(x - x^*)$ as ordinate (not shown) between $x = 0.30$ and $x = 0.02$ is determined to be 8.40. In these solutions, the mutual solubility of water and isopropyl ether is very small, so that r can be taken as $18/60 = 0.30$. Eq. (10.110):

$$N_{IOR} = 8.40 + \frac{1}{2} \ln \frac{1 - 0.02}{1 - 0.30} + \frac{1}{2} \ln \frac{0.02(0.3 - 1) + 1}{0.30(0.3 - 1) + 1} = 8.46$$

The operating and equilibrium curves are nearly parallel in this case, so that N_{IOR} and N_p are nearly the same. The curvature of the lines makes the simplified methods for N_{IOR} inapplicable, however.

Illustration 10.10 Determine the number of transfer units N_{IOR} for the extraction of Illustration 10.4 if 1150 kg/h of kerosene are used.

SOLUTION Use weight-ratio concentrations, as in Illustration 10.4. $x'_1 = x'_F = 0.0101$; $y'_2 = 0$; $x'_2 = 0.001001$; $y'_1 = 0.0782$. The calculation can be done through Eq. (10.116) or the equivalent, Fig. 8.20.

$$\frac{x'_2 - y'_2/m'}{x'_1 - y'_2/m'} = \frac{0.001001}{0.0101} = 0.0909$$

The average $mE/R = m'B/A = 1.01$ (Illustration 10.4). From Fig. 8.20, $N_{IOR} = 8.8$.

Performance of Continuous-Contact Equipment

While over the past 40 years a considerable number of data has been accumulated, taken almost entirely from laboratory-size equipment of a few centimeters diameter, no satisfactory correlation of them has as yet been possible owing to the very large number of variables which influence extraction rates. For the design of new extractors it is essential that pilot-plant experiments be performed under conditions as nearly like those expected in the large-scale equipment as possible [75]. This discussion will therefore be limited to a brief consideration of the important variables, but no data for design will be presented.

The principal difficulty in obtaining an understanding of extractor performance lies with the very large number of variables which influence the performance. The following at least have influence.

1. The liquid system.
 - a. Chemical identity and corresponding physical properties. In this category may be included the presence or absence of surface-active agents, finely divided solids, and the like.
 - b. Concentration of solute, since this influences physical properties.
 - c. Direction of extraction, whether from aqueous to organic, from continuous to dispersed phase.
 - d. Total flow rate of the liquids.
 - e. Ratio of liquid flows.
 - f. What liquid is dispersed.

2. The equipment.

- Design, which includes not only the gross and obvious design such as whether the extractor is a packed or mechanically agitated tower, but also such details as size and shape of packing, arrangement of baffles, and the like.
- Nature and extent of mechanical agitation, whether rotary or pulsating, fast or slow.
- Materials of construction, which influence the relative wetting by the liquids.
- Height of the extractor and the end effects.
- Diameter of extractor and extent of axial mixing.

It is only recently that a beginning has been made in systematizing axial-mixing data, and reliable correlations of mass-transfer coefficients must wait upon this. Practical applications are advanced far ahead of sound design data.

NOTATION FOR CHAPTER 10

Any consistent set of units may be used, except as noted.

a	specific interfacial surface, L^2/L^3
A	component A, mass/time (continuous), M/Θ ; mass (batch), M
A_a	active area of tower devoted to perforations, L^2
A_d	cross-sectional area of downspouts (or upspouts), L^2
A_n	net tower cross-sectional area = $A_t - A_d$, L^2
A_p	surface of drops, L^2
A_r	surface of rising (or falling) drops between trays, L^2
A_t	tower cross-sectional area, L^2
b	constant, dimensionless
B	component B, mass/time (continuous), M/Θ ; mass (batch), M
B_n	eigenvalue, dimensionless
c	concentration, especially of solute, mole/ L^3
c_D^*	dispersed-phase concentration in equilibrium with bulk continuous phase, mole/ L^3
C	component C, mass/time (continuous), M/Θ ; mass (batch), M
C_D	drag coefficient = $4\Delta\rho d_p g / 3\rho_C V_t^2$, dimensionless
d_i	impeller diameter, L
d_j	jet diameter, L
d_o	orifice diameter, L
d_p	drop diameter, L
$d_{p,trans}$	transition-size drop diameter, L
D	component D, mass/time (continuous), M/Θ ; diffusivity, L^2/Θ
E	extract solution, mass/time (staged extractors), M/Θ ; mass (batch), M ; moles/(area)(time) (continuous-contact extractors), mole/ $L^2\Theta$
E'	solvent (B)-free extract, mass/time (continuous), M/Θ ; mass (batch), M
E_{MD}	Murphree dispersed-phase stage efficiency, fractional
E_{ME}	Murphree extract stage efficiency, fractional
E_{MR}	Murphree raffinate stage efficiency, fractional
E_O	overall stage efficiency, fractional
F	feed, mass/time (continuous), M/Θ ; mass (batch), M
F'	feed, solvent (B)-free basis, mass/time (continuous), M/Θ ; mass (batch), M
g	acceleration of gravity, L^2/Θ
g_c	conversion factor, $ML/F\Theta^2$
h	depth of coalesced dispersed liquid accumulating on a tray, L
h_C	depth of dispersed liquid accumulating on a tray owing to flow of continuous liquid, L

h_D	depth of dispersed liquid accumulating on a tray owing to flow of dispersed liquid, L
h_o	head required to cause flow through an orifice, L
h_R	head required to cause flow through a downspout restriction, L
h_o	head required to overcome effect of interfacial tension, L
H_i	height of a transfer unit, L
k	mass-transfer coefficient, mole/L ² Θ(mole fraction)
k_L	mass-transfer coefficient, mole/L ² Θ(mole/L ³)
K	overall mass-transfer coefficient, mole/L ² Θ(mole fraction)
K_L	overall mass-transfer coefficient, mole/L ² Θ(mole/L ³)
ln	natural logarithm
m	equilibrium distribution coefficient, concentration in extract/concentration in raffinate, dimensionless
m_{CD}	equilibrium distribution coefficient, concentration in continuous phase/concentration in dispersed phase, (mole/L ³)/(mole/L ³)
M	mixture M , mass/time (continuous), M/Θ; mass (batch), M
M'	solvent (B)-free mixture M , mass/time (continuous), M/Θ; mass (batch), M
N	solvent concentration, solvent (B)-free basis, mass B/mass (A + C), M/M
N_f	time-average flux of mass transfer during drop formation, mole/L ² Θ
N_o	number of perforations, dimensionless
N_p	number of theoretical stages, dimensionless
N_t	number of transfer units, dimensionless
P	power, FL/Θ
P'_E	solvent (B)-free extract product, mass/time, M/Θ
q	volumetric rate, L ³ /Θ
r	ratio of molecular weight, nonsolute to solute, dimensionless
R	raffinate, mass/time (staged extraction), M/Θ; mass (batch), M; moles/(area)(time) (continuous-contact extraction), mole/L ² Θ
R'	solvent (B)-free raffinate, mass/time (continuous), M/Θ; mass (batch), M
Re	drop Reynolds number, $d_p V_p \rho_C / \mu_C$, dimensionless
S	solvent, mass/time (continuous), M/Θ; mass (batch), M
S'	B-free solvent, mass/time (continuous), M/Θ; mass (batch), M
Sc	Schmidt number, $\mu / \rho D$, dimensionless
Sh	Sherwood number, $k d_p / D$, dimensionless
T	tower diameter, L
T_S	settler diameter, L
U	dimensionless group defined by Eq. (10.75)
v_L	liquid volume, L ³
V	superficial velocity, L/Θ
V_d	velocity of liquid in downspout, L/Θ
V_n	velocity based on A_n , L/Θ
V_o	velocity through orifice or perforation, L/Θ
$V_{o, \max}$	velocity through perforation leading to maximum specific interfacial area, L/Θ
V_R	velocity based on a restricted area, L/Θ
V_S	slip velocity [Eqs. (10.83) and (10.84)], L/Θ
V_t	terminal settling velocity, L/Θ
We	drop Weber number, $d_p V_p^2 \rho_C / \sigma_{g_c}$, dimensionless
x	concentration of C in A-rich (raffinate) phase, mass fraction (staged extractors); mole fraction (continuous-contact extractors)
x'	concentration of C in A-rich (raffinate) phase, mass C/mass non-C, M/M concentration of solute in D-rich phase (fractional extraction), mass solute/mass nonsolute, M/M
X	concentration of C in A-rich (raffinate) phase, B-free basis, mass C/mass (A + C), M/M

y	concentration of C in B-rich (extract) phase, mass fraction (staged extractors); mole fraction (continuous extractors)
y'	concentration of C in B-rich (extract) phase, mass C/mass non-C, M/M ; concentration of solute in A-rich phase (fractional extraction), mass solute/mass nonsolute, M/M
Y	concentration of C in B-rich (extract) phase, B-free basis, mass C/mass (A + C), M/M
Z	tray spacing, L; depth of liquid in agitated vessel, L
β	selectivity, dimensionless
δ	amplitude factor, dimensionless
Δ'_E	difference in flow, B-free basis [Eq. (10.40)]
Δ_R	difference in flow rate [Eq. (10.25)]
Δ'_R	difference in flow rate, B-free basis [Eq. (10.31)]
θ	time, Θ
θ_f	time for drop formation, Θ
λ_n	eigenvalue, dimensionless
μ	viscosity, $M/L\Theta$
ρ	density, M/L^3
$\Delta\rho$	difference in densities, M/L^3
σ	interfacial tension, F/L
ϕ	volume fraction
ω	oscillation frequency, Θ^{-1}

Subscripts

c	during coalescence
C	continuous phase
D	dispersed phase
e	stage e
E	extract
f	formation
F	feed
m	stage m ; minimum
M	mixture M ; mean
n	stage n
o	organic; orifice
O	overall
R	raffinate
s	stage s
S	solvent
w	water
1	stage 1; that end of a continuous-contact tower where feed enters
2	stage 2; that end of a continuous-contact tower where solvent enters

Superscripts

$*$	in equilibrium with bulk concentration in the other phase
$'$	solvent-free

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PROBLEMS

Problems 10.1 to 10.7 refer to the system water (A)–chlorobenzene (B)–pyridine (C) at 25°C. Equilibrium tie-line data, interpolated from those of Peake and Thompson, *Ind. Eng. Chem.*, **44**,

2439 (1952), in weight percent are:

Pyridine	Chlorobenzene	Water	Pyridine	Chlorobenzene	Water
0	99.95	0.05	0	0.08	99.92
11.05	88.28	0.67	5.02	0.16	94.82
18.95	79.90	1.15	11.05	0.24	88.71
24.10	74.28	1.62	18.90	0.38	80.72
28.60	69.15	2.25	25.50	0.58	73.92
31.55	65.58	2.87	36.10	1.85	62.05
35.05	61.00	3.95	44.95	4.18	50.87
40.60	53.00	6.40	53.20	8.90	37.90
49.0	37.8	13.2	49.0	37.8	13.2

10.1 Plot the equilibrium data on the following coordinate systems: (a) triangular; (b) x and y against weight fraction B; (c) x against y .

10.2 Compute the selectivity of chlorobenzene for pyridine at each tie line, and plot selectivity against concentration of pyridine in water.

10.3 It is desired to reduce the pyridine concentration of 2000 kg of an aqueous solution from 50 to 2% in a single batch extraction with chlorobenzene. What amount of solvent is required? Solve on triangular coordinates.

10.4 A 2000-kg batch of pyridine-water solution, 50% pyridine, is to be extracted with an equal weight of chlorobenzene. The raffinate from the first extraction is to be reextracted with a weight of solvent equal to the raffinate weight, and so on ($B_2 = R_1$, $B_3 = R_2$, etc.). How many theoretical stages and what total solvent will be required to reduce the concentration of pyridine to 2% in the final raffinate? Solve on triangular coordinates.

10.5 A pyridine-water solution, 50% pyridine, is to be continuously and countercurrently extracted at the rate 2.25 kg/s (17 800 lb/h) with chlorobenzene to reduce the pyridine concentration to 2%. Using the coordinate systems plotted in (b) and (c) of Prob. 10.1:

(a) Determine the minimum solvent rate required.

(b) If 2.3 kg/s (18 250 lb/h) is used, what are the number of theoretical stages and the saturated weights of extract and raffinate?

Ans.: 3 theoretical stages.

(c) Determine the number of transfer units N_{OR} for the extraction of part (b).

Ans.: 4.89

10.6 The properties of the solutions have not been completely studied, but from those of the pure constituents, the following properties are estimated:

	Density		Viscosity		Interfacial tension	
	kg/m ³	lb/ft ³	kg/m · s	cP	N/m	dyn/cm
Feed	994.8	62.1	0.001	1.0	0.008	8
Extract	1041	65	0.0013	1.3		
Raffinate	996.4	62.2	0.00089	0.89	0.035	35
Solvent	1097	68.5	0.00125	1.25		

For the solvent-entrance end of the cascade of Prob. 10.5b, specify the dimensions of a mixing vessel and the size and speed of the impeller and estimate the stage efficiency. Disperse the solvent, and use an average holding time of 30 s.

10.7 Can the separation of Prob. 10.5 be made by distillation at atmospheric pressure?

10.8 Water-dioxane solutions form a minimum-boiling azeotrope at atmospheric pressure and cannot be separated by ordinary distillation methods. Benzene forms no azeotrope with dioxane and can be used as an extraction solvent. At 25°C, the equilibrium distribution of dioxane between water and benzene [*J. Am. Chem. Soc.*, **66**, 282 (1944)] is as follows:

Wt % dioxane in water	5.1	18.9	25.2
Wt % dioxane in benzene	5.2	22.5	32.0

At these concentrations water and benzene are substantially insoluble, and 1000 kg of a 25% dioxane-75% water solution is to be extracted with benzene to remove 95% of the dioxane. The benzene is dioxane-free.

(a) Calculate the solvent requirement for a single batch operation.

(b) If the extraction were done with equal amounts of solvent in five crosscurrent stages, how much solvent would be required?

10.9 A 25% solution of dioxane in water is to be continuously extracted at a rate of 1000 kg/h in countercurrent fashion with benzene to remove 95% of the dioxane. Equilibrium data are given in Prob. 10.8.

(a) What is the minimum solvent requirement, kg/h?

(b) If 900 kg/h of solvent is used, how many theoretical stages are required?

Ans.: 6.1 theoretical stages.

(c) How many transfer units N_{OR} correspond to the extraction of part (b)?

10.10 An aqueous solution contains 25% acetone by weight together with a small amount of an undesired contaminant. For the purpose of a later process, it is necessary to have the acetone dissolved in water without the impurity. To accomplish this, the solution will be extracted countercurrently with trichloroethane, which extracts the acetone but not the impurity. The extract will then be countercurrently extracted with pure water in a second extractor to give the desired product water solution, and the recovered solvent will be returned to the first extractor. It is required to obtain 98% of the acetone in the final product. Water and trichloroethane are insoluble over the acetone-concentration range involved, and the distribution coefficient (kg acetone/kg trichloroethane)/(kg acetone/kg water) = 1.65 = const.

(a) What is the largest concentration of acetone possible in the recovered solvent?

(b) How many stages would be required in each extractor to obtain the acetone in the final water solution at the same concentration as in the original solution?

(c) If recovered solvent contains 0.005 kg acetone/kg trichloroethane, if 1 kg trichloroethane/kg water is used in the first extractor, and if the same number of stages is used in each extractor, what concentration of acetone in the final product will result?

10.11 A sieve-tray tower is to be designed for the extraction of 90% of the acetic acid from a water solution containing 4.0% acid, using methyl isobutyl ketone, initially free of acid, as solvent, at 25°C. The flow rates are to be 1.6×10^{-3} m³/s aqueous (203.4 ft³/h), 3.2×10^{-3} m³/s organic. The ketone is to be dispersed.

Physical properties are, for the aqueous solution, viscosity = 0.001 kg/m · s, density = 998 kg/m³, diffusivity of acetic acid = 1.00×10^{-9} m²/s; for the organic solution, viscosity = 5.70×10^{-4} kg/m · s, density = 801 kg/m³, diffusivity of acetic acid = 1.30×10^{-9} m²/s; distribution coefficient, (c in ketone)/(c in water) = 0.545; interfacial tension = 9.1×10^{-3} N/m.

Specify the tray dimensions and arrangement, estimate the values of E_{MD} and E_O , and specify the number of real trays required.

Problems 10.12 to 10.14 refer to the system cottonseed oil (A)-liquid propane (B)-oleic acid (C) at 98.5°C, 625 lb/in² abs. Smoothed equilibrium tie-line data of Hixson and Bockelmann, *Trans.*

AIChE, 38, 891 (1942), in weight percent, are as follows:

Cottonseed oil	Oleic acid	Propane	Cottonseed oil	Oleic acid	Propane
63.5	0	36.5	2.30	0	97.7
57.2	5.5	37.3	1.95	0.76	97.3
52.0	9.0	39.0	1.78	1.21	97.0
46.7	13.8	39.5	1.50	1.90	96.6
39.8	18.7	41.5	1.36	2.73	95.9
31.0	26.3	42.7	1.20	3.8	95.0
26.9	29.4	43.7	1.10	4.4	94.5
21.0	32.4	46.6	1.0	5.1	93.9
14.2	37.4	48.4	0.8	6.1	93.1
8.3	39.5	52.2	0.7	7.2	92.1
4.5	41.1	54.4	0.4	6.1	93.5
0.8	43.7	55.5	0.2	5.5	94.3

10.12 Plot the equilibrium data on the following coordinate systems: (a) N against X and Y ; (b) X against Y .

10.13 If 100 kg of a cottonseed oil-oleic acid solution containing 25% acid is to be extracted twice in crosscurrent fashion, each time with 1000 kg of propane, determine the compositions, percent by weight, and the weights of the mixed extracts and the final raffinate. Determine the compositions and weights of the solvent-free products. Make the computations on the coordinates plotted in part (a) of Prob. 10.12.

10.14 If 1000 kg/h of a cottonseed oil-oleic acid solution containing 25% acid is to be continuously separated into products containing 2 and 90% acid (solvent-free compositions) by countercurrent extraction with propane, make the following computations on the coordinate systems of parts (a) and (b) of Prob. 10.12:

(a) What is the minimum number of theoretical stages required?

Ans.: 5.

(b) What is the minimum external extract-reflux ratio required?

Ans.: 3.08.

(c) For an external extract-reflux ratio of 4.5, determine the number of theoretical stages, the position of the feed stage, and the quantities, in kg/h, of the following streams: E_1 , B_E , E' , R_0 , R_{Np} , P_E , and S .

Ans.: 10.5 theoretical stages.

(d) What do the equilibrium data indicate as to the maximum purity of oleic acid that could be obtained?

Problems 10.15 and 10.16 refer to the system oxalic and succinic acids distributed between water and n -amyl alcohol. The acids distribute practically independently of each other, and for present purposes they will be considered to do so. Equilibrium concentrations, expressed as g acid/l, are:

Either acid in water	0	20	40	60
Oxalic in alcohol	0	6.0	13.0	21.5
Succinic in alcohol	0	12.0	23.5	35.0

The water and amyl alcohol are essentially immiscible. In the calculations, express acid concentrations as g/l and solvent rates as l/time.

10.15 A water solution containing 50 g each of oxalic and succinic acids per liter of water is to be extracted countercurrently with n -amyl alcohol (free of acid) to recover 95% of the oxalic acid.

(a) Compute the minimum solvent required, alcohol/l water.

(b) If an alcohol/water ratio of 3.8 l/l^2 is used, compute the number of theoretical stages required and the solvent-free analysis of the extract product.

Ans.: 9 theoretical stages.

10.16 A mixture of succinic (B) and oxalic (C) acids containing 50% of each is to be separated into products each 90% pure on a solvent-free basis, using *n*-amyl alcohol (A) and water (D) by a fractional extraction.

(a) Calculate the number of theoretical stages and the location of the feed stage of 9 l of alcohol and 4 l of water are used per 100 g of acid feed.

Ans.: 12.75 stages.

(b) Investigate the effect of solvent ratio on the number of stages, keeping the total liquid flow constant. What is the least number of stages required and the corresponding solvent ratio?

PART FOUR

SOLID-FLUID OPERATIONS

The mass-transfer operations in this category are adsorption-desorption, drying, and leaching. Adsorption involves contact of solids with either liquids or gases and mass transfer in the direction fluid to solid. Mass transfer is in the opposite direction for its companion operation, desorption, and the combination is analogous to the gas-absorption-stripping pair of Chap. 8. Drying involves gas-solid, and leaching liquid-solid, contact, with mass transfer in each case in the direction solid to fluid, so that these are special cases of desorption.

Theoretically, at least, the same apparatus and equipment useful for gas-solid or liquid-solid contact in adsorption should also be useful in the corresponding operations of drying and leaching. In practice, however, we find special types of apparatus in all three categories. This is probably the result of many years of development of the practical applications of these operations without the realization that basically they are very similar. For example, we find considerable inventive genius applied to the development of equipment for the continuous gas-solid operations of adsorption but little application of the results to the problem of drying. Many clever devices have been developed for continuous leaching of solids with liquids, but there is little application of these to the practical problems of adsorption from liquids. Ideal devices have not yet been invented. But we may reasonably expect a reduction in the number of equipment types and greater interapplication between the three operations as the many problems of solids handling are eventually solved.

It helps considerably in understanding these operations if the strong resemblances to the gas and liquid operations of earlier chapters are kept in mind, although the fixed-bed and fluidized-bed operations of which solids are capable have no gas-liquid counterparts.

ADSORPTION AND ION EXCHANGE

The adsorption operations exploit the ability of certain solids preferentially to concentrate specific substances from solution onto their surfaces. In this manner, the components of either gaseous or liquid solutions can be separated from each other. A few examples will indicate the general nature of the separations possible and at the same time demonstrate the great variety of practical applications. In the field of gaseous separations, adsorption is used to dehumidify air and other gases, to remove objectionable odors and impurities from industrial gases such as carbon dioxide, to recover valuable solvent vapors from dilute mixtures with air and other gases, and to fractionate mixtures of hydrocarbon gases containing such substances as methane, ethylene, ethane, propylene, and propane. Typical liquid separations include the removal of moisture dissolved in gasoline, decolorization of petroleum products and aqueous sugar solutions, removal of objectionable taste and odor from water, and the fractionation of mixtures of aromatic and paraffinic hydrocarbons. The scale of operations ranges from the use of a few grams of adsorbent in the laboratory to industrial plants with an adsorbent inventory exceeding 135 000 kg [59].

These operations are all similar in that the mixture to be separated is brought into contact with another insoluble phase, the adsorbent solid, and the unequal distribution of the original constituents between the adsorbed phase on the solid surface and the bulk of the fluid then permits a separation to be made. All the techniques previously found valuable in the contact of insoluble fluids are useful in adsorption. Thus we have batchwise single-stage and continuous multistage separations and separations analogous to countercurrent absorption and stripping in the field of gas-liquid contact and to rectification and extraction with the use of reflux. In addition, the rigidity and immobility of a bed of solid

adsorbent particles make possible useful application of semicontinuous methods which are not at all practicable when two fluids are contacted.

Another solid-liquid operation of great importance is ion exchange, the reversible exchange of ions between certain solids and an electrolyte solution, which permits the separation and fractionation of electrolytic solutes. It is, of course, chemical in nature but involves not only the interaction of the ions with the solid but also diffusion of ions within the solid phase. Although the phenomenon may be more complex than adsorption, the general techniques and the results obtained are very similar. The special features of ion exchange are considered separately at the end of this chapter.

Types of Adsorption

We must distinguish at the start between two types of adsorption phenomena, physical and chemical.

Physical adsorption, or *van der Waals adsorption*, a readily reversible phenomenon, is the result of intermolecular forces of attraction between molecules of the solid and the substance adsorbed. When, for example, the intermolecular attractive forces between a solid and a gas are greater than those existing between molecules of the gas itself, the gas will condense upon the surface of the solid even though its pressure may be lower than the vapor pressure corresponding to the prevailing temperature. Such a condensation will be accompanied by an evolution of heat, in amount usually somewhat larger than the latent heat of vaporization and of the order of the heat of sublimation of the gas. The adsorbed substance does not penetrate within the crystal lattice of the solid and does not dissolve in it but remains entirely upon the surface [82]. If, however, the solid is highly porous, containing many fine capillaries, the adsorbed substance will penetrate these interstices if it wets the solid. The equilibrium vapor pressure of a concave liquid surface of very small radius of curvature is lower than that of a large flat surface, and the extent of adsorption is correspondingly increased. In any case, at equilibrium the partial pressure of the adsorbed substance equals that of the contacting gas phase, and by lowering the pressure of the gas phase or by raising the temperature the adsorbed gas is readily removed or desorbed in unchanged form. Industrial adsorption operations of the type we shall consider depend upon this reversibility for recovery of the adsorbent for reuse, for recovery of the adsorbed substance, or for the fractionation of mixtures. Reversible adsorption is not confined to gases but is observed with liquids as well.

Chemisorption, or activated adsorption, is the result of chemical interaction between the solid and the adsorbed substance [13]. The strength of the chemical bond may vary considerably, and identifiable chemical compounds in the usual sense may not actually form, but the adhesive force is generally much greater than that found in physical adsorption. The heat liberated during chemisorption is usually large, of the order of the heat of chemical reaction. The process is frequently irreversible, and on desorption the original substance will often be found to have undergone a chemical change. The same substance which, under

conditions of low temperature, will undergo substantially only physical adsorption upon a solid will sometimes exhibit chemisorption at higher temperatures, and both phenomena may occur at the same time. Chemisorption is of particular importance in catalysis but will not be considered here.

Nature of Adsorbents

Adsorbent solids are usually used in granular form, varying in size from roughly 12 mm in diameter to as small as 50 μm . The solids must possess certain engineering properties depending upon the application to which they are put. If they are used in a fixed bed through which a liquid or gas is to flow, for example, they must not offer too great a pressure drop for flow nor must they easily be carried away by the flowing stream. They must have adequate strength and hardness so as not to be reduced in size during handling or crushed in supporting their own weight in beds of the required thickness. If they are to be transported frequently in and out of bins, they should be free-flowing. These are properties which are readily recognized.

The adsorptive ability of solids is quite another matter. Adsorption is a very general phenomenon, and even common solids will adsorb gases and vapors at least to a certain extent. For example, every student of analytical chemistry has observed with annoyance the increase in weight of a dried porcelain crucible on a humid day during an analytical weighing which results from the adsorption of moisture from the air upon the crucible surface. But only certain solids exhibit sufficient specificity and adsorptive capacity to make them useful as industrial adsorbents. Since solids are frequently very specific in their ability to adsorb certain substances in large amounts, the chemical nature of the solid evidently has much to do with its adsorption characteristics. But mere chemical identity is insufficient to characterize its usefulness. In liquid extraction, all samples of pure butyl acetate will extract acetic acid from a water solution with identical ability. The same is not true for the adsorption characteristics of silica gel with respect to water vapor, for example. Much depends on its method of manufacture and on its prior history of adsorption and desorption.

Large surface per unit weight seems essential to all useful adsorbents. Particularly in the case of gas adsorption, the significant surface is not the gross surface of the granular particles which are ordinarily used but the very much larger surface of the internal pores of the particles. The pores are usually very small, sometimes of the order of a few molecular diameters in width, but their large number provides an enormous surface for adsorption. It is estimated, for example, that a typical gasmask charcoal has an effective surface of 1 000 000 m^2/kg [32]. There are many other properties evidently of great importance which are not at all understood, and we must depend largely on empirical observation for recognition of adsorptive ability. The following is a list of the principal adsorbents in general use.

1. *Fuller's earths*. These are natural clays, the American varieties coming largely from Florida and Georgia. They are chiefly magnesium aluminum silicates in the form of the minerals attapulgite

- and montmorillonite. The clay is heated and dried, during which operation it develops a porous structure, ground, and screened. Commercially available sizes range from coarse granules to fine powders. The clays are particularly useful in decolorizing, neutralizing, and drying such petroleum products as lubricating oils, transformer oils, kerosenes, and gasolines, as well as vegetable and animal oils. By washing and burning the adsorbed organic matter accumulating upon the clay during use, the adsorbent can be reused many times.
2. *Activated clays.* These are bentonite or other clays which show essentially no adsorptive ability unless activated by treatment with sulfuric or hydrochloric acid. Following such treatment, the clay is washed, dried, and ground to a fine powder. It is particularly useful for decolorizing petroleum products and is ordinarily discarded after a single application.
 3. *Bauxite.* This is a certain form of naturally occurring hydrated alumina which must be activated by heating to temperatures varying from 230 to 815°C in order to develop its adsorptive ability. It is used for decolorizing petroleum products and for drying gases and can be reactivated by heating.
 4. *Alumina.* This is a hard, hydrated aluminum oxide which is activated by heating to drive off the moisture. The porous product is available as granules or powders, and it is used chiefly as a desiccant for gases and liquids. It can be reactivated for reuse.
 5. *Bone char.* This is obtained by the destructive distillation of crushed, dried bones at temperatures in the range 600 to 900°C. It is used chiefly in the refining of sugar and can be reused after washing and burning.
 6. *Decolorizing carbons.* These are variously made by (1) mixing vegetable matter with inorganic substances such as calcium chloride, carbonizing, and leaching away the inorganic matter, (2) mixing organic matter such as sawdust, etc., with porous substances such as pumice stone, followed by heating and carbonizing to deposit the carbonaceous matter throughout the porous particles, and (3) carbonizing wood, sawdust, and the like, followed by activation with hot air or steam. Lignite and bituminous coal are also raw materials. They are used for a great variety of purposes, including the decolorizing of solutions of sugar, industrial chemicals, drugs, and dry-cleaning liquids, water purification, refining of vegetable and animal oils, and in recovery of gold and silver from cyanide ore-leach solutions.
 7. *Gas-adsorbent carbon.* This is made by carbonization of coconut shells, fruit pits, coal, lignite, and wood. It must be activated, essentially a partial oxidation process, by treatment with hot air or steam. It is available in granular or pelleted form and is used for recovery of solvent vapors from gas mixtures, gas masks, collection of gasoline hydrocarbons from natural gas, and the fractionation of hydrocarbon gases. It is revived for reuse by evaporation of the adsorbed gas.
 8. *Molecular-screening activated carbon.* This is a specially made form with pore openings controlled from 5 to 5.5 Å (those of most activated carbons range from 14 to 60 Å.† The pores can admit paraffin hydrocarbons, for example, but reject isoparaffins of large molecular diameters. The product is useful in fractionating acetylene compounds, alcohols, organic acids, ketones, aldehydes, and many others. A good general review of activated carbon is available [37].
 9. *Synthetic polymeric adsorbents* [95]. These are porous spherical beads, 0.5 mm diameter, each bead a collection of microspheres, 10^{-4} mm diameter. The material is synthetic, made from polymerizable monomers of two major types. Those made from unsaturated aromatics such as styrene and divinylbenzene are useful for adsorbing nonpolar organics from aqueous solution. Those made from acrylic esters are suitable for more polar solutes. They are used principally for treating water solutions and are regenerated by leaching with low-molecular-weight alcohols or ketones.
 10. *Silica gel.* This is a hard, granular, very porous product made from the gel precipitated by acid treatment of sodium silicate solution. Its moisture content before use varies from roughly 4 to 7 percent, and it is used principally for dehydration of air and other gases, in gas masks, and for fractionation of hydrocarbons. It is revived for reuse by evaporation of the adsorbed matter.
 11. *Molecular sieves* [41, 58]. These are porous, synthetic zeolite crystals, metal aluminosilicates.

† The angstrom (Å) is 10^{-10} m.

The "cages" of the crystal cells can entrap adsorbed matter, and the diameter of the passageways, controlled by the crystal composition, regulates the size of the molecules which can enter. The sieves can thus separate according to molecular size, but they also separate by adsorption according to molecular polarity and degree of unsaturation. There are some nine types industrially available, with nominal pore diameters ranging from 3 to 10 Å, in the form of pellets, beads, and powders. They are used for dehydration of gases and liquids, separation of gas and liquid hydrocarbon mixtures, and in a great variety of processes. They are regenerated by heating or elution.

ADSORPTION EQUILIBRIA

The great bulk of the experimental data pertaining to adsorption represents equilibrium measurements. Many of them were gathered in an attempt to provide corroboration for one or another of the many theories which have been advanced in an attempt to explain the adsorption phenomena. No one theory has yet been devised which satisfactorily explains even a majority of the observations, and this discussion is therefore limited simply to a description of the more commonly observed adsorption characteristics. The theories are reviewed elsewhere [13, 18, 44, 82, 102].

SINGLE GASES AND VAPORS

In many respects the equilibrium adsorption characteristics of a gas or vapor upon a solid resemble the equilibrium solubility of a gas in a liquid. Figure 11.1 shows several equilibrium adsorption isotherms for a particular activated carbon as adsorbent, where the concentration of adsorbed gas (the adsorbate) on the solid is plotted against the equilibrium partial pressure $\bar{p}^*$ of the vapor or gas at constant temperature. Curves of this sort are analogous to those of Fig. 8.1. At 100°C, for example, pure acetone vapor at a pressure of 190 mmHg is in equilibrium with an adsorbate concentration of 0.2 kg adsorbed acetone/kg carbon, point *A*. Increasing the pressure of the acetone will cause more to be adsorbed, as the rising curve indicates, and decreasing the pressure of the system at *A* will cause acetone to be desorbed from the carbon. While not determined experimentally in this case, it is known that the 100°C isotherm for acetone will continue to rise only to a pressure of 2790 mmHg, the saturation vapor pressure of acetone at this temperature. At higher pressures, no acetone can exist in the vapor state at this temperature but instead will condense entirely to a liquid. It will thus be possible to obtain indefinitely large concentrations of the substance on the solid at pressures higher than the vapor pressure, as at point *B* (Fig. 11.2). However, concentrations in excess of that corresponding to point *B* indicate liquefaction but not necessarily adsorption of the vapor. Gases above their critical temperature, of course, do not show this characteristic.

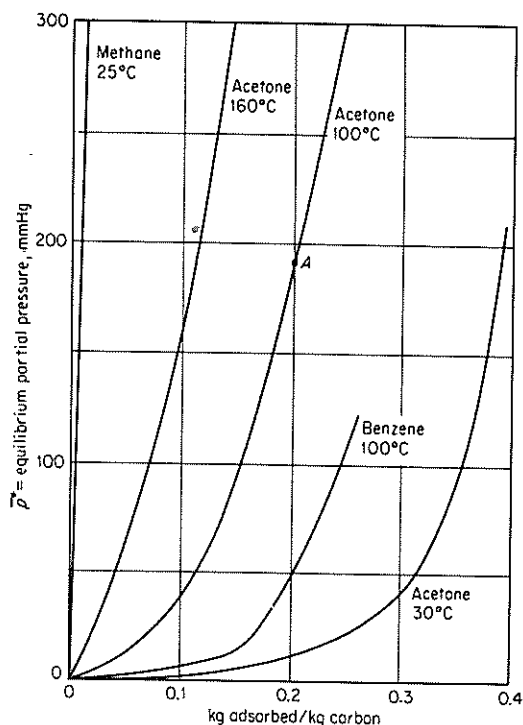


Figure 11.1 Equilibrium adsorption of vapors on activated carbon.

Different gases and vapors are adsorbed to different extents under comparable conditions. Thus benzene (Fig. 11.1) is more readily adsorbed than acetone at the same temperature and gives a higher adsorbate concentration for a given equilibrium pressure. As a general rule, vapors and gases are more readily adsorbed the higher their molecular weight and the lower their critical temperature, although chemical differences such as the extent of unsaturation in the

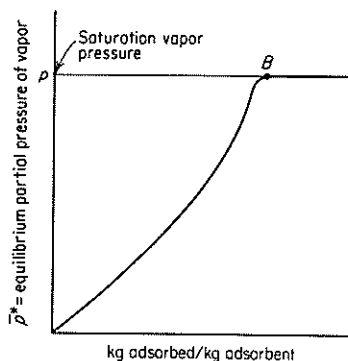


Figure 11.2 Typical complete vapor-adsorption isotherm.

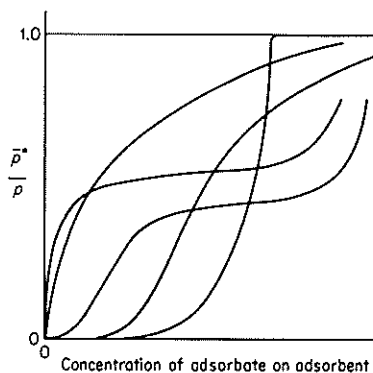


Figure 11.3 Types of adsorption isotherms for vapors.

molecule also influence the extent of adsorption. The so-called permanent gases are usually adsorbed only to a relatively small extent, as the methane isotherm of Fig. 11.1 indicates.

Adsorption isotherms are not always concave to the pressure axis. The shapes shown in Fig. 11.3 have all been observed for various systems. Here the ordinate is plotted as equilibrium partial pressure $\bar{p}^*$ divided by the saturation vapor pressure p of the adsorbed substance (actually, the relative saturation) in order to place all the curves on a comparable basis.

It will be recalled that a change of liquid solvent profoundly alters the equilibrium solubility of a gas except in the case of ideal liquid solutions. In a similar fashion the equilibrium curves for acetone, benzene, and methane on silica gel as adsorbent would be entirely different from those of Fig. 11.1. Indeed, differences in the origin and method of preparation of an adsorbent will result in significant differences in the equilibrium adsorption as well. For this reason, many of the data gathered years ago are no longer of practical value, since methods of preparing adsorbents, and consequently the corresponding adsorbent capacities, have improved greatly over the years. Repeated adsorption and desorption will also frequently alter the characteristics of a particular adsorbent, perhaps as a result of progressive changes in the pore structure within the solid.

There are three commonly used mathematical expressions to describe vapor adsorption equilibria: the Langmuir, the Brunauer-Emmett-Teller (BET), and the Freundlich isotherms [13, 18, 44, 82, 102]. All except the last were derived with a theory in mind, but none is applicable universally, nor can it be predicted which, if any, will apply to a particular case. Since we shall make no application of them, they are not detailed here.

Adsorption Hysteresis

The curves of Fig. 11.1 are true equilibrium curves and therefore represent completely reversible phenomena. The conditions corresponding to point A on

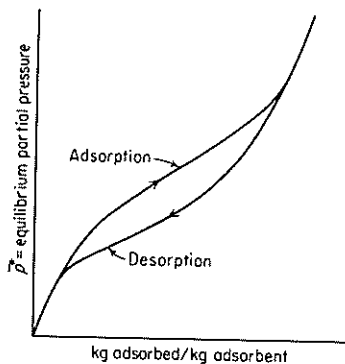


Figure 11.4 Adsorption isotherm showing hysteresis.

the figure, for example, can be obtained either by adsorption onto fresh carbon or by desorption of a sample with an initially higher adsorbate concentration. Occasionally, however, different equilibria result, at least over a part of an isotherm, depending upon whether the vapor is adsorbed or desorbed, and this gives rise to the hysteresis phenomenon indicated in Fig. 11.4. This may be the result of the shape of the openings to the capillaries and pores of the solid or of complex phenomena of wetting of the solid by the adsorbate. In any case, when

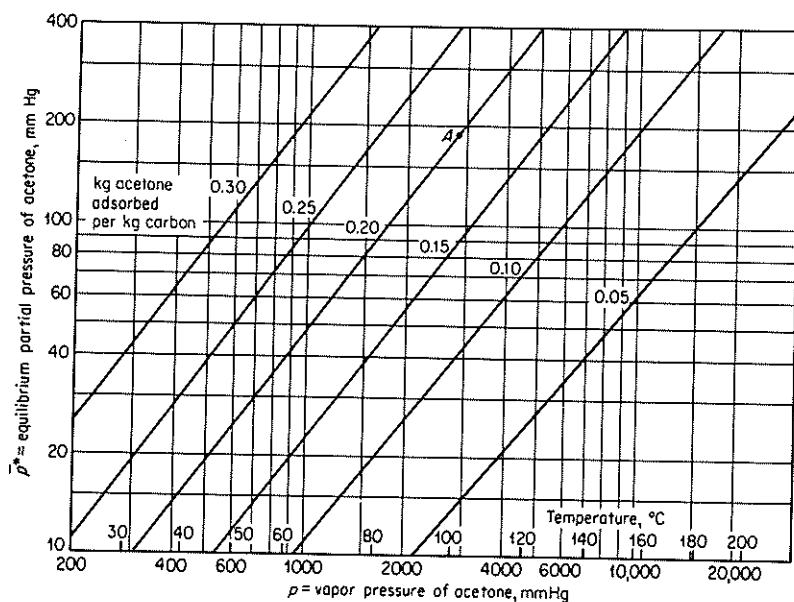


Figure 11.5 Reference-substance plot of equilibrium adsorption of acetone on an activated carbon. [Data of Josefowitz and Othmer, *Ind. Eng. Chem.*, 40, 739 (1948).]

hysteresis is observed, the desorption equilibrium pressure is always lower than that obtained by adsorption.

Effect of Temperature

Since adsorption is an exothermic process, the concentration of adsorbed gas decreases with increased temperature at a given equilibrium pressure, as the several acetone isotherms of Fig. 11.1 indicate.

The reference-substance method of plotting, described in Chap. 8 for gas-liquid solubilities, is conveniently applicable also to adsorption data [81]. As reference substance it is best to use the pure substance being adsorbed, unless the temperatures are above the critical temperature. Figure 11.5, for example, was prepared for the adsorption of acetone vapor on activated carbon with

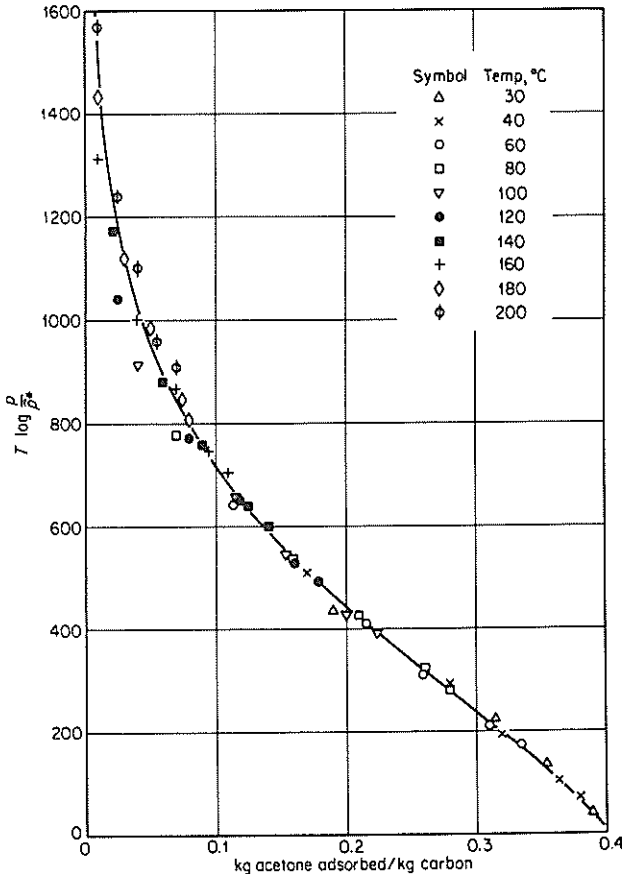


Figure 11.6 Adsorption of acetone on an activated carbon, 30 to 200°C.

acetone as reference substance. The abscissa of the logarithmic coordinates was marked with the vapor pressure of pure acetone, and the corresponding saturation temperatures were also marked. The equilibrium partial pressure of adsorbate was plotted on the ordinate. Points of equal temperature for the vapor pressure of pure acetone and the partial pressure of adsorbate were then plotted. Thus, point *A* of Fig. 11.1 ($\bar{p}^* = 190$ mmHg, 100°C) was plotted on Fig. 11.5 at *A*, where $p = 2790$ mmHg, the vapor pressure of acetone at 100°C . Points of constant adsorbate concentration (*isosteres*) form straight lines with few exceptions, and thus only two points are required to establish each.

Figure 11.6 shows all the same data plotted in such a manner as to reduce all the measurements to a single curve, thereby permitting considerable extension of meager data. The free energy of compression of 1 mol of a gas from the equilibrium adsorption pressure $\bar{p}^*$ to the vapor pressure p , the *adsorption potential*, is $RT \ln (p/\bar{p}^*)$. In the case of single substances, when this quantity is plotted against adsorbate concentration, a single curve results for all temperatures, at least over a moderate temperature range [31]. The ordinate of Fig. 11.6 is proportional to this quantity. More complex methods of expressing the adsorbate concentration can be used to provide improved and extended correlations of the data.

Heat of Adsorption

The *differential heat of adsorption* ($-\bar{H}$) is defined as the heat liberated at constant temperature when unit quantity of vapor is adsorbed upon a large quantity of solid already containing adsorbate. Such a large quantity of solid is used that the adsorbate concentration is unchanged. The *integral heat of adsorption* at any concentration X of adsorbate upon the solid is defined as the enthalpy of the adsorbate-adsorbent combination minus the sum of the enthalpies of unit weight of pure solid adsorbent and sufficient pure adsorbed substance (before adsorption) to provide the required concentration X , all at the same temperature. These are both functions of temperature and adsorbate concentration for any system.

Othmer and Sawyer [81] have shown that plots of the type shown in Fig. 11.5 are useful in estimating the heat of adsorption, which can be calculated in a manner similar to that for the latent heat of vaporization of a pure liquid (see Chap. 7). Thus, the slope of an isostere of Fig. 11.5 is

$$\frac{d \ln \bar{p}^*}{d \ln p} = \frac{(-\bar{H})M}{\lambda_r M_r} \quad (11.1)$$

where $\bar{H}$, energy per mass of vapor adsorbed, is referred to the pure vapor and λ_r is the latent heat of vaporization of the reference substance at the same temperature, energy per mass. M and M_r are the molecular weights of the vapor and reference substance, respectively. If $\bar{H}$ is computed at constant temperature for each isostere, the integral heat of adsorption at this temperature can be

computed from the relation

$$\Delta H'_A = \int_0^X \bar{H} dX \quad (11.2)$$

$\Delta H'_A$, energy per mass of adsorbate-free solid, is referred to the pure vapor, and X is the adsorbate concentration, mass adsorbate/mass solid. The integral can be evaluated graphically by determining the area under a curve of $\bar{H}$ vs. X . The integral heat of adsorption referred to solid and the adsorbed substance in the liquid state is $\Delta H_A = \Delta H'_A + \lambda X$, energy per mass solid. The quantities $\bar{H}$, ΔH_A , and $\Delta H'_A$ are negative quantities if heat is evolved during adsorption.

Illustration 11.1 Estimate the integral heat of adsorption of acetone upon activated carbon at 30°C as a function of adsorbate concentration.

SOLUTION Refer to Fig. 11.5. The isosteres for various concentrations are straight on this diagram, and their slopes are measured with the help of a millimeter rule. In the accompanying table, column 1 lists the adsorbate concentration of each isostere and column 2 the corresponding slope.

X , kg acetone kg carbon (1)	Slope of isostere (2)	Differential heat of adsorption $\bar{H}$, kJ/kg acetone (3)	Integral heat of adsorption, kJ/kg carbon	
			$\Delta H'_A$, referred to acetone vapor (4)	ΔH_A , referred to acetone liquid (5)
0.05	1.170	-640	-29.8	-2.1
0.10	1.245	-686	-63.0	-7.9
0.15	1.300	-716	-97.9	-15.1
0.20	1.310	-721	-134.0	-23.7
0.25	1.340	-740	-170.5	-32.6
0.30	1.327	-730	-207.2	-41.9

Since in this case the adsorbate and reference substance are the same, $M = M_r$ and consequently $\bar{H} = -\lambda_r$ (slope of isostere). At 30°C, $\lambda_r = \lambda$, the latent heat of vaporization of acetone = 551 kJ/kg. Column 3 of the table lists values of $\bar{H}$ calculated in this manner.

Column 3 was plotted as ordinate against column 1 as abscissa (not shown). The area under the curve between $X = 0$ and any value of X is listed in column 4 as the corresponding integral heat of adsorption, referred to acetone vapor. Thus the area under the curve between $X = 0$ and $X = 0.20$ is -134.0. If 0.20 kg acetone vapor at 30°C is adsorbed on 1 kg fresh carbon at 30°C and the product brought to 30°C, 134 kJ will be evolved.

Column 5, the integral heat of adsorption referred to liquid acetone, is computed from the relation $\Delta H_A = \Delta H'_A + \lambda X$. Thus, at $X = 0.20$, $\Delta H_A = -134 + 551(0.20) = -23.7$ kJ/kg carbon.

VAPOR AND GAS MIXTURES

It is necessary to distinguish between mixtures depending upon whether one or several of the components are adsorbed.

One Component Adsorbed

In the case of many mixtures, particularly vapor-gas mixtures, only one component is appreciably adsorbed. This would be the circumstance for a mixture of acetone vapor and methane in contact with activated carbon (Fig. 11.1), for example. In such instances, the adsorption of the vapor will be substantially unaffected by the presence of the poorly adsorbed gas, and the adsorption isotherm for the pure vapor will be applicable provided the equilibrium pressure is taken as the *partial* pressure of the vapor in the vapor-gas mixture. The isotherms for acetone (Fig. 11.1) thus apply for mixtures of acetone with any poorly adsorbed gas such as nitrogen, hydrogen, and the like. This is similar to the corresponding case of gas-liquid solubility.

Binary Gas or Vapor Mixtures, Both Components Appreciably Adsorbed

When both components of a binary gas or vapor mixture are separately adsorbed to roughly the same extent, the amount of either one adsorbed from the mixture will be affected by the presence of the other. Since such systems are composed of three components when the adsorbent is included, the equilibrium data are conveniently shown in the manner used for ternary liquid equilibria in Chap. 10. For this purpose it is convenient to consider the solid adsorbent as being analogous to liquid solvent in extraction operations. However, adsorption is greatly influenced by both temperature and pressure, unlike liquid solubility, which is scarcely affected by pressure under ordinary circumstances. Equilibrium diagrams are consequently best plotted at constant temperature and constant total pressure, and they are therefore simultaneously *isotherms* and *isobars*.

A typical system is shown in Fig. 11.7 on triangular and rectangular coordinates. The properties of these coordinate systems and the relations between them were considered in detail in Chap. 10. Even though mole fraction is generally a more convenient concentration unit in dealing with gases, the figure is plotted in terms of weight-fraction compositions since the molecular weight of the adsorbent is uncertain.[†] Since the adsorbent is not volatile and does not appear in the gas phase, the equilibrium gas compositions fall upon one axis of either graph, as shown. Points *G* and *H* represent the adsorbate concentration for the individual pure gases and the curve *GEH* that of gas mixtures. Tie lines such as line *RE* join equilibrium compositions of the gas and adsorbate. The fact that the tie lines do not, when extended, pass through the adsorbent apex indicates that under the prevailing conditions the adsorbent can be used to separate the binary gas mixture into its components. The separation factor, or *relative adsorptivity*, similar to relative volatility in distillation or selectivity in liquid extraction, is obtained by dividing the equilibrium ratio of gas compositions in the adsorbate (as at point *E*) by the ratio in the gas (as at *R*). The relative adsorptivity must be larger than unity if the adsorbent is to be useful for

[†] Alternatively, some arbitrary molecular weight could be assigned to the adsorbent.

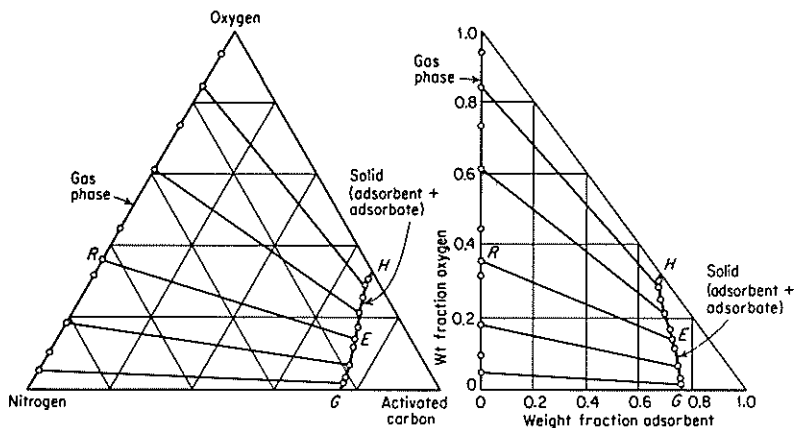


Figure 11.7 System oxygen-nitrogen-activated carbon, -150°C , 1 std atm, concentrations expressed as weight fractions. [Data of Maslan, Altman, and Aberth, *J. Phys. Chem.*, 57, 106 (1953).]

separating the components of the gas mixture. In the system of Fig. 11.7 the more strongly adsorbed of the two pure gases (oxygen) is also selectively adsorbed from any mixture. This appears to be true for most mixtures, although an inversion of the relative adsorptivity in some systems (analogous to azeotropism in distillation) is certainly a possibility.

Especially when the extent of adsorption is small, it will be more convenient to express compositions on an adsorbent-free basis and to plot them in the manner of Fig. 11.8. Such diagrams are also analogous to those used in liquid extraction (Fig. 10.10, for example), where adsorbent solid and extraction solvent play an analogous role, and to the enthalpy-concentration diagrams of distillation, where heat is analogous to adsorbent. The adsorption characteristics of the binary gas mixture acetylene-ethylene on silica gel for one temperature and pressure are shown in Fig. 11.8*a*. In the upper portion of this figure, the gas phase appears entirely along the abscissa of the plot owing to the absence of adsorbent in the gas. The adsorbent-free equilibrium compositions corresponding to the tie lines can be plotted in the lower half of the diagram, as at point (*R*, *E*), to produce a figure analogous to the *xy* diagram of distillation. Silica gel selectively adsorbs acetylene from these gas mixtures.

The powerful influence of the adsorbent on the equilibrium is demonstrated with the same gas mixture by Fig. 11.8*b*, where activated carbon is the adsorbent. Not only is the extent of adsorption greater than for silica gel, so that the curve *GH* is lower than the corresponding curve for silica gel, but in addition the relative adsorptivity is reversed: ethylene is selectively adsorbed on activated carbon. In both cases, however, that gas which is separately more strongly adsorbed on each adsorbent is selectively adsorbed from mixtures. If a condition corresponding to azeotropism should arise, the curve of the lower half of these figures would cross the 45° diagonal line. In cases like Fig. 11.8*b*, it will

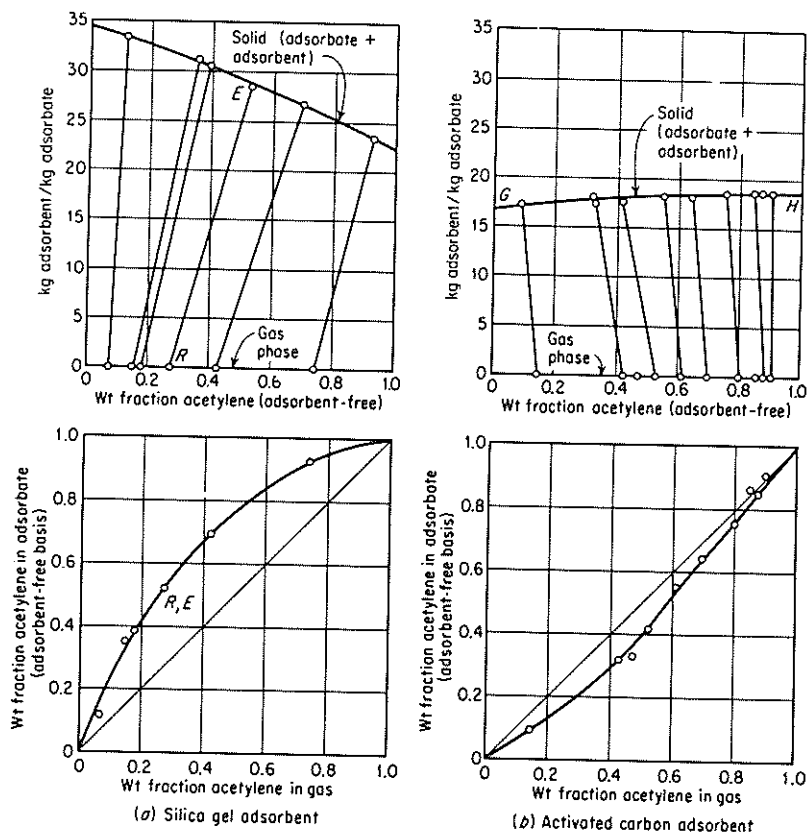


Figure 11.8 Adsorption of acetylene-ethylene on (a) silica gel and (b) activated carbon, at 25°C, 1 std atm. [Data of Lewis, et al., *J. Am. Chem. Soc.*, **72**, 1157 (1950).]

generally be preferable to plot compositions in terms of the more strongly adsorbed gas (ethylene), to keep the appearance of the diagram similar to those used previously in liquid extraction.

A beginning has been made on the estimation of adsorbate concentrations from the isotherms of pure vapors, assuming that the adsorbate is ideal and follows an analog of Raoult's law [49, 74, 75]. Some systems follow such a law, such as ethane-propane-carbon black [26] and *n*-butane-ethane-molecular sieves [29]. Others do not. Thus, in the case of benzene-*n*-heptane-silica gel, adsorbates deviate negatively from ideal solutions, whereas in the vapor-liquid equilibria for these hydrocarbons the deviation is positive [90]. An excellent review of the thermodynamics is available [97], including the concept of two-dimensional equations of state for the adsorbate layer, strongly reminiscent of Abbott's "Flatland" [1].

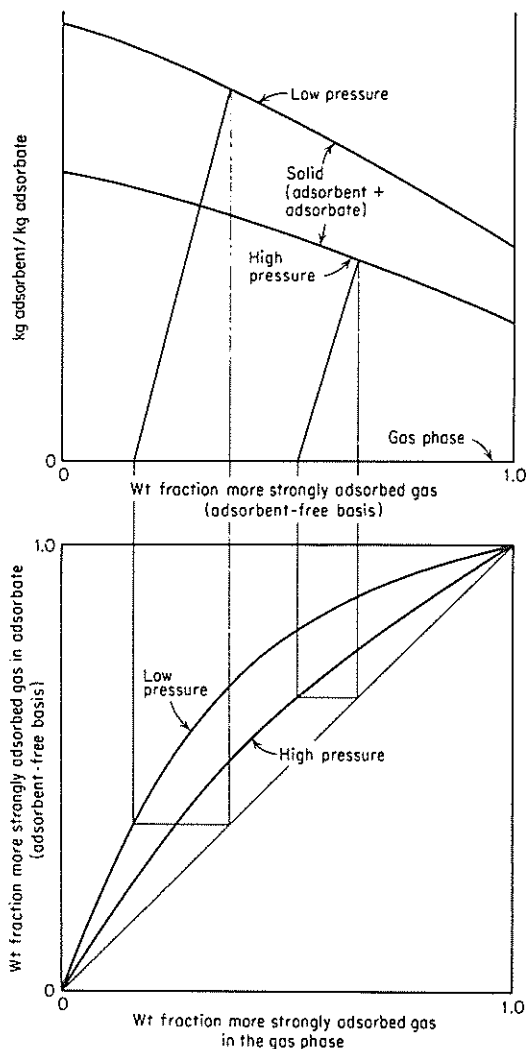


Figure 11.9 Effect of pressure on adsorption isotherms for binary gas mixtures.

Effect of Change of Pressure or Temperature

The available data are meager, and generalizations are very difficult to make. Lowering the pressure will of course reduce the amount of adsorbate upon the adsorbent, as shown in the upper half of Fig. 11.9. In several cases investigated over any appreciable range [24, 62] the relative adsorptivity of paraffin hydrocarbons on carbon decreased at increased pressure, as shown in the lower part

of this figure, just as it does in distillation. Owing to the increased tendency toward liquid condensation in the absorbent capillaries at higher pressures, the equilibrium may simply be shifting toward the ordinary vapor-liquid equilibrium with increased pressure, and in each of the investigated cases this corresponded to lower separation factor. Increasing the temperature at constant pressure will decrease the amount adsorbed from a mixture and will influence the relative adsorptivity as well but in a manner for which no generalizations can now be made.

LIQUIDS

When an adsorbent solid is immersed in a pure liquid, the evolution of heat, known as the *heat of wetting*, is evidence that adsorption of the liquid does occur. But immersion does not provide an effective method of measuring the extent of adsorption. No appreciable volume change of the liquid which might be used as a measure of adsorption is ordinarily observed, while withdrawal of the solid and weighing it will not distinguish between the adsorbed liquid and that which is mechanically occluded. This problem does not exist in the case of adsorption of gases, where the change in weight of the solid due to adsorption is readily measured.

Adsorption of Solute from Dilute Solution

When an adsorbent is mixed with a binary solution, adsorption of both solute and solvent occurs. Since the total adsorption cannot be measured, the relative or apparent adsorption of solute is determined instead. The customary procedure is to treat a known volume of solution with a known weight of adsorbent, v volume solution/mass adsorbent. As a result of preferential adsorption of solute, the solute concentration of the liquid is observed to fall from the initial value c_0 to the final equilibrium value c^* mass solute/volume liquid. The apparent adsorption of solute, neglecting any volume change in the solution, is then $v(c_0 - c^*)$ mass solute adsorbed/mass adsorbent. This is satisfactory for dilute solutions when the fraction of the original solvent which can be adsorbed is small.

Correction is sometimes made for the volume of the solute apparently adsorbed. Thus, the initial solvent content of this solution is $v(1 - c_0/\rho)$, and on the assumption that no solvent is adsorbed, the volume of residual solution is $v(1 - c_0/\rho)/(1 - c^*/\rho)$. The apparent solute adsorption is then the difference between initial and final solute content of the liquid, $vc_0 - [v(1 - c_0/\rho)/(1 - c^*/\rho)]c^*$ or $v(c_0 - c^*)/(1 - c^*/\rho)$. This of course, still neglects solvent adsorption.

The apparent adsorption of a given solute depends upon the concentration of solute, the temperature, the solvent, and the type of adsorbent. Typical isotherms are shown in Fig. 11.10. Isotherms of all the indicated forms have been observed, for example, when a given solute is adsorbed on the same

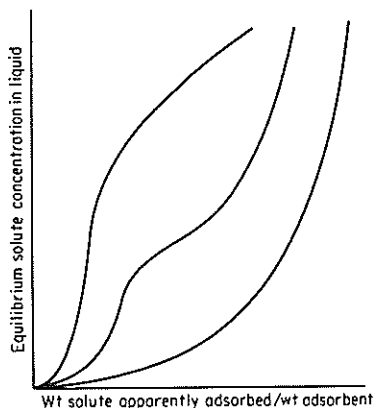


Figure 11.10 Typical adsorption isotherms for dilute solutions.

adsorbent, but from different solvents. The extent of adsorption of a given solute practically always decreases at increased temperature and usually is greater the smaller the solubility in the solvent. It is usually reversible, so that the same isotherm results whether solute is desorbed or adsorbed.

The Freundlich Equation

Over a small concentration range, and particularly for dilute solutions, the adsorption isotherms can frequently be described by an empirical expression usually attributed to Freundlich,

$$c^* = k[v(c_0 - c^*)]^n \quad (11.3)$$

where $v(c_0 - c^*)$ is the apparent adsorption per unit mass of adsorbent and k and n are constants. Other concentration units are frequently used also, and while they will result in different values of k , for the dilute solutions for which the equation is applicable the value of n will be unaffected. The form of the equation indicates that plotting the equilibrium solute concentration as ordinate against adsorbate content of the solid as abscissa on logarithmic coordinates will provide a straight line of slope n and intercept k . Several typical isotherms are plotted in this manner in Fig. 11.11. The effect of the nature of the solvent on adsorption of benzoic acid on silica gel is shown by curves *a* and *b*, which follow Eq. (11.3) excellently over the concentration range shown. The adsorption is less strong from benzene solutions, which is to be expected in view of the higher solubility of the acid in this solvent. Curve *c* of this figure shows the deviation from linearity to be expected over large concentration ranges, although Eq. (11.3) is applicable for the lower concentration ranges. Failure of the data to follow the equation at high solute concentrations may be the result of appreciable adsorption of the solvent which is not taken into account or simply general inapplicability of the expression.

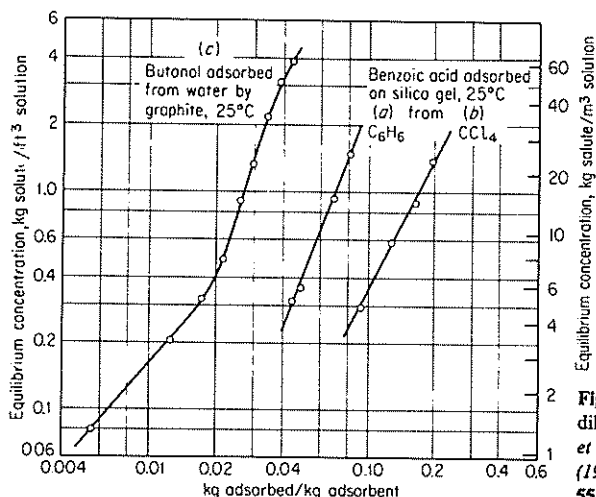


Figure 11.11 Adsorption from a dilute solution. [Data of Bartell, et al., *J. Phys. Chem.*, 33, 676 (1929); *J. Phys. Colloid Chem.*, 55, 1456 (1951).]

The Freundlich equation is also frequently useful in cases where the actual identity of the solute is not known, as in the adsorption of colored substances from such materials as sugar solutions and mineral or vegetable oils. In such cases, the concentration of solute can be measured by means of a colorimeter or spectrophotometer and expressed in terms of arbitrary units of color intensity, provided the color scale used varies linearly with the concentration of the responsible solute. Figure 11.12 illustrates the application of this method of plotting the adsorption of colored substances from a petroleum fraction on two adsorbent clays. Here the color concentrations of the solutions are measured on an arbitrary scale, and the concentration of adsorbate on the clay is determined by measuring the change in color expressed in these terms when 100 g of oil is treated with various amounts of clay.

Adsorption from Concentrated Solutions

When the apparent adsorption of solute is determined over the entire range of concentrations from pure solvent to pure solute, curves like those of Fig. 11.13 will result. Curves of the shape marked *a* occur when at all concentrations the solute is adsorbed more strongly relative to the solvent. At increasing solute concentrations, the extent of solute adsorption may actually continue to increase; yet the curve showing apparent solute adsorption necessarily returns to point *E*, since in a liquid consisting of pure solute alone there will be no concentration change on addition of adsorbent. In cases where both solvent and solute are adsorbed to nearly the same extent, the S-shaped curves of type *b* are produced. In the range of concentrations from *C* to *D*, solute is more strongly adsorbed than solvent. At point *D*, both are equally well adsorbed, and the

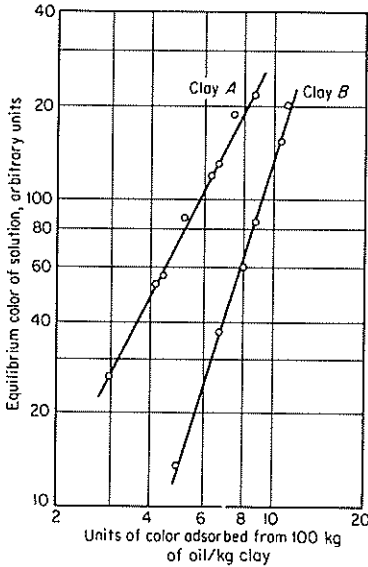


Figure 11.12 Decolorization of cylinder oil with clay. [Data of Rogers, *et al.*, *Ind. Eng. Chem.*, **18**, 164 (1926).]

apparent adsorption falls to zero. In the range of concentrations from *D* to *E*, solvent is more strongly adsorbed. Consequently, on addition of adsorbent to such solutions, the solute concentration of the liquid increases, and the quantity $v(c_0 - c^*)$ indicates an apparent *negative* solute adsorption.

The true adsorption of the substances can be estimated from apparent adsorption data if some mechanism for the process, such as the applicability of the Freundlich isotherm, separately for each

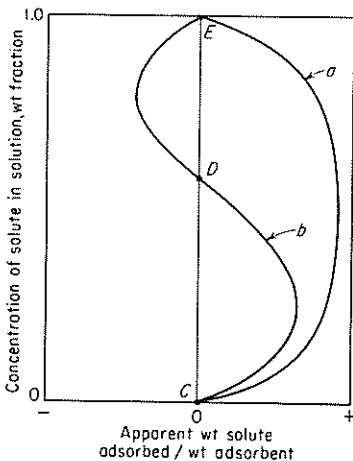


Figure 11.13 Apparent adsorption of solute from solutions.

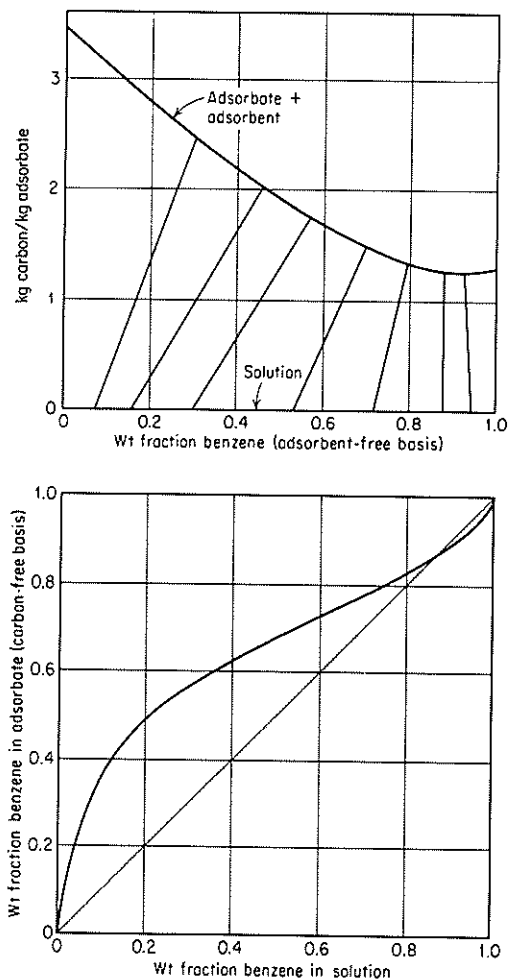


Figure 11.14 Calculated adsorption of benzene and ethanol by activated carbon. [Bartell and Sloan, *J. Am. Chem. Soc.* 51, 1643 (1929).]

component, is assumed [6]. In this manner, curves like those of Fig. 11.14 can be computed from the S-shaped apparent adsorption data (note that such data characteristically show a situation analogous to azeotropism in distillation, with relative adsorptivity equal to unity at some liquid concentration). It has been shown, however, that such isotherms are not always applicable, and another approach is to determine the adsorption of a liquid solution by measuring that of the vapor in equilibrium with the liquid [50, 76]. The thermodynamics of binary liquid adsorption has been established [90].

For adsorption of multiple solutes from solution, there has been a beginning on the work of establishing a theoretical background [72, 83]. In this case, the adsorption is a function not only of the relative, but also of the total, solute concentration, just as for vapor mixtures, where the relative and total pressure are both important.

ADSORPTION OPERATIONS

Adsorption is unique in the very diverse nature of its applications. For example, it is applied to a wide variety of processes such as recovery of vapors from dilute mixture with gases, solute recovery and removal of contaminants from solution, and the fractionation of gas and liquid mixtures. The techniques used include both stagewise and continuous-contacting methods, and these are applied to batch, continuous, and semicontinuous operations. Within each of these categories it is possible to recognize operations which are exactly analogous to those already discussed in previous chapters of this book. Thus, when only one component of a fluid mixture (either a gas or a liquid) is strongly adsorbed, the separation of the mixture is analogous for purposes of calculation to gas absorption, where the added insoluble phase is adsorbent in the present case and liquid solvent in the case of absorption. When both components of the fluid (either gas or liquid) are adsorbed strongly, the separation requires a fractionation procedure. The operation is then conveniently considered as being analogous to liquid extraction, where the added insoluble adsorbent corresponds to the use of solvent in extraction. By this means many simplifications in the treatment become possible.

STAGewise OPERATION

Liquid solutions, where the solute to be removed is adsorbed relatively strongly compared with the remainder of the solution, are treated in batch, semicontinuous, or continuous operations in a manner analogous to the mixer-settler operations of liquid extraction (*contact filtration*). Continuous countercurrent cascades can be simulated or actually realized by use of such techniques as fluidized beds.

Gases are treated for solute removal or for fractionation, usually with fluidized-bed techniques.

Contact Filtration of Liquids

Typical process applications include (1) the collection of valuable solutes from dilute solutions e.g., the adsorption onto carbon of iodine from brines after liberation of the element from its salts by oxidation and the collection of insulin from dilute solutions and (2) the removal of undesirable contaminants from a solution.

The extremely favorable equilibrium distribution of solute toward the adsorbent which is frequently possible makes adsorption a powerful tool for the latter purpose, and most industrial applications of stagewise techniques fall into this category. Adsorption of colored substances from aqueous sugar solutions

onto carbon, in order to provide a pure product and to assist the crystallization, is a typical example. Similarly, carbon is sometimes used to adsorb odorous substances from potable water, and grease is adsorbed from dry-cleaning liquids. The colors of petroleum and vegetable oils are lightened by treatment with clay.

Equipment and methods As pointed out in Chap. 1, each stage requires the intimate contact of two insoluble phases for a time sufficient for a reasonable approach to equilibrium, followed by physical separation of the phases. The equipment used in applying these principles to adsorption is varied, depending upon the process application. That shown in Fig. 11.15 is very typical of many installations operated in a batchwise fashion. The liquid to be processed and the adsorbent are intimately mixed in the treating tank at the desired temperature for the required period of time, following which the thin slurry is filtered to separate the solid adsorbent and accompanying adsorbate from the liquid. Air agitation, as with sparged vessels (Chap. 6) is also used, particularly in ion exchange. The equipment is readily adaptable to multistage operation by providing additional tanks and filters as necessary. If the operation is to be made continuous, which is sometimes done in the decolorizing of petroleum lubricating oils, for example, centrifuges or a continuous rotary filter can be substituted for the filter press, or the solid can be allowed to settle out by virtue of its higher density when the mixture is passed through a large tank.

The type of adsorbent used depends upon the solution to be treated. Aqueous solutions are frequently treated with activated carbon especially prepared for the purpose at hand, whereas organic liquids such as oils are usually treated with inorganic adsorbents such as clays. Occasionally mixed adsorbents

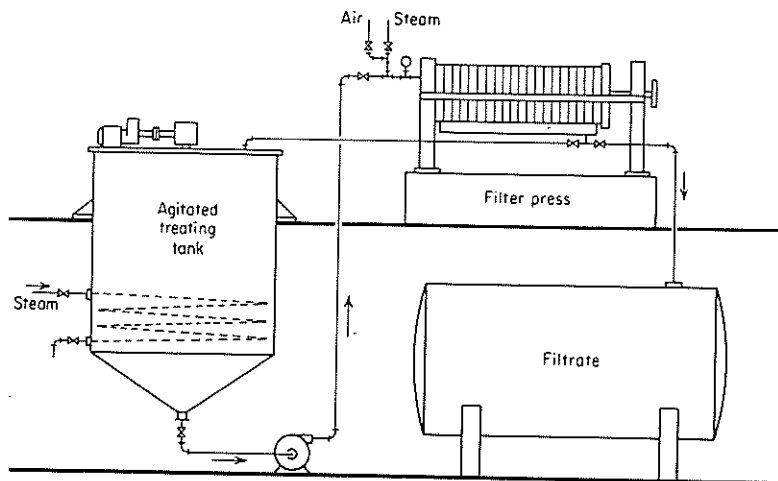


Figure 11.15 Contact filtration. Schematic arrangement for single-stage batch treatment of liquids.